



AbbasPour **Student Center**
High Performance Building Design

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Winter 2020

Contents

Preliminary
Evaluation and
design, Early
Investigations

Concept Design

Introduction

Predesign Studies

Benchmarking & Goal-Setting

Programming, Massing and Orientation

Form Finding and Design

Façade Design

Definition	5
Benefits	5
Case Studies	5
Location	6
Public Transportation	6
Site Dimensions and Adjacent buildings	6
Views From the Site	6
Location and Accessibility	7
Temperature and Comfort	8
Wind Analysis	8
Radiation Analysis	9
Photovoltaic Production Potential	9
Rainwater Harvesting Potential	9
Solar Envelope	9
Benchmarking	10
Goals and Targets	10
Site Orientation Study	11
Massing Alternatives	12
Orientation	12
Form Alternatives	14
Form Selection Process	15
Form Alternatives Analysis	16
Form Alternatives Analysis Results	19
Selected Forms From Step 1	20
Form Analysis(Step 2)	21
Form Analysis(Step 2)	22
Ventilation Analysis	23
Thermal Comfort Analysis	24
Final Results	25
Plans	26
Facade Design Process	30
Facade Alternatives	32
Daylight Analysis	34
Glare Analysis	36
Facade Alternatives Analysis Results(Step1)	38
Facade Design Alternatives (Step2)	39
Daylight Analysis	40
Glare Potential (Eye Level)	41
Horizontal 60° Cone of Vision Analysis (Eye Level)	42
Natural Ventilation Strategy Selection Process	43
Natural Ventilation Strategies results	44
Concept Stage Final Performance Results	47

49	Constructions Alternatives
50	Constructions Selection (Step1)
51	Constructions Selection (Step2)
52	HVAC Selection
53	HVAC Selection Results
53	HVAC System Sizing
53	Lighting System
54	HVAC System Diagram And Specifications
55	Solar DHW system design
56	Construction and Systems Final Performance Results
57	Photovoltaic System design
58	Final System and Results
59	Step2 - Orientation Improvement
60	Water Management
60	Water Management Case Studies
61	Outdoor Water Management
62	Landcape Layout
63	Indoor Water Management
65	Facade Heat Transfer And Details(Thermal Bridges)
67	Natural Ventilation Analysis (CFD)
69	Natural Ventilation Analysis Results
70	Lighting Design
72	Circadian lighting Analysis
73	Annual Glare Analysis
74	Site Acoustical Assessment
75	Labelling Acoustic Zones
77	Acoustic Requirements
78	Sample Spaces
79	Noise control
79	Air-borne sound insulation
79	β-Impact noise Control
82	Sound Transmission Class Calculation for Multiple Partitions
83	Improvement options details
84	Room Acoustics
84	reververation Time Calculations
86	Acoustic tiles details for Ceilings
87	Acoustic tiles details for walls
88	Conference Hall reflector panels design
89	Sound Masking
91	Building Type Determination
92	Allowed exit Distances
93	Occupancy Loads - Exits
97	Exits
101	Travel distances
105	Site Accessibility
106	Required Fire Ratings-Walls
107	Required Fire Ratings-Doors
108	Construction details Drawings
112	
117	
118	
128	

Construction Selection
Design Development
and analysis

HVAC and Lighting system Selection and
design

Renewable System Design

Water Management

Detailed Design and Analysis

Acoustic Design

Fire Protection

Design Process and Final Performance Data
Project evaluation according to LEED
Final Architecture Documents
References and tools

Final Performance
Results, Assessments
and Documentations

01 | Preliminary Evaluation and design, Early Investigations

Definition

Definition

Cost-Effectiveness

Safety and security

Sustainability

Accessibility

Functionality

Productivity

Historic Preservation

Aesthetics

Benefits

Green Building

>

Green University

GSA data-driven study examined 200 a three-year period:

%23



Energy Use

%28



Water Use

%23



Operation Expenses

%9



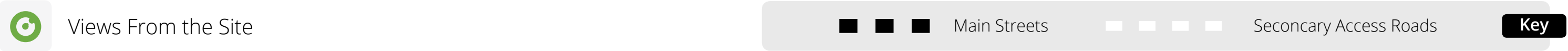
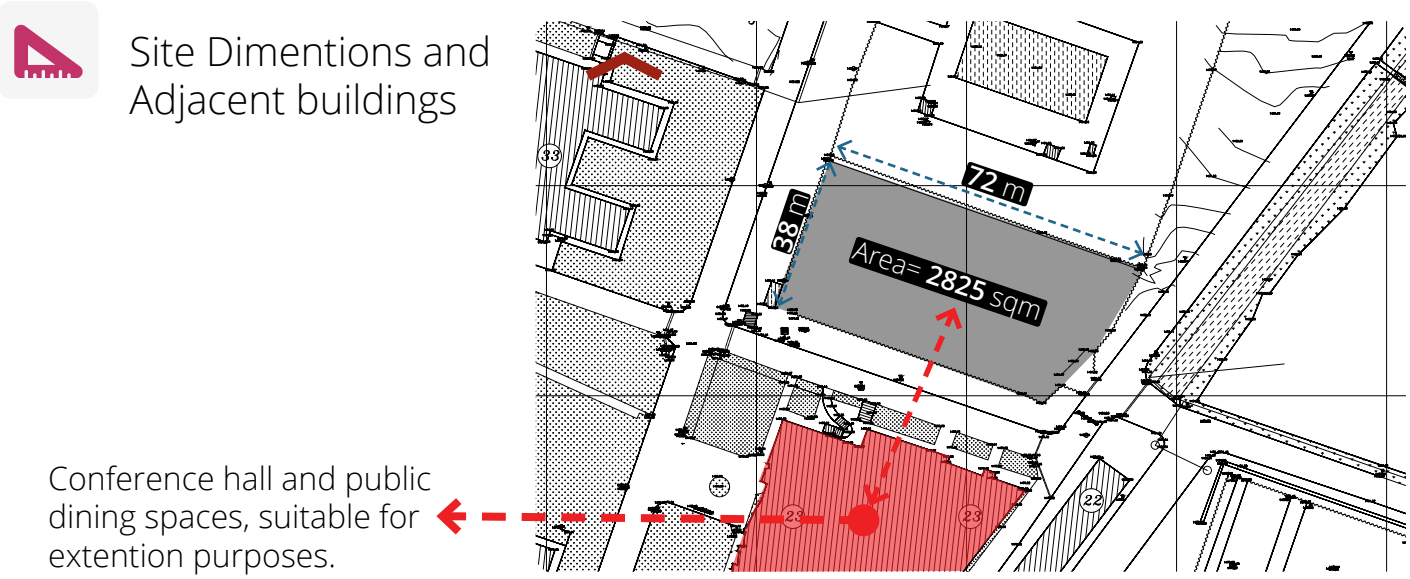
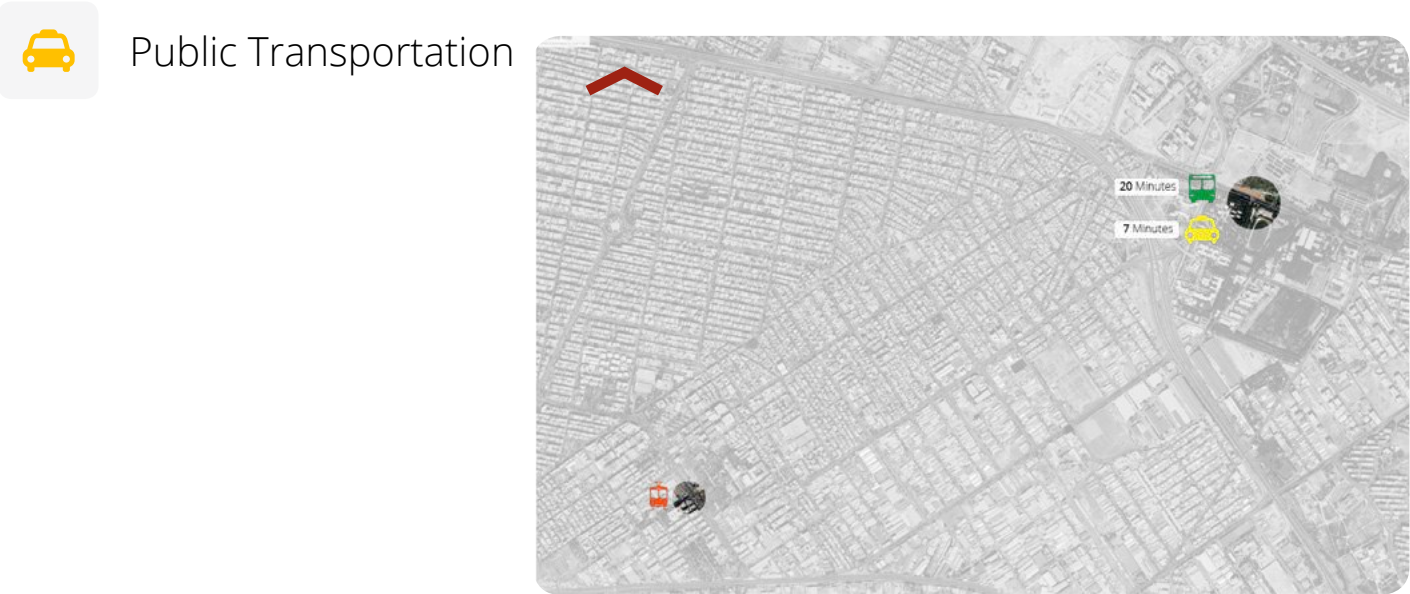
Historic Preservation

%2



Tenant Satisfaction

Case Studies					
		Name/Location	Climate (Koppen)	Highlights	Pictures
1		Green Center for Student Success at Southern New Hampshire University/ New Hampshire	Humid Continental Climate(Dbf)	• LEED Silver certification. • replacing the entire envelope with high-performance metal panels, curtain wall and storefronts systems, skylights and insulated and reflective roof, improved thermal performance and minimized energy costs. • New HVAC units improved performance drastically while minimizing energy costs and eliminating harmful refrigerants. • Low flow fixtures and water efficient landscaping. • sustainable finishes and wood products • Low VOC adhesives and paints • specified materials that minimized waste of natural resources and maximized comfort.	
		Emory University Student Center/ Atlanta	Humid Subtropical Climate(Cfa)	• LEED Platinum Certification. • Geothermal system • Solar-powered water systems • Exterior shading through a sunshade canopy that unifies the Student Center complex • Energy-efficient mechanical heating/cooling systems • Window shade sensors • Occupancy sensors • Chilled beams in the majority of student life areas • Ceiling fans set to automatically increase thermal comfort during hot months • Daylight harvesting, with electric lighting automatically adapting to interior light levels • Improved wall insulation to minimize heat gain/loss	
		New Mexico HightLands University/ Las Vegas	Subtropical Hot Desert (BWh)	• LEED Gold Certified • A large three-storey atrium • The -100seat student and governance chamber council that floats above this central space; to enhance its presence, natural light washes the curved surface of the chamber drum. • Motorized sun tracking louver systems • The ground floor has interior blinds rather than the louver system • Green roofs • Geothermal well field	

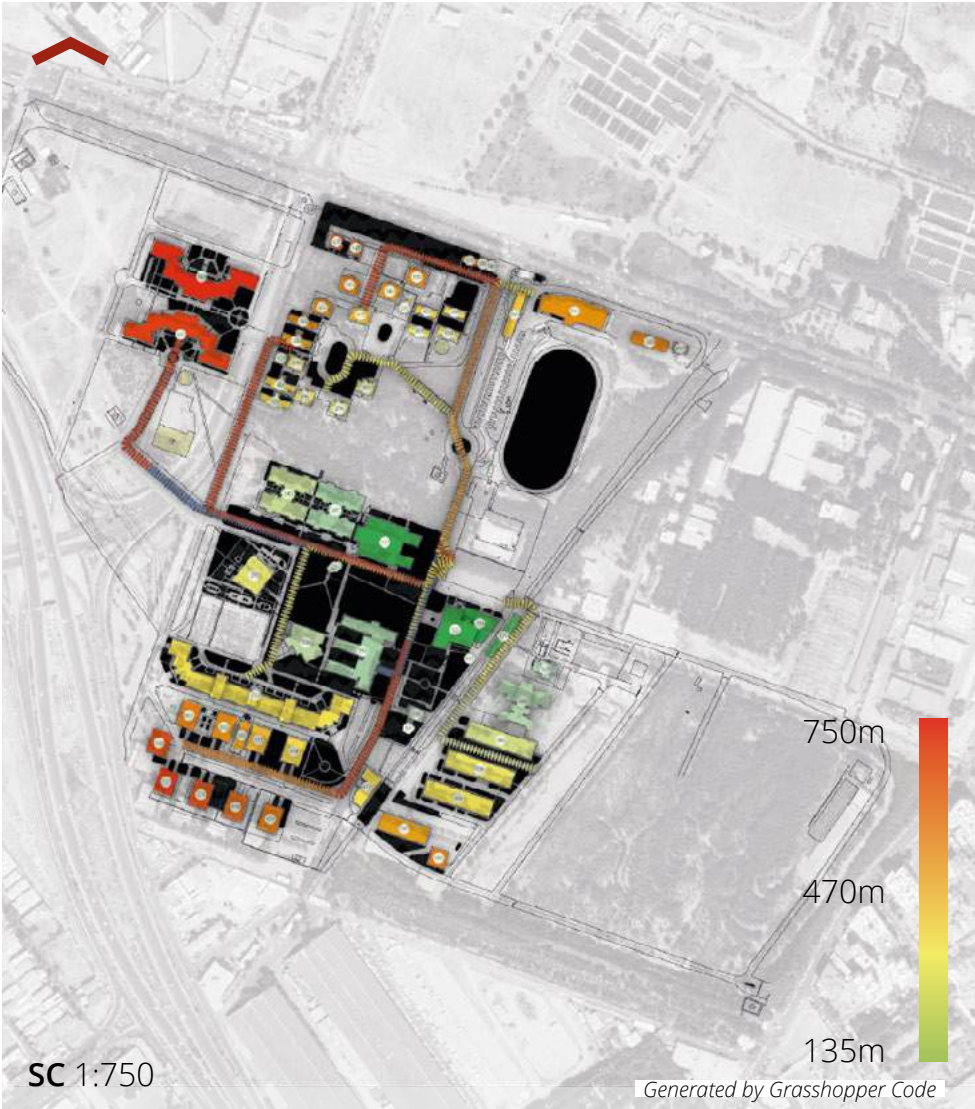
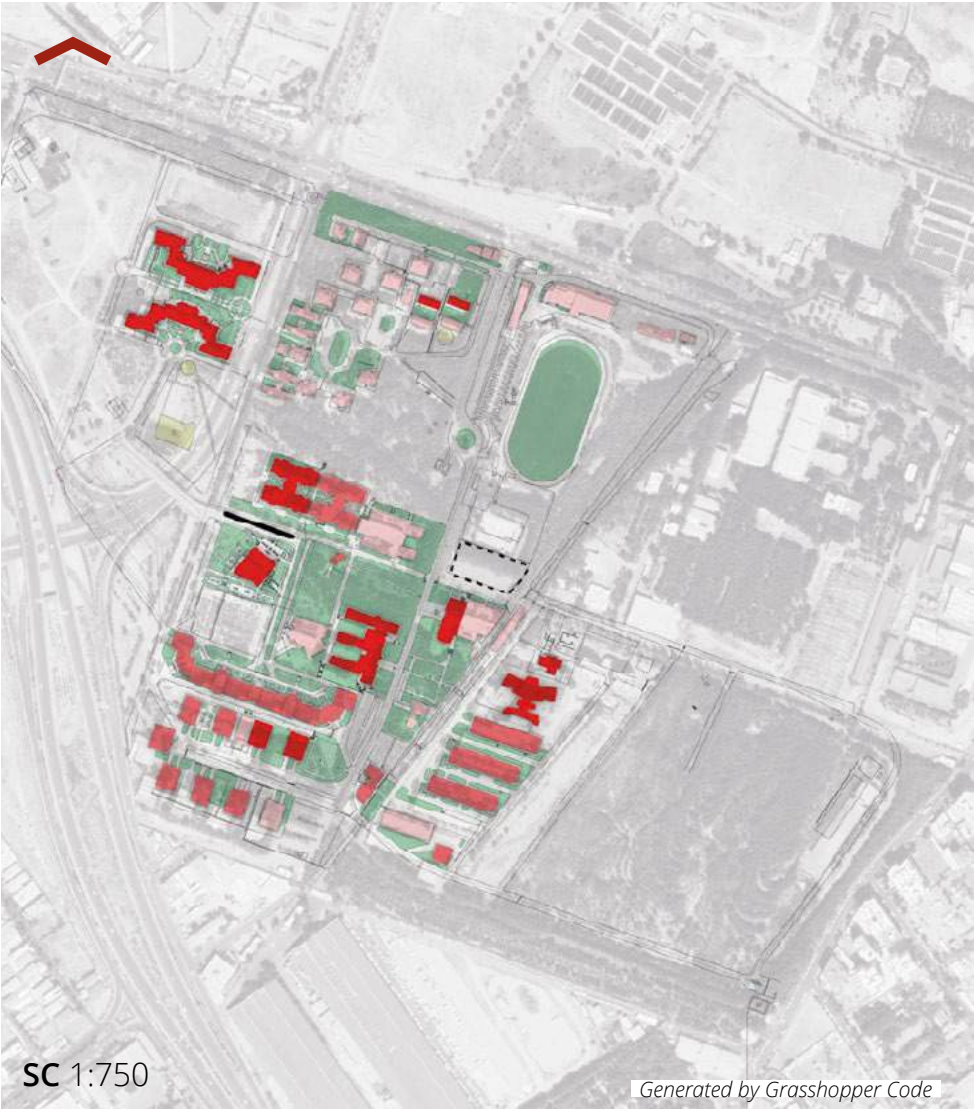
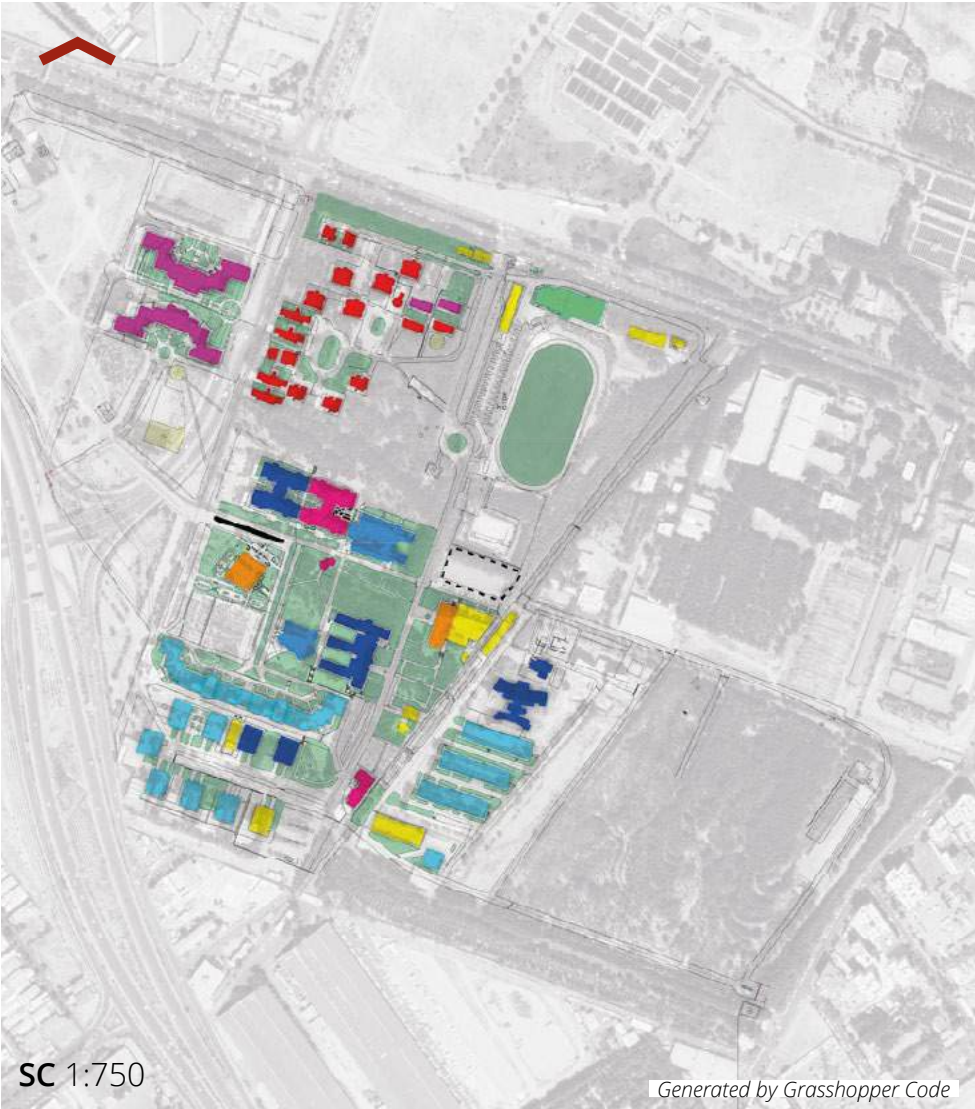


Location and Accessibility

Buildings Use

Frequency of Use

University Buildings
Accessibility to the Site



- | | |
|--|--|
|  Lecture Classrooms |  Equipements and Services |
|  Workshops |  Cultural |
|  Classrooms-Staff |  Dormitory |
|  Administration |  Temporary Residence |
|  Sport Hall |  Green Space |

 High  Medium  Low

		
Average Distance	470m	5.5 minutes
Minimum Distance	135m	1.5 Minutes
Maximum Distance	750m	9 Minutes

Temperature and Comfort

Tehran in Köppen
Climate Classification

Cold and Semi-arid

Bsk

HDD Cooling Degree-days and Heating
CDD Degree-days in Occupancy days

HDD
1779

CDD
459

Thermal Stress (hrs)

Heat
Stress
1021

Neutral
Stress
2683

Cold
Stress
1171

Common Strategies and TC

Measures list

No Measures

% of Time
Comfortable

12.1

Active
Evaporative Cooling
Occupant Use of fans

35.23

23.37

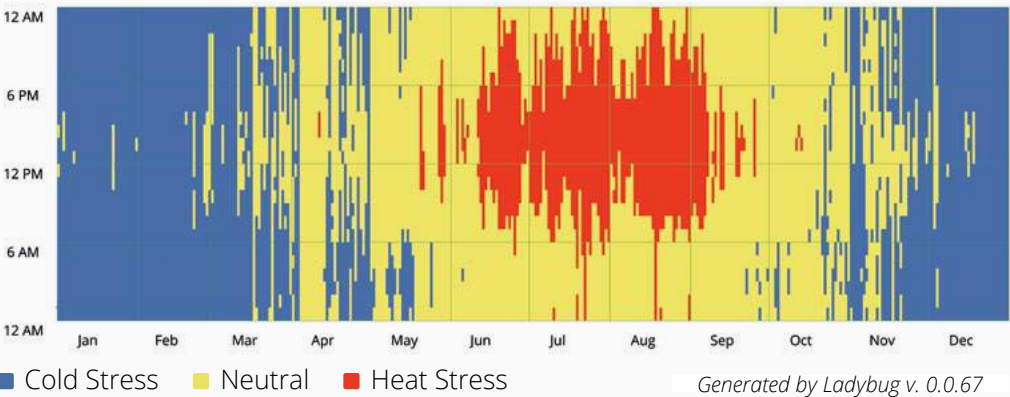
Passive
Thermal Mass + Night Ventilation
Capture Internal Heat Gain
Passive Solar Heating

18.14

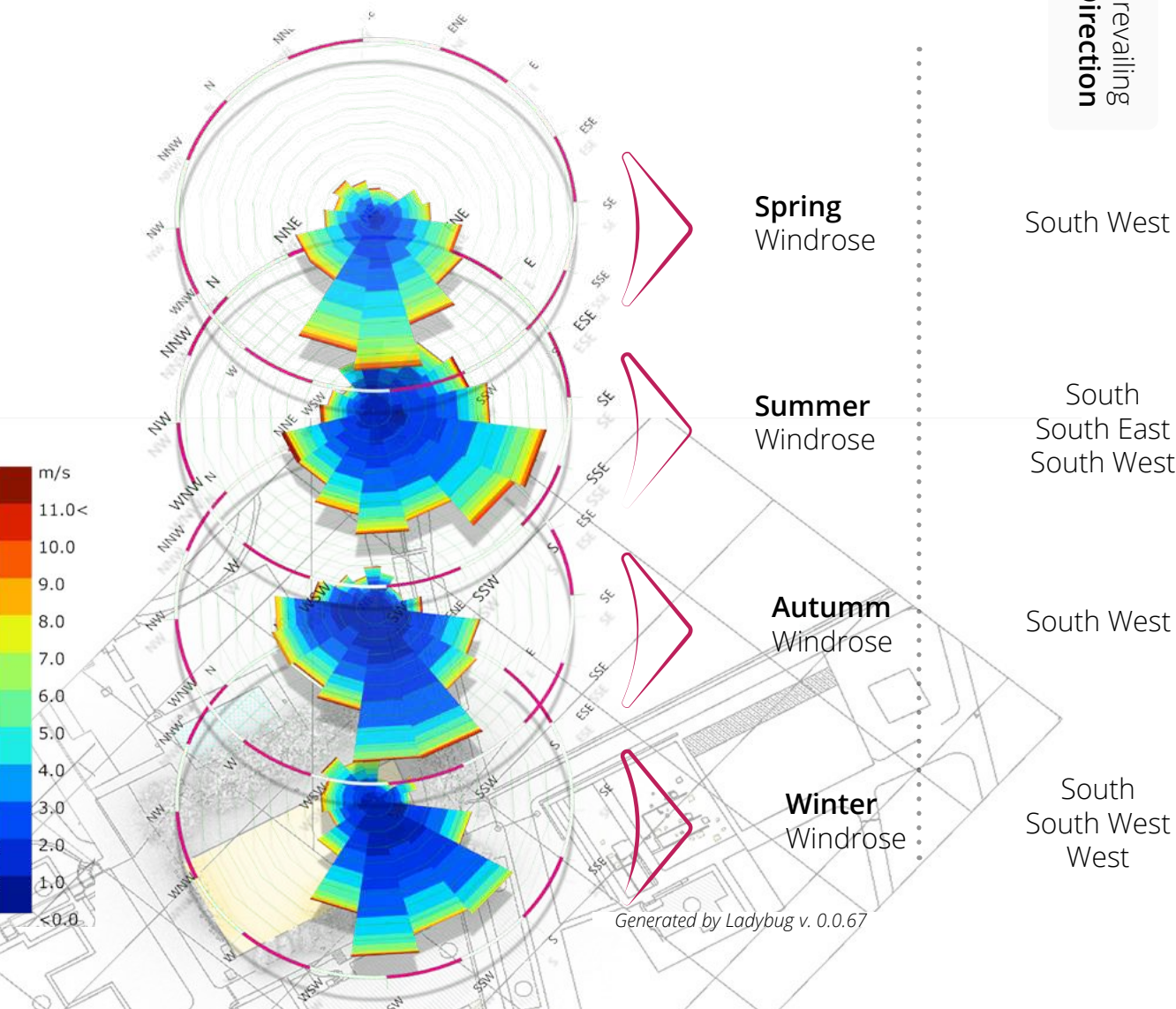
37.2

39.2

Annual Thermal Stress



Wind Analysis



Prevailing
Direction

Average
Velocity (m/s)

Frequency

Spring
Windrose

South West

3.5

12

Summer
Windrose

South
South East
South West

2.88

10

Autumm
Windrose

South West

2.11

12

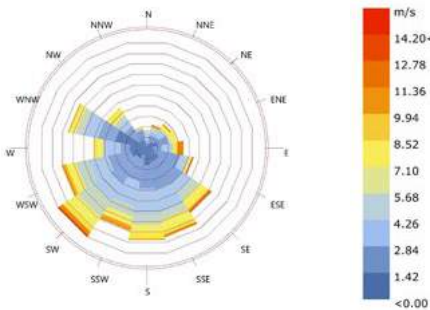
Winter
Windrose

South
South West
West

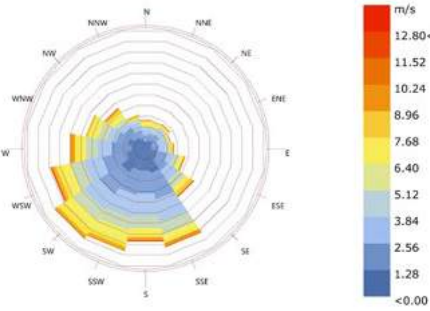
2.03

12

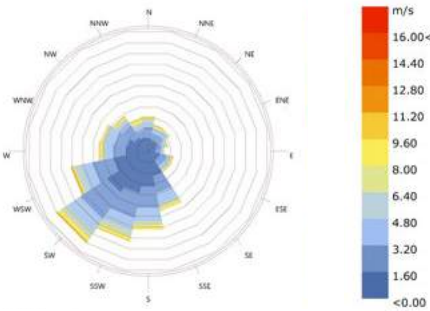
Windrose in
Hot Hours
(DBT <28 °C)



Windrose in
Cold Hours
(DBT >20 °C)



Windrose in
Desirable
Temperatures
(20< DBT <28 °C)

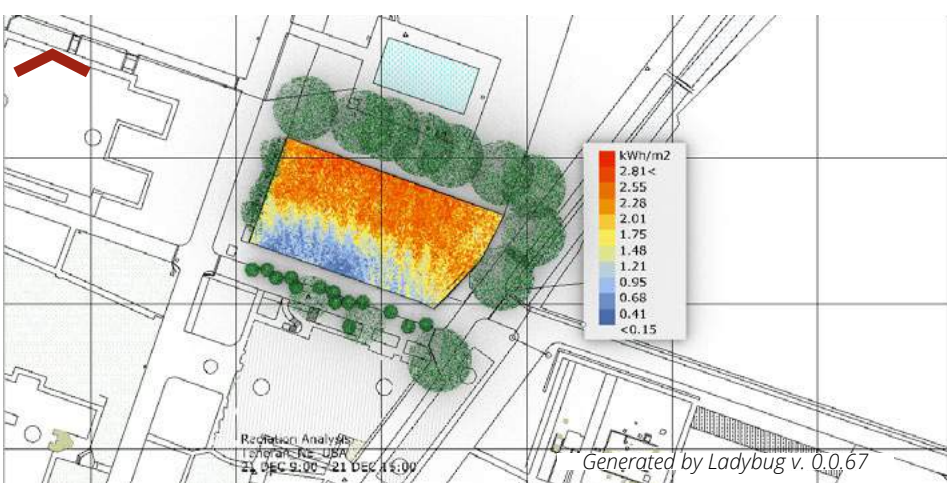


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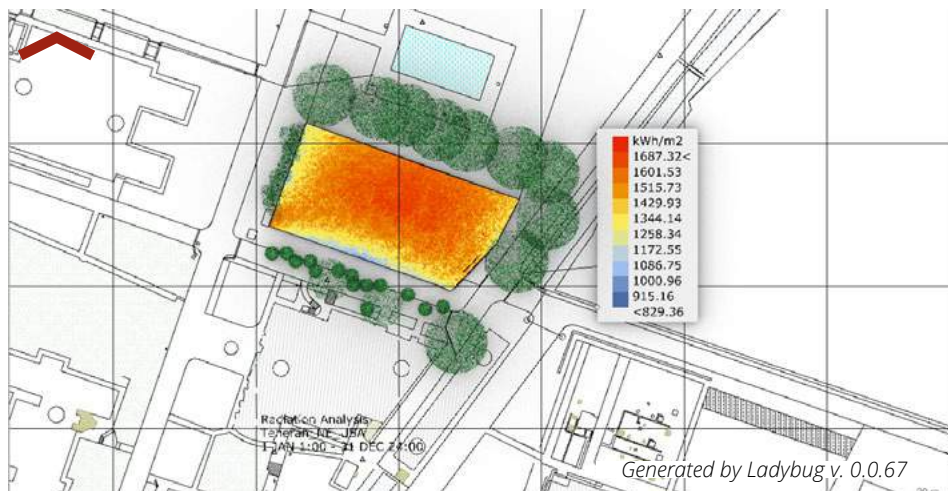
Windroses in Occupancy months



Radiation Analysis



■ Radiation Analysis, 21 DEC. , 9:00-15:00

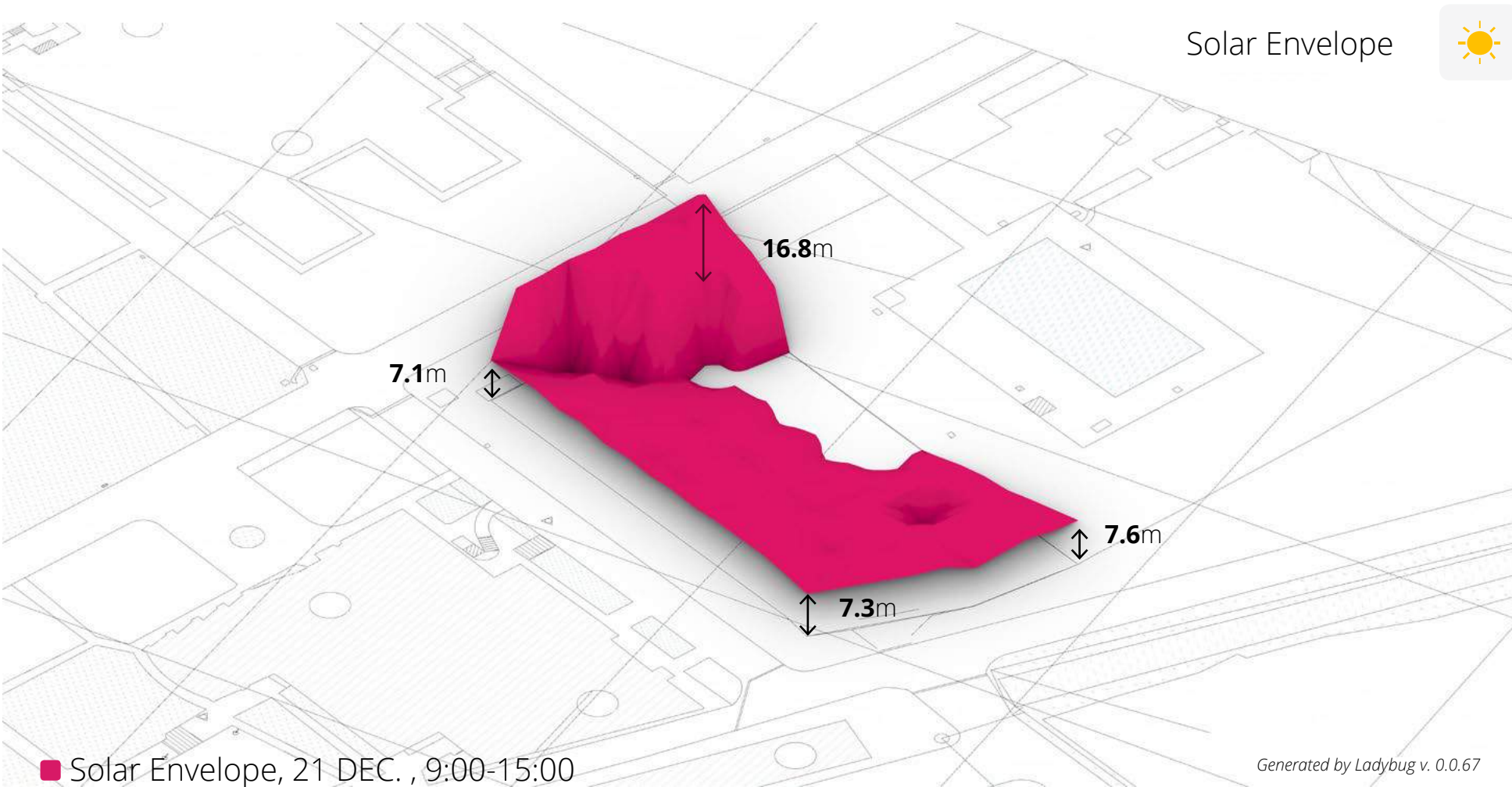


■ Radiation Analysis, Annual

Photovoltaic Production Potential

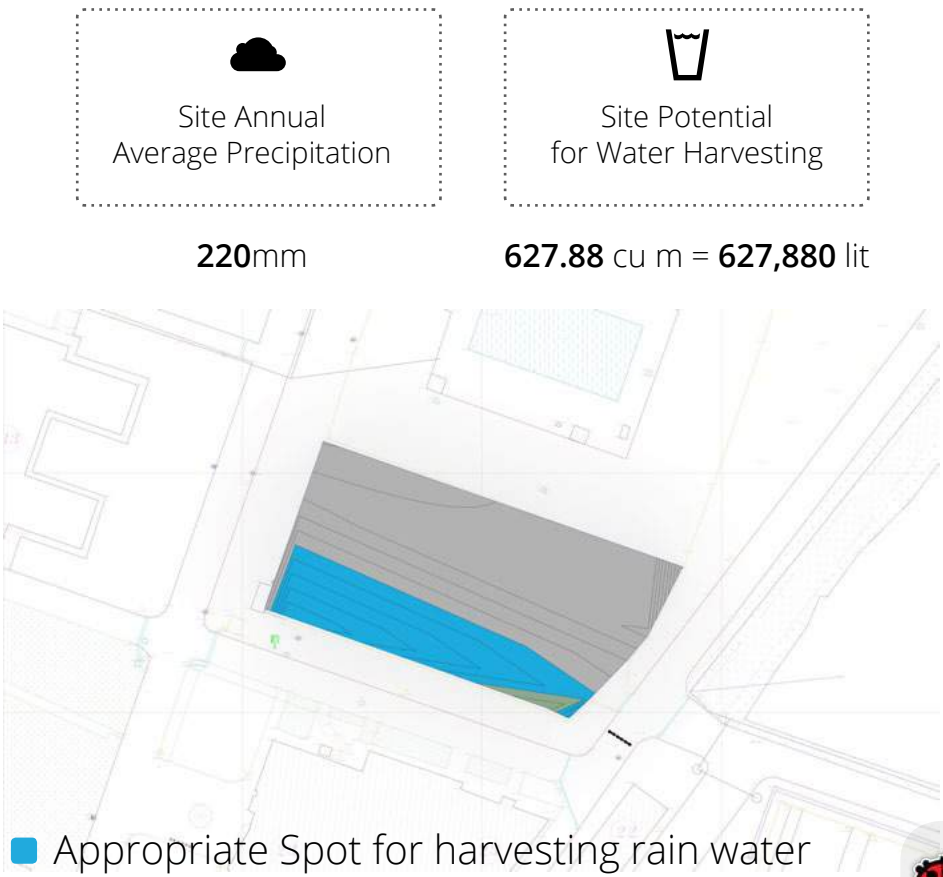
 Area (s.q.m.)	 Annual Energy(kWh)	 % of Baseline	 Co2 Emmission Reduction	 Annual Cost Saving (IRR)
50	2908.5	% 0.5		9,412,000
100	5817	% 1		18,824,000
150	8725.5	% 1.5		28,236,000
200	11634	% 2		37,648,000
250	14542.5	% 2.5		47,060,000
300	17451	% 3		56,472,000
350	20359.5	% 3.5		65,884,000
400	23268	% 4		75,296,000

Generated by Ladybug v. 0.0.67



■ Solar Envelope, 21 DEC. , 9:00-15:00

Rainwater Harvesting Potential



■ Appropriate Spot for harvesting rain water

**Benchmarking**

**Energy**

Gas Consumption

39.56 kWh/sqm

Electricity Consumption

161 kWh/sqm

Total Site Energy Consumption

200 kWh/sqm

Primary Energy Consumption

622 kWh/sqm

**Carbon**

Operational Carbon Emmiton

270,400 kg

=

58.7 X 

**Water**

Gas Consumption

753,275.75 (liters/year)

**Goals and Targets**

**Main Goals**

**Energy & atmosphere**

• %50-32 Energy Use Reduction from Baseline (Baseline: 130kWh/sqm)

• %3-1Renewable energy production

**Indoor Environmental Quality(IEQ)**

• Ventilation Requierements

• Tobacco Smoke Control

• Enhanced IAQ Strategies

• Low-emitting materials

• Thermal comfort

• Interior lighting

• Daylighting

**Water Efficiency**

• Reduce Water Consumption

• Harvest Rain Water

• Water recycling

**Acoustics**

• Sound Mapping

• Maximum Noise Levels

• Sound Barriers

• Sound Absorption

• Sound Masking

• β - Impact Noise Management

USGBC LEED

WELL

**Secondary Goals**

**Mind**

**Thermal Comfort**

**Water**

**Light**

**Movement**

**Materials**

**Air**

**Resiliency**

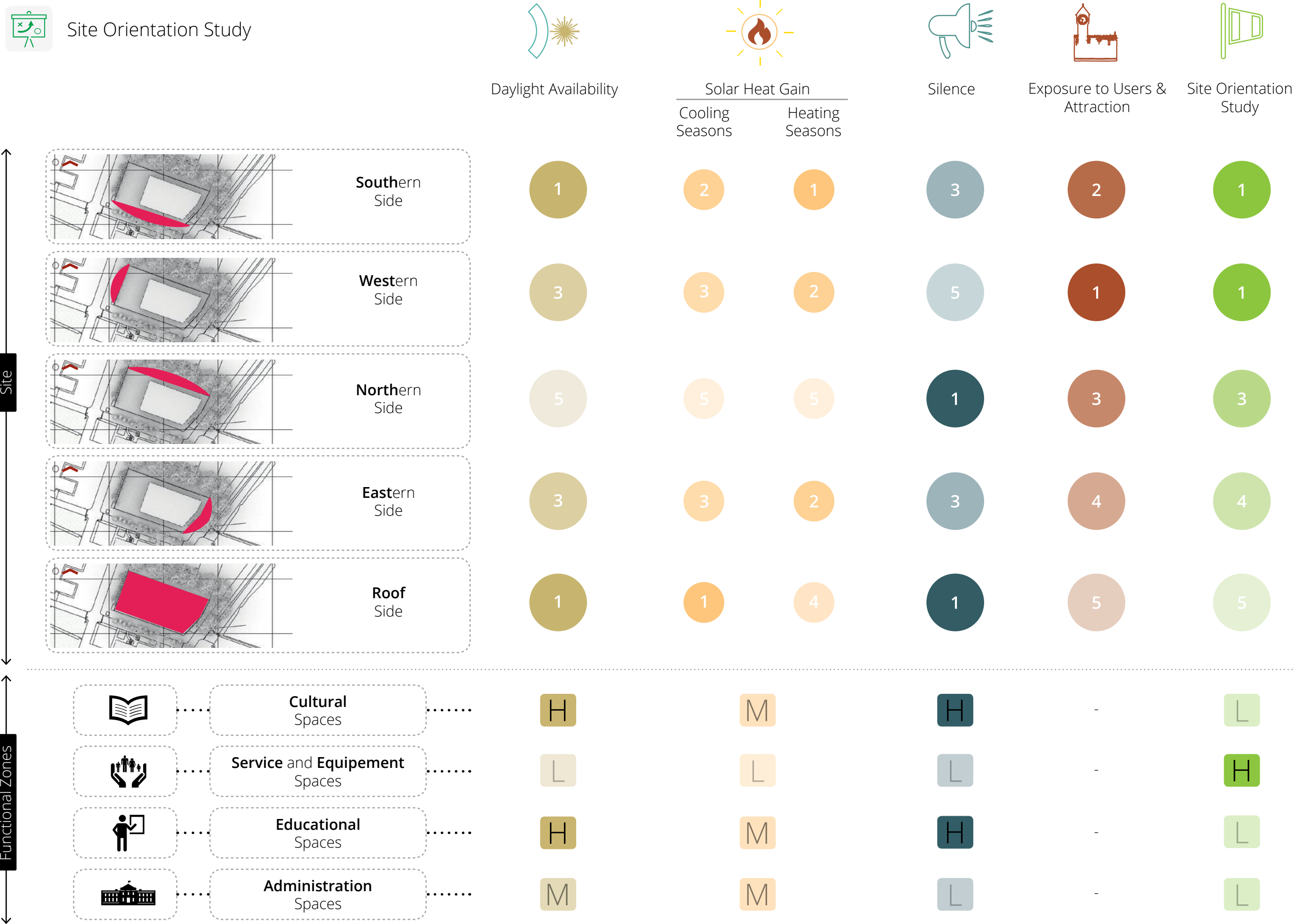
**Community**

**Innovation**

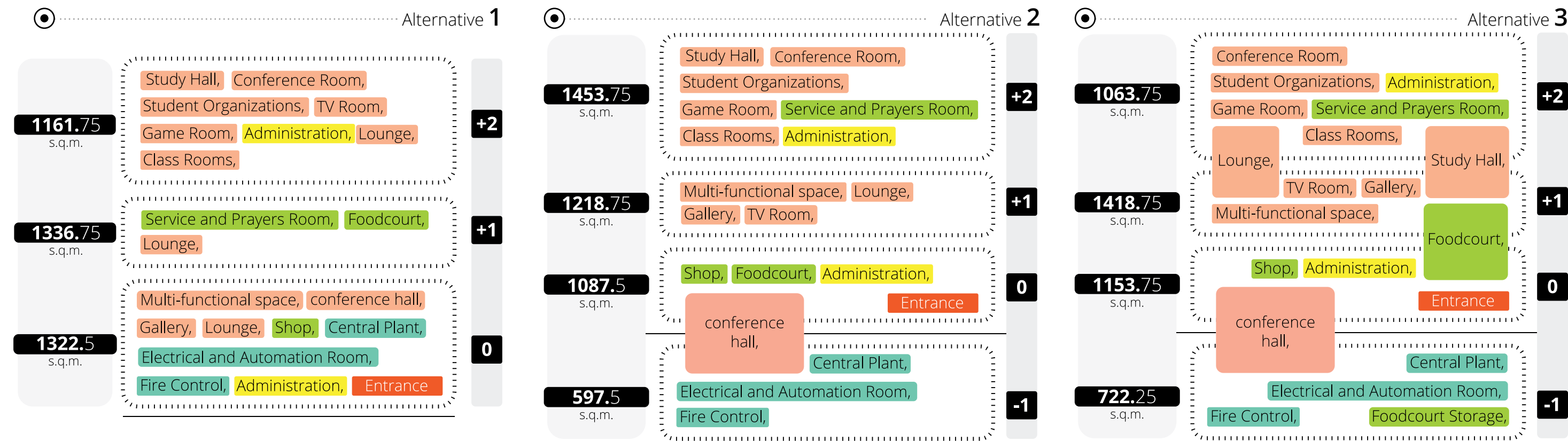
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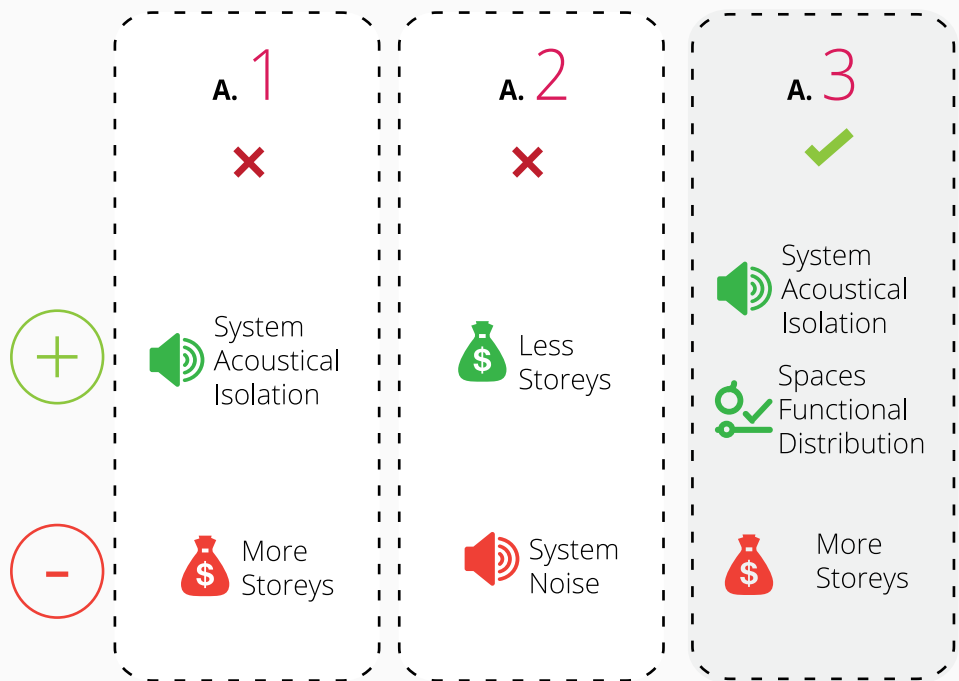
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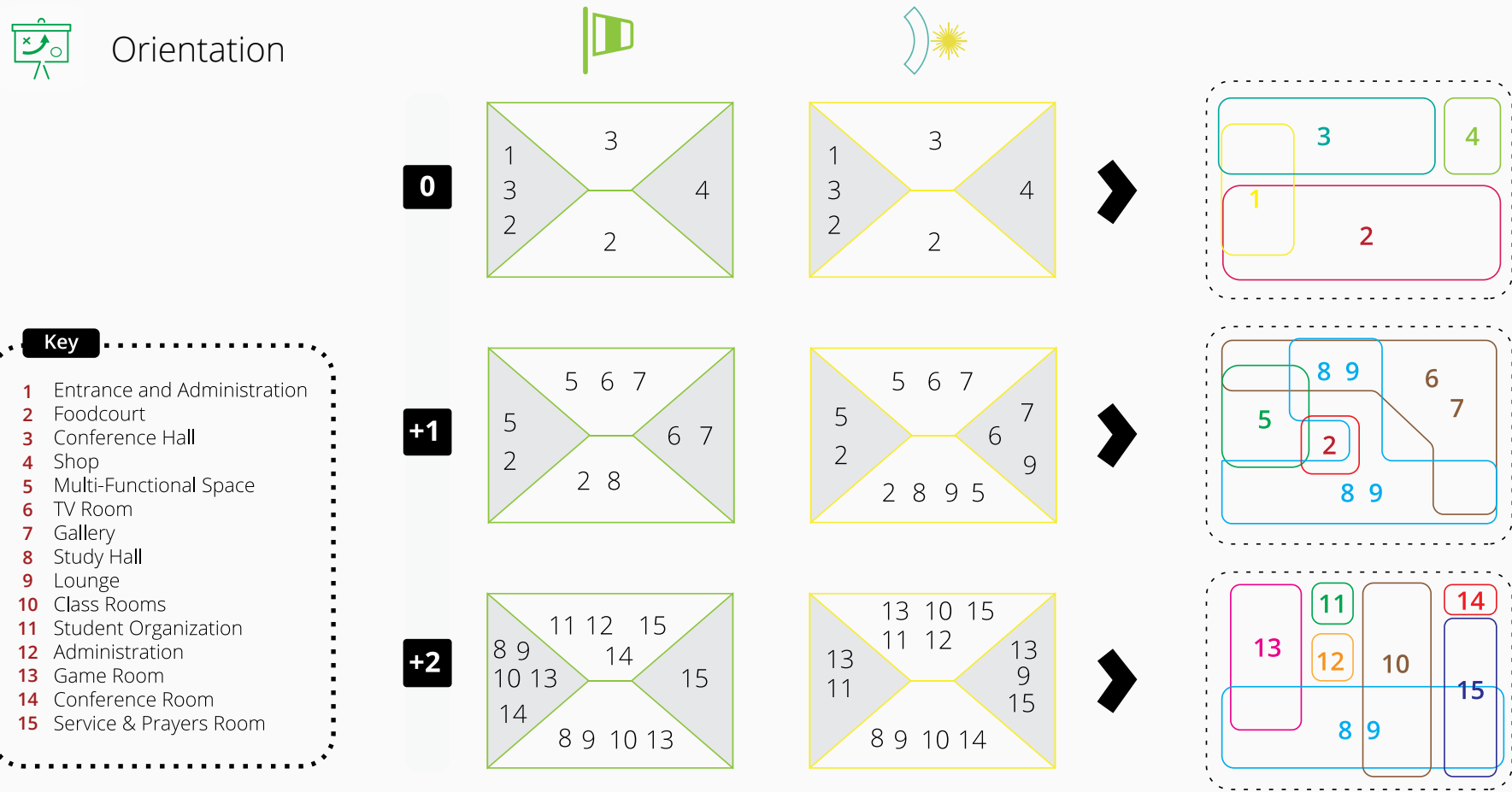
Massing and Orientation



Massing Alternatives

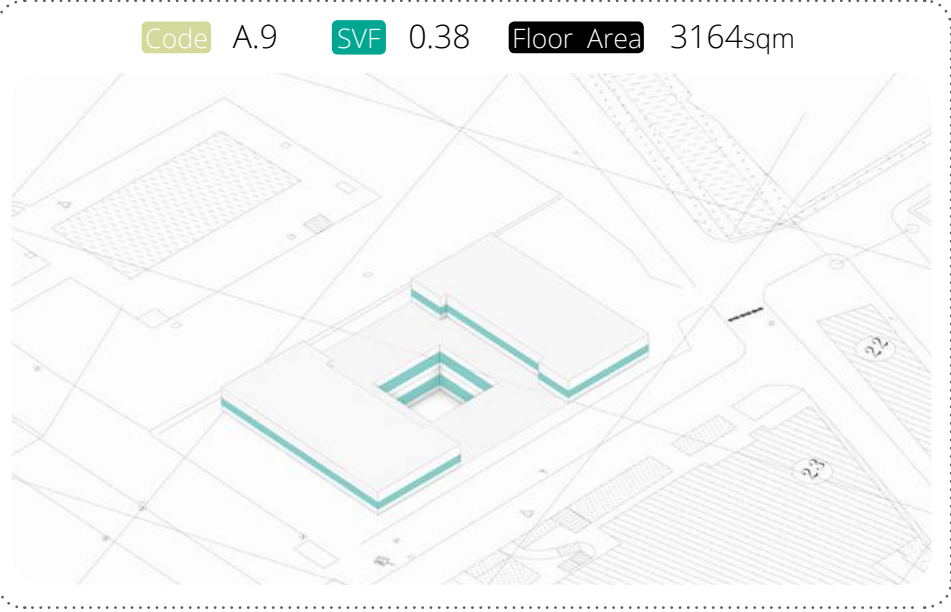
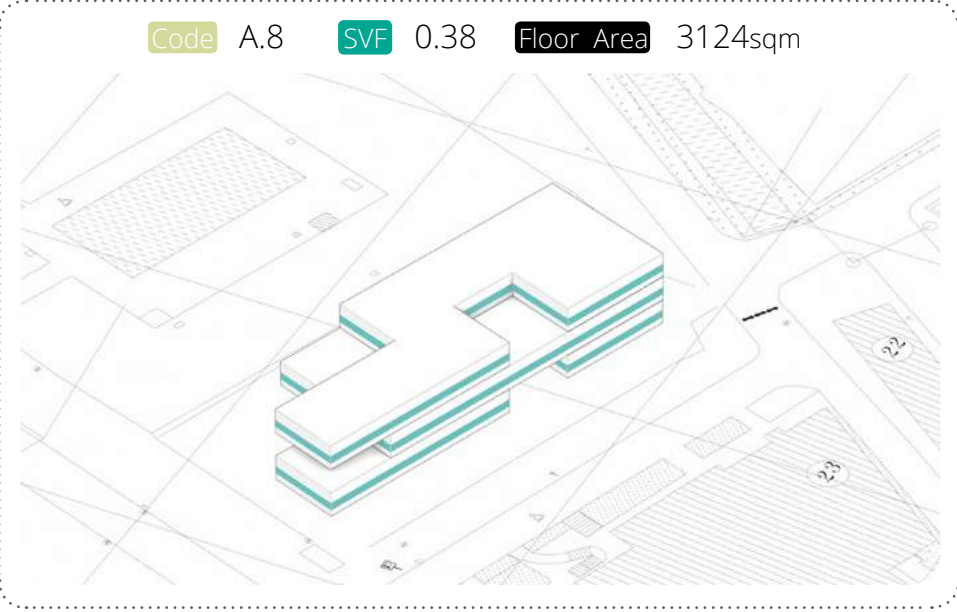
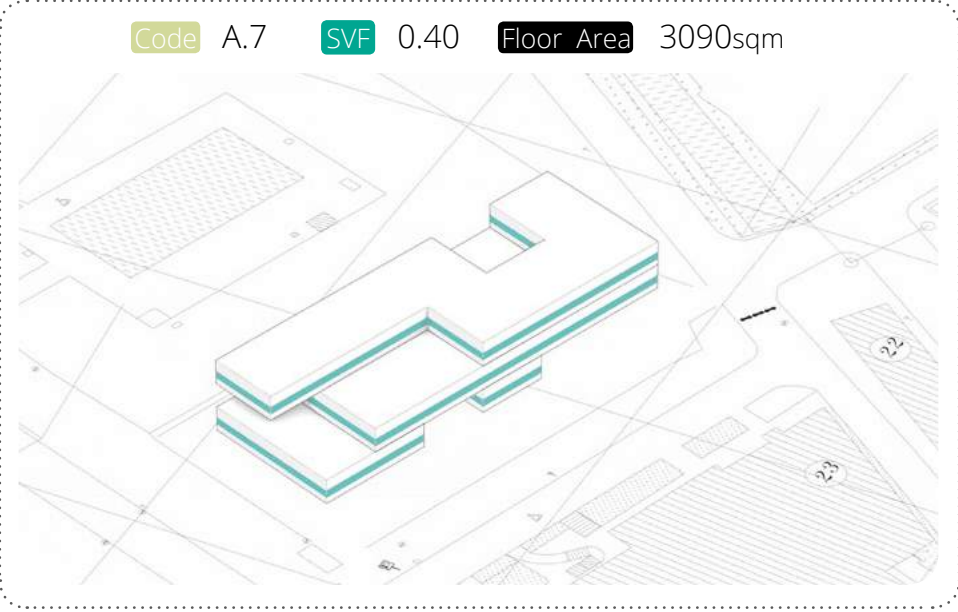
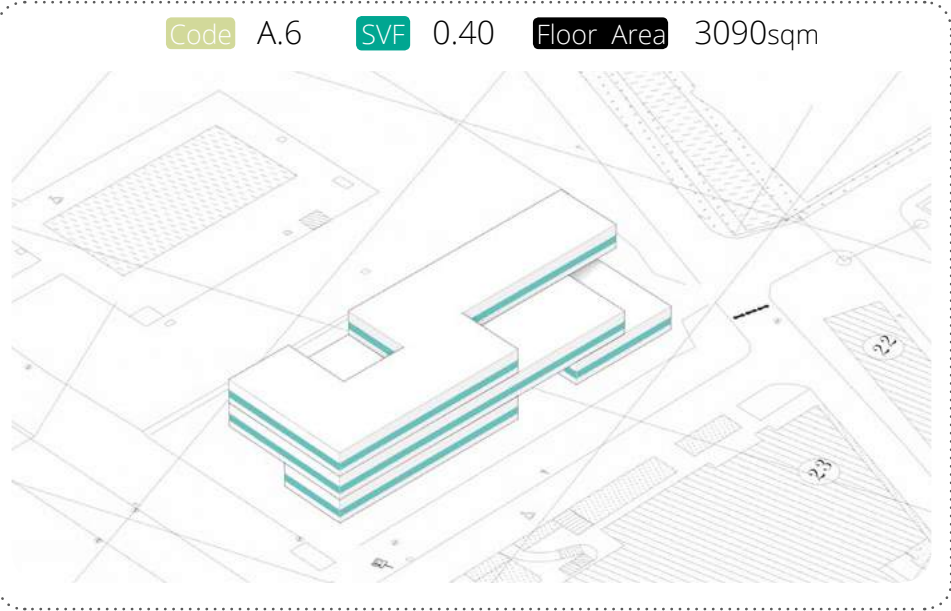
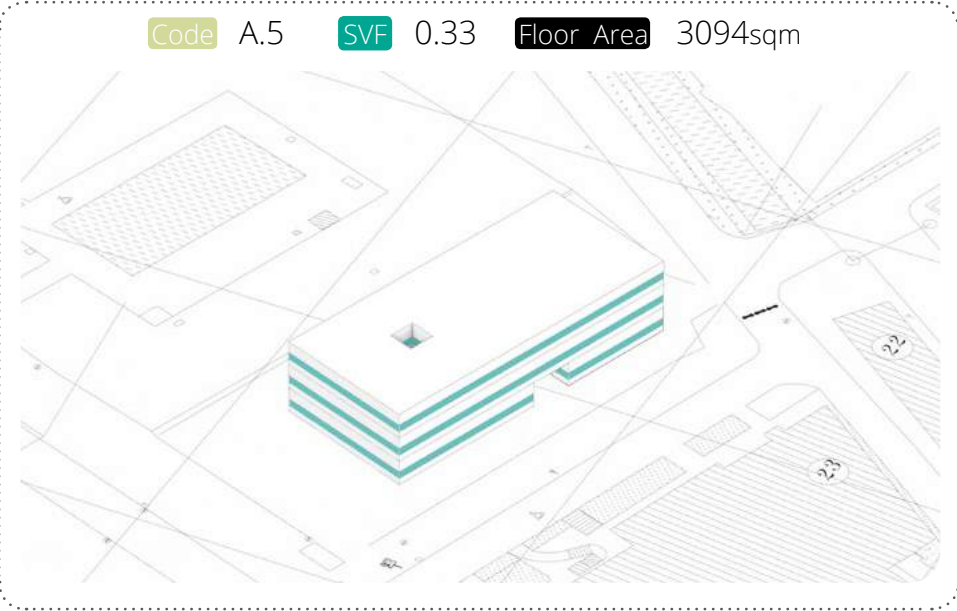
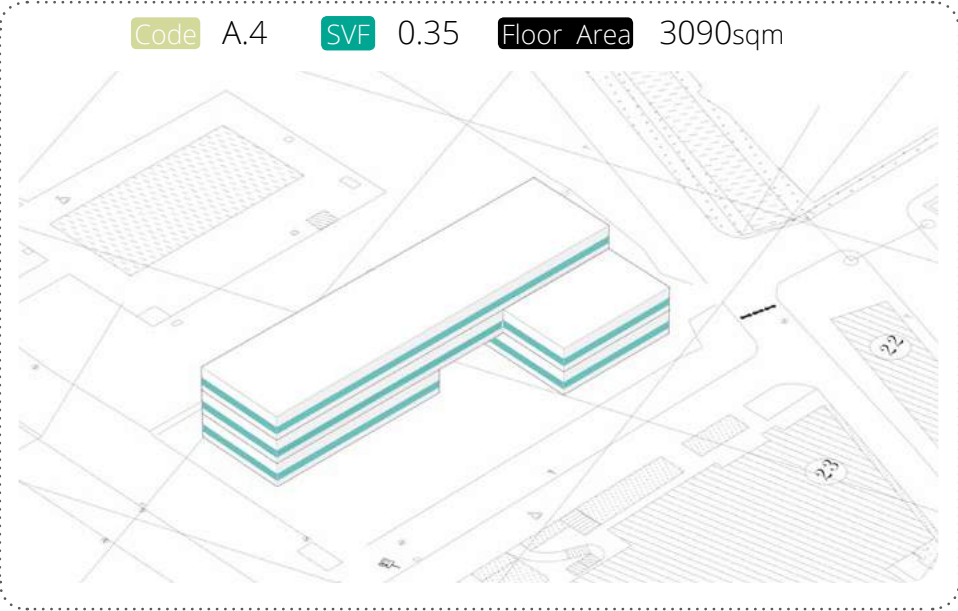
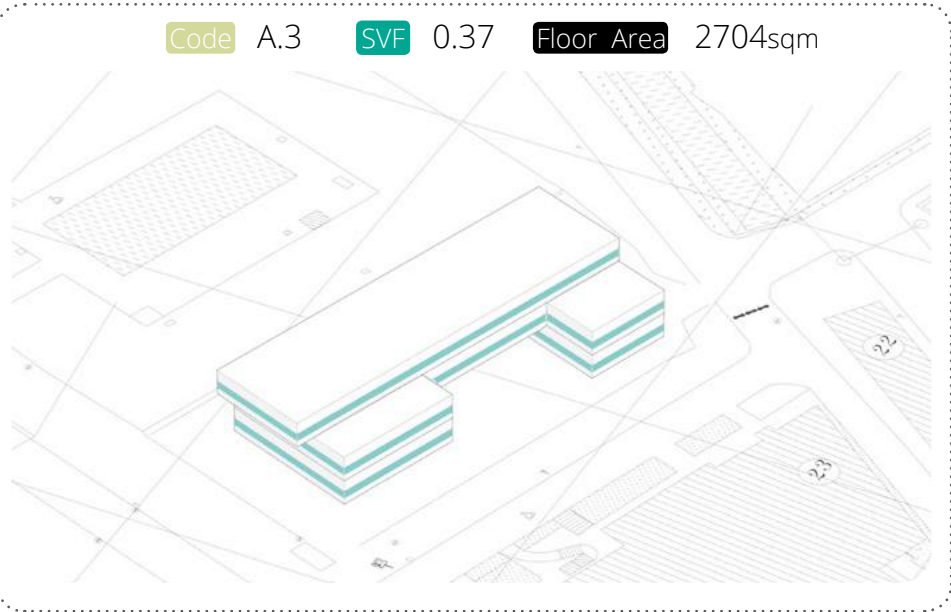
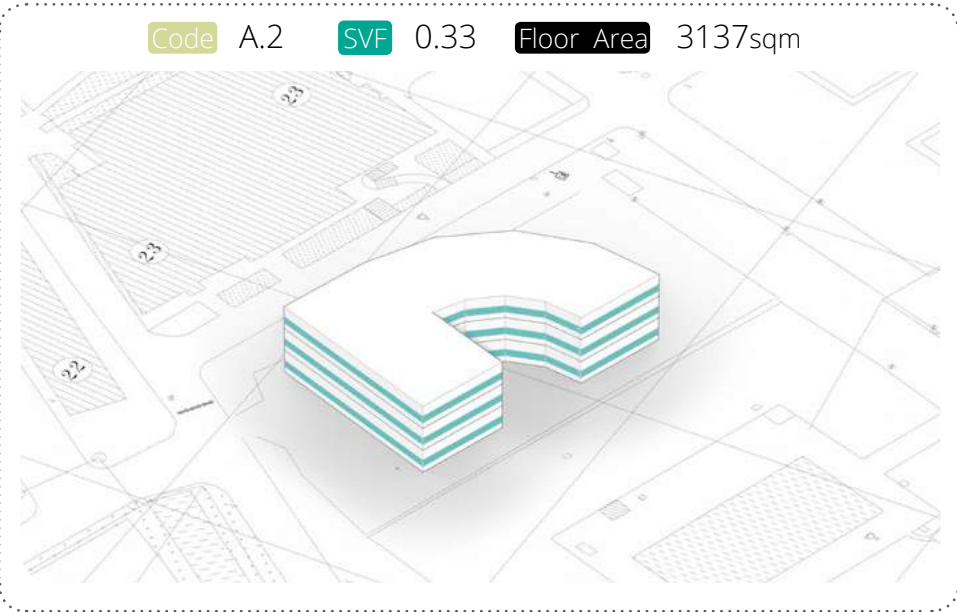
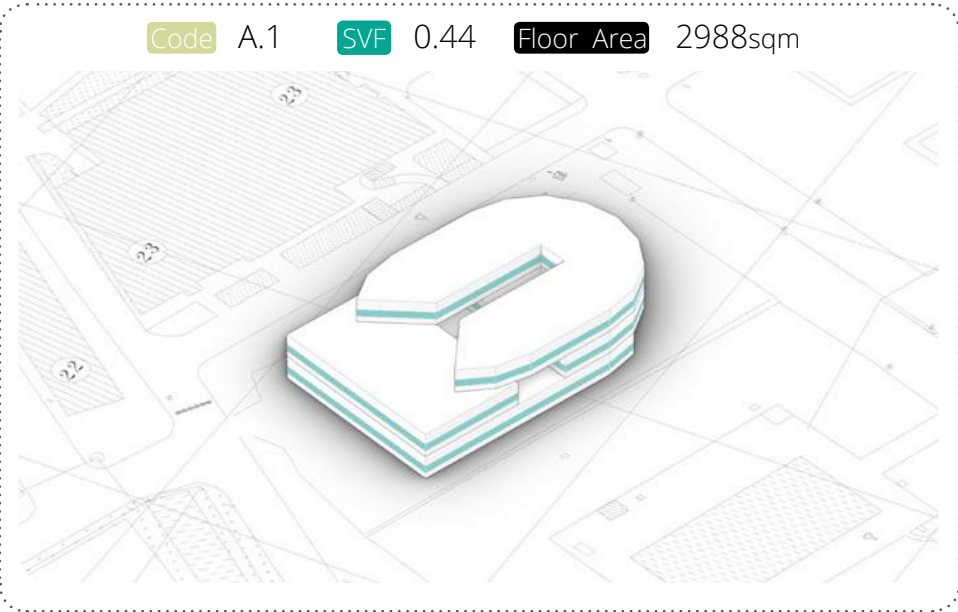


Orientation



02 | **Concept** Design

Form Alternatives



Modelling Key Assumptions

Ventilation

- Scheduled Ventilation Off
- Natural Ventilation On
- Setpoint 18
- Min Outdoor Temperature 0
- Max Outdoor Temperature 30
- Max Rel. Humidity 90
- Infiltration On
- Infiltration ACH 0.1
- Constant Term Coefficient 1
- Temperature Term Coefficient 0
- Velocity Term Coefficient 0
- Velocity Squared Term Coefficient 0

Constructions

- Window Conductivity [W/m.K] 0.9
- Window Visible Transmittance 0.913
- Window SHGF 0.905
- Wall U-value [W/m-2k] 9.2(Default)
- Floor U-value [W/m-2k] 9.2(Default)
- Ceiling U-value [W/m-2k] 9.2(Default)
- Roof U-value[W/m-2k] 9.2(Default)

Loads

- WWR[%] 30
- People On
- Schedule 8 to 6

Conditioning

- density [p/m2] 0.2
- Heating On
- Constant Setpoint 20
- Availability Schedule AllOn
- Heating Limit[W/m2] 100
- Flow Limit[W/m2] 100
- Cooling On
- Constant Setpoint 26
- Availability Schedule AllOn
- Cooling Limit[W/m2] 100
- Flow Limit[W/m2] 100
- Humidity Control Off

Analysis Grid

- Mechanical Ventilation Off
- Spacing[m] 1.5
- Height[m] 0.76
- Offset from walls[m] 0.6

Surfaces

- Ceiling reflectance[%] 70
- Wall reflectance[%] 70
- Window Transmittance[%] 80
- Floor reflectance[%] 30
- Ground reflectance[%] 20



Form Selection Process

A. 1

A. 2

A. 3

A. 4

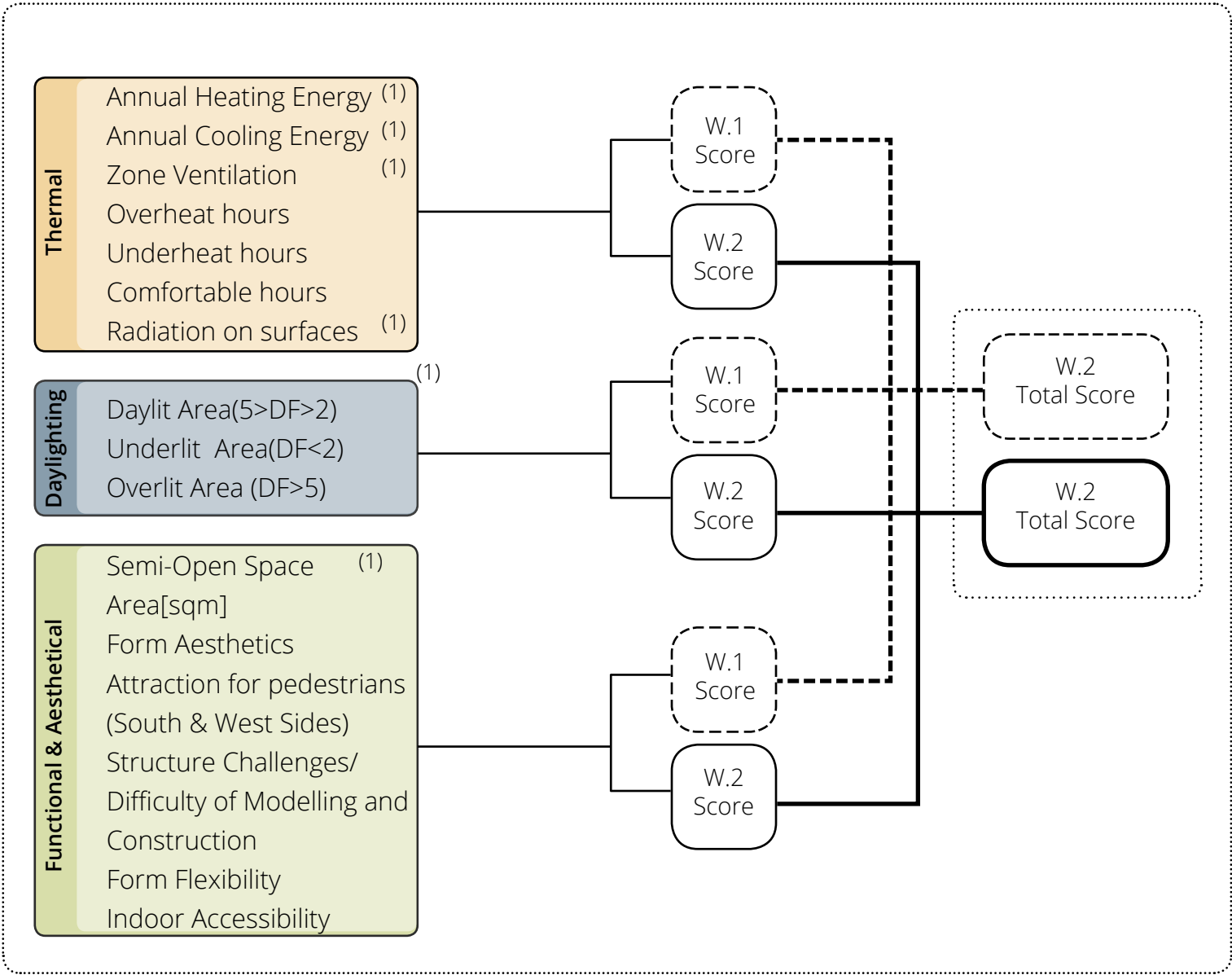
A. 5

A. 6

A. 7

A. 8

A. 9



A. 6

A. 7

A. 8

(1) Score Calculation Equations

$$Score_{Heating/Cooling\ Energy} = \frac{E - E_{min}}{E_{max} - E_{min}} \times 100$$

$$Score_{Natural\ Ventilation} = \frac{V - V_{min}}{V_{max} - V_{min}} \times 100$$

$$Score_{Daylight} = \frac{1}{5} \times (Overlit\ \% \ of\ times) + \frac{4}{5} \times (Daylit\ \% \ of\ times)$$

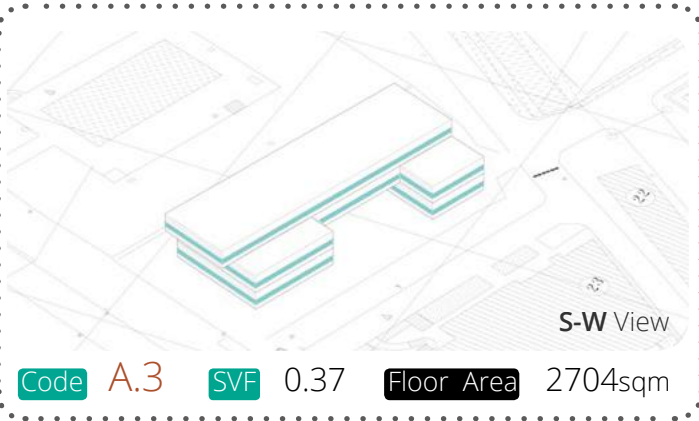
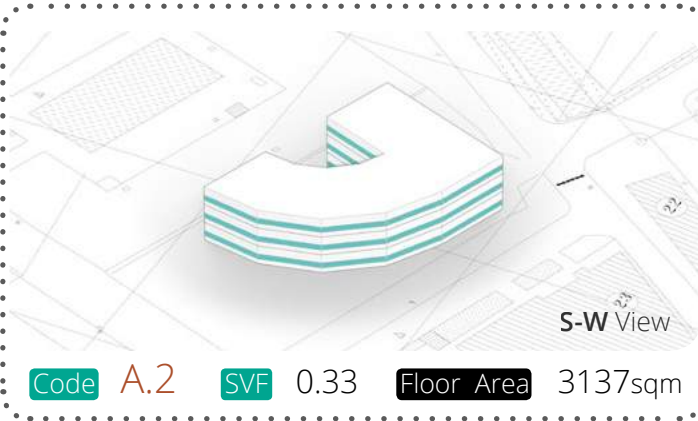
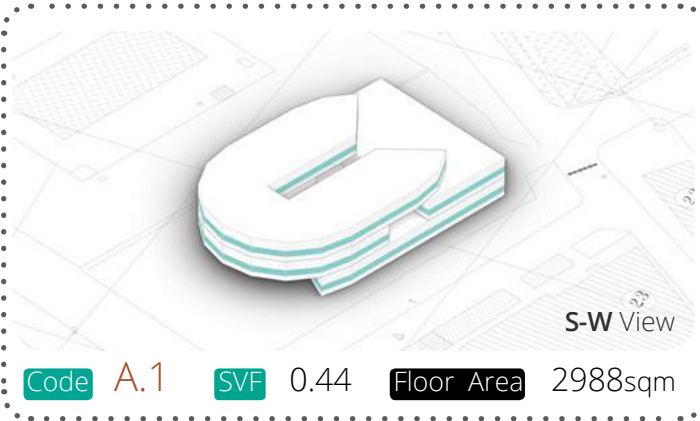
$$Score_{Radiation} = \frac{R - R_{min}}{R_{max} - R_{min}} \times 100$$

$$Score_{Open\ and\ Semi-open\ Areas} = \frac{A - A_{min}}{A_{max} - A_{min}} \times 100$$



Form Alternatives Analysis

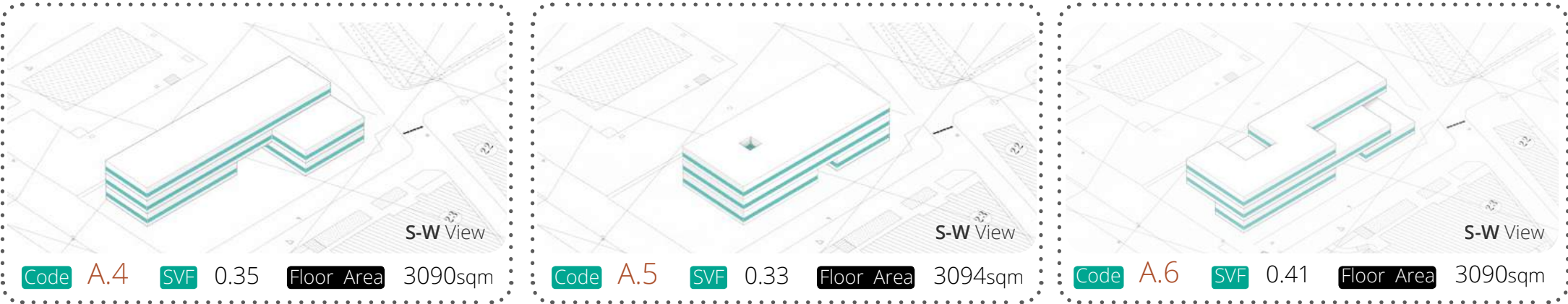
	Weighting I	Weighting II
Thermal	X 1	X 3
Daylighting	X 1	X 2
Functional	X 1	X 4



Key Parameters		Detailed Weightings	Values/ Raw Scores	Scores	Overall Scores	Values/ Raw Scores	Scores	Overall Scores	Values/ Raw Scores	Scores	Overall Scores
Thermal	Annual Heating Energy [Jouls/sqm]	20/100	4.87E+ 9	0	I 28.85 II 28.85	3.53E +9	28	I 35.10 II 35.10	4.83E+09	1	I 25.52 II 25.52
	Annual Cooling Energy [Jouls/sqm]	20/100	4.57E+ 7	0		3.71E+07	19		4.32E+07	6	
	Zone Ventilation [ACH]	15/100	7.54	100		6.69	61		7.22	85	
	Overheat hours [% of times]		2.12			1.82			1.56		
	Underheat hours [% of times]		73.02			68.98			73.90		
	Comfortable hours [% of times]	40/100	24.86	24.9		29.19	29.2		24.53	24.5	
	Radiation on surfaces [W/sqm](BIPV Potential)	5/100	334512.09	78		382492.72	100		238364.67	33	
Daylighting	Daylit Area(5>DF>2)	80/100	19.51	22	I 21.55 II 14.44	17.22	19	I 19.49 II 13.06	20.16	21	I 21.04 II 14.10
	Underlit Area(DF<2)		51.19			54.47			32.27		
	Overlit Area (DF>5)	20/100	27.64			26.31			23.66		
Functional & Aesthetical	Semi-Open Space Area[sqm]	20/100	645	56	I 37.71 II 32.25	60	0	I 25.00 II 66.04	616	53	I 49.65 II 63.4
	Form Aesthetics	10/100	40	40		40	40		60	60	
	Attraction for pedestrians (South & West Sides)	20/100	25	25		25	25		50	50	
	Structure Challenges/ Difficulty of Modelling and Construction	10/100	25	25		60	60		30	30	
			25	25		25	25				
	Form Flexibility	20/100							50	50	
	Indoor Accessibility	20/100	50	50		75	75		50	50	
Total Scores:				I 88.1 II 76.5	Total Scores:		I 79.6 II 114.2	Total Scores:		I 96.2 II 102.7	

Form Alternatives Analysis

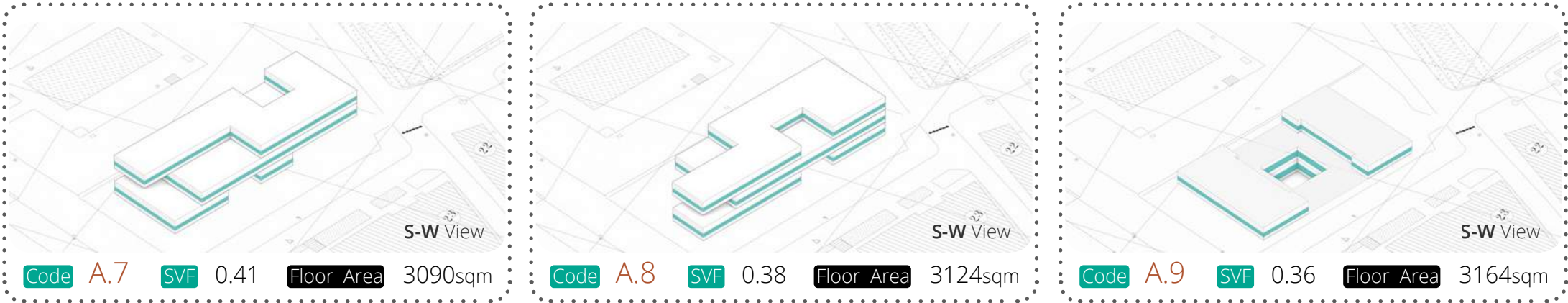
	Weighting I	Weighting II
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Functional	X 1	X 4



Key Parameters		Detailed Weightings	Values/ Raw Scores	Scores	Overall Scores	Values/ Raw Scores	Scores	Overall Scores	Values/ Raw Scores	Scores	Overall Scores		
Thermal	Annual Heating Energy [Jouls/sqm]	20/100	4.18E+09	14	I 27.53 II 27.53	4.24E+09	13	I 29.45 II 29.45	4.56E+09	6	I 27.78 II 27.78		
	Annual Cooling Energy [Jouls/sqm]	20/100	4.08E+07	11		3.97E+07	13		4.32E+07	5			
	Zone Ventilation [ACH]	15/100	6.934869	72		7.21	85		6.96	73			
	Overheat hours [% of times]		2.14			2.05			2.02				
	Underheat hours [% of times]		71.55			71.88			73.14				
	Comfortable hours [% of times]	40/100	26.30	26.3		26.07	26.1		24.83	24.8			
	Radiation on surfaces [W/sqm](BIPV Potential)	5/100	220460.08	24		211583.3	20		358619.62	89			
	Daylighting	Daylit Area(5>DF>2)	80/100	23.57		24	I 24.39 II 16.34		5.12	4		I 3.85 II 2.58	27.55
Underlit Area(DF<2)			48.23		93.93			16.97					
Overlit Area (DF>5)		20/100	26.82		0			35.20					
Functional & Aesthetical	Semi-Open Space Area[sqm]	20/100	420	34	I 47.40 II 55.34	144	8	I 41.61 II 90.61	302	81	I 68.13 II 102.41		
	Form Aesthetics	10/100	60	60		50	50		80	80			
	Attraction for pedestrians (South & West Sides)	20/100	50	50		25	25		75	75			
	Structure Challenges/ Difficulty of Modelling and Construction	10/100	45	45		100	100		80	80			
	Form Flexibility	20/100	50	50		25	25		80	80			
	Indoor Accessibility	20/100	50	50		75	75		25	25			
Total Scores:				I 99.3	II 99.2	Total Scores:		I 74.9	II 122.6	Total Scores:		I 125.4	II 149.9

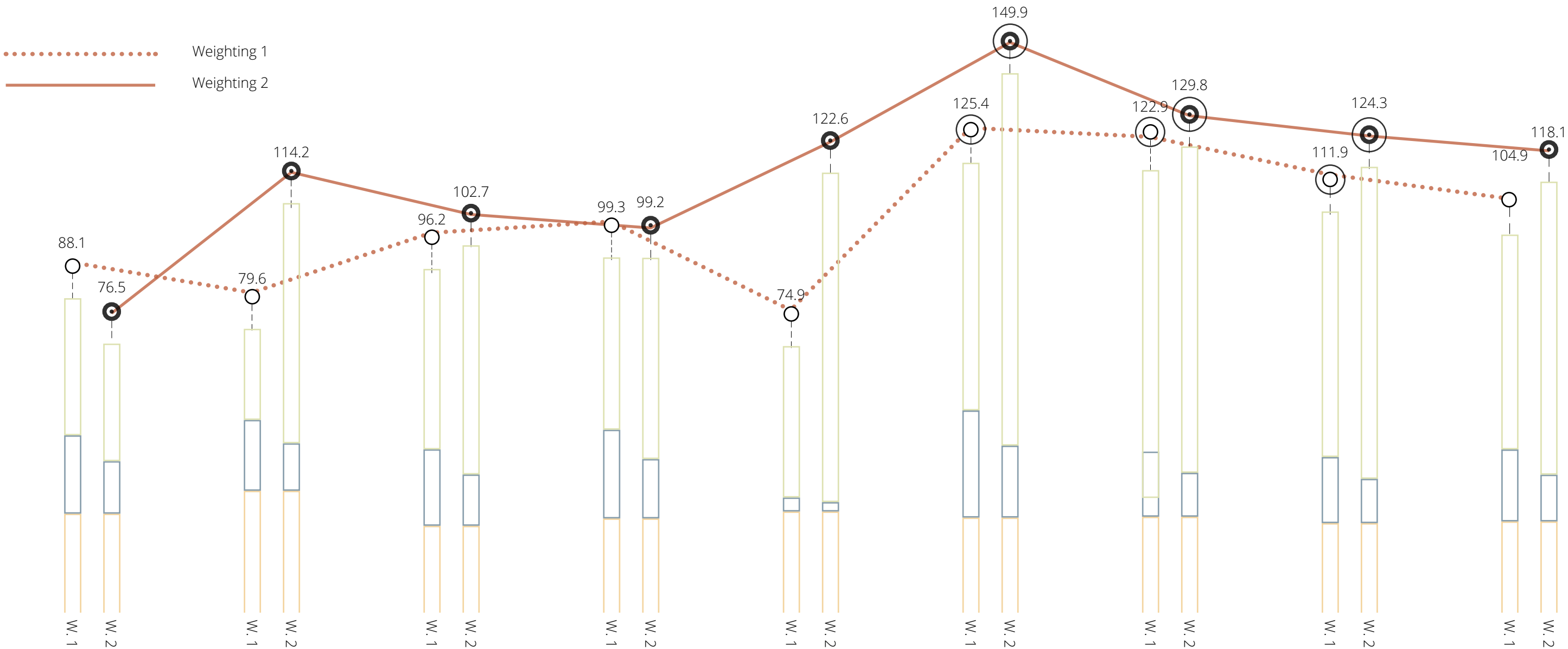
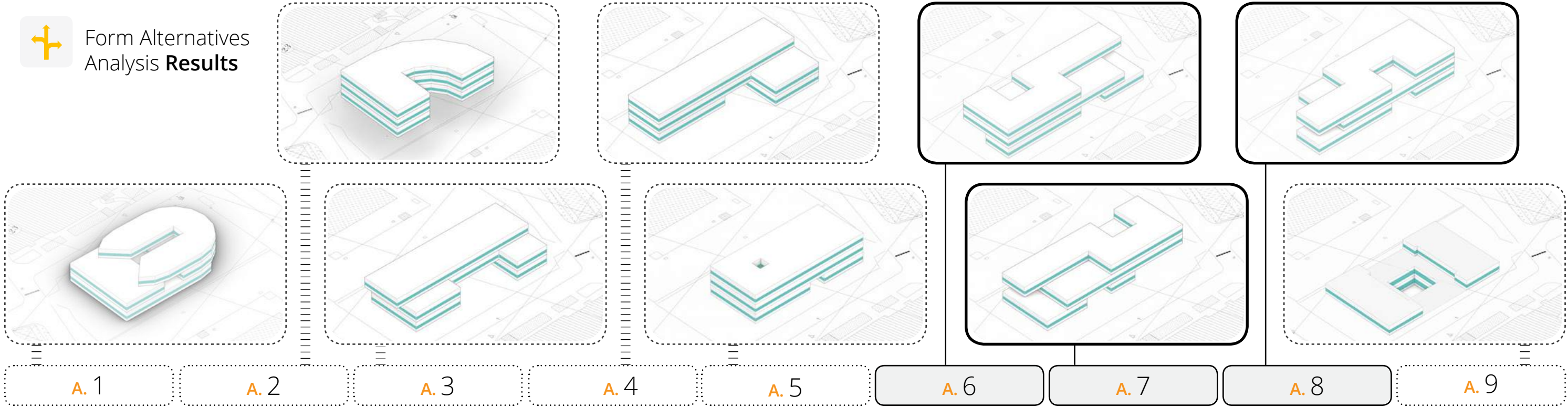
Form Alternatives Analysis

	Weighting I	Weighting II
Thermal	X 1	X 3
Daylighting	X 1	X 2
Functional	X 1	X 4



Key Parameters		Detailed Weightings	Values/ Raw Scores	Scores	Overall Scores	Values/ Raw Scores	Scores	Overall Scores	Values/ Raw Scores	Scores	Overall Scores				
Thermal	Annual Heating Energy [Jouls/sqm]	20/100	4.57E+09	6	I 28.01 II 28.01	4.50E+09	8	I 26.24 II 26.24	4.87E+09	64	I 26.70 II 26.70				
	Annual Cooling Energy [Jouls/sqm]	20/100	4.20E+07	8		4.30E+07	6		4.57E+07	17					
	Zone Ventilation [ACH]	15/100	6.95	73		6.77	65		7.54	0					
	Overheat hours [% of times]		1.94			2.00			2.12						
	Underheat hours [% of times]		73.41			73.19			73.02						
	Comfortable hours [% of times]	40/100	24.65	24.6		24.81	24.8		24.86	25.8					
	Radiation on surfaces [W/sqm](BIPV Potential)	5/100	354517.25	87		334122.21	77		334512.09	0					
	Daylighting	Daylit Area(5>DF>2)	80/100	17.75		18	I 17.91 II 12.00		18.18	18		I 18.18 II 12.18	19.51	8	I 19.18 II 12.85
Underlit Area(DF<2)			63.46		63.4			51.19							
Overlit Area (DF>5)		20/100	18.36		18.18			27.64							
Functional & Aesthetical	Semi-Open Space Area[sqm]	20/100	1104	100	I 77.00 II 89.77	608	52	I 67.50 II 85.84	558	48	I 59.4 II 78.52				
	Form Aesthetics	10/100	80	80		80	80		40	40					
	Attraction for pedestrians (South & West Sides)	20/100	100	100		100	100		25	25					
	Structure Challenges/ Difficulty of Modelling and Construction	10/100	80	80		80	80		45	45					
	Form Flexibility	20/100	80	80		80	80		80	80					
	Indoor Accessibility	20/100	25	25		25	25		100	100					
	Total Scores:		I	122.9		II	129.8		Total Scores:	I		111.9	II	124.3	Total Scores:

Form Alternatives
Analysis Results



Selected Forms From Step 1

Highlights and Features

- High Comfortable hours
High Solar Radiation on Envelope
High Daylit Floor Area
- Open and Semi-open Areas
Form Aesthetics
Form Flexibility

Weighting I

Weighting II

• Total Scores

• Final Ranking

125.4

149.9

1st

1st

- High Comfortable hours
High Solar Radiation on Envelope
- Attraction for pedestrians
Open and Semi-open Areas
Form Aesthetics
Form Flexibility
Low Structure Challenges/ Difficulty of
Modelling and Construction

Weighting I

Weighting II

• Total Scores

• Final Ranking

122.8

129.8

2nd

2nd

- High Comfortable hours
High Solar Radiation on Envelope
- Attraction for pedestrians
Open and Semi-open Areas
Form Aesthetics
Form Flexibility
Low Structure Challenges/ Difficulty of
Modelling and Construction

Weighting I

Weighting II

• Total Scores

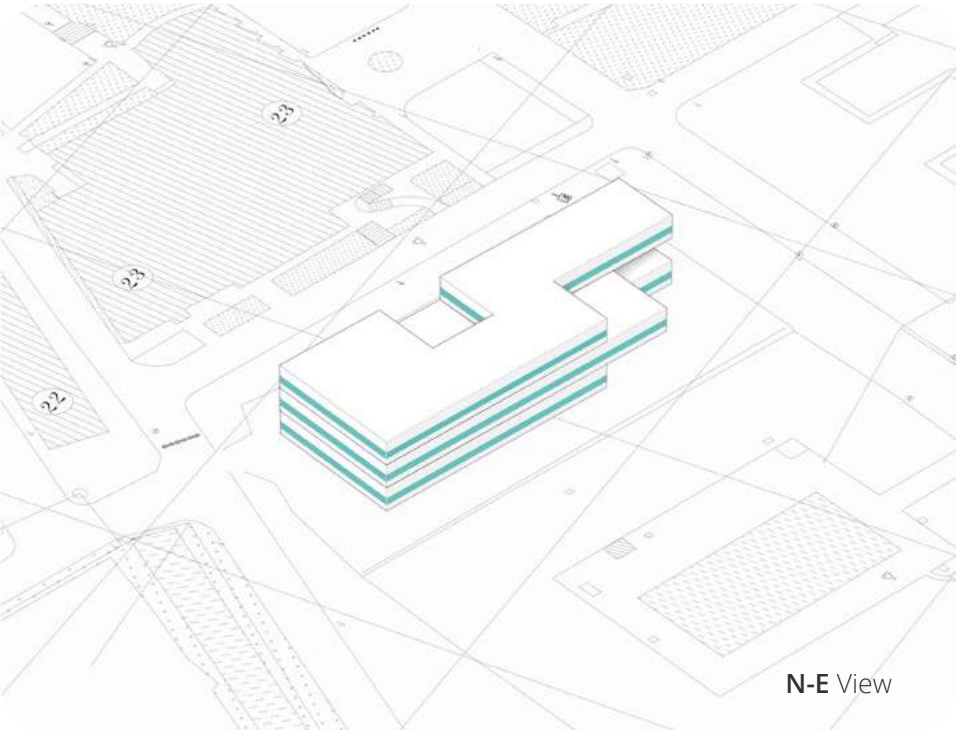
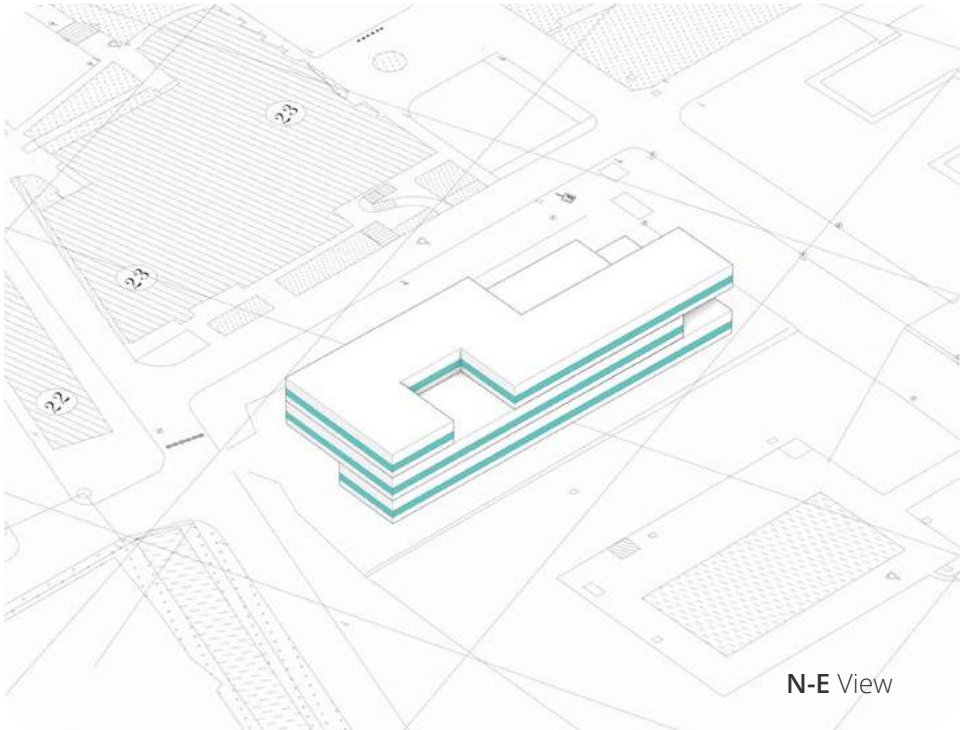
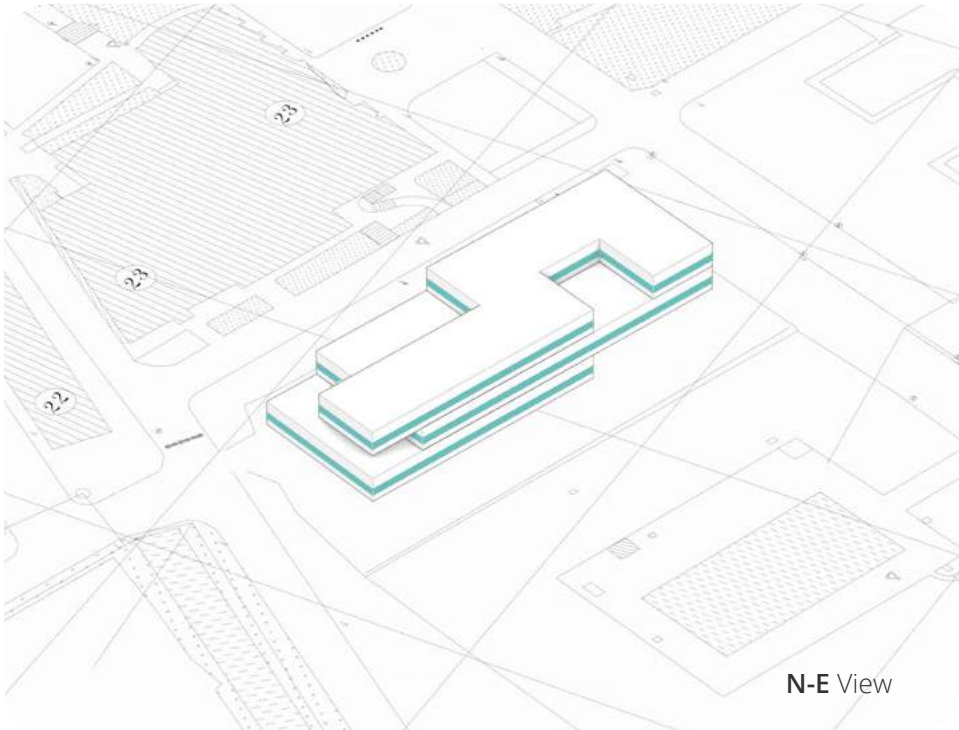
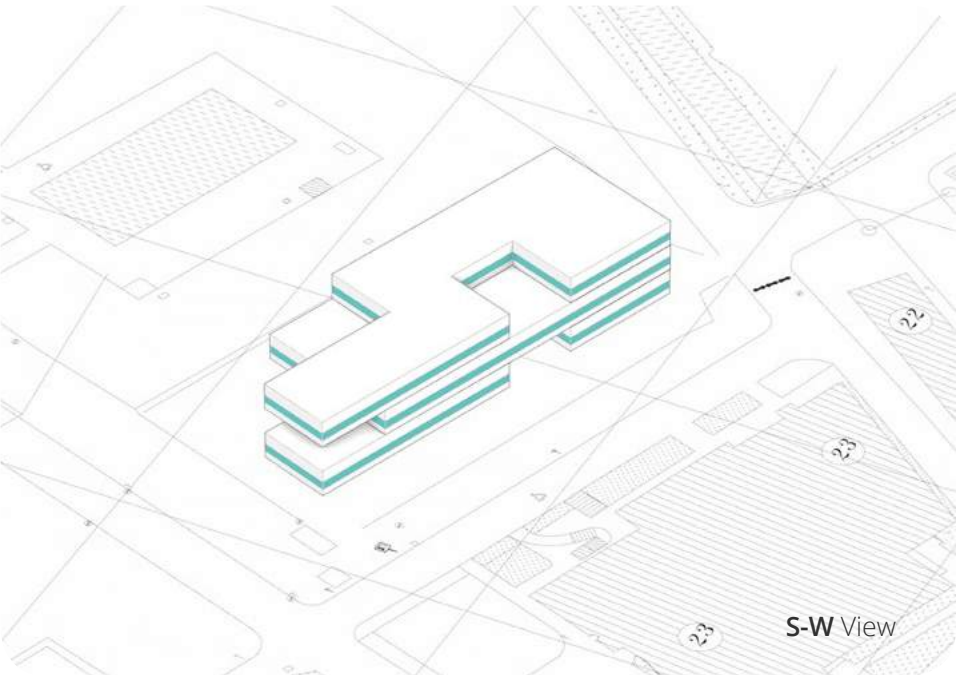
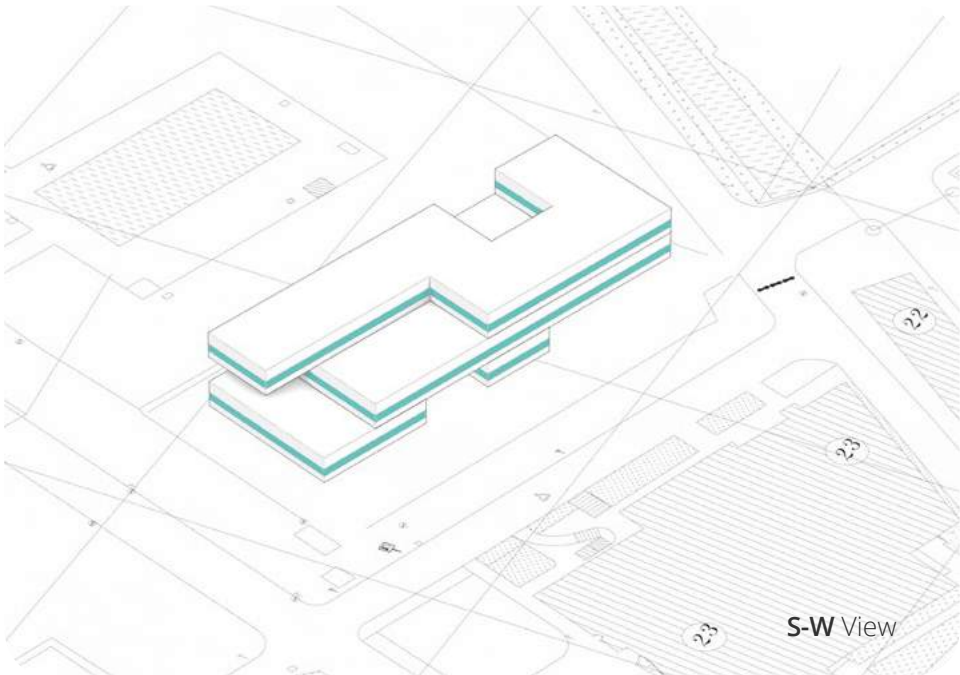
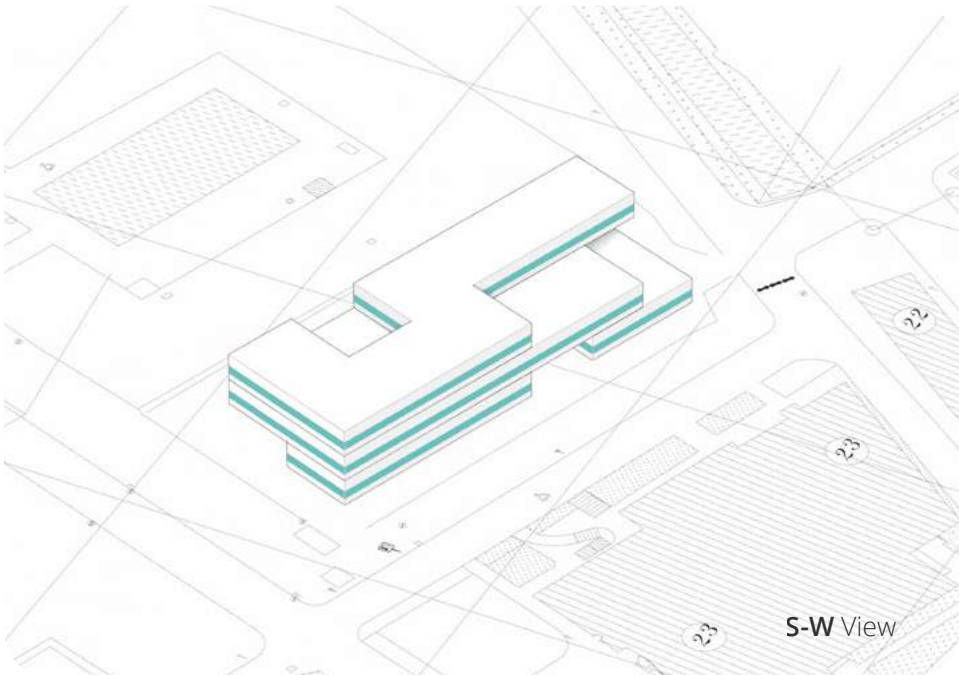
• Final Ranking

111.9

124.3

3rd

3rd



Code A.6

SVF 0.40

Floor Area 3090sqm

Code A.7

SVF 0.41

Floor Area 3090sqm

Code A.8

SVF 0.38

Floor Area 3124sqm

Modelling Key Assumptions

Ventilation

- Scheduled Ventilation Off
- Natural Ventilation On
- Setpoint 18
- Min Outdoor Temperature 0
- Max Outdoor Temperature 30
- Max Rel. Humidity 90
- Infiltration On
- Infiltration ACH 1.1
- Constant Term Coefficient 1
- Temperature Term Coefficient 0
- Velocity Term Coefficient 0
- Velocity Squared Term Coefficient 0

Constructions

- Window U-Value [W/m²K] 2.72
- Window Visible Transmittance 0.813
- Window SHGF 0.76
- Wall U-value [W/m²K] 0.7(Default)
- Floor U-value [W/m²K] 0.7(Default)
- Ceiling U-value [W/m²K] 0.7(Default)
- Roof U-value[W/m-2k] 0.7(Default)
- WWR[%] 30

Loads

- People On
- Schedule 8 to 6
- density [p/m2] 0.2
- Equipement On
- density [W/m2] 6
- Lighting On
- density [W/m2] 5

Conditioning

- Heating Off
- Cooling Off
- Humidity Control Off
- Mechanical VentilationOff

Analysis Grid

- Spacing[m] 1.0
- Height[m] 0.76
- Offset from walls[m] 0.6

Surfaces

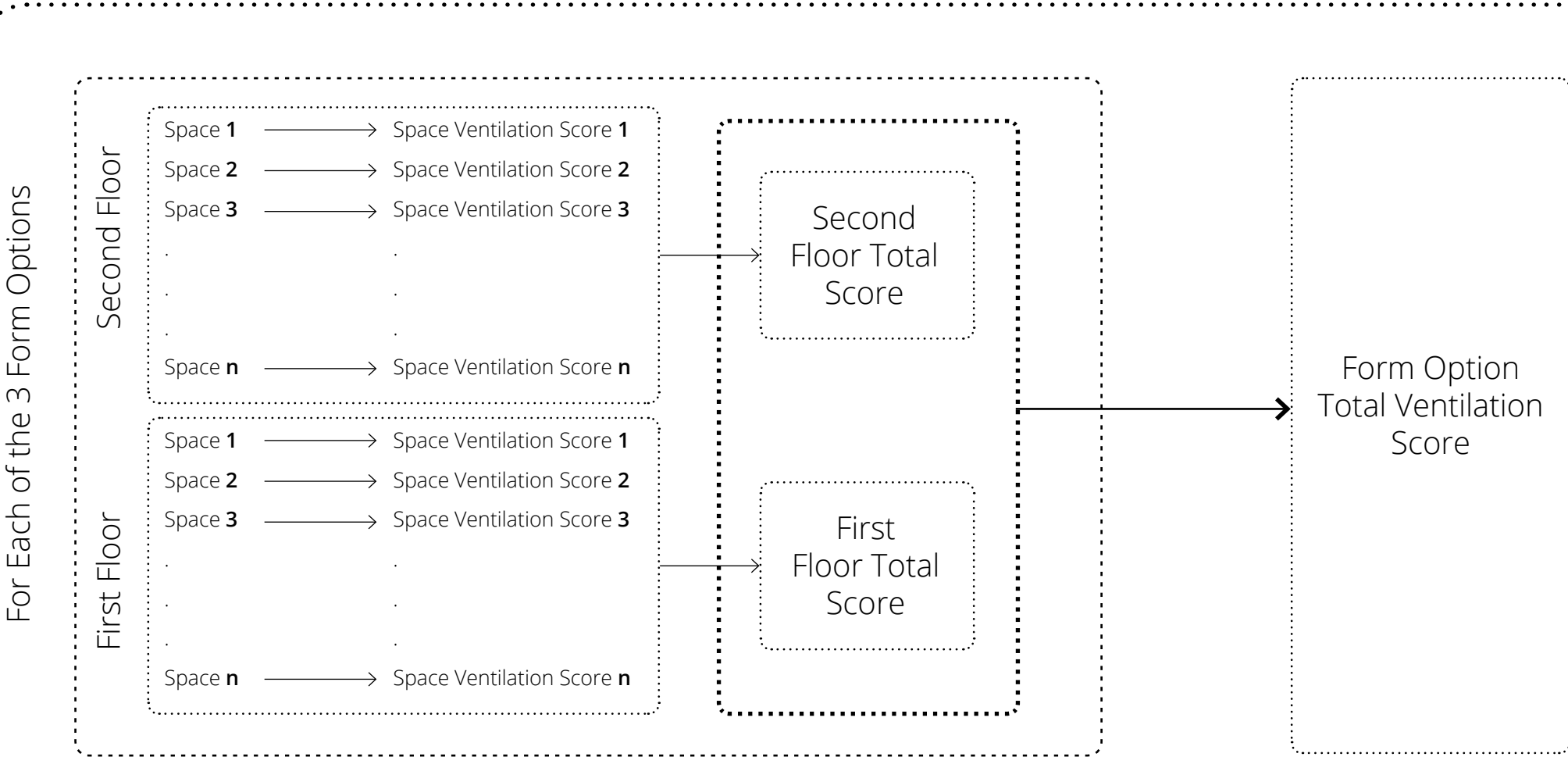
- Ceiling reflectance[%] 70
- Wall reflectance[%] 70
- Window Transmittance[%] 80
- Floor reflectance[%] 30
- Ground reflectance[%] 20
- Tree Transmittance[%] 19



Form Analysis(Step 2)



Ventilation Analysis Process

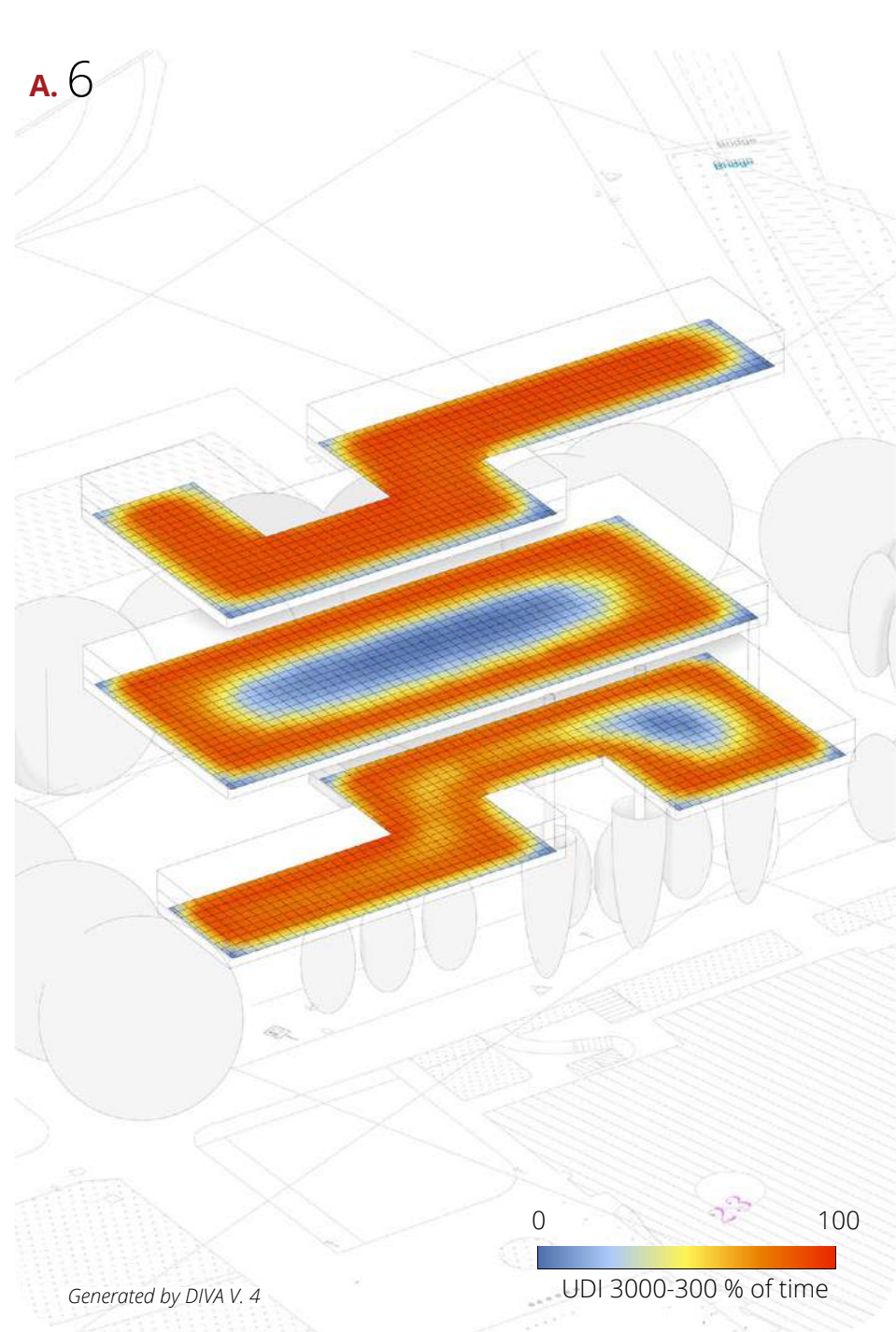


$$\text{Space Ventilation Score} = \frac{A \times K \times v \times 3600}{\text{Vol.}} \times \text{FA}$$

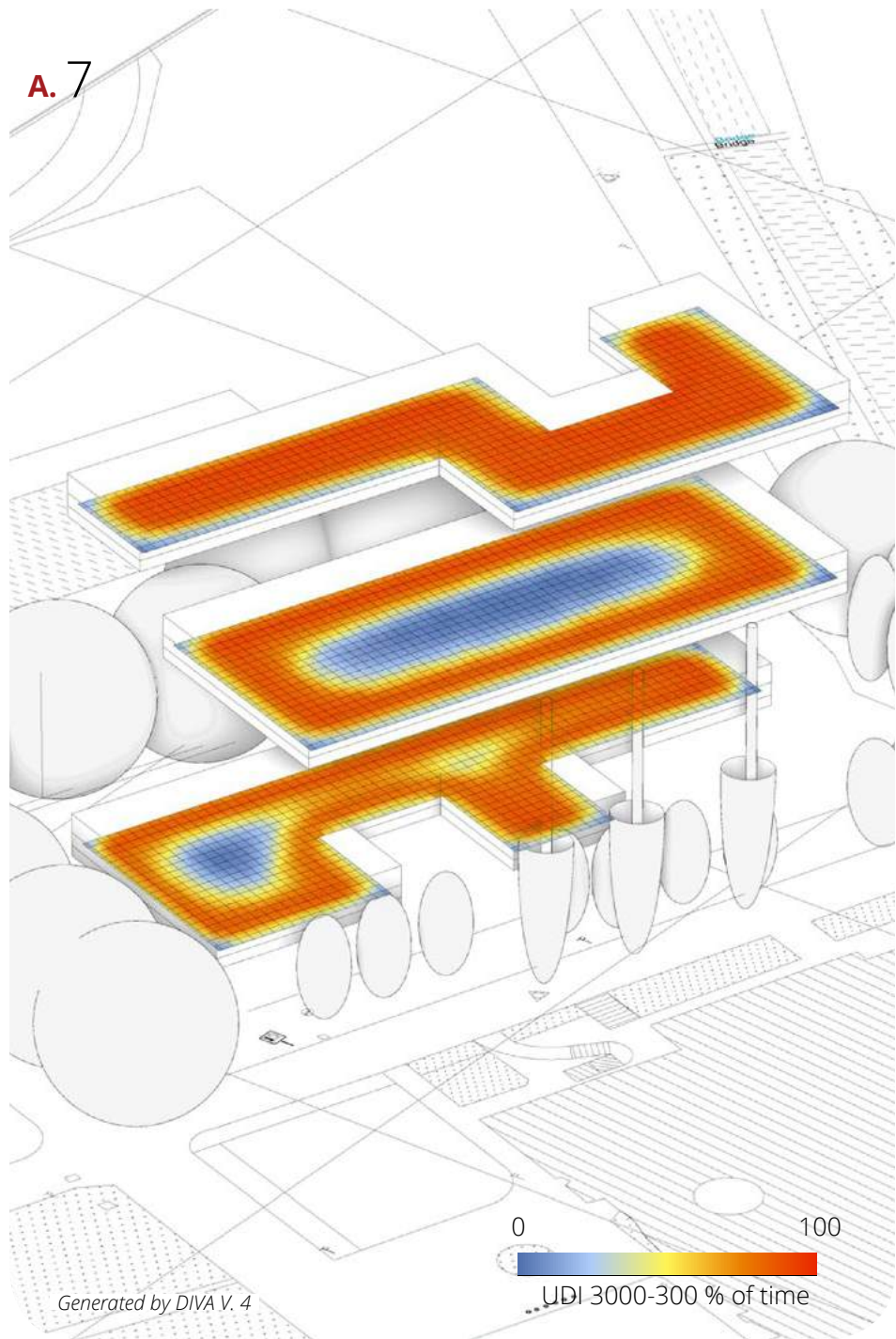
(Ventilation Factor x Acoustic Factor)

A : Operable Window Area [s.q.m.]
K : Coefficient of Effectiveness
v : Average Wind Speed (2 m/s)
Vol. : Space Volume [Cubic meters]
Ventilation Factor : Level of space need for Ventilation [1,2 or 3]
Acoustic Factor : Level of Space need for Silence(less Background noise level) [1,2 or 3]
FA : Floor Area

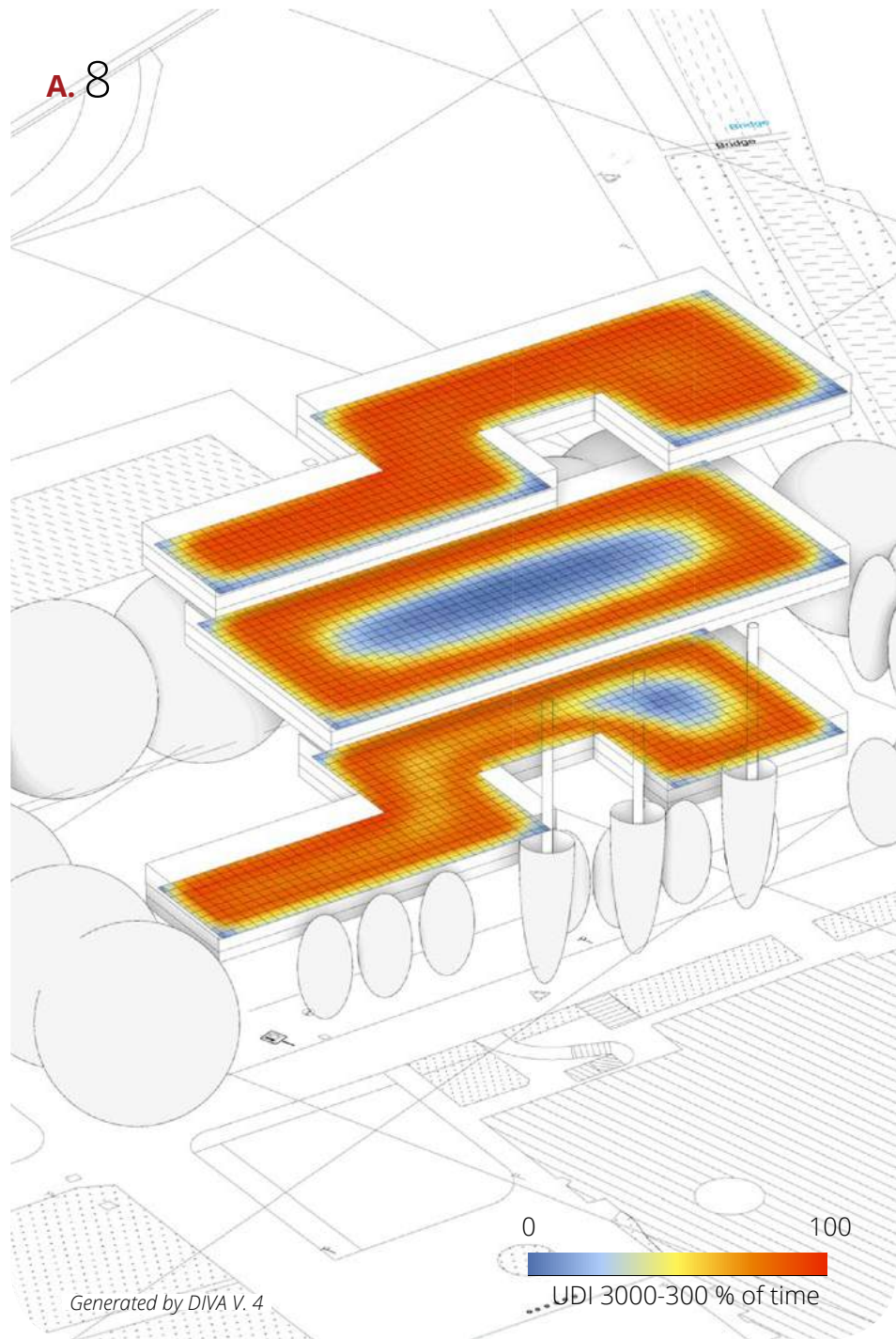
 Form Analysis(Step 2)



UDI % of Area more than %50 of times Between 3000-300	
Second Floor	76.9
First Floor	60.03
Ground Floor	78.52

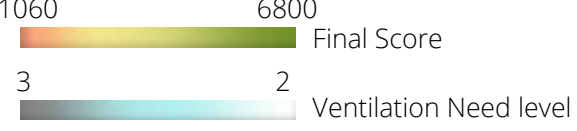
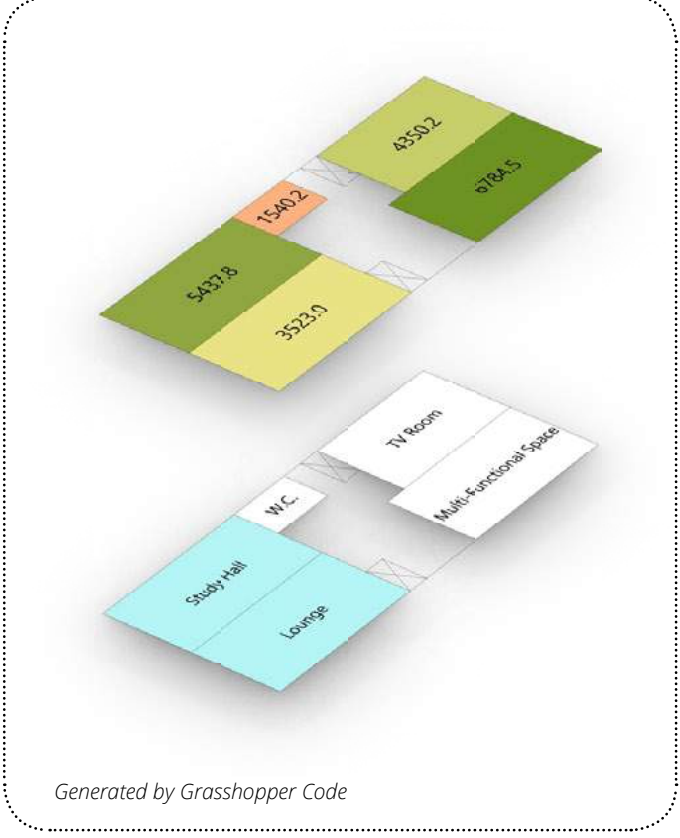
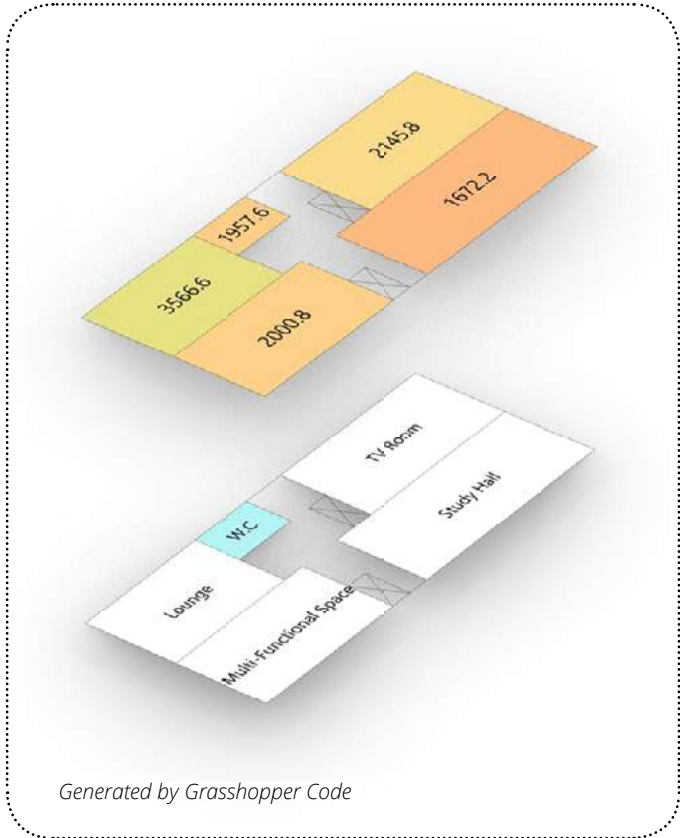
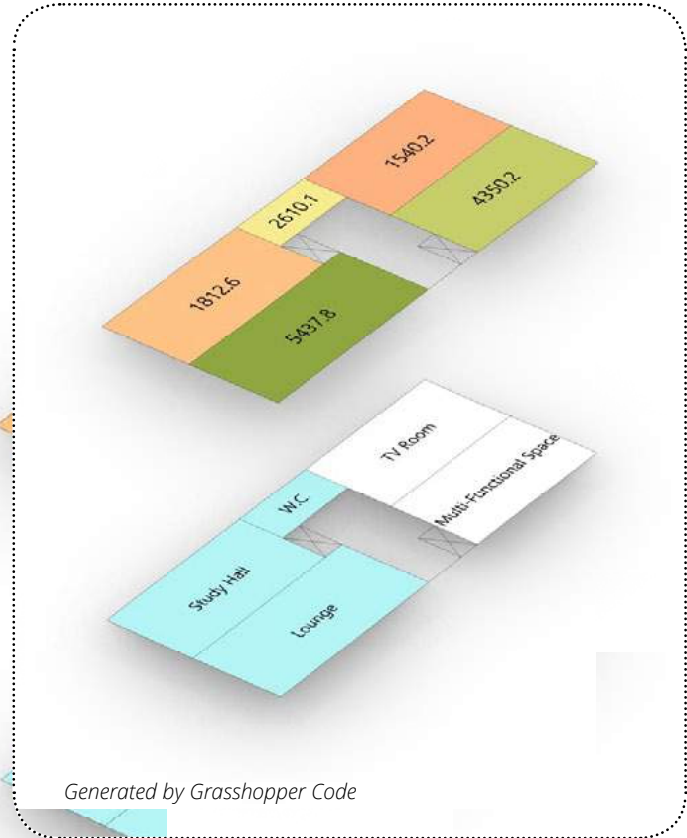
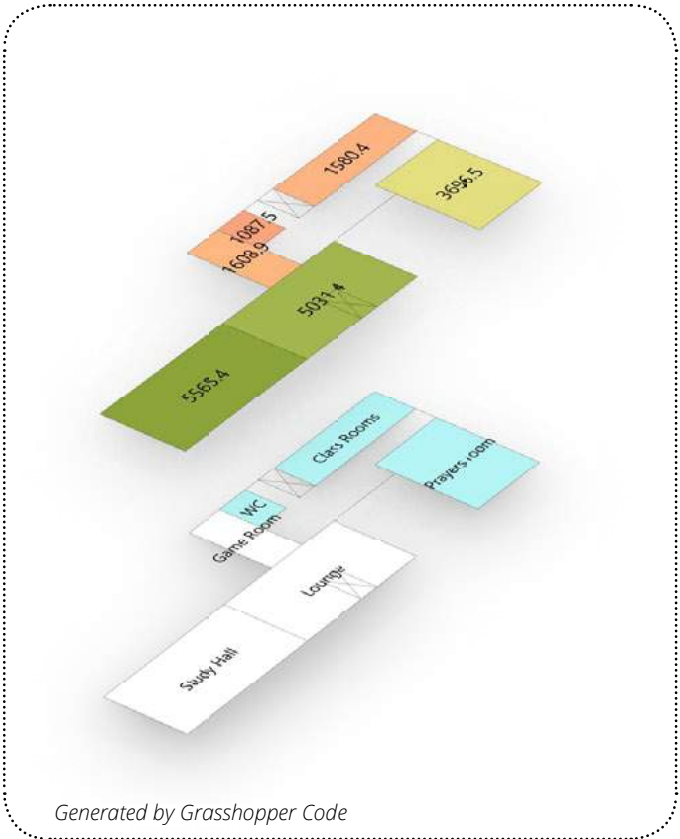
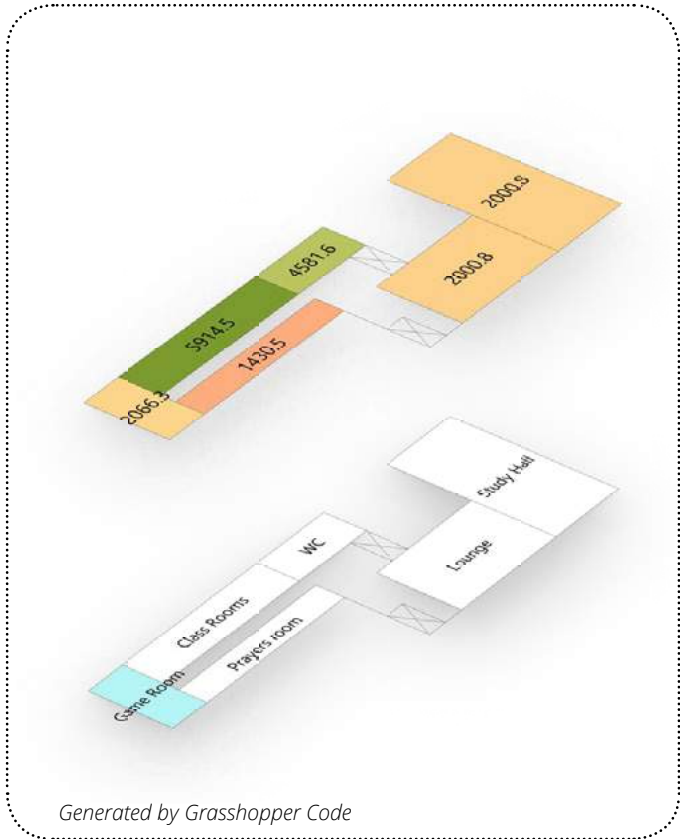
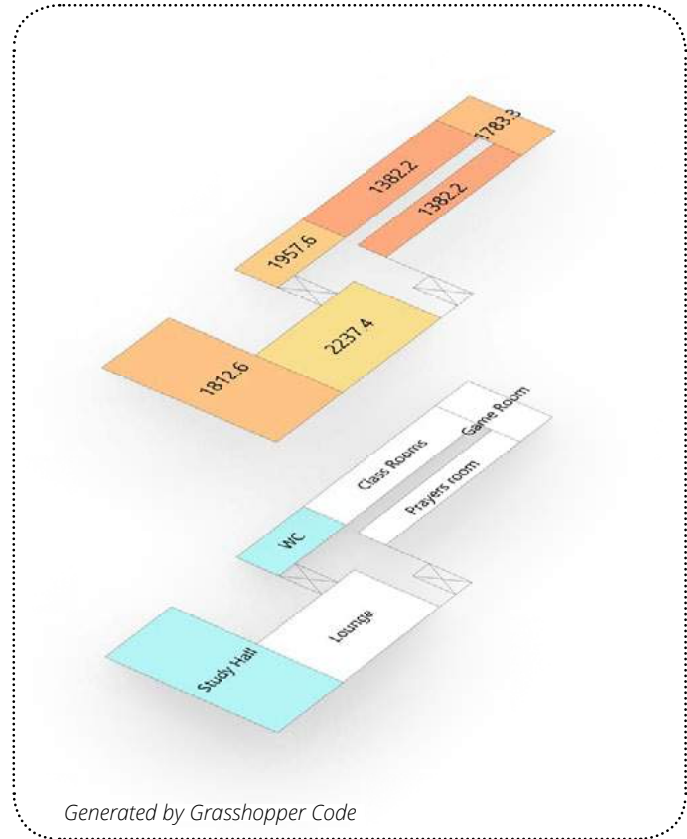


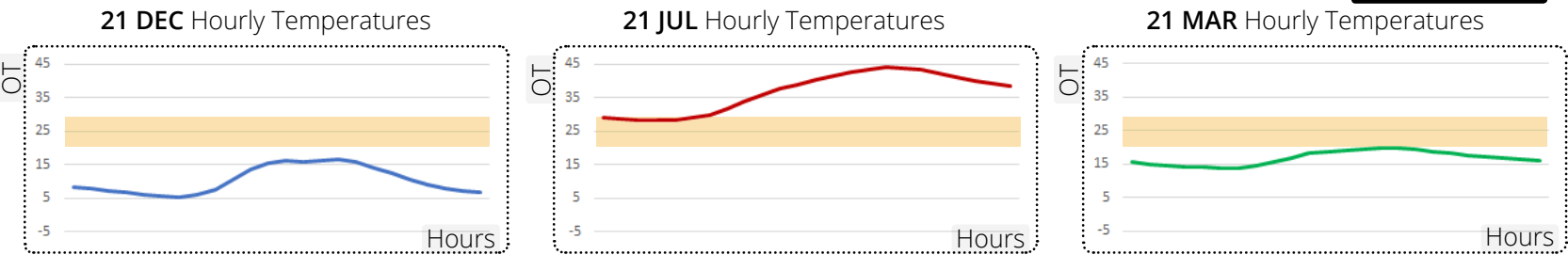
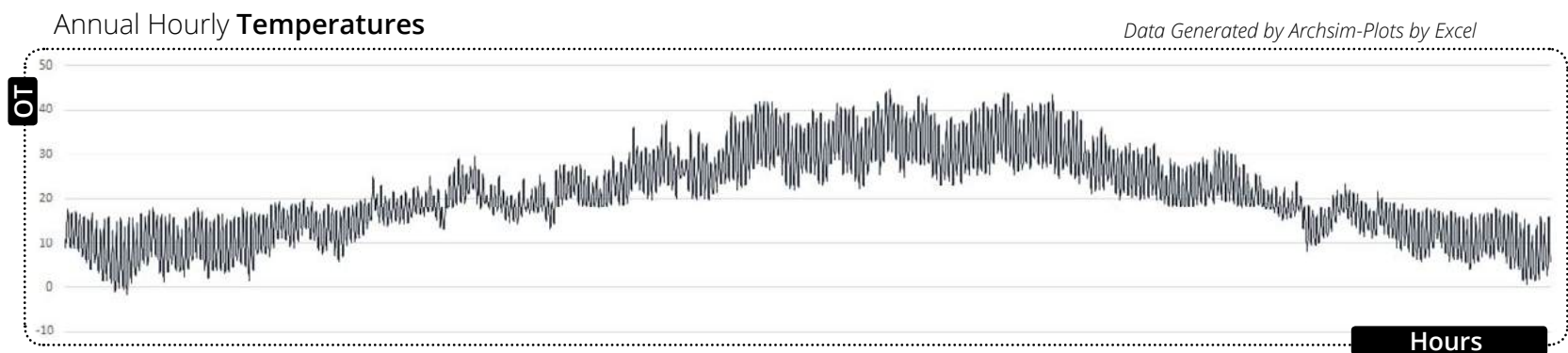
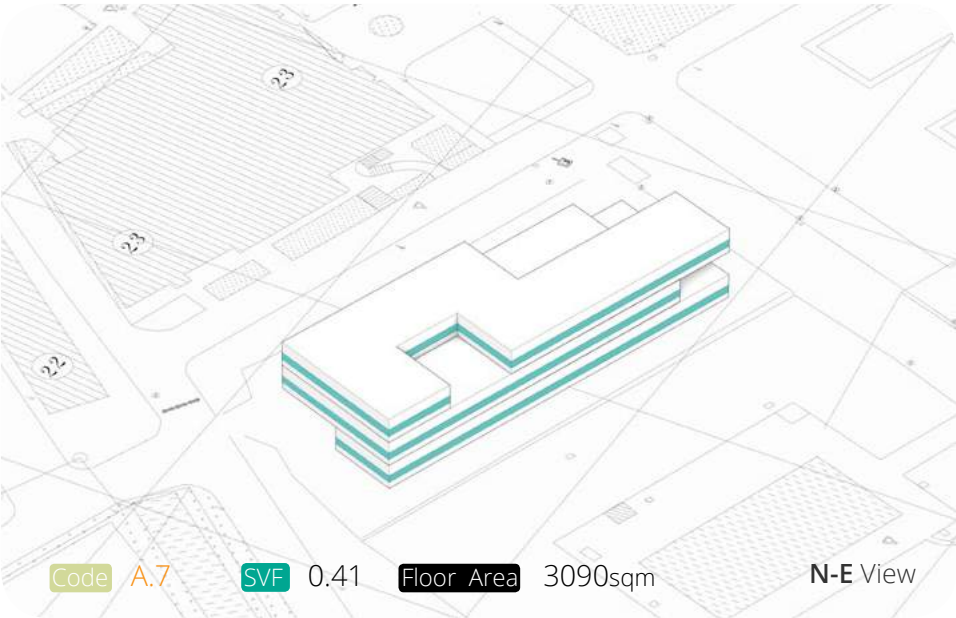
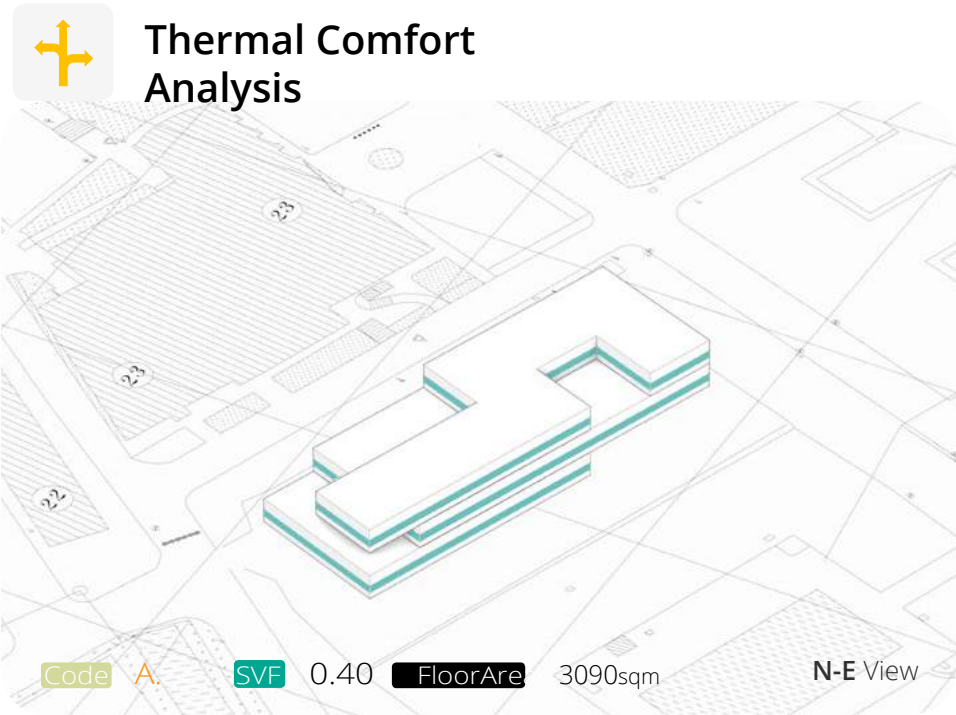
UDI % of Area more than %50 of times Between 3000-300	
Second Floor	74.3
First Floor	59.8
Ground Floor	77.58



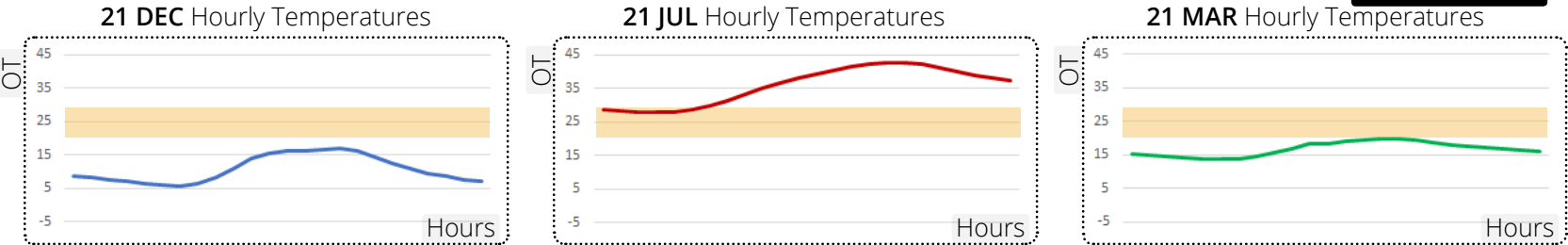
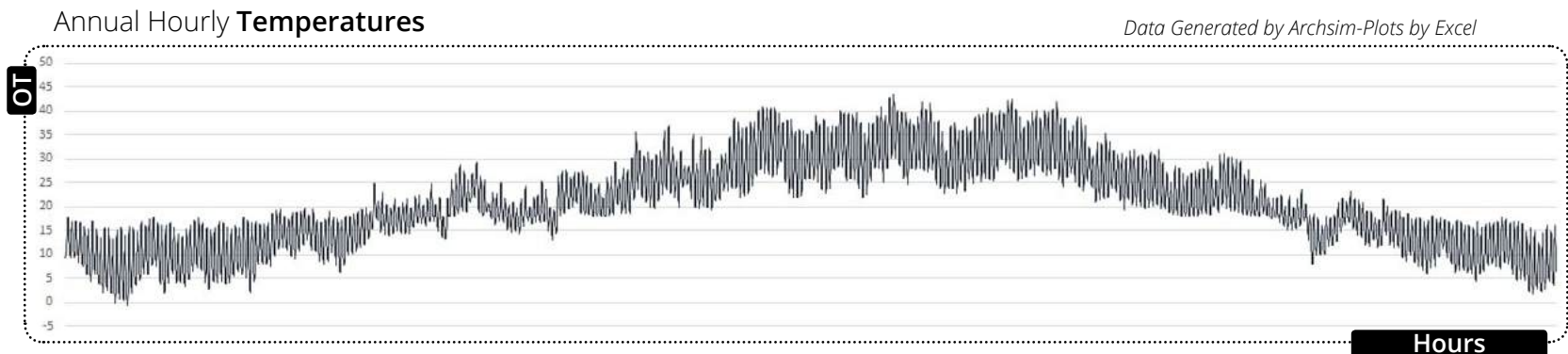
UDI % of Area more than %50 of times Between 3000-300	
Second Floor	75.8
First Floor	61.02
Ground Floor	74.7

Ventilation Analysis

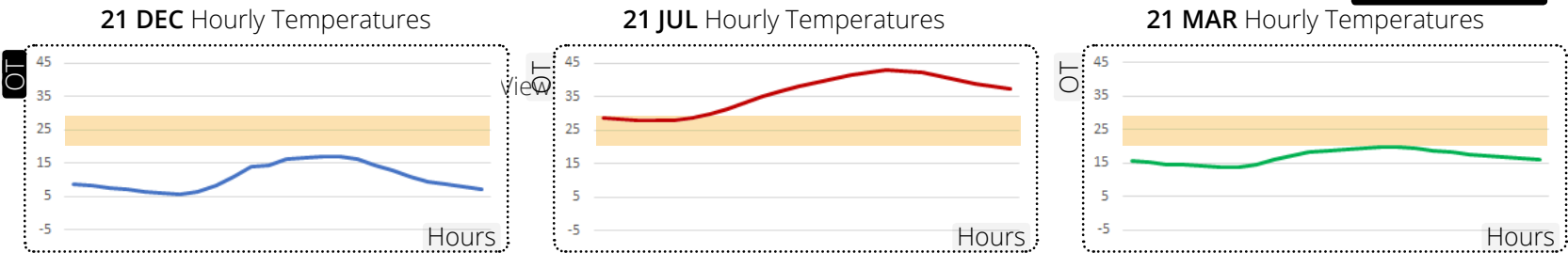
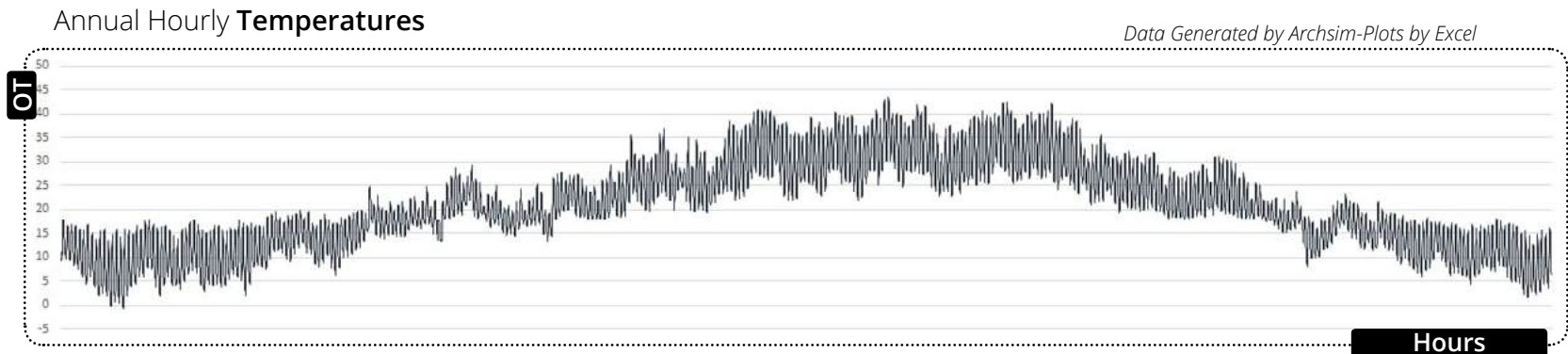




Maximum OT	44.7
Minimum OT	-1.6
Average OT	20.8
Total Air change rate[ACH]	7.23
Overheat % of times	23.42
Underheat % of times	54.38
Comfortable % of times	22.21

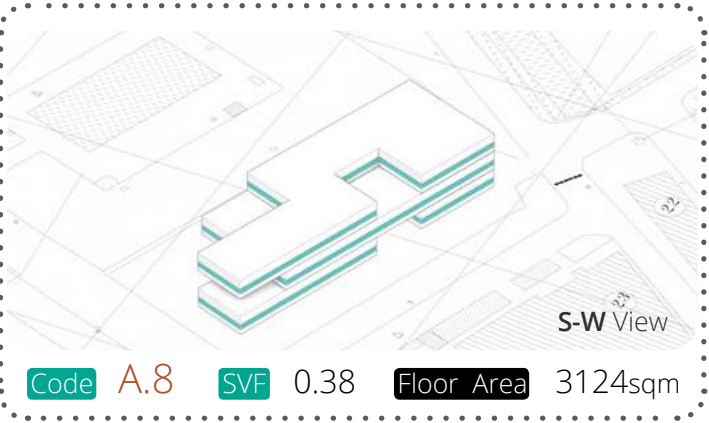
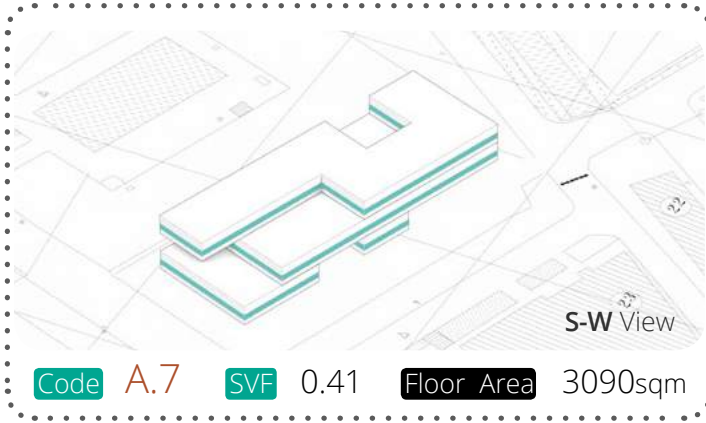
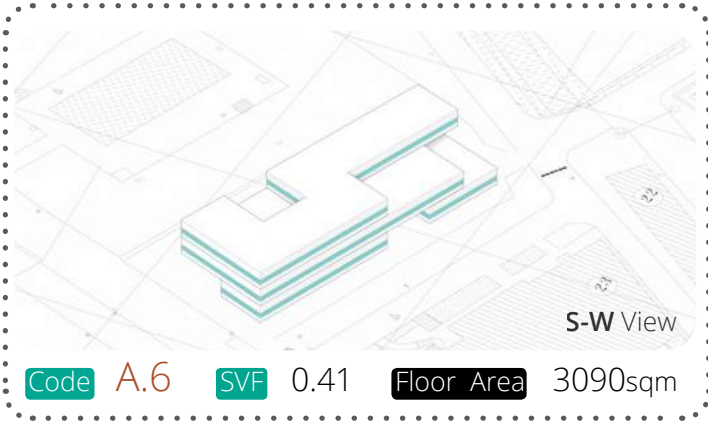


Maximum OT	43.3
Minimum OT	-0.7
Average OT	20.7
Total Air change rate[ACH]	6.69
Overheat % of times	22.50
Underheat% of times	54.86
Comfortable % of times	22.56



Maximum OT	43.3
Minimum OT	-0.7
Average OT	20.7
Total Air change rate[ACH]	6.67
Overheat % of times	22.62
Underheat% of times	54.73
Comfortable% of times	22.66

Final Results



	Key Parameters	Detailed Weightings	Values/ Raw Scores	Scores (Unweighted)	Overall Weighted Scores	Values/ Raw Scores	Scores (Unweighted)	Overall Weighted Scores	Values/ Raw Scores	Scores (Unweighted)	Overall Weighted Scores
Ventilation Potential	Ventilation Score(Rules of Thumb)	80/100	33746	88.8	x 3/13	34791	80.4	x 3/13	27313	0.0	x 3/13
	Average Annual ACR(Simulated[ACH])	20/100	7.23			6.69			6.68		
Loads/Energy	Zone Ideal Loads Total		2.85E+13	0.0	x 2/13	2.85E+13	51.5	x 2/13	2.83E+13	89.4	x 2/13
	Heating Energy	80/100									
	Cooling Energy	20/100	8.05E+11			7.35E+11			7.50E+11		
Thermal Autonomy	Overheat hours [% of times]	10/100	23.4	10	x 2/13	22.5	89.0	x 2/13	22.6	83.2	x 2/13
	Underheat hours [% of times]	30/100	54.3			54.8			54.7		
	Comfortable hours [% of times]	60/100	22.2			22.6			22.7		
Daylighting	Daylit Area(5>DF>2)	80/100	71.5	78.3	x 3/13	68.3	20.0	x 3/13	72.5	80.0	x 3/13
	Underlit Area(DF<2)		11.6			16.1			12.0		
	Overlit Area (DF>5)	20/100	36.7			36.9			34.5		
Functional & Aesthetical	Semi-Open Space Area[sqm]	20/100	608	63.9	x 3/13	608	77.0	x 3/13	558	57.0	x 3/13
	Form Aesthetics	10/100	80			80			40		
	Attraction for pedestrians (South & West Sides)	20/100	100			100			25		
	Structure Challenges/ Difficulty of Modelling and Construction	10/100	80			80			45		
	Form Flexibility	20/100	80			80			80		
	Indoor Accessibility	20/100	25			25			100		



Ground Floor Plan

- KEY
- 1

Entrance
- 2

Information
- 3

Conference Hall
- 4

W.C.
- 5

Kitchen
- 6

Food Stall
- 7

Shop
- 8

FoudCourt Sitting Area
- 9

Fire Control Room
- 10

Electrical/ Systems Control
- 11

Facility Room
- 12

Facility Access/Parking
- 13

Gallery
- 14

Gallery-Storage
- 15

Multi-Functional Space
- 16

TV Room
- 17

Lounge
- 18

Study Hall (Individual)
- 19

Study Hall (Group)
- 20

Game Room
- 21

Class Room
- 22

Prayers Room
- 23

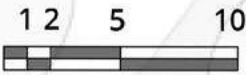
Management
- 24

Administration
- 25

Conference Room
- 26

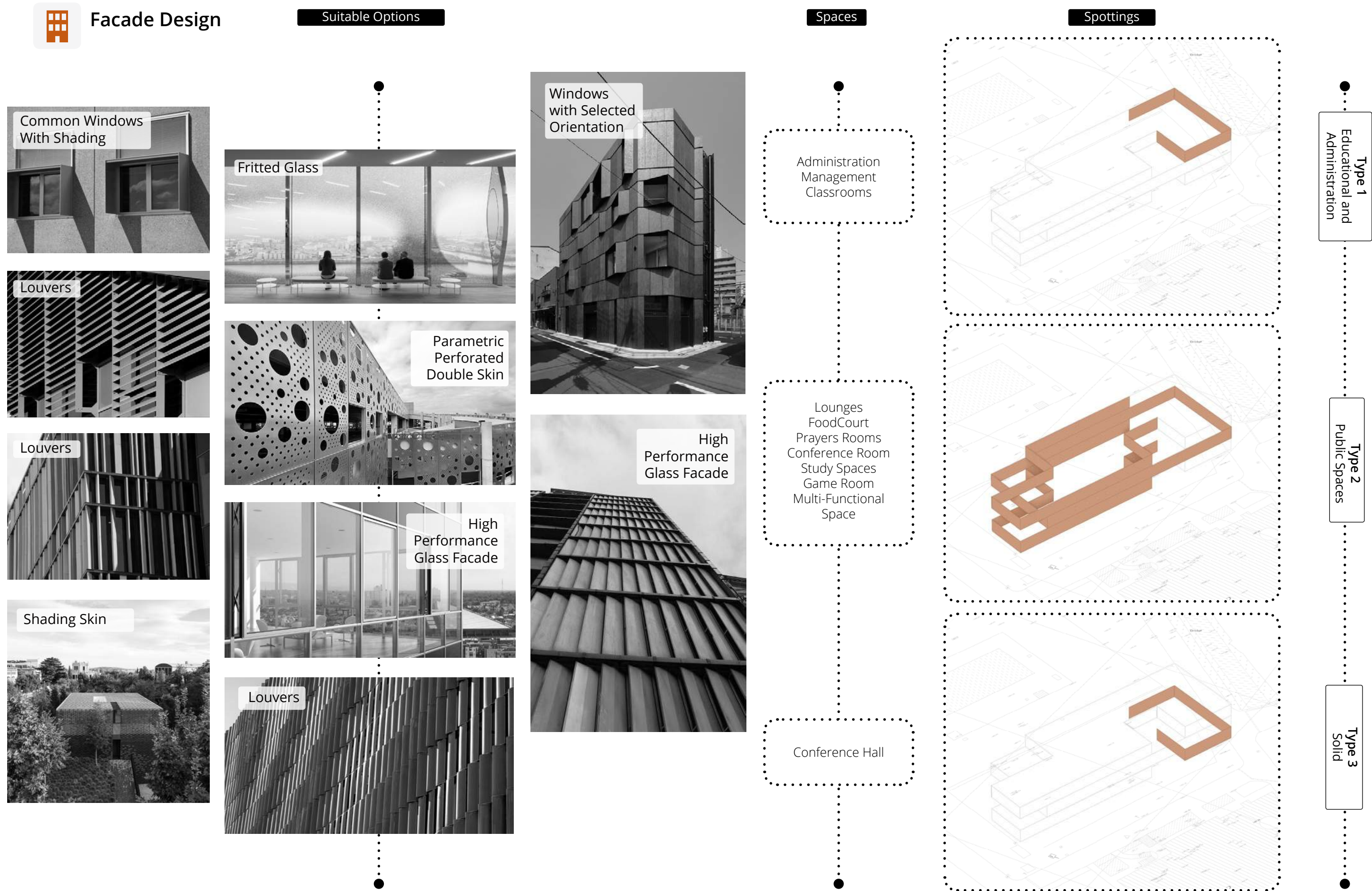
Student Association
- 27

Book Cafe









Modelling Key Assumptions- Step1

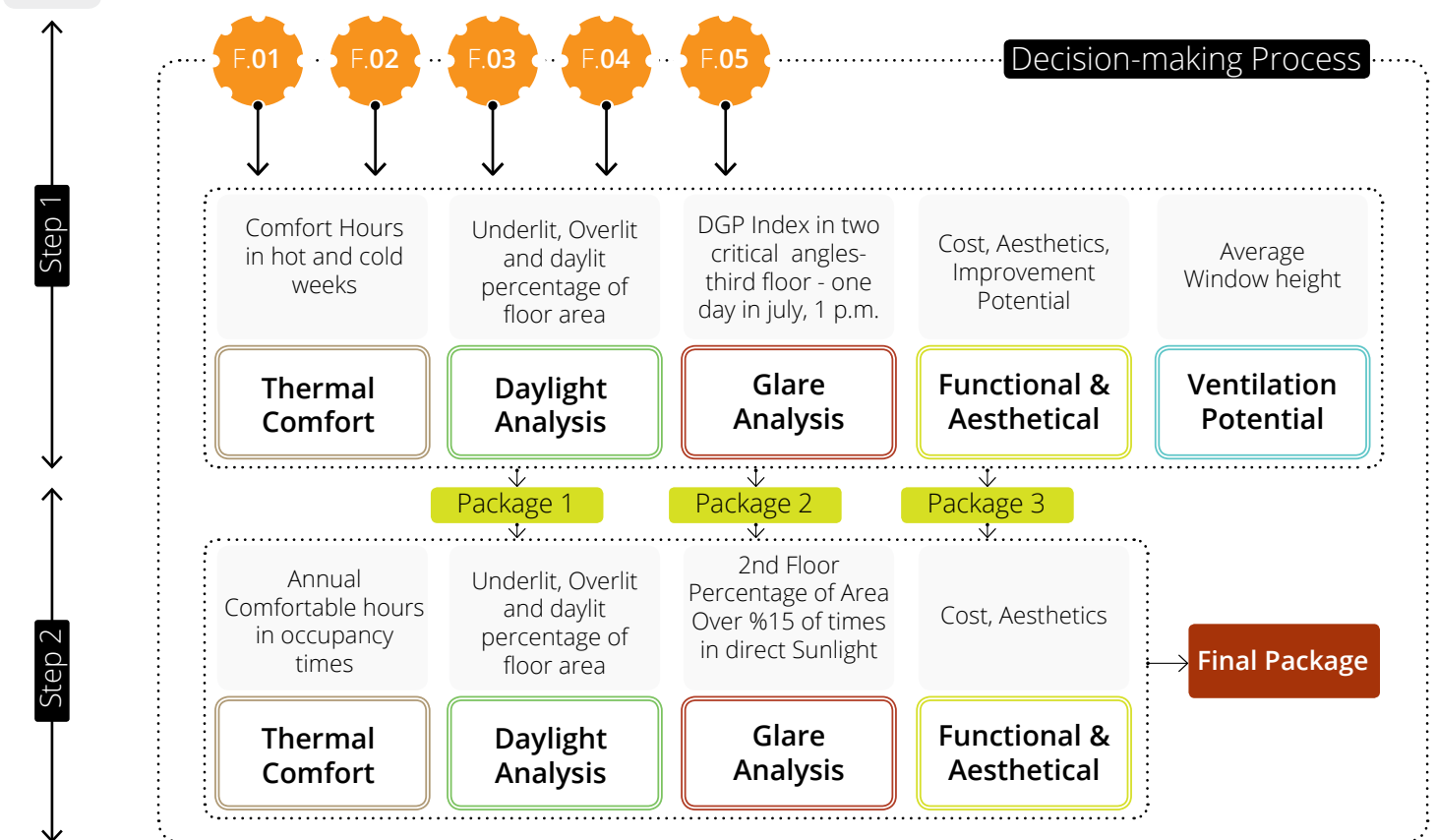
Ventilation	- Scheduled Ventilation Off - Natural Ventilation On - Setpoint 18 - Min Outdoor Temperature 0 - Max Outdoor Temperature 30 - Max Rel. Humidity 90 - Infiltration On - Infiltration ACH 1.1 - Constant Term Coefficient 1 - Temperature Term Coefficient 0 - Velocity Term Coefficient 0 - Velocity Squared Term Coefficient 0
Constructions	- Window U-Value [W/m²K] 2.72 - Window Visible Transmittance 0.813 - Window SHGF 0.76 - Wall U-value [W/m²K] 0.7(Default) - Floor U-value [W/m²K] 0.7(Default) - Ceiling U-value [W/m²K] 0.7(Default) - Roof U-value[W/m-2k] 0.7(Default) - WWR[%] 30
Loads	- People On - Schedule 8 to 6 - density [p/m2] 0.2 - Equipment On - density [W/m2] 6 - Lighting On - density [W/m2] 5
Conditioning	- Heating Off - Cooling Off - Humidity Control Off - Mechanical Ventilation Off
Analysis Grid	- Spacing[m] 1.0 - Height[m] 0.76 - Offset from walls[m] 0.6
Surfaces	- Ceiling reflectance[%] 70 - Wall reflectance[%] 70 - Window Transmittance[%] 80 - Floor reflectance[%] 30 - Ground reflectance[%] 20
Glare Analysis	- Sky Clear Sky with sun - Time July 21st, 12:01 P.M. - Quality Lowest - Tree Transmittance[%] 19

Modelling Key Assumptions- Step2

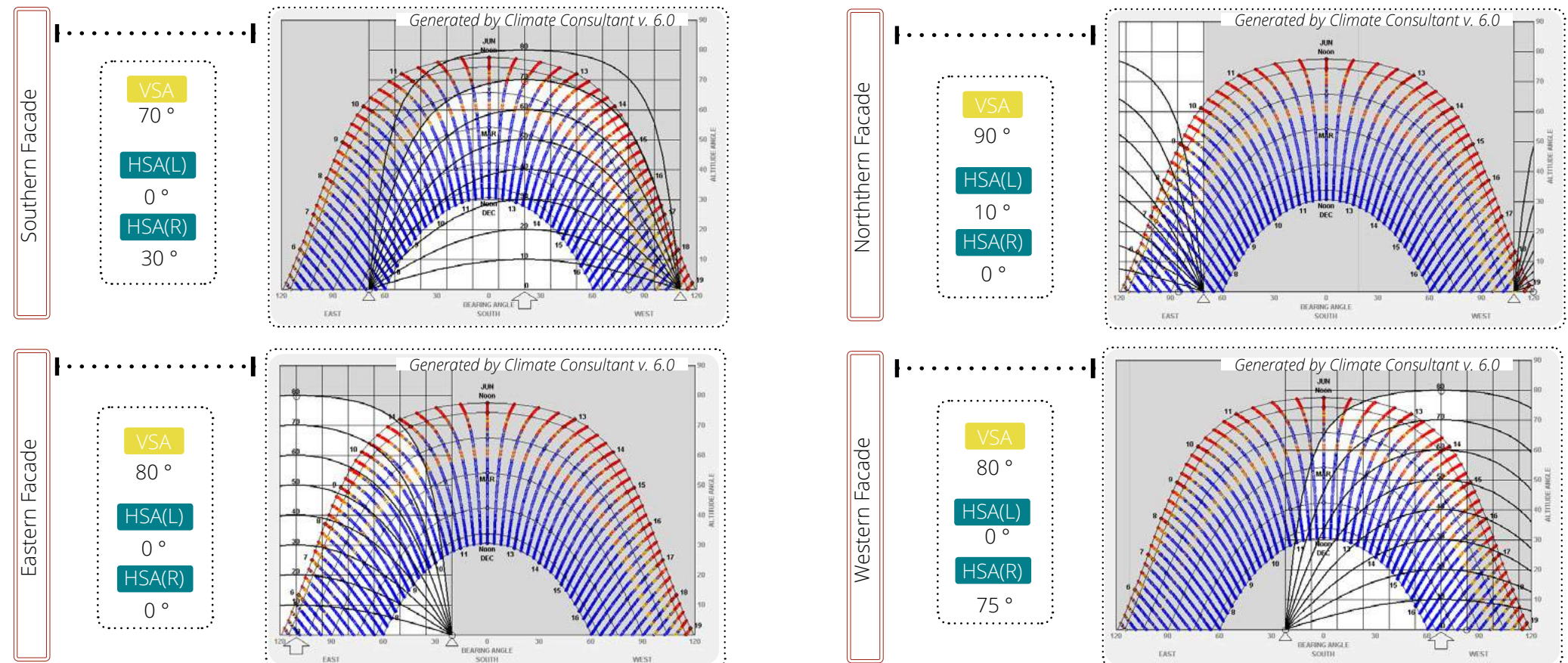
Thermal	- Infiltration Constant Rate(Scheduled method): 1 ACH - Infiltration Crack Template: Medium - Heating: Off - Cooling: Off - Mechanical Ventilation: Off - Natural Ventilation: On - Natural Ventilation Method: Calculated - Occupancy density[people/s.q.m.]: 0.5
HVAC	- WWR-South: %60 ,%30 - WWR-North: %50 ,%30 - WWR-East/West: %25 - Windows U-value[W/m-2K]: 1.96 - WindowsSHGC[%]: 0.691 - WindowsDirect Solar Transmission[%]: 0.624 - Windows Light Transmission[%]: 65 - External Walls U-Value[W/m-2K]: 0.95 - Roofs U-Value[W/m-2K]: 0.5
Constructions	
Analysis Grid	- Spacing[m] 1.0 - Height[m] 0.76 - Offset from walls[m] 0.6
Surfaces	- Ceiling reflectance[%] 70 - Wall reflectance[%] 70 - Window Transmittance[%] 80 - Floor reflectance[%] 30 - Ground reflectance[%] 20



Facade Design



Simple Shading calculations





Facade Alternatives

- Horizontal and Vertical Shadings
- Double pane Windows with %65 Transmittance
- shadings for solid Surfaces
- Window to Wall Ratio of %40

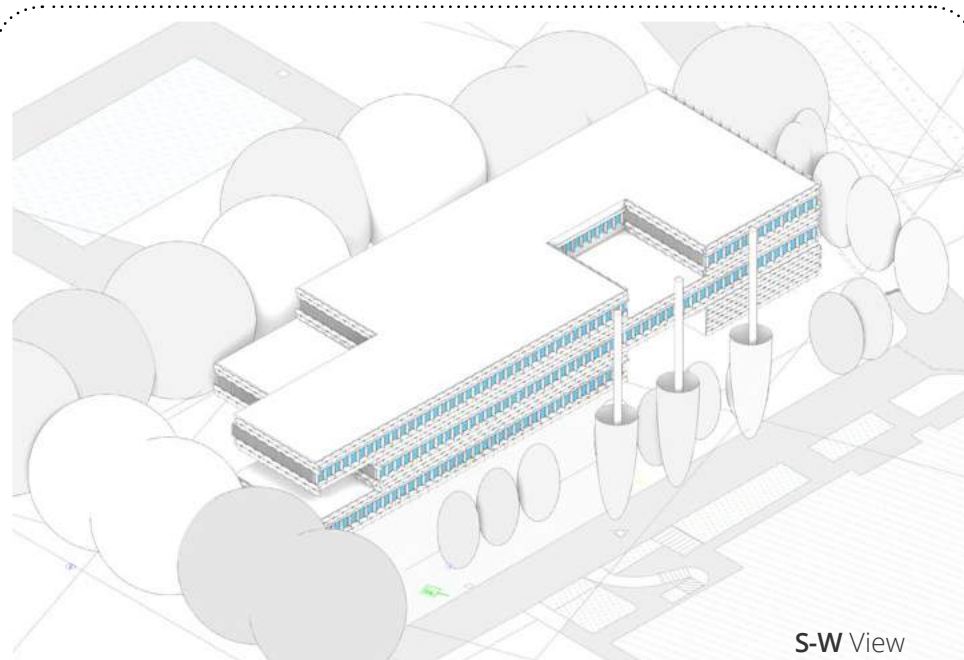
Code **F.1**

- No Shadings
- High performance Windows in South, East and West Facades With %47 Transmittance
- Double pane Windows in northern side With %65 transmittance
- Window to Wall Ratio of %90

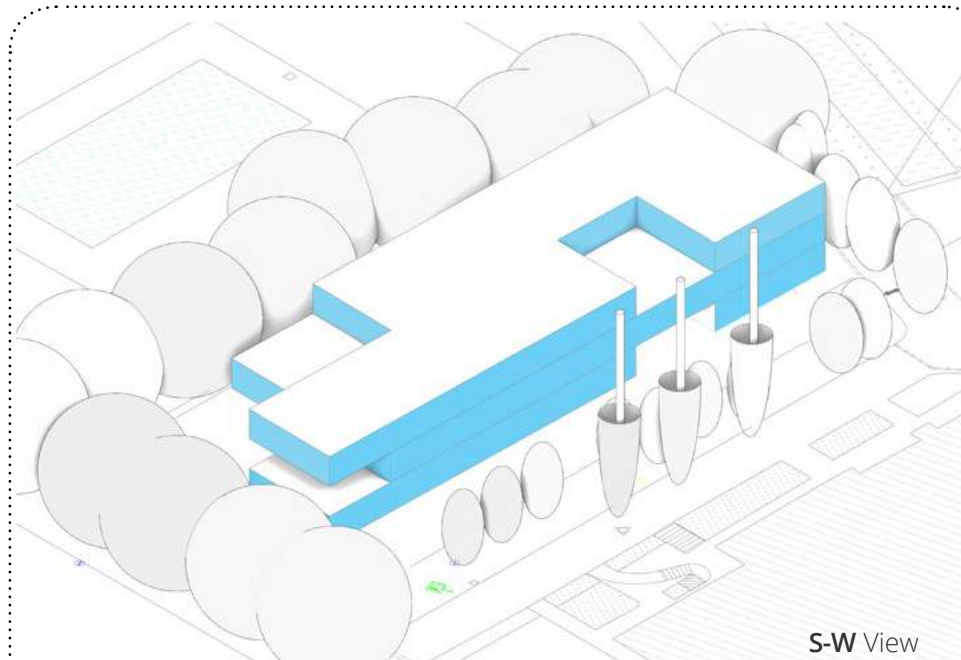
Code **F.2**

- Parametric Shading (South, East, West, North)
- Double pane Windows With %65 transmittance
- Window to Wall Ratio of %90

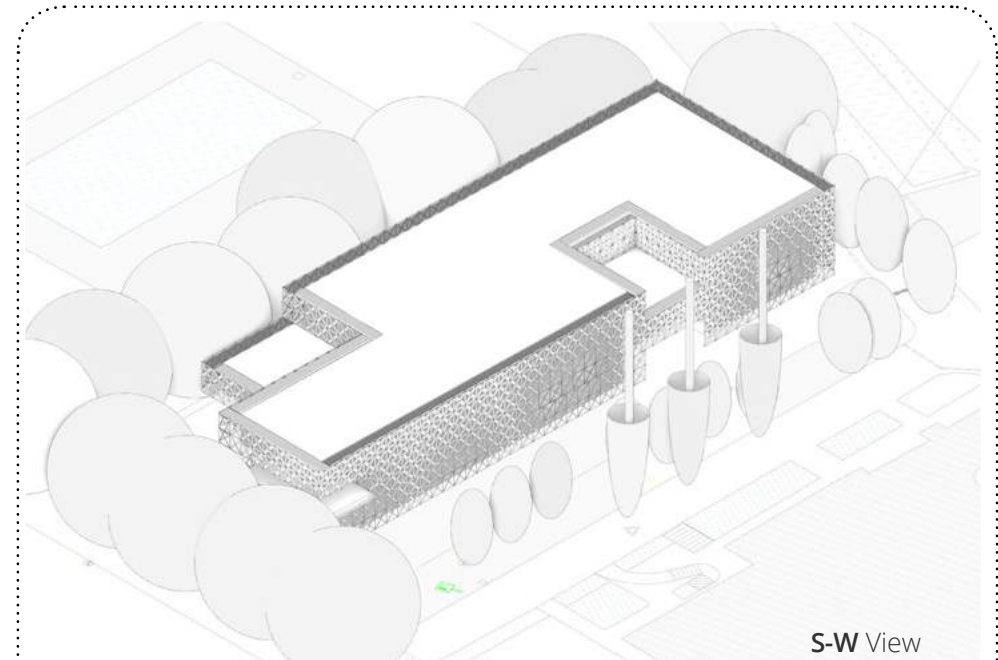
Code **F.3**



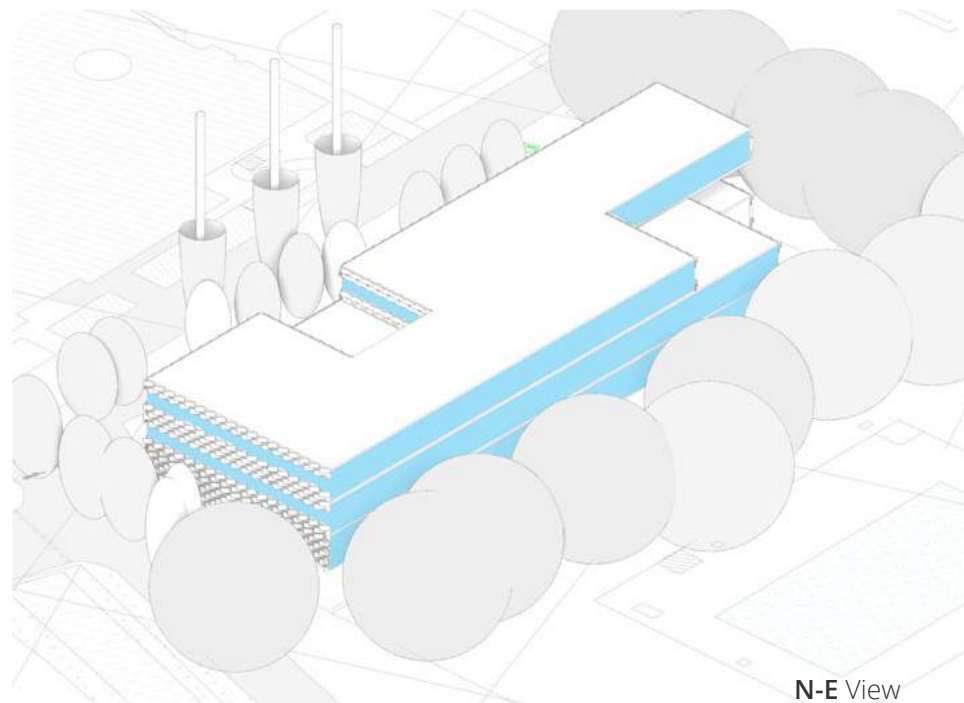
S-W View



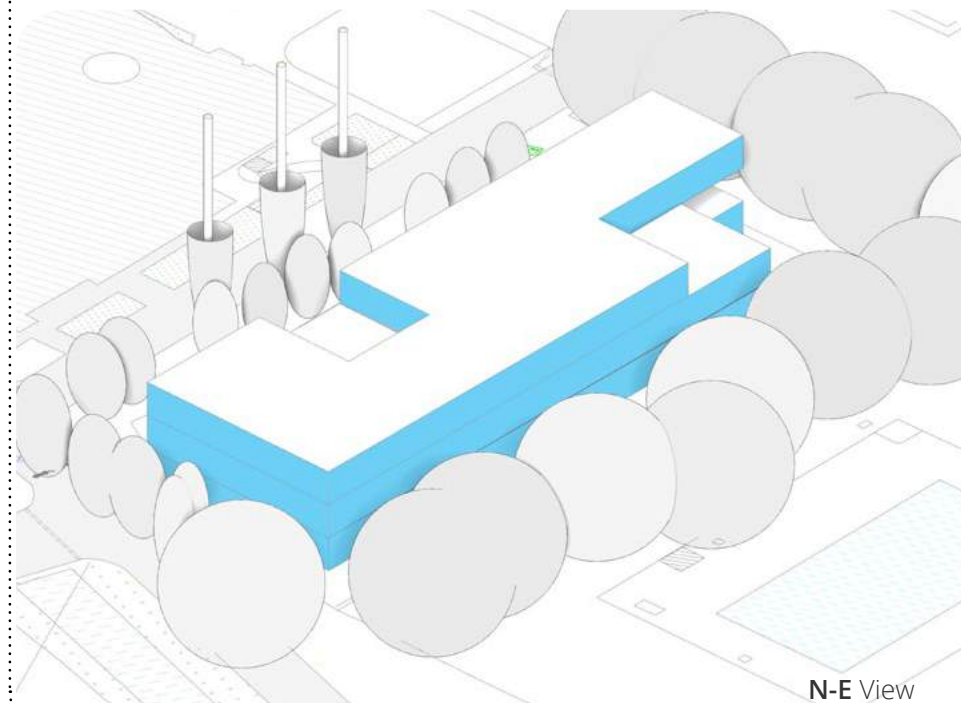
S-W View



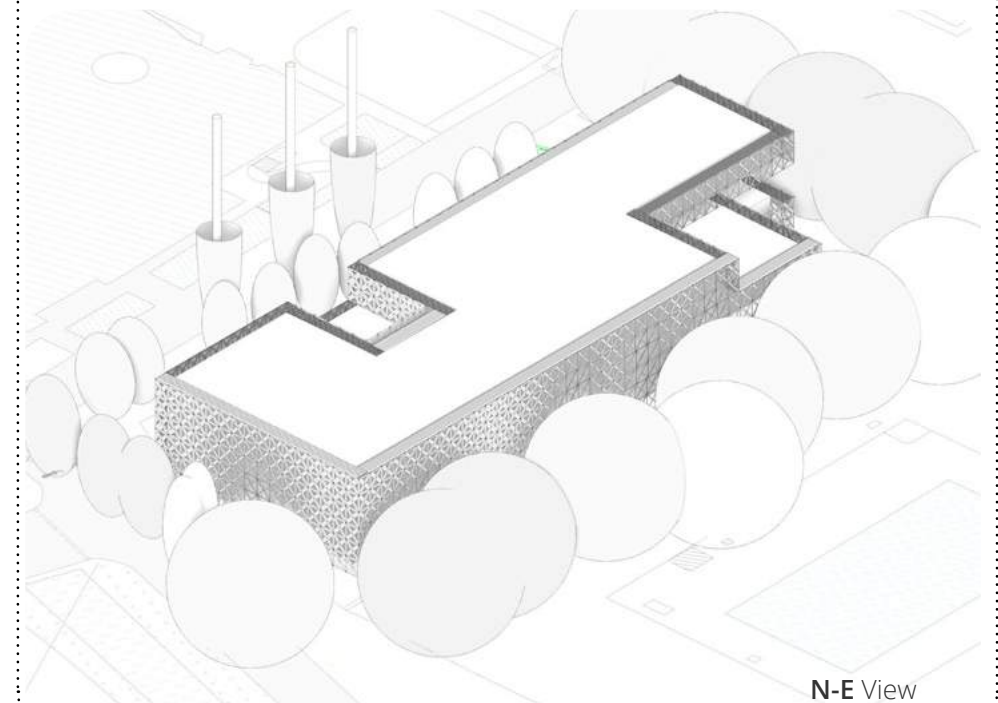
S-W View



N-E View



N-E View



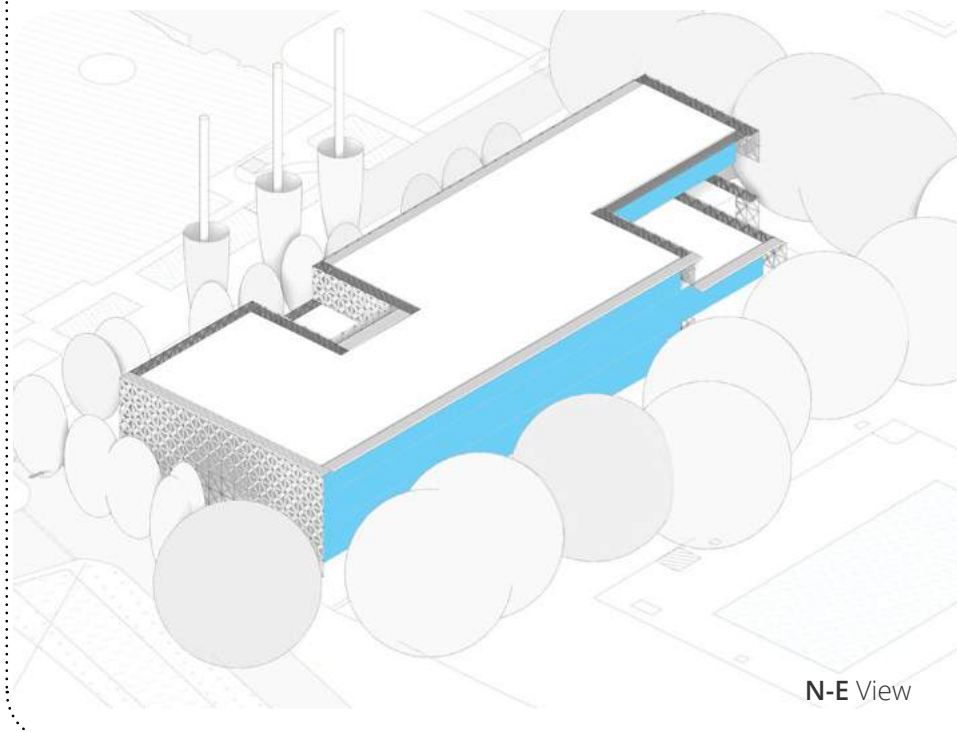
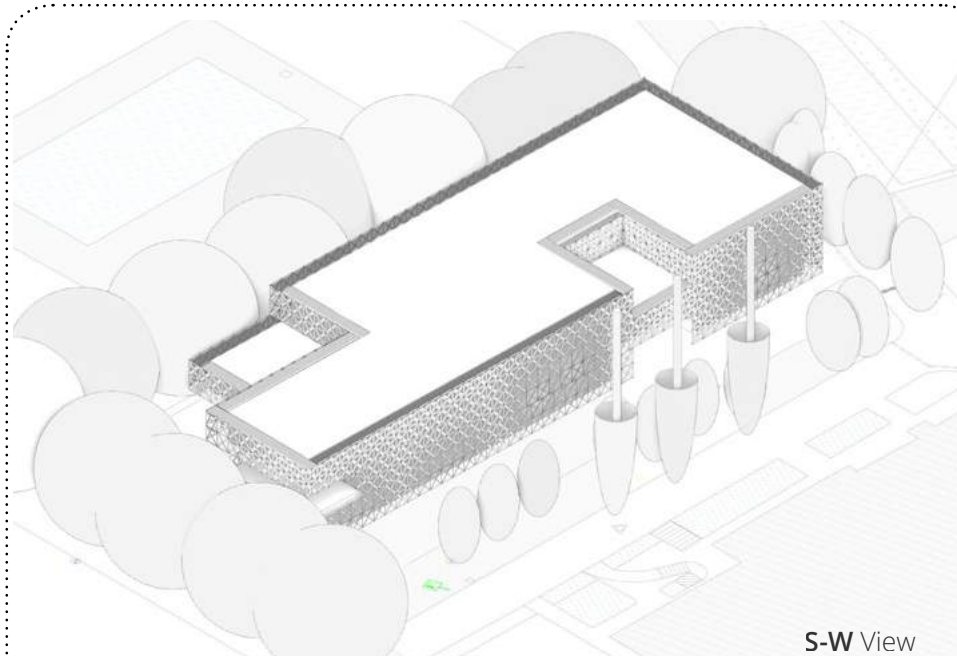
N-E View



Facade Alternatives

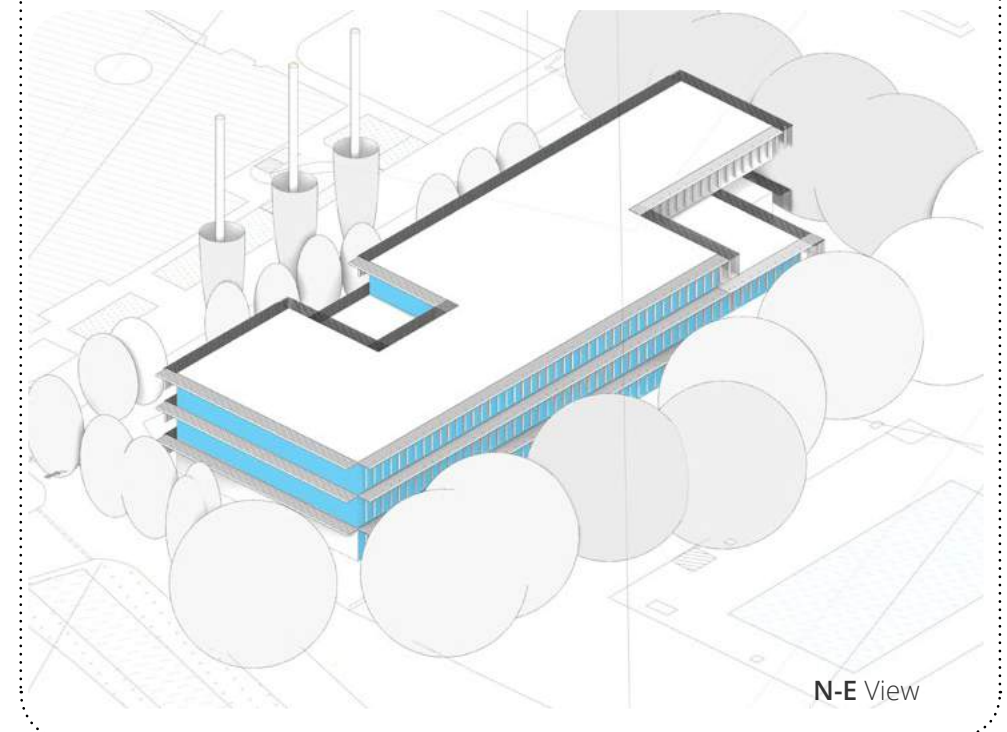
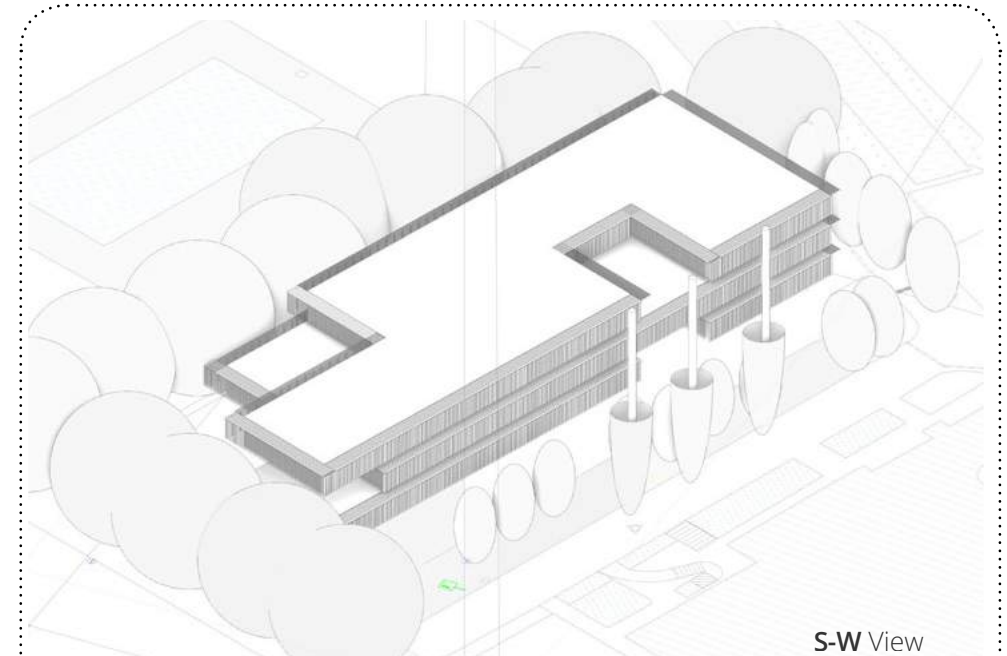
- Parametric Shading (South, East, West)
- No Shading (North)
- Double pane Windows With %65 transmittance
- Window to Wall Ratio of %90

Code **F.5**



- Horizontal and Vertical Louvers (South, East, West)
- No Shading (North)
- Double pane Windows With %65 transmittance
- Window to Wall Ratio of %90

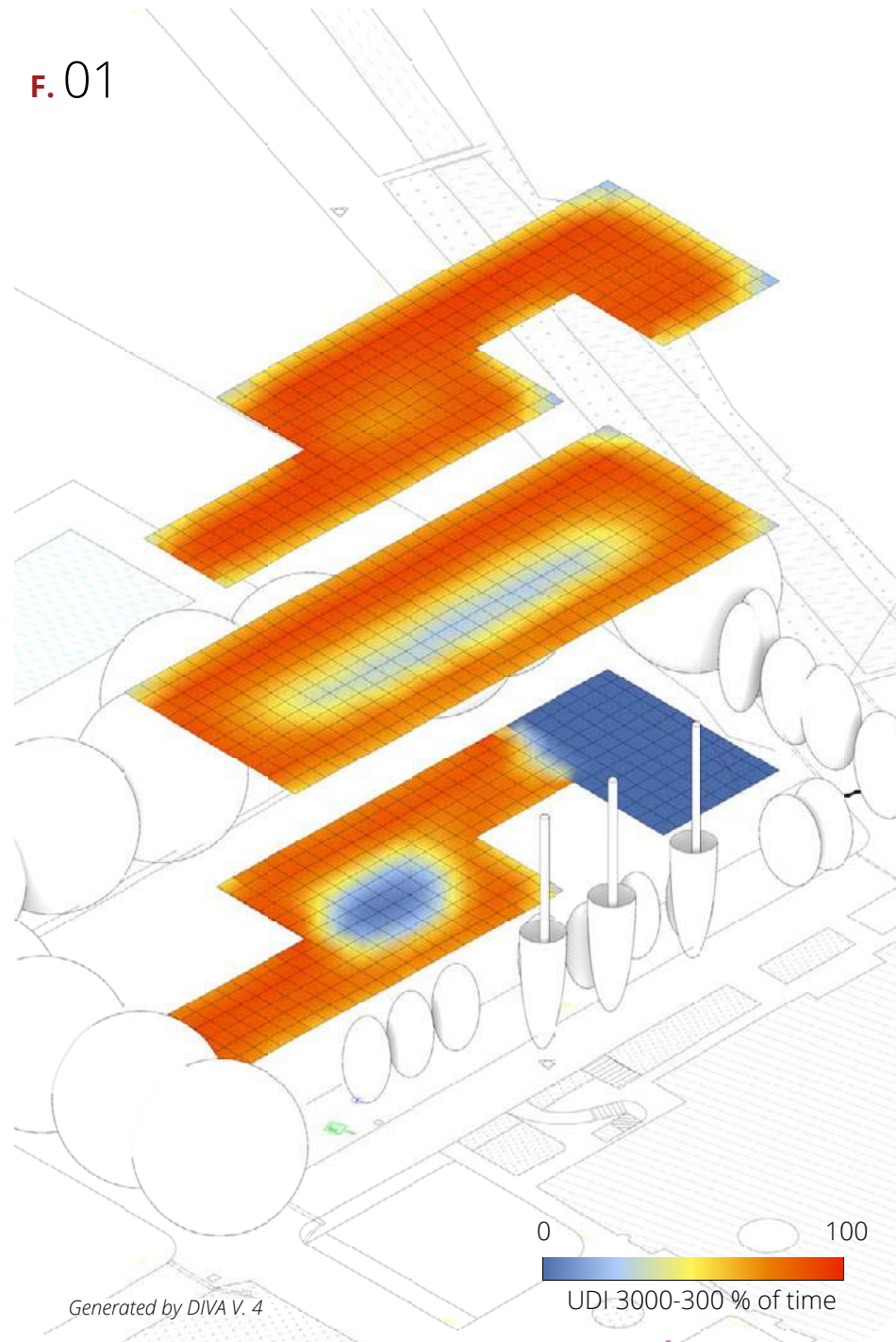
Code **F.6**





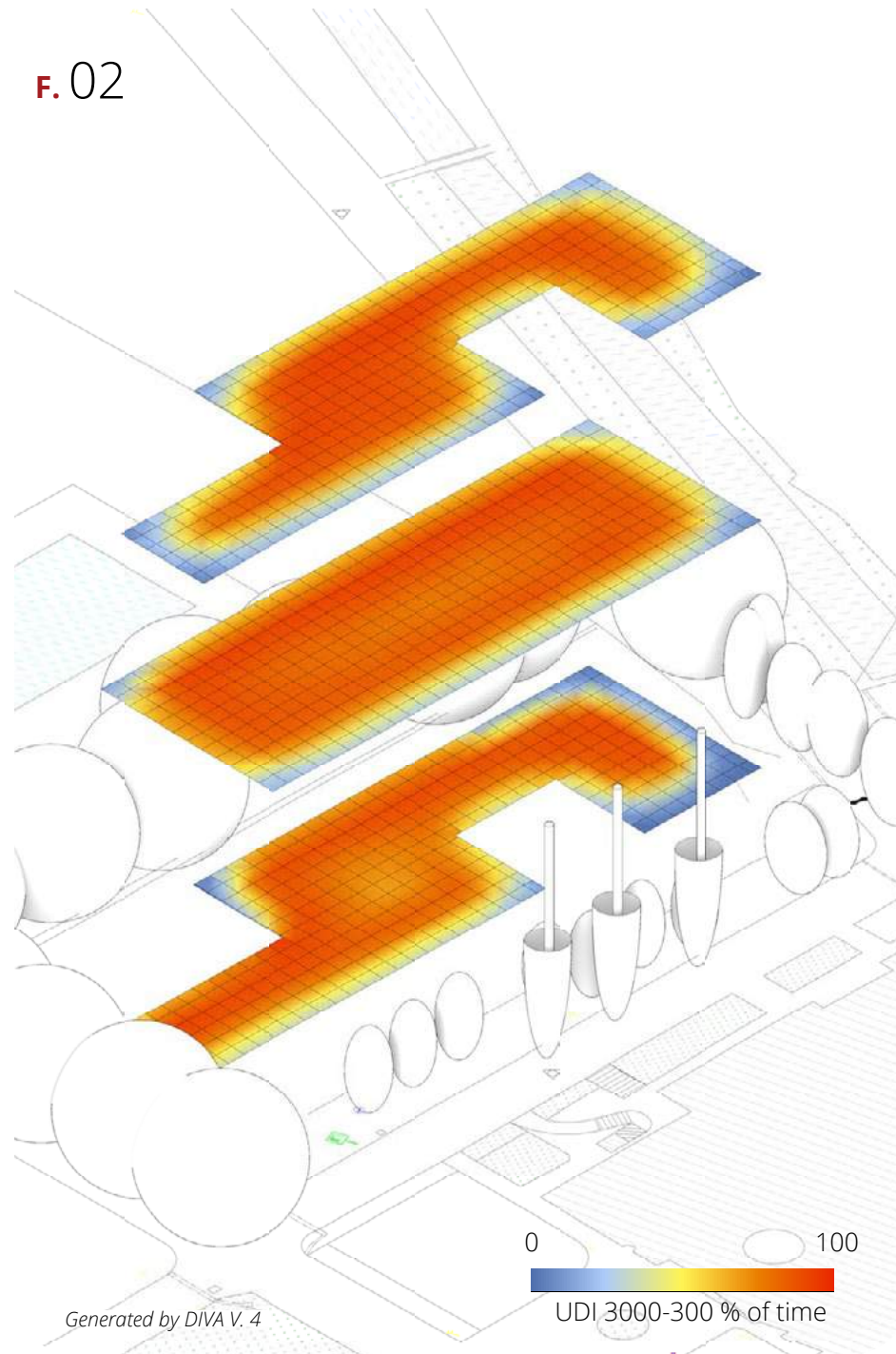
Daylight Analysis

F.01



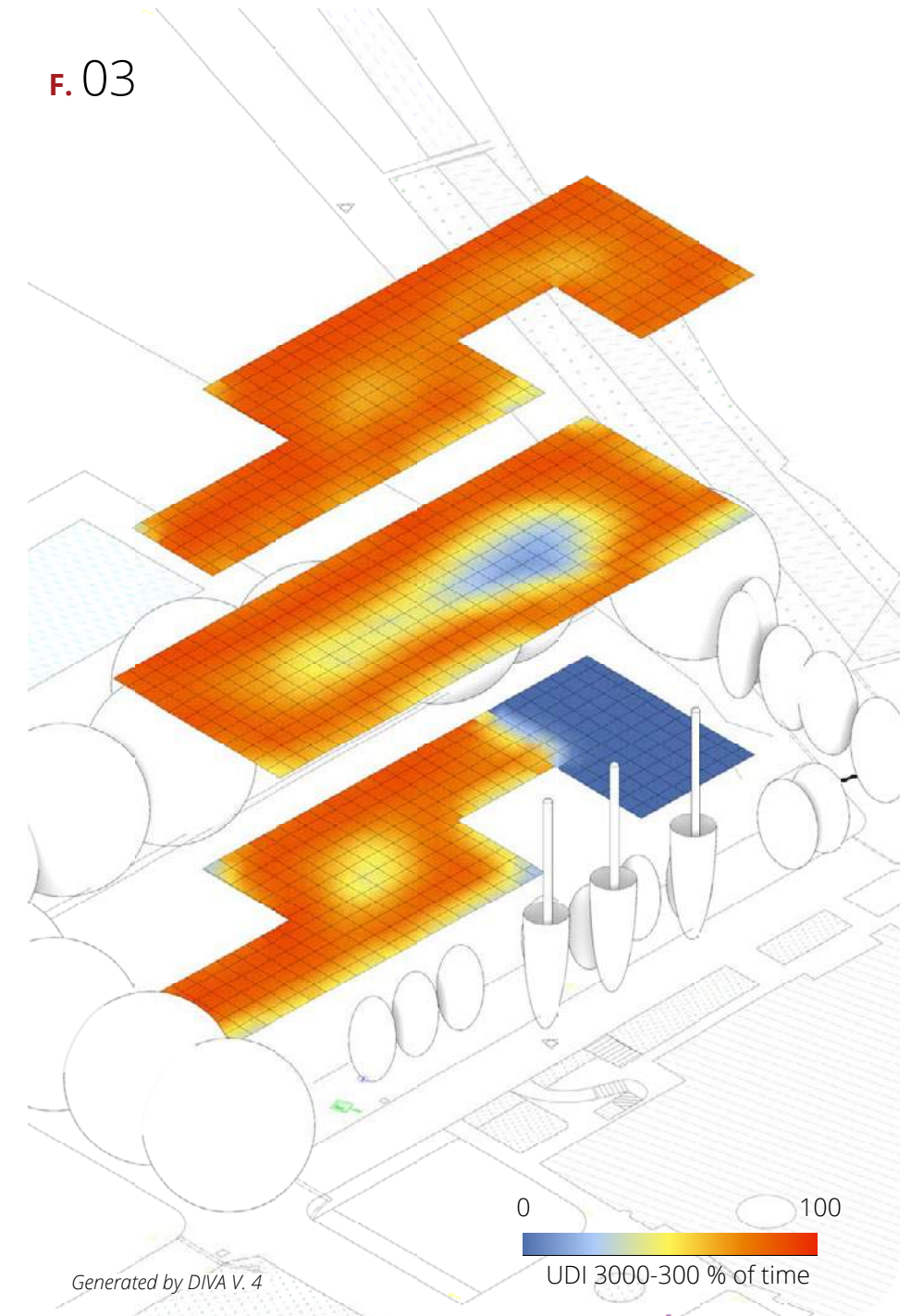
UDI % of Area more than %50 of times Between 3000-300	
Second Floor	76.57
First Floor	83.24
Ground Floor	87

F.02



UDI % of Area more than %50 of times Between 3000-300	
Second Floor	81
First Floor	86
Ground Floor	79

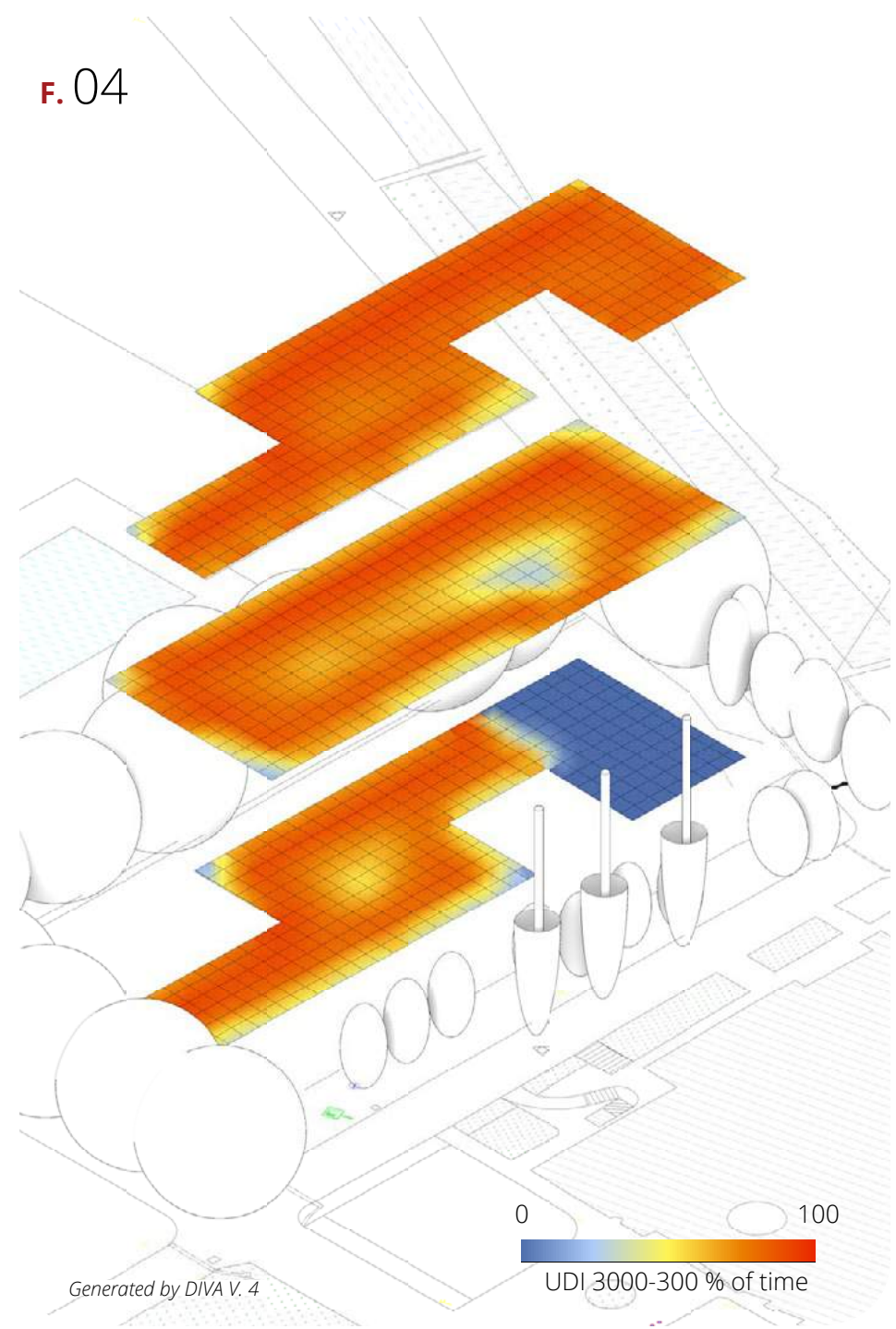
F.03



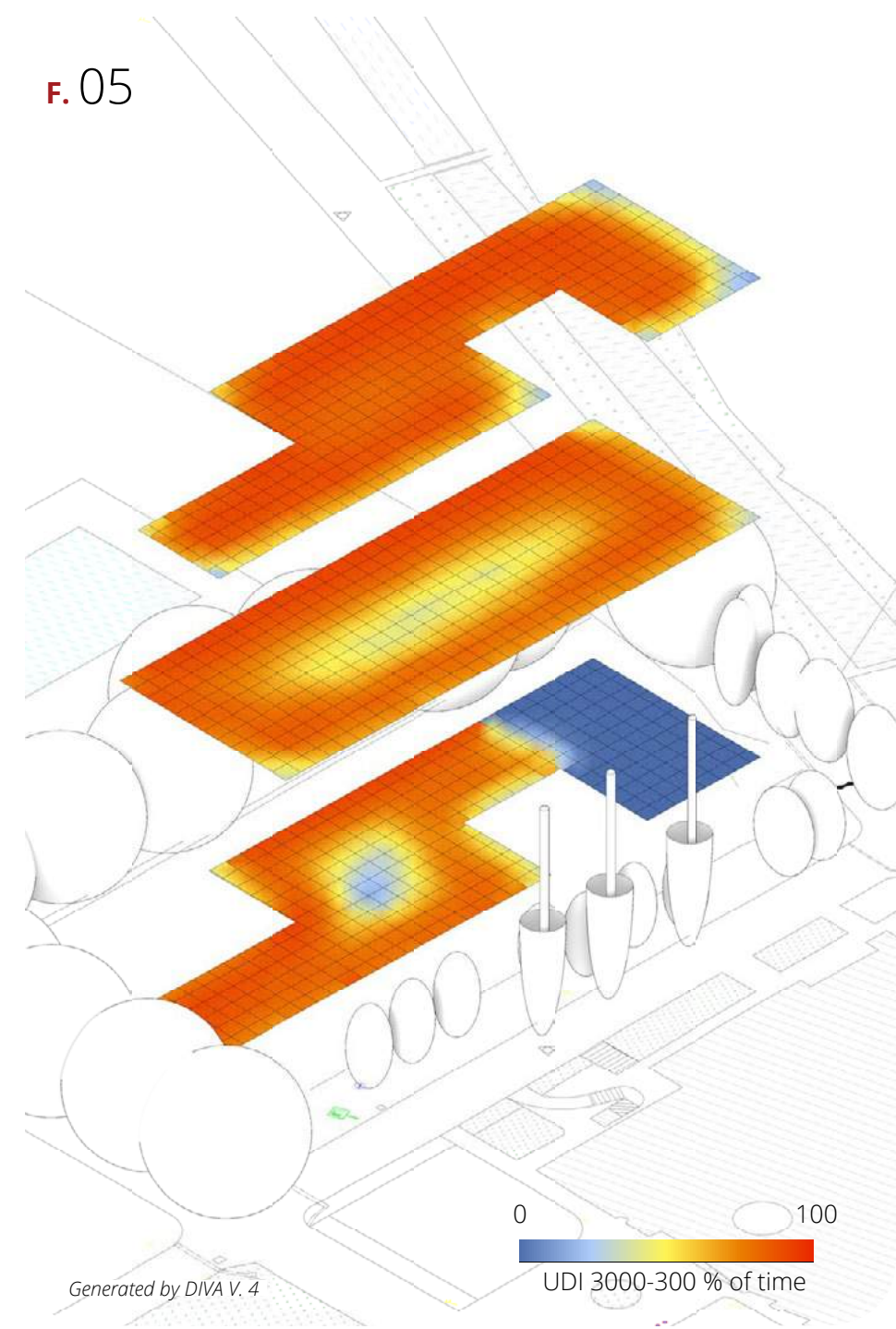
UDI % of Area more than %50 of times Between 3000-300	
Second Floor	78
First Floor	82
Ground Floor	89

Daylight Analysis

F. 01	UDI % of Area more than %50 of times Between 3000-300	
	Second Floor	76.57
	First Floor	83.24
	Ground Floor	87
F. 02	UDI % of Area more than %50 of times Between 3000-300	
	Second Floor	81
	First Floor	86
	Ground Floor	79
F. 03	UDI % of Area more than %50 of times Between 3000-300	
	Second Floor	78
	First Floor	82
	Ground Floor	89
F. 04	UDI % of Area more than %50 of times Between 3000-300	
	Second Floor	78
	First Floor	86
	Ground Floor	89
F. 05	UDI % of Area more than %50 of times Between 3000-300	
	Second Floor	78
	First Floor	87
	Ground Floor	87



UDI % of Area more than %50 of times Between 3000-300	
Second Floor	78
First Floor	86
Ground Floor	89



UDI % of Area more than %50 of times Between 3000-300	
Second Floor	78
First Floor	87
Ground Floor	87

F.01

F.02

F.03

DGP

27.5 %

DGP

31.1 %

DGP

25.9 %

False Color

Glare Spots

View 1

View 1

View 1

DGP

25.8 %

DGP

29.7 %

DGP

25.1 %

False Color

Glare Spots

View 2

View 2

View 2

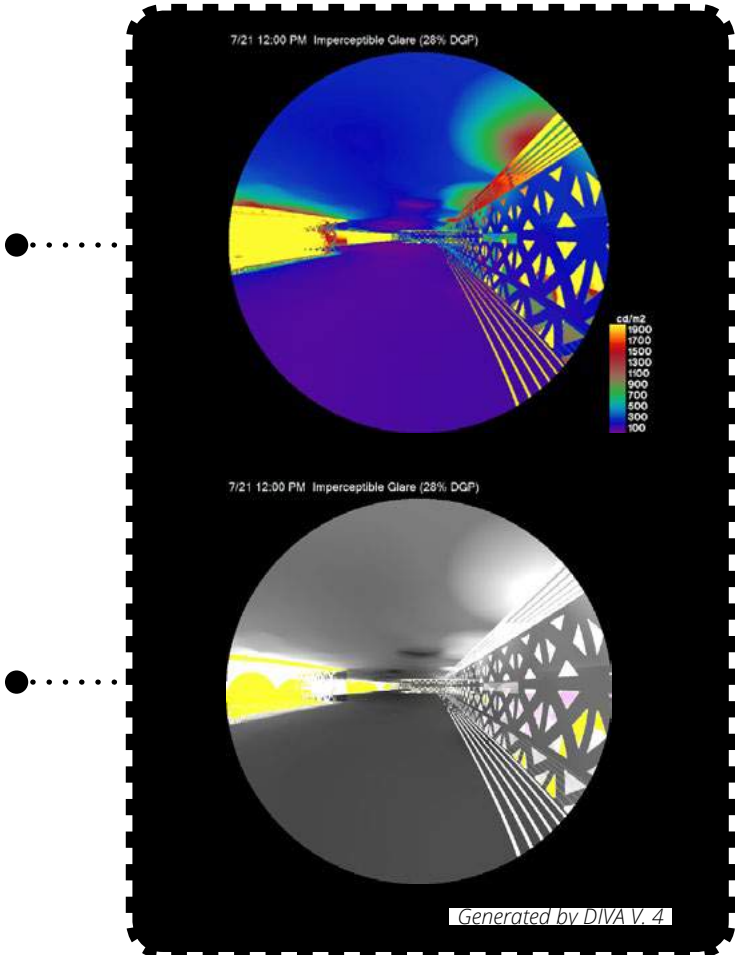
View 1

F.04

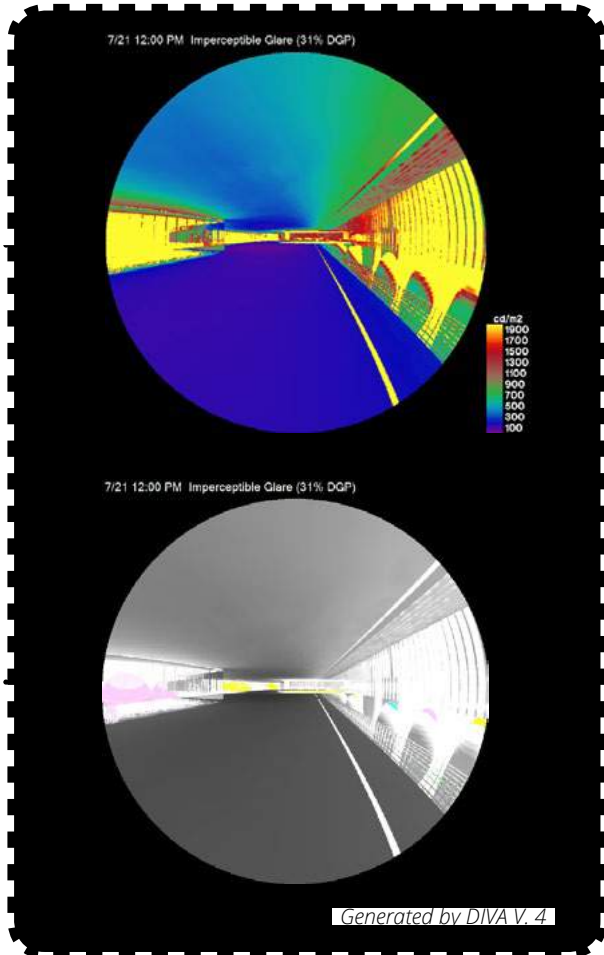
F.05

Glare Analysis

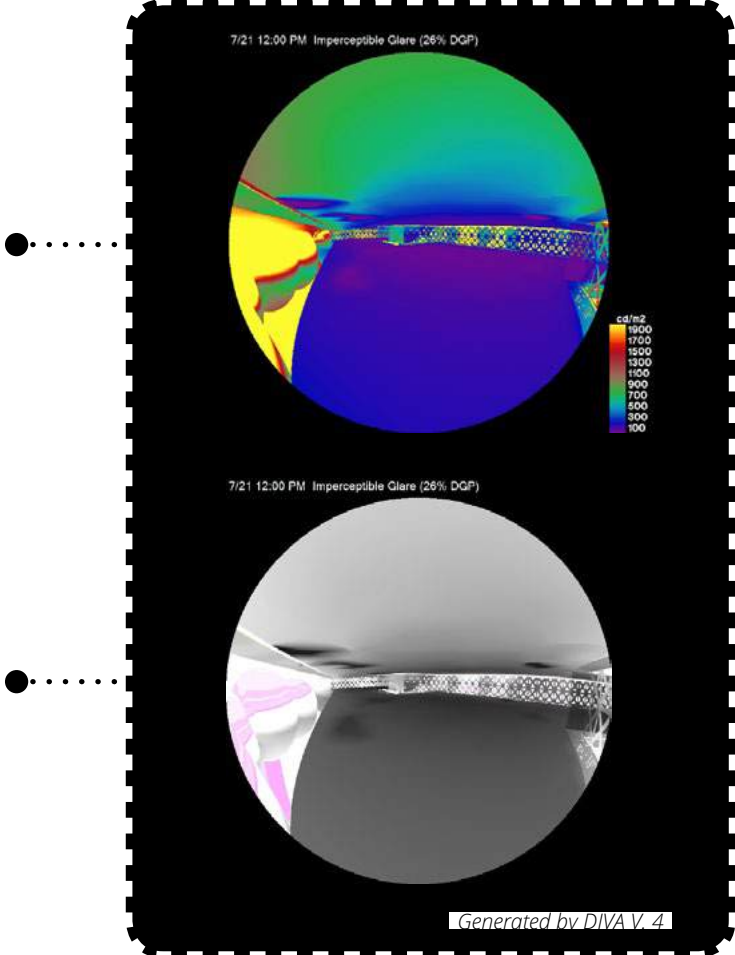
F.01	View 1	
	DGP Index[%] DGR Index	27.5 Imperceptible
	View 2	
	DGP Index[%] DGR Index	25.8 Imperceptible
F.02	View 1	
	DGP Index[%] DGR Index	31.1 Imperceptible
	View 2	
	DGP Index[%] DGR Index	29.7 Imperceptible
F.03	View 1	
	DGP Index[%] DGR Index	25.9 Imperceptible
	View 2	
	DGP Index[%] DGR Index	25.1 Imperceptible
F.04	View 1	
	DGP Index[%] DGR Index	27.5 Imperceptible
	View 2	
	DGP Index[%] DGR Index	26 Imperceptible
F.05	View 1	
	DGP Index[%] DGR Index	31.3 Imperceptible
	View 2	
	DGP Index[%] DGR Index	28.6 Imperceptible



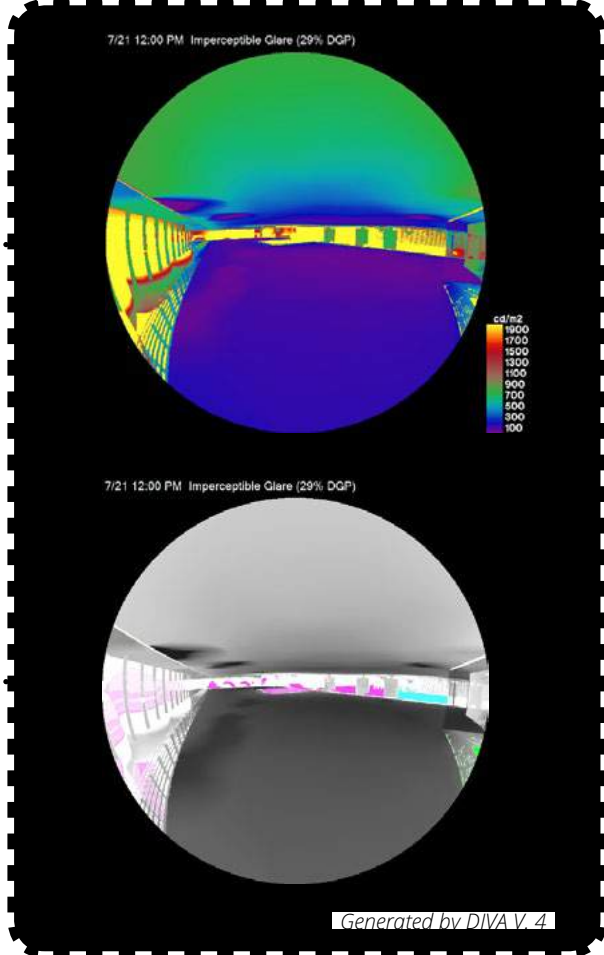
DGP
27.5 %



DGP
31.3 %



DGP
26.0 %



DGP
28.6 %

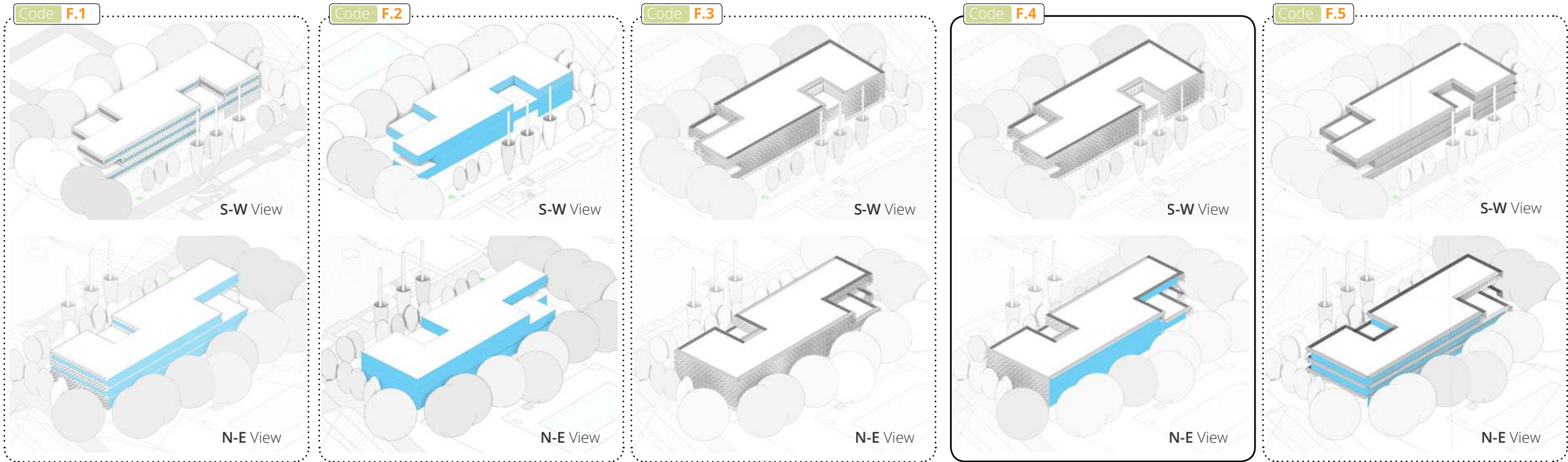
False Color

Glare Spots

False Color

Glare Spots

Facade Alternatives Analysis Results(Step1)

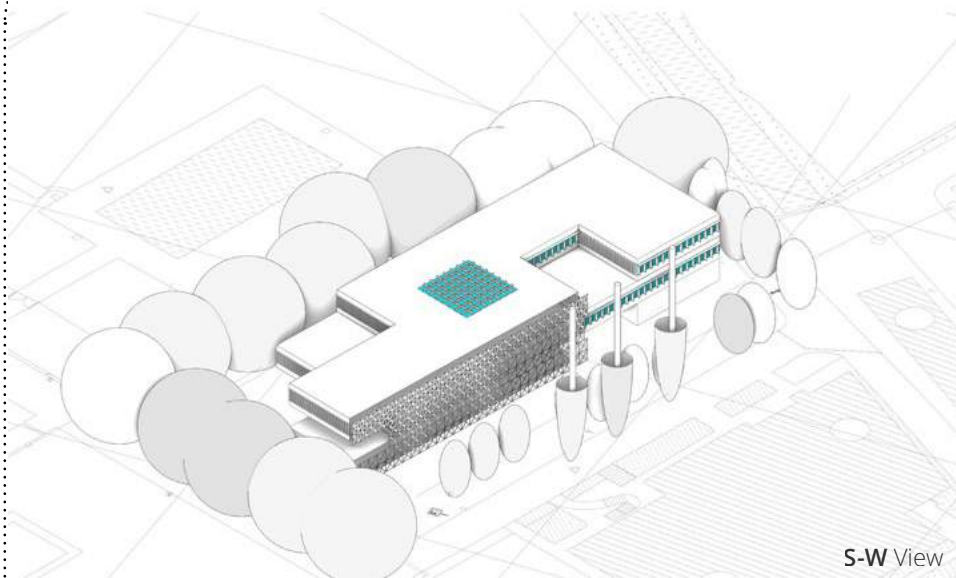


	Key Parameters	Detailed Weightings	Values	Scores (Unweighted)	Overall Scores	Values	Scores (Unweighted)	Overall Scores	Values	Scores (Unweighted)	Overall Scores	Values	Scores (Unweighted)	Overall Scores
Thermal Comfort	Comfortable [% of time](Hot Week)	50/100	0.3	0.0	324	3.30	87	631	0.5	100	763	0.6	54	520
	Comfortable [% of time](Cold Week)	50/100	13.0	0.0 x 1/13		7.00	0.0 x 1/13		18.0	0.0 x 1/13		15	0.0 x 1/13	
Daylighting	Daylit	80/100	76.5	0.0	310	74.91	23	310	79.35	171	631	83.24	400	763
	Underlit		18.5	0.0		0.00	0.0		14.81	0.0		10.09	0.0	
	Overlit	20/100	24.2	0.0 x 4/13		57.13	0.0 x 4/13		29.72	0.0 x 4/13		34.44	0.0 x 4/13	
Glare	DGP View 1 [%]	50/100	27.6	153	324	31.1	0	310	25.9	200	631	27.5	149	763
	DGP View 2 [%]	50/100	25.8	153 x 2/13		29.7	0 x 2/13		25.1	200 x 2/13		26.0	149 x 2/13	
Functional & Aesthetical	Cost Improvement Potential	25/100	80	109	310	70	0	310	50	0	631	50	0	763
	Aesthetics	50/100	40	0.0 x 4/13		30	0.0 x 4/13		80	0.0 x 4/13		80	0.0 x 4/13	
						40	0.0 x 4/13		80	0.0 x 4/13		80	0.0 x 4/13	
Ventilation Potential	Average Window Height		40	80 x 2/13		100	200 x 2/13		80	160 x 2/13		80	160 x 2/13	

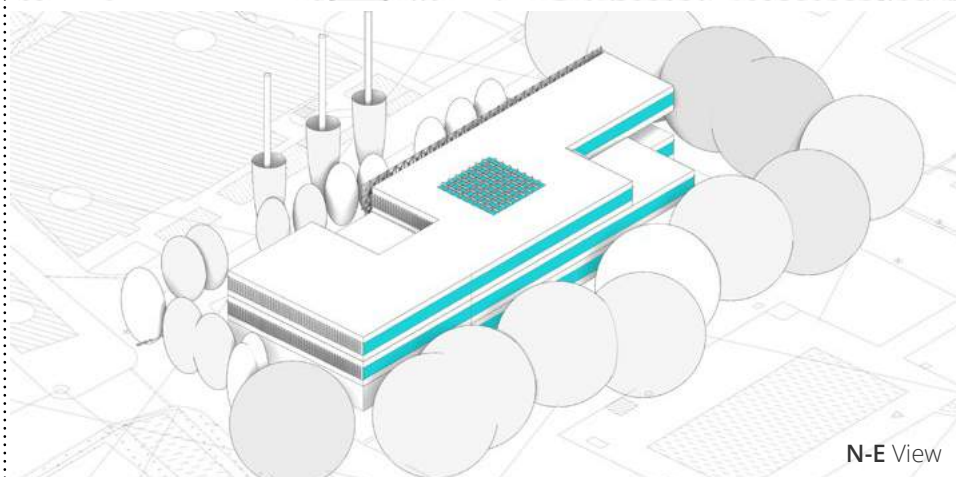


Facade Design Alternatives (Step2)

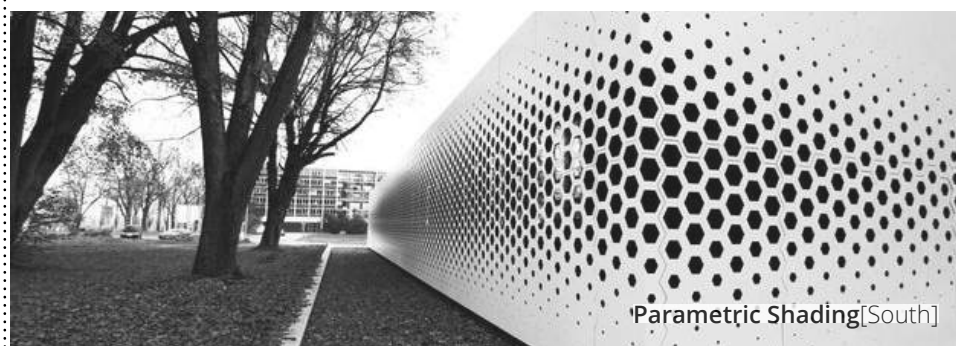
Code P.1



S-W View



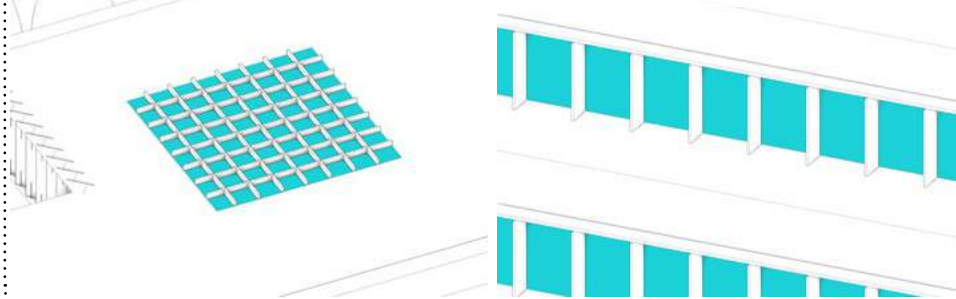
N-E View



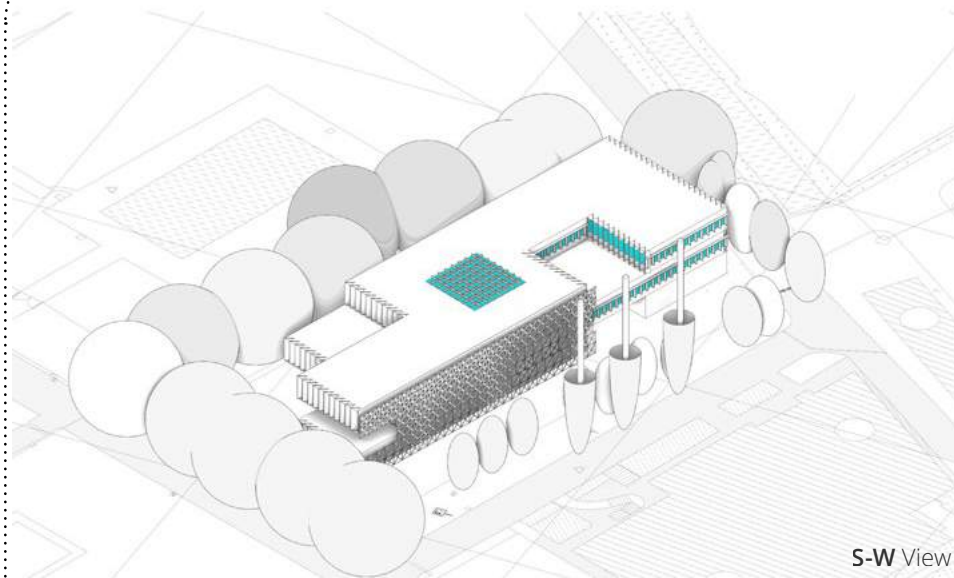
Parametric Shading[South]

Skylight

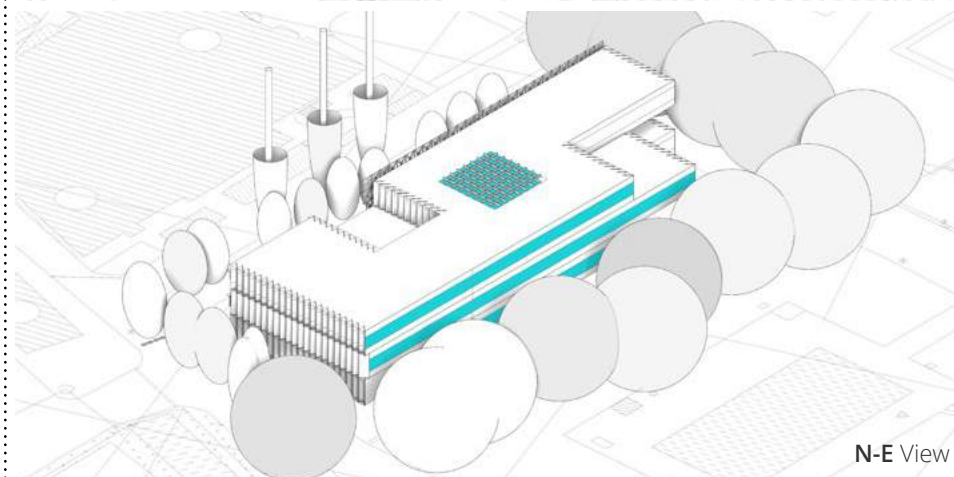
Vertical+Horizontal Shadings[South]



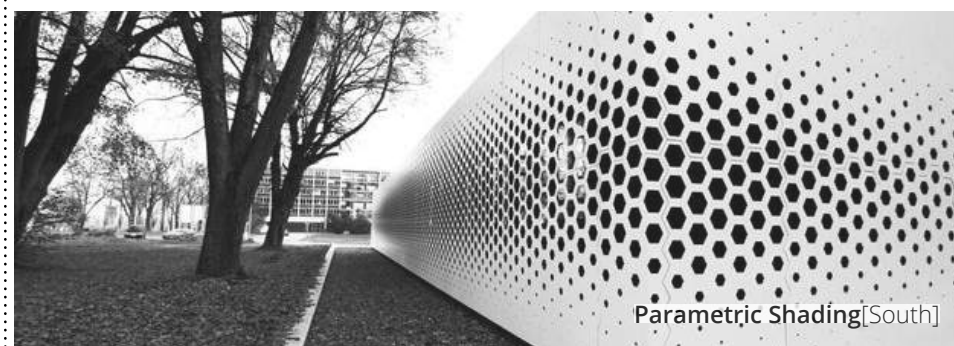
Code P.2



S-W View



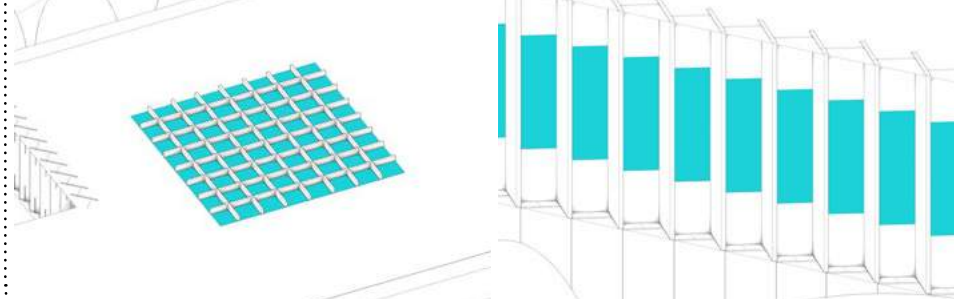
N-E View



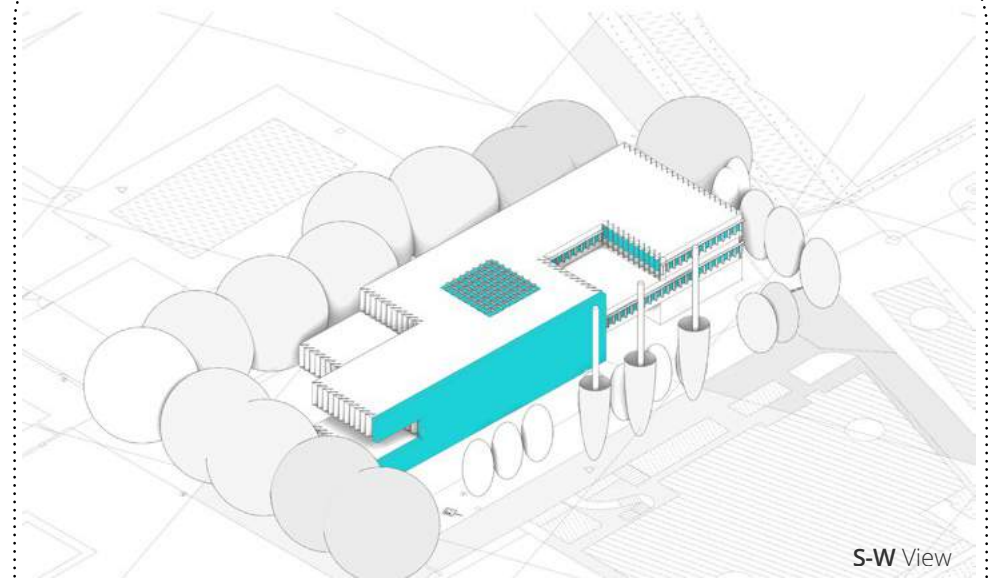
Parametric Shading[South]

Skylight

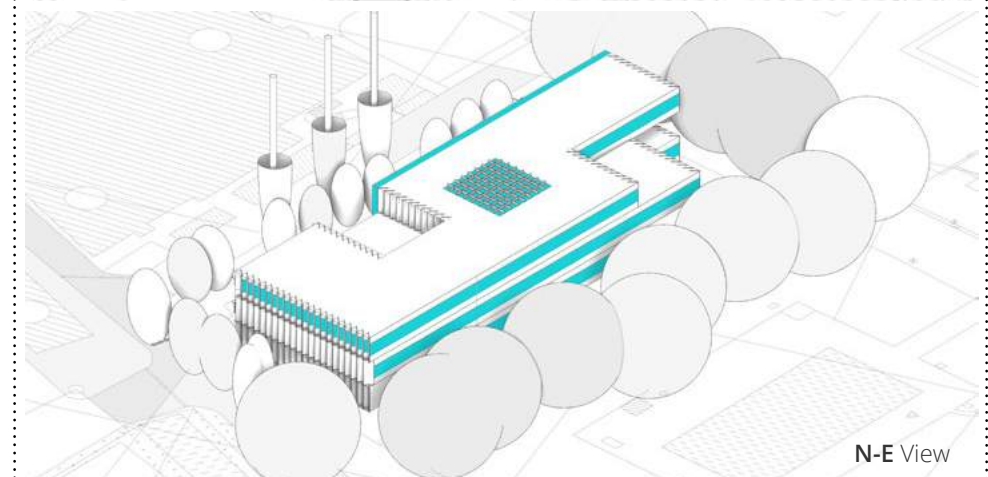
East/West side



Code P.3



S-W View



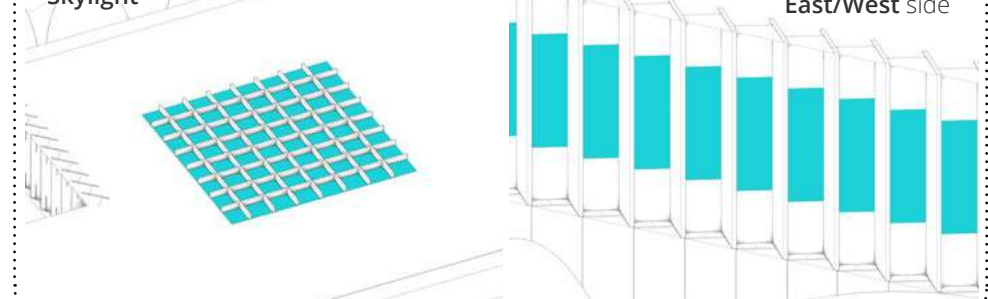
N-E View



Double-Skin Facade[South]

Skylight

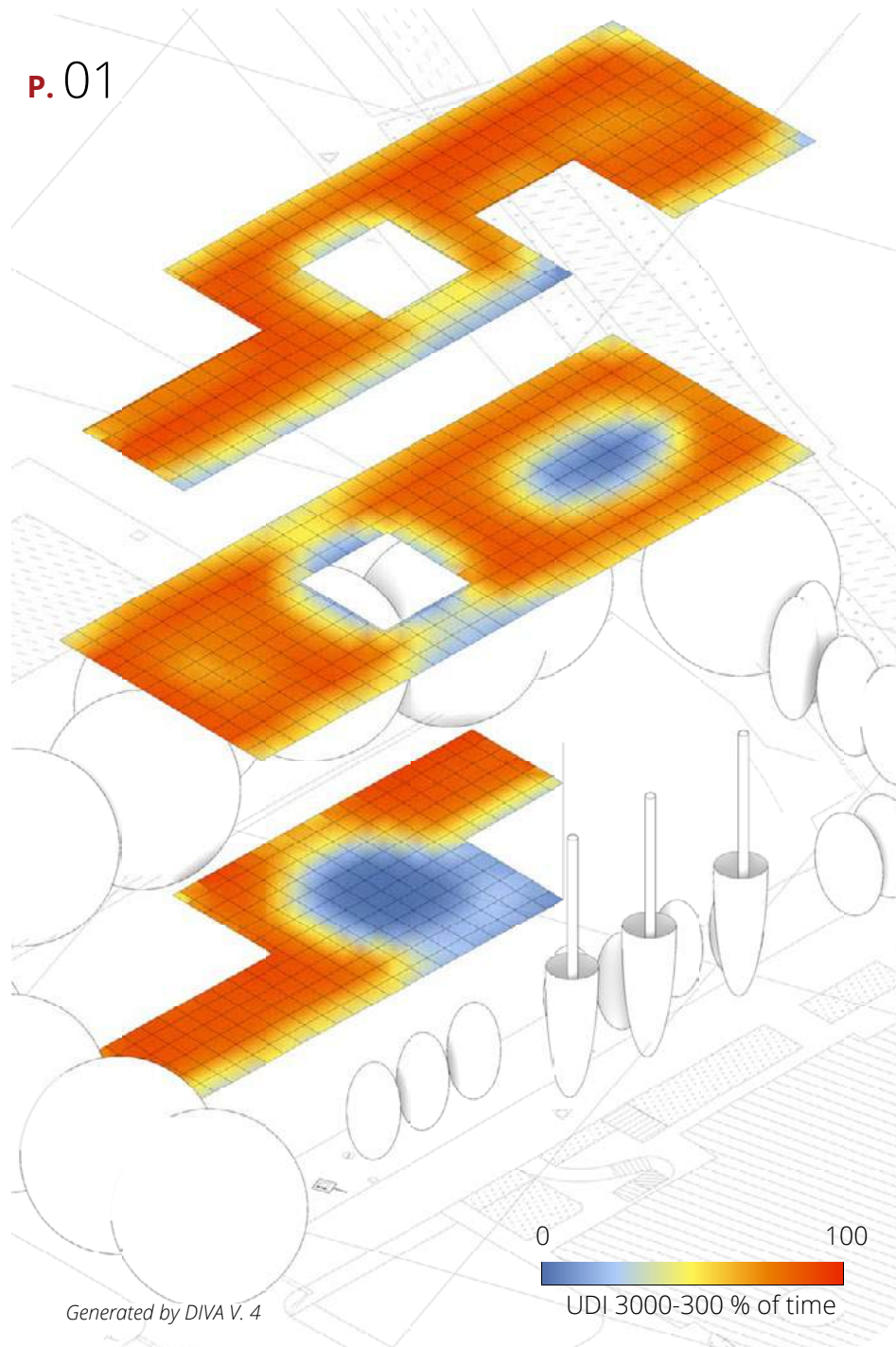
East/West side





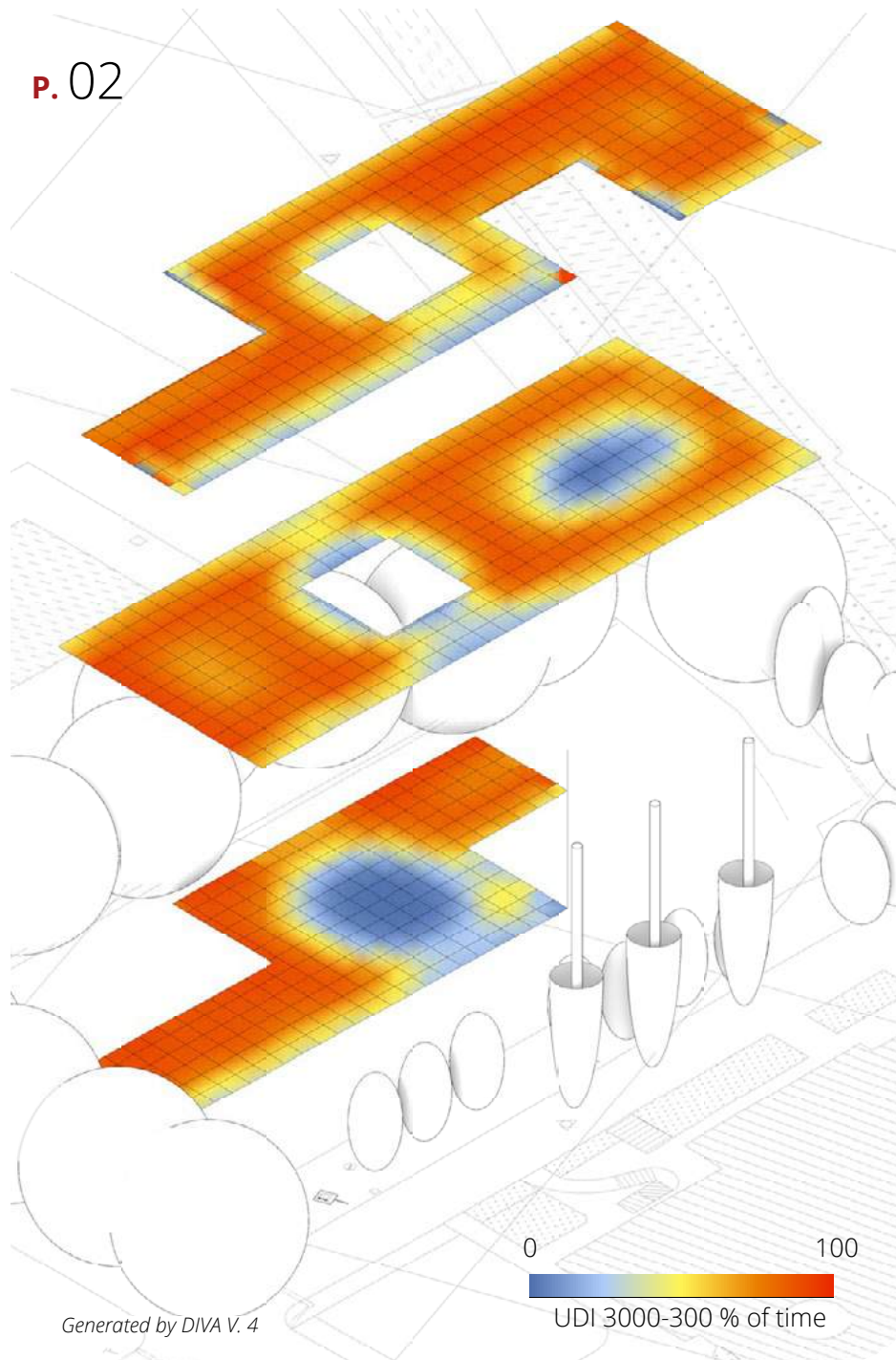
Daylight Analysis

P.01



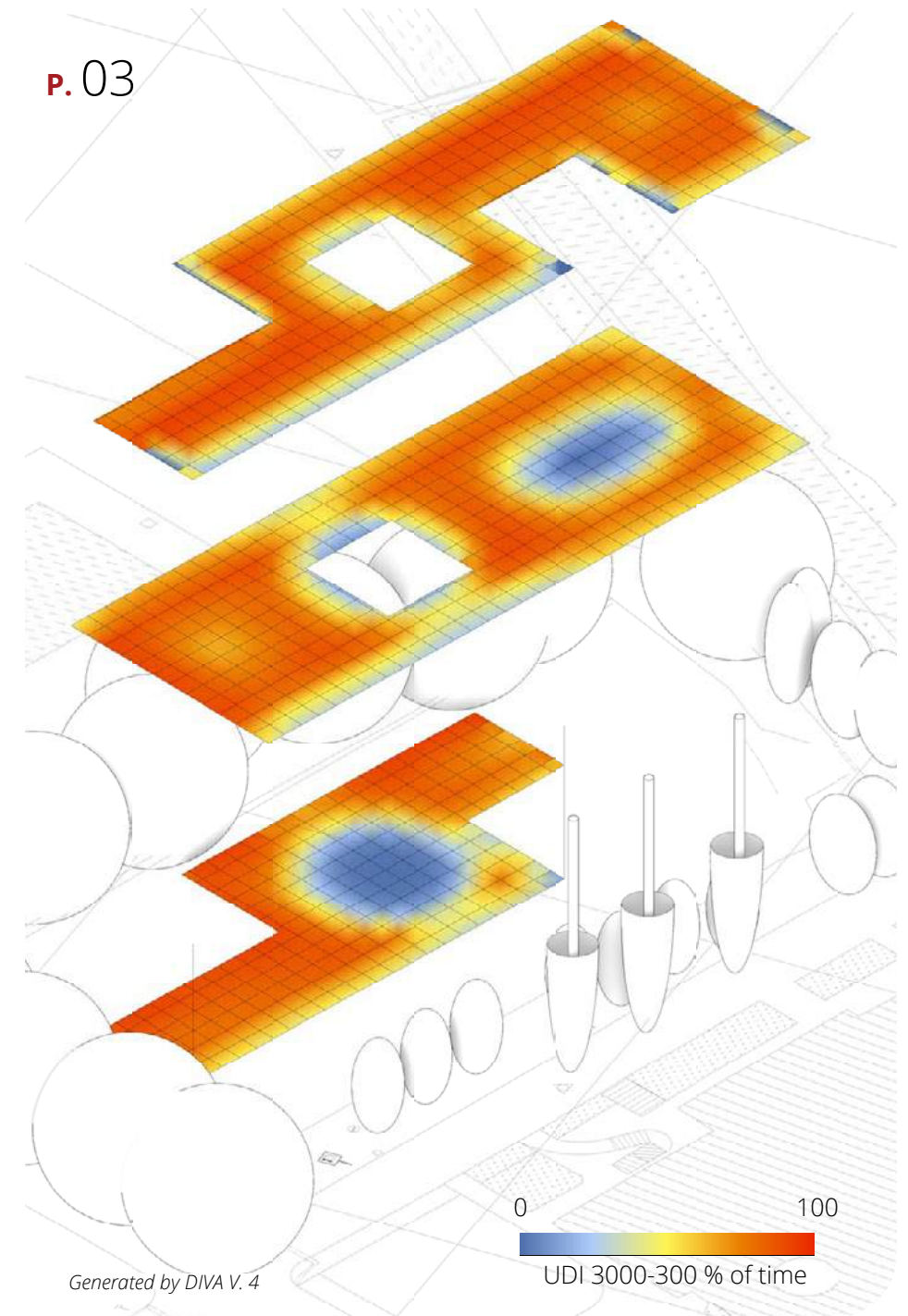
UDI % of Area more than %50 of times Between 3000-300	
Second Floor	59.58
First Floor	72.83
Ground Floor	81.87

P.02



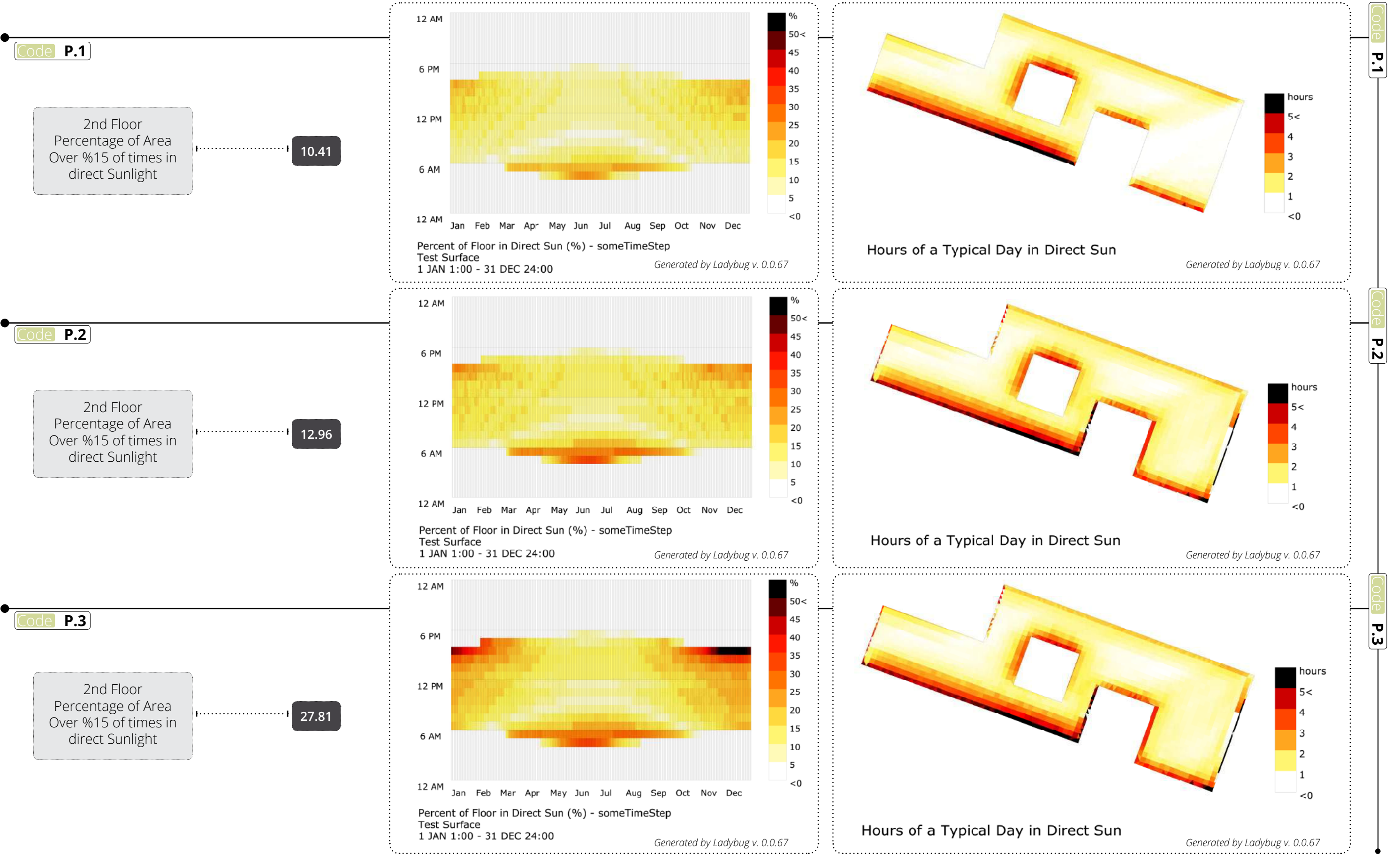
UDI % of Area more than %50 of times Between 3000-300	
Second Floor	62.50
First Floor	72.01
Ground Floor	79.60

P.03

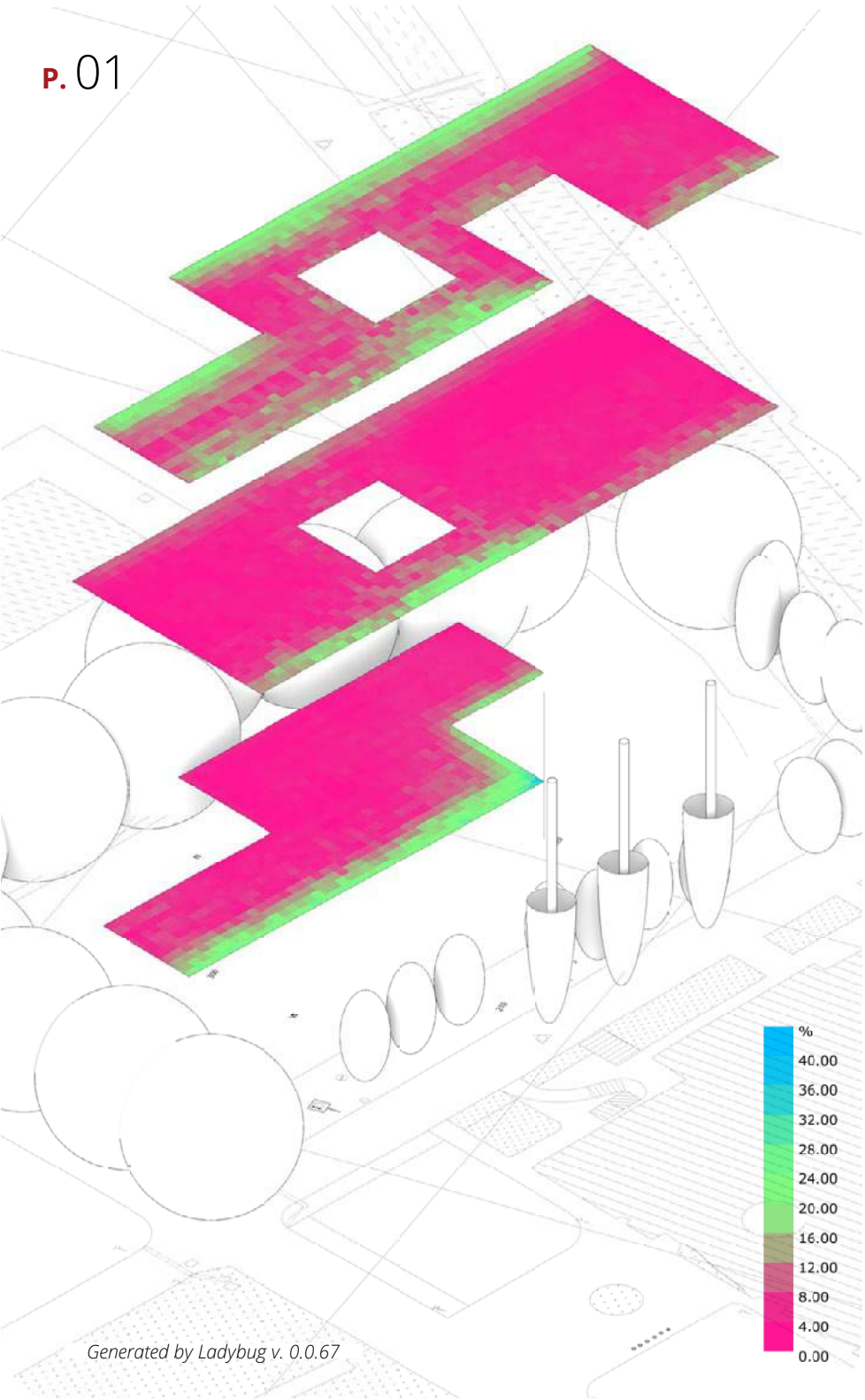


UDI % of Area more than %50 of times Between 3000-300	
Second Floor	68.75
First Floor	73.10
Ground Floor	80.17

 Glare Potential (Eye Level)

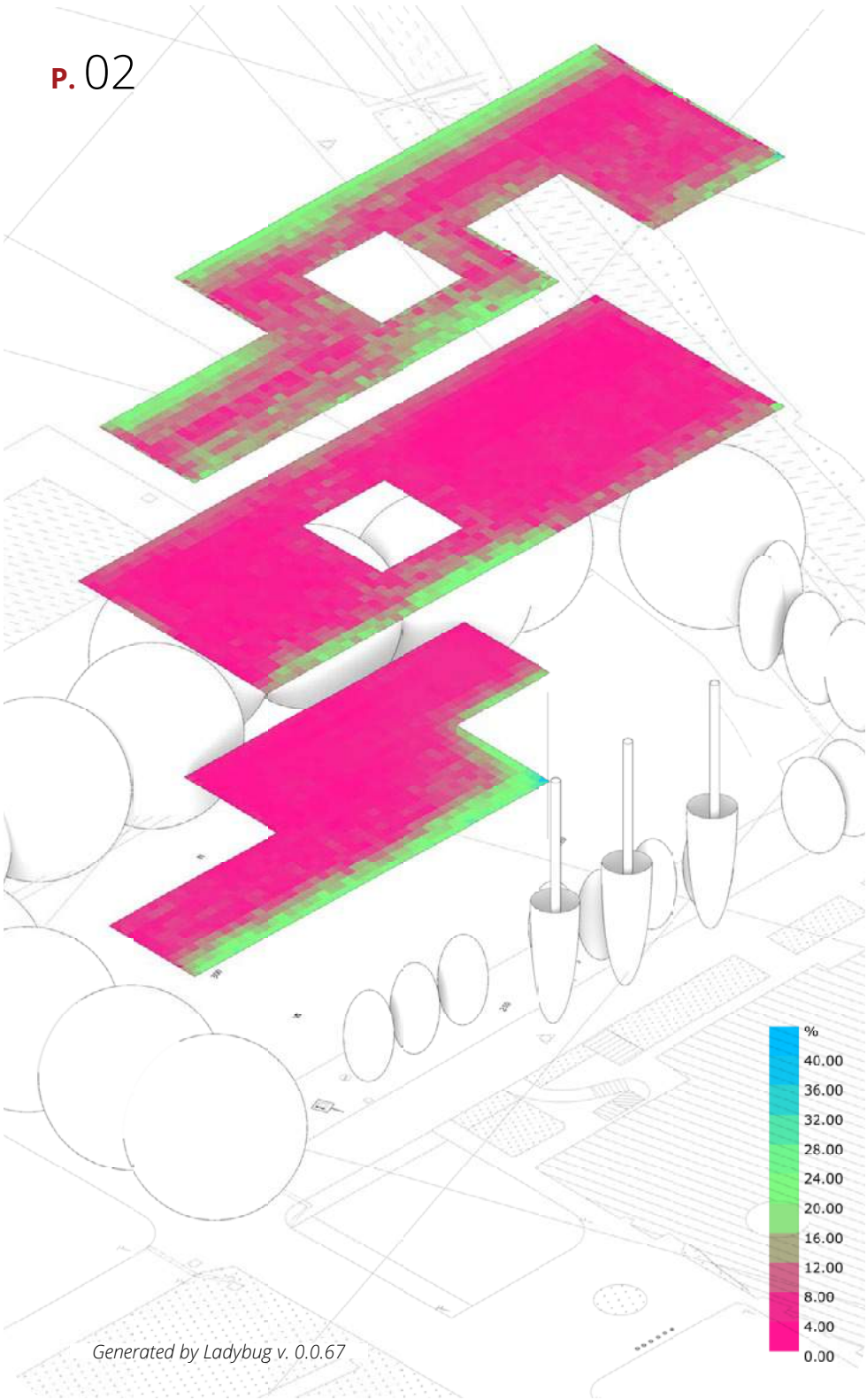


Horizontal 60° Cone of Vision Analysis (Eye Level)



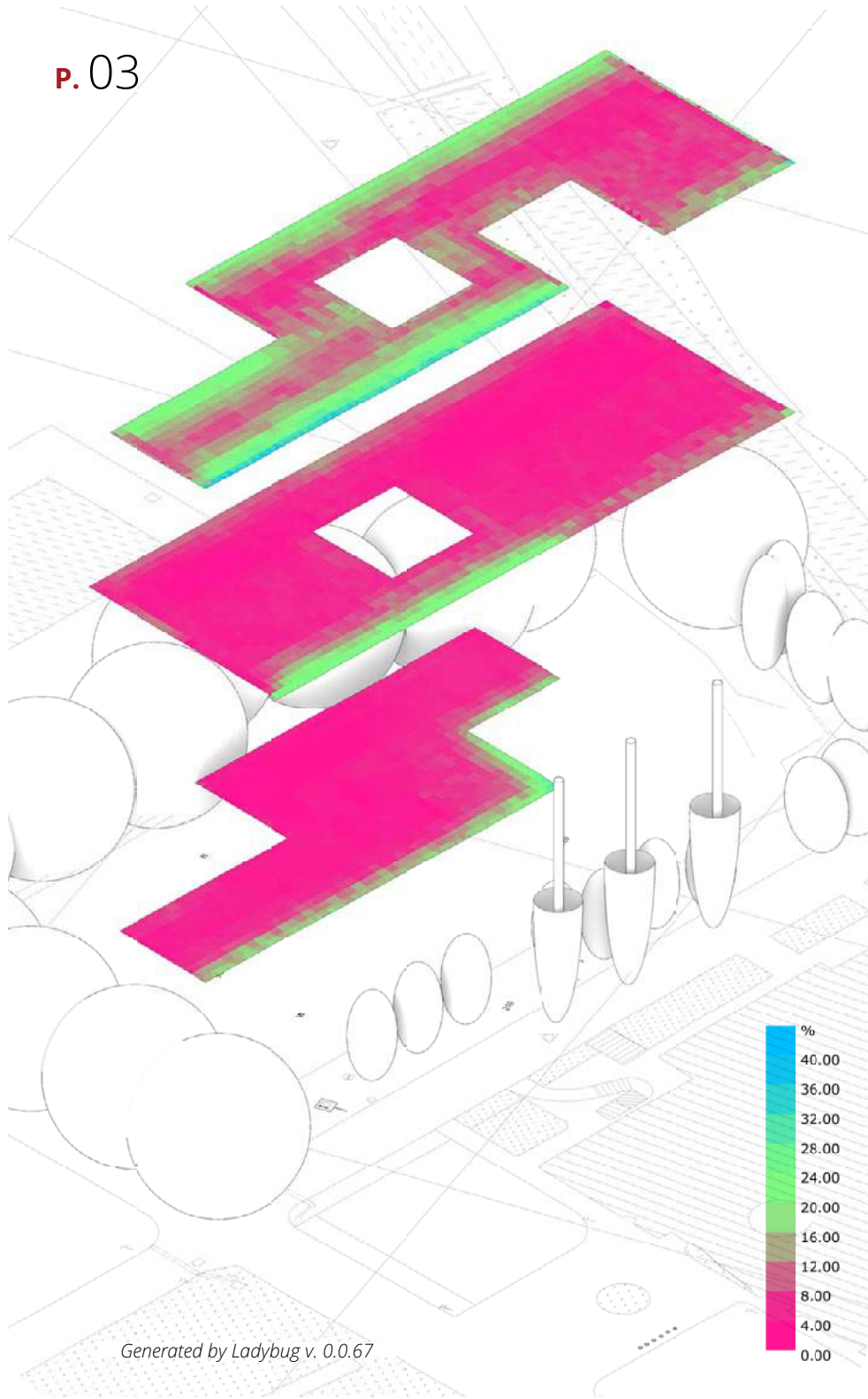
%of Area more than %15 of outdoor space is visible

8.15



%of Area more than %15 of outdoor space is visible

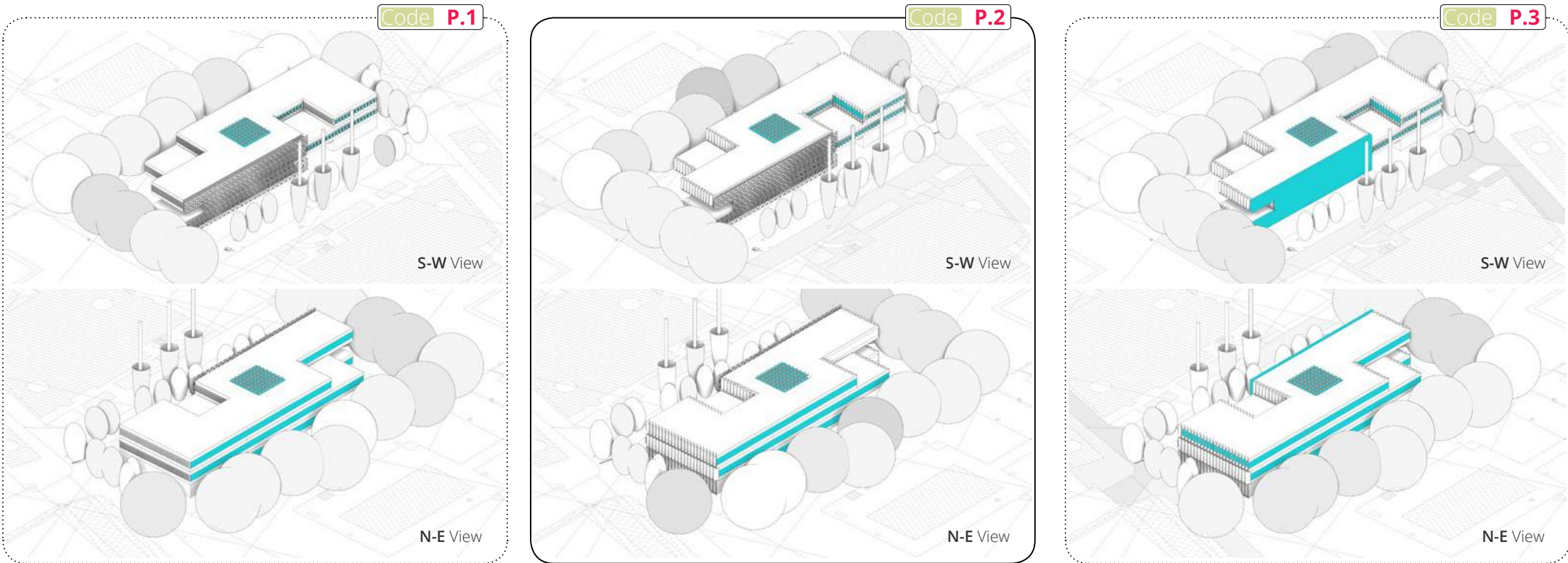
9.22



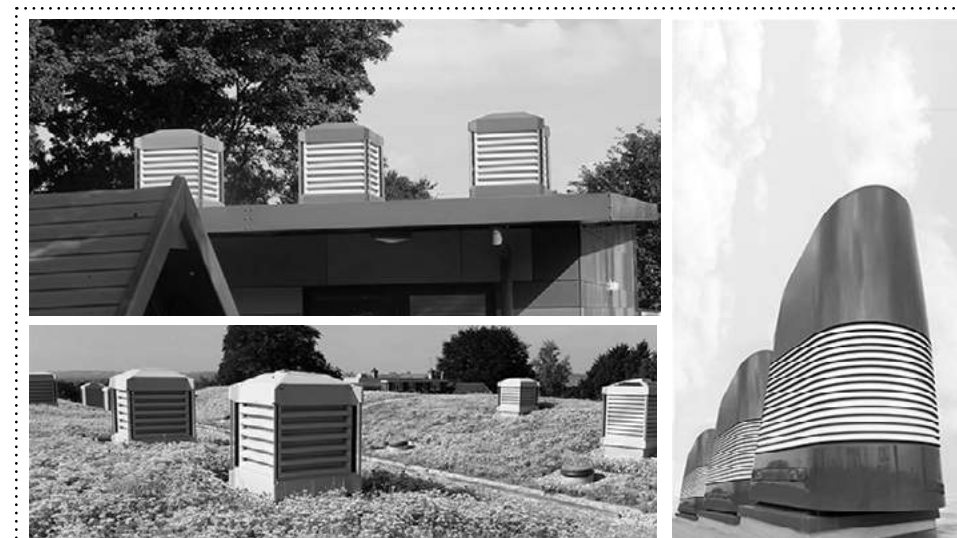
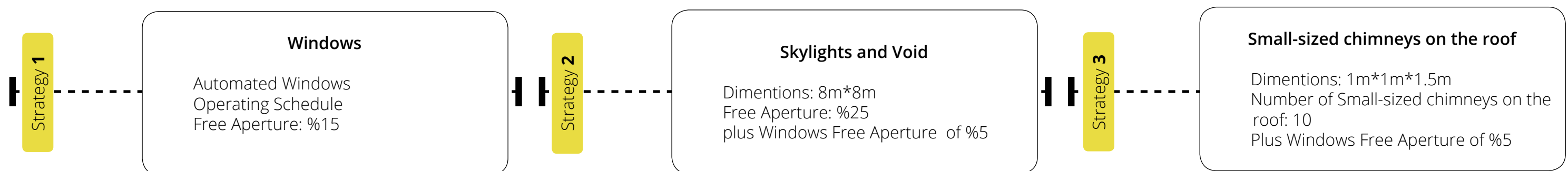
%of Area more than %15 of outdoor space is visible

12.18

Facade Alternatives
Analysis Results(Step2)



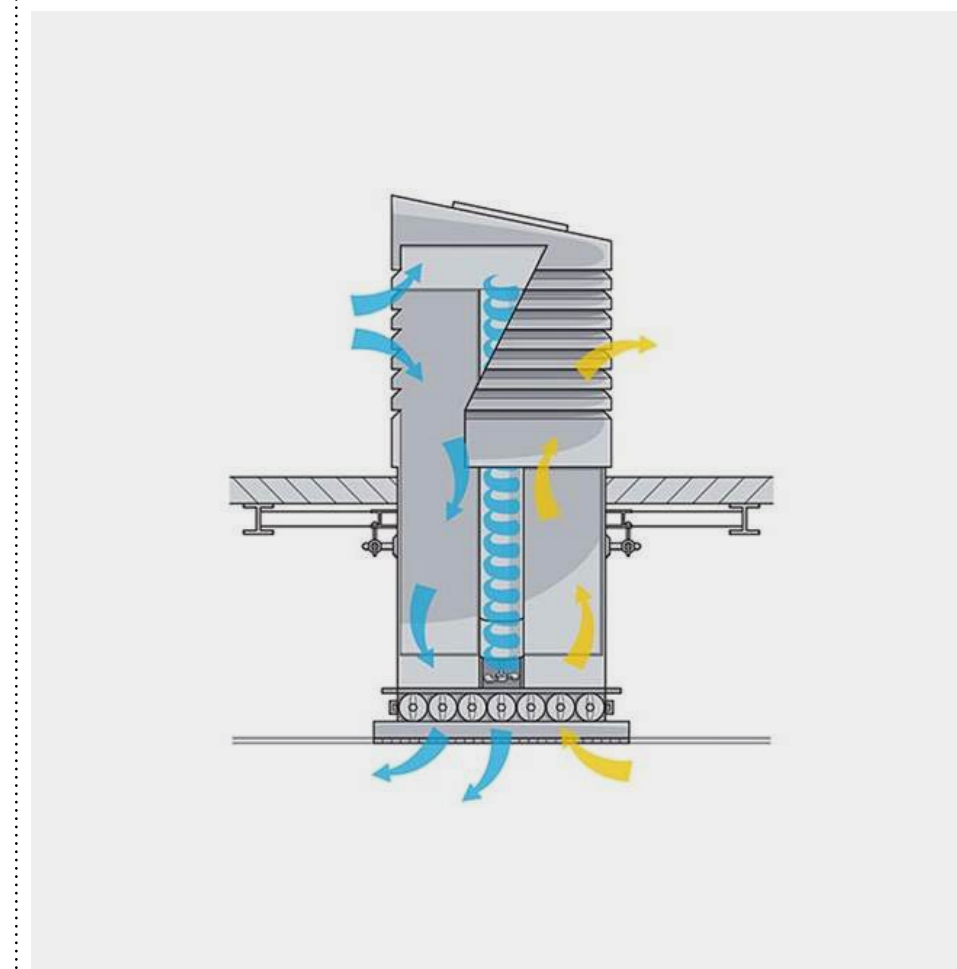
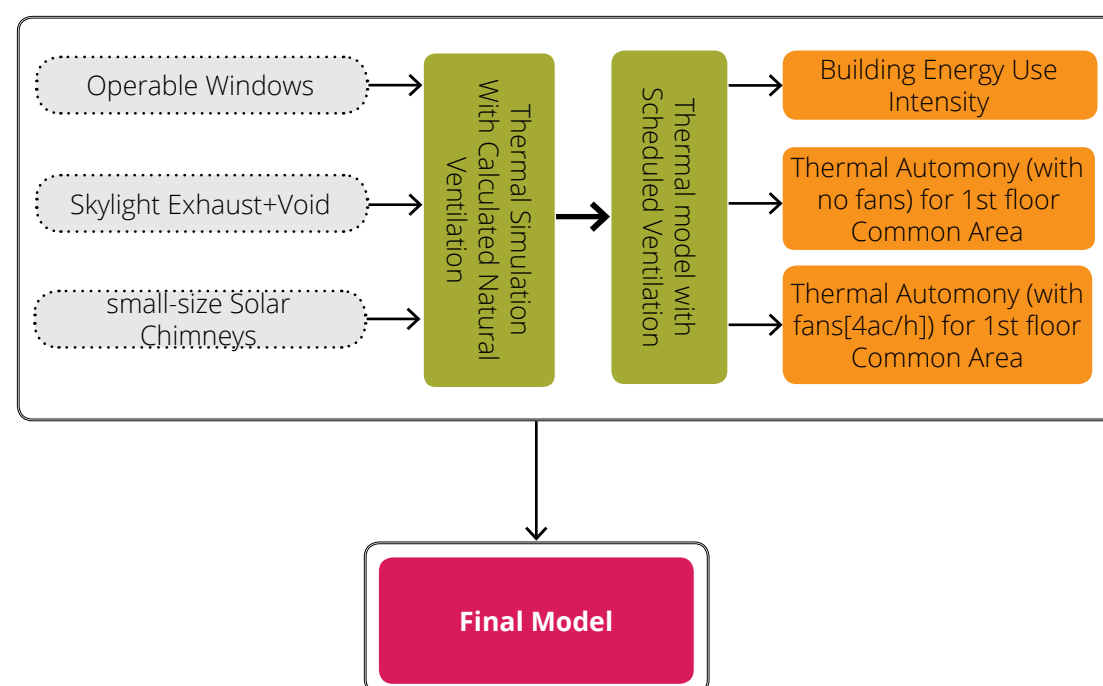
Key Parameters		Detailed Weightings	Values/ Raw Scores	Scores (Unweighted)	Overall Weighted Scores	Values/ Raw Scores	Scores (Unweighted)	Overall Weighted Scores	Values/ Raw Scores	Scores (Unweighted)	Overall Weighted Scores
Thermal Autonomy	Overheat hours() [% of Occ times>28]		49.9			47.4			49.6		
	Underheat hours [% of Occ times<21]		5.7	44		2.4	50		1.8	48	
	Comfortable hours [% of Occ times]	100/100	44.3			50			48.4		
				x4/12			x4/12	1033		x4/12	
Daylighting	UDI Daylit [300-3000]	100/100	75.3			74.4			77.3		
	UDI Underlit [300>]		2.8	75		2.9	74		3.8	77	
	UDI Overlit [3000<]		54.6			57			51.8		
Glare Potential	2nd Floor Percentage of Area Over %15 of times in direct Sunlight	100/100	10.41	89		12.96	87		27.81	72	
				x1/12			x1/12			x1/12	
Cone of Vision Analysis	%of Area more than %15 of outdoor space is visible	100/100	8.15	65		9.22	73		12.18	97	
Functional & Aesthetical	Cost	50/100	80			50			0		
	Aesthetics	50/100	0	80		100	150		80		
				x3/12			x3/12			x3/12	



Modelling Key Assumptions- Step2

Thermal	Infiltration Constant Rate(Scheduled method): 1 ACH Infiltration Crack Template: Medium
HVAC	Heating: Off Cooling: Off Mechanical Ventilation: Off Natural Ventilation: On Natural Ventilation Method: Calculated Occupancy density[people/s.q.m.]: 0.5
Constructions	WWR-South: %60 ,%30 WWR-North: %50 ,%30 WWR-East/West: %25 Wdows U-value[W/m-2K]: 1.96 WdowsSHGC[%]: 0.691 WdowsDirect Solar Transmission[%]: 0.624 Wdows Light Transmission[%]: 65 External Walls U-Value[W/m-2K]: 0.95 Roofs U-Value[W/m-2K]: 0.5

Natural Ventilation Strategy Selection Process



Base Case

No Natural Ventilation Strategies

% of Occupied Hours

Overheat[>28]69.7

Underheat[<21]0.5

Comfortable29.7

Weighted Overheat Hours7056

EUI [kWh/m2]

231.4

6

4

2

0

JANFEBMARAPRMAYJUNJULYAUGSEPOCTNOVDEC

Windows

Automated Windows Operating Schedule Free Aperture: %15

% of Occupied Hours

Overheat[>28]41.42

Underheat[<21]18.33

Comfortable40.25

Weighted Overheat Hours8470

EUI [kWh/m2]

181.0

6

4

2

0

JANFEBMARAPRMAYJUNJULYAUGSEPOCTNOVDEC

Skylights and Void

Dimentions: 8m*8m Free Aperture: %25 plus Windows Free Aperture of %5

% of Occupied Hours

Overheat[>28]41.39

Underheat[<21]18.92

Comfortable39.70

Weighted Overheat Hours9519

EUI [kWh/m2]

182.8

6

4

2

0

JANFEBMARAPRMAYJUNJULYAUGSEPOCTNOVDEC

Small-sized chimneys on the roof

Dimentions: 1m*1m*1.5m Number of Small-sized chimneys on the roof: 10 Plus Windows Free Aperture of %5

% of Occupied Hours

Overheat[>28]41.1

Underheat[<21]19.99

Comfortable38.90

Weighted Overheat Hours9824

EUI [kWh/m2]

185.5

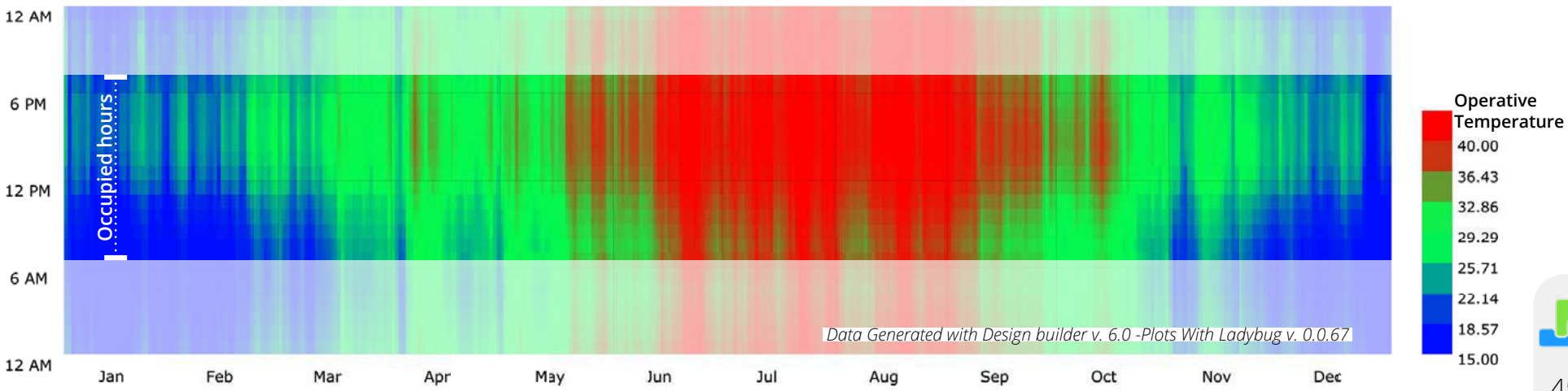
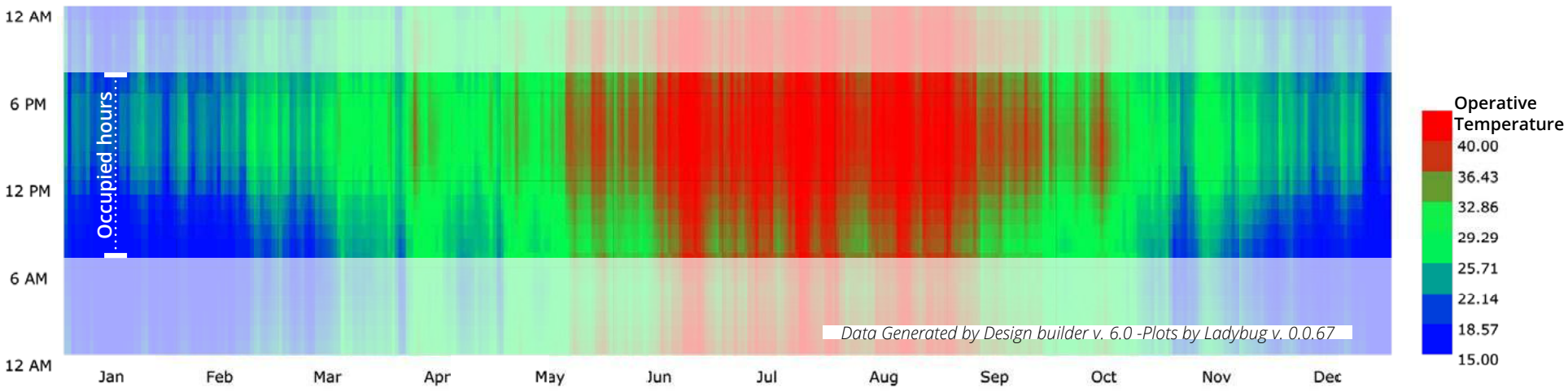
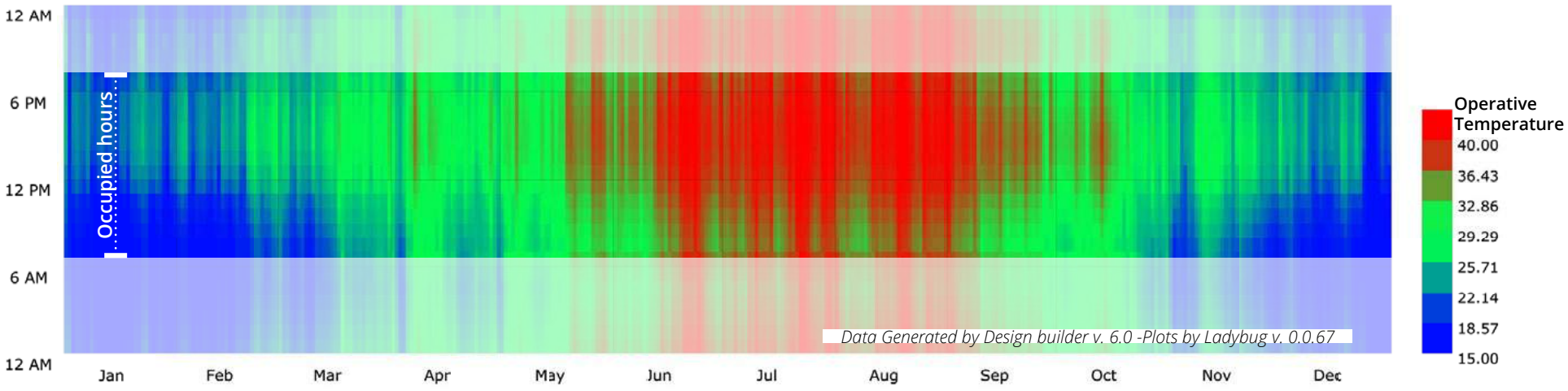
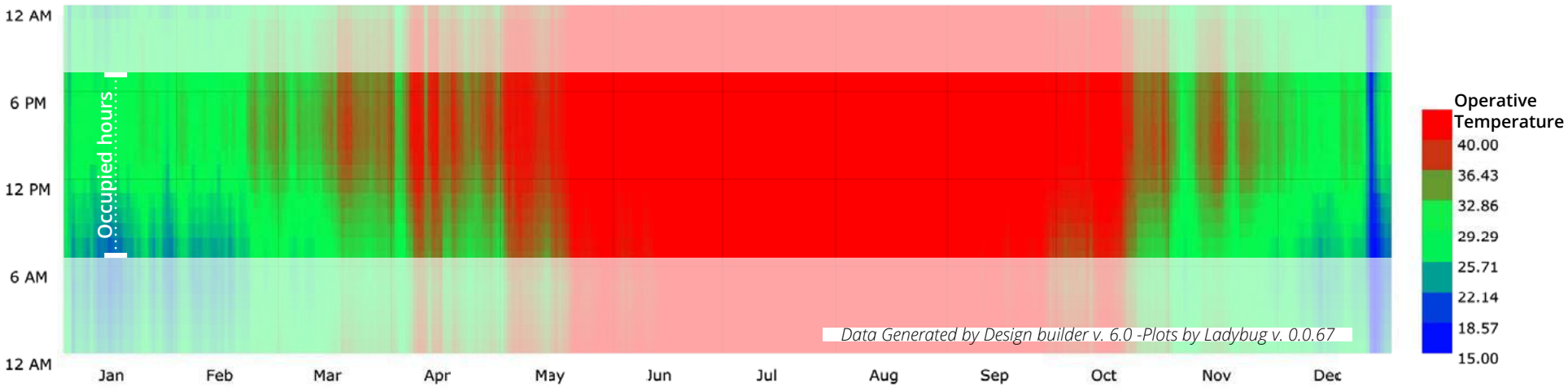
6

4

2

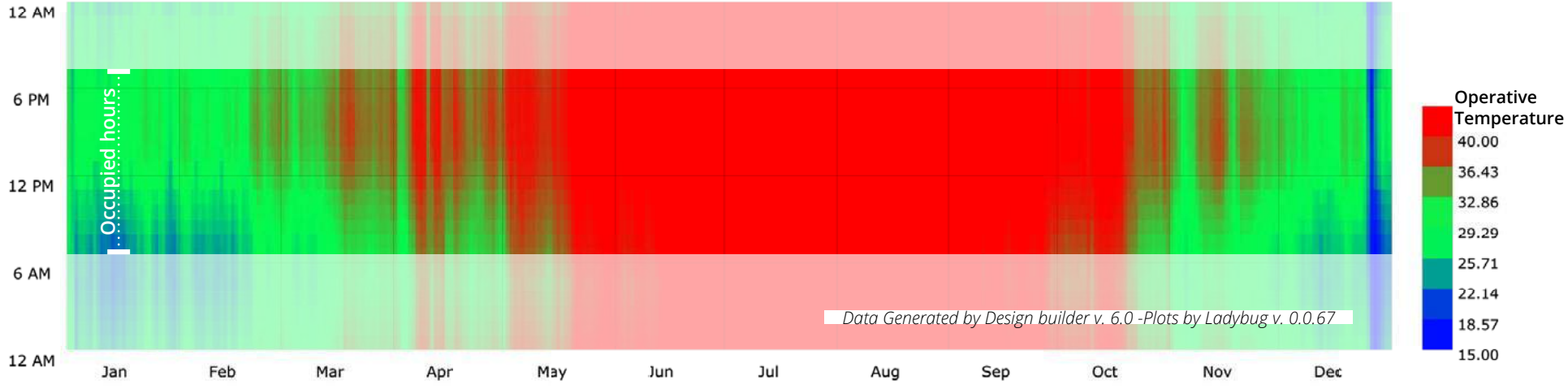
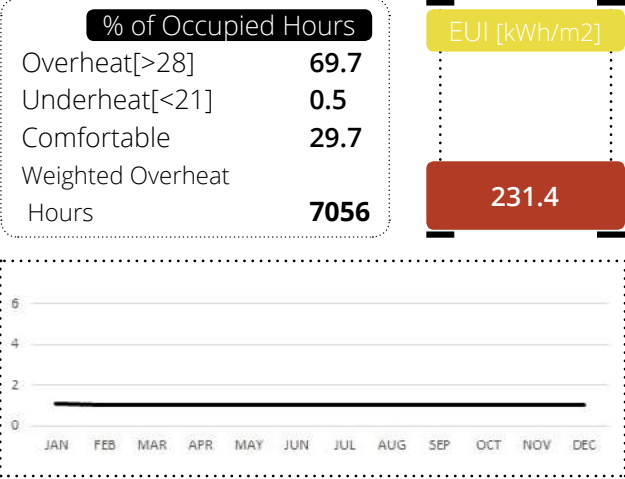
0

JANFEBMARAPRMAYJUNJULYAUGSEPOCTNOVDEC



Base Case

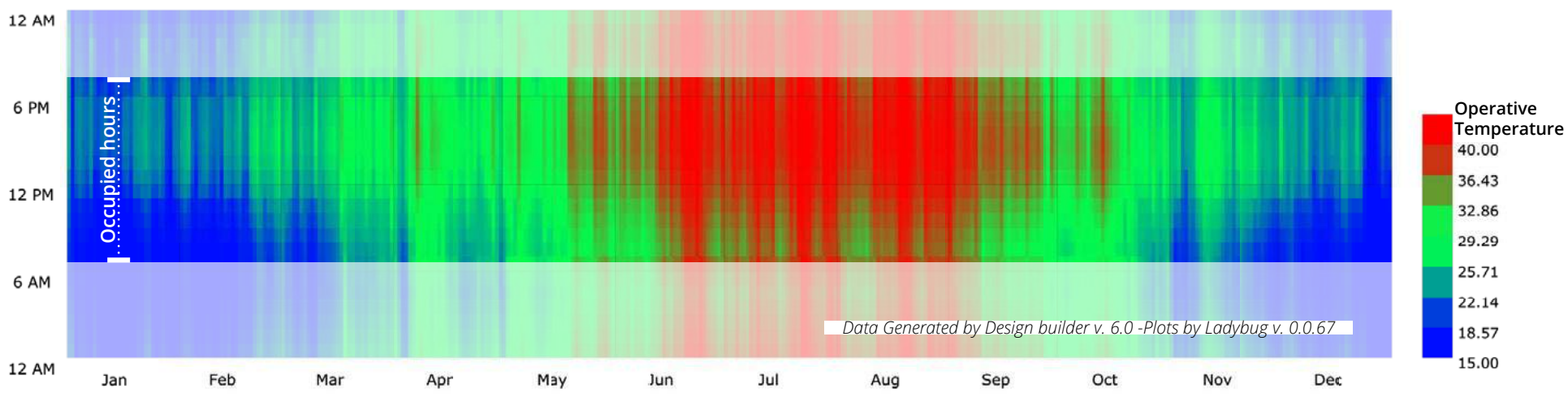
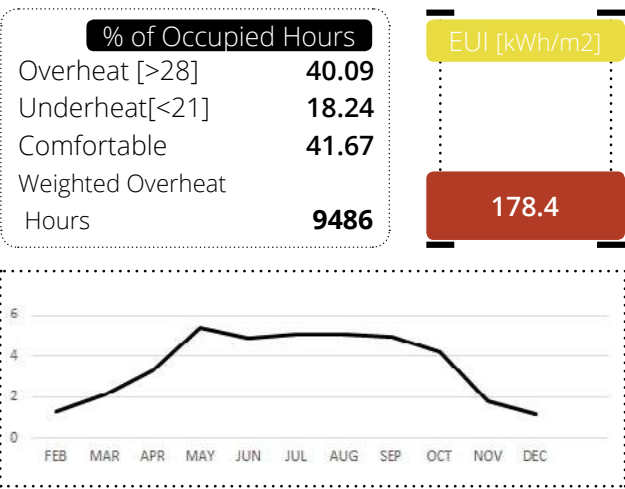
No Natural Ventilation Strategies



+ FAN
(4 ac/h)

Windows

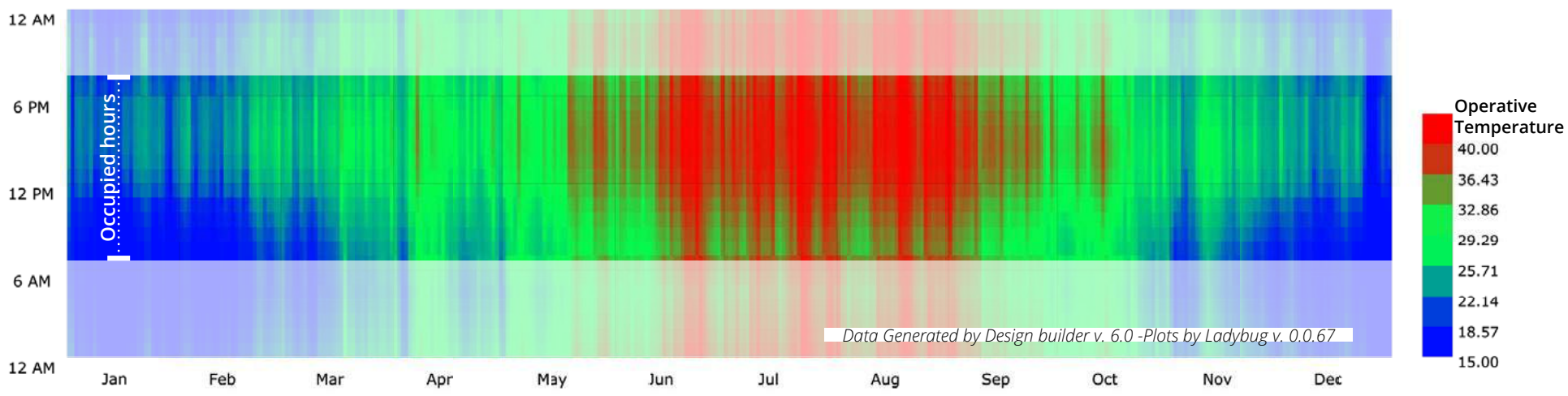
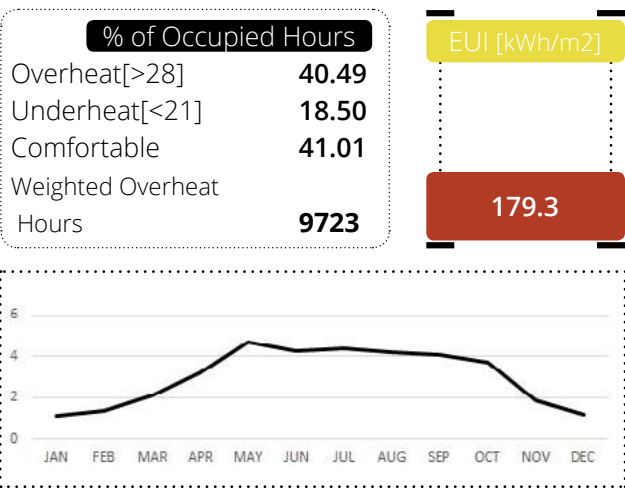
Automated Windows Operating Schedule
Free Aperture: %15



+ FAN
(4 ac/h)

Skylights and Void

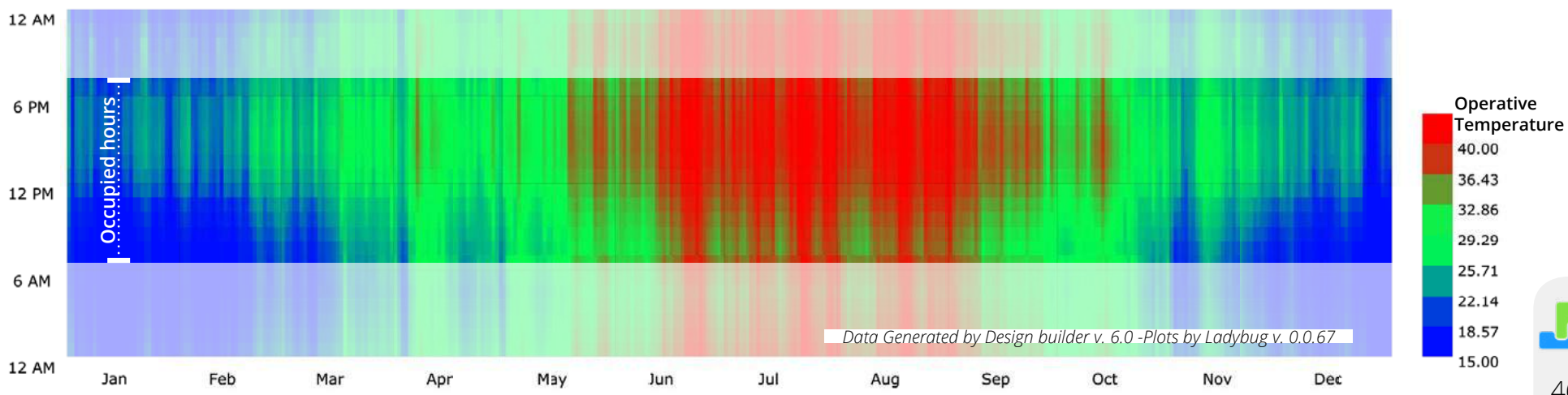
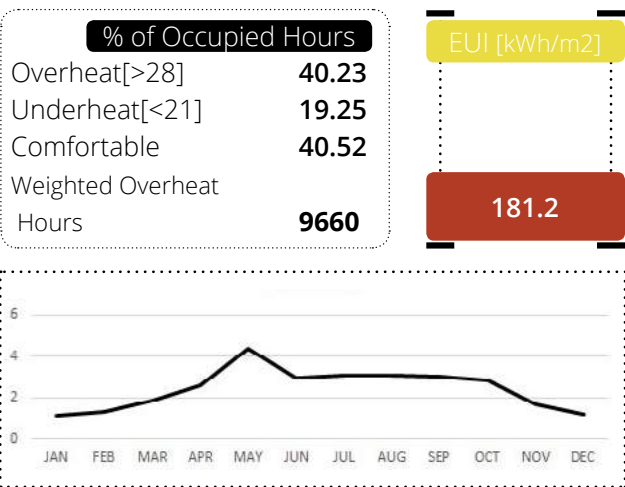
Dimentions: 8m*8m
Free Aperture: %25
plus Windows Free Aperture of %5



+ FAN
(4 ac/h)

Small-sized chimneys on the roof

Dimentions: 1m*1m*1.5m
Number of Small-sized chimneys on the roof: 10
Plus Windows Free Aperture of %5





Concept Stage Final Results

Modelling Key Assumptions

Thermal	Infiltration Constant Rate(Scheduled method): 1 ACH
HVAC	Template: General Energy Code
	Mechanical Ventilation: Off
	Natural Ventilation: On
	Natural Ventilation Method: Scheduled
	Occupancy density[people/s.q.m.]: 0.5

Surface

External walls
Below grade walls
Flat Roof
Pitched roof(occupied)
Pitched roof(unoccupied)
Partitions
Semi-exposed walls
Semi-exposed Ceilings
Semi-exposed floor
Ground floor
External floor
Internal floor
Glazing

Construction

Brick air l/w concrete block & l/w plaster	0.95
Brick air l/w concrete block & l/w plaster	0.95
19mm felt/bitumen on 25mm eps slab on 3mm steel	1.245
19mm felt/bitumen on 25mm eps slab on 3mm steel	1.24
19mm felt/bitumen on 25mm eps slab on 3mm steel	1.245
Project partition	1.6
Brick air l/w concrete block & l/w plaster	0.95
19mm felt/bitumen on 25mm eps slab on 3mm steel	1.245
Floors, Steel-Joist, Spray-On Ins, R0.7) 4-	0.84
Floors, Steel-Joist, Spray-On Ins, R0.7) 4-	0.84
Floors, Steel-Joist, Spray-On Ins, R0.7) 4-	0.84
Project internal floor	2.9
Project external glazing	1.96

U-value

Space

Lounge/Foodcourt seating area/Study halls
Game Room
Multi-functional
Conference hall
Gallery
Elevator shafts/ Staircase/ facilities/parkings,...
Classrooms
Conference room
Prayers room
Administration+ student associations

Category

ASHRAE
ASHRAE
ASHRAE
ASHRAE
ASHRAE
ASHRAE
ASHRAE
ASHRAE
ASHRAE
ASHRAE

Name

Educational Facilities - Multi-use assembly (%200 population of conference/meeting room)
Sports and Entertainment - Health club/aerobics room
Educational Facilities - Multi-use assembly (%200 population of conference/meeting room)
Exh/w Assumed Occ/Rp/Ra - WC-public (continous operation>25 - (%50 L/(s*unit)
Public Assembly Spaces - Museum/gallery
Circulation area (corridors and stairways) - non public
Educational Facilities - Classroom (+9)
General - Conference/Meeting
General - Conference/Meeting
Office Buildings - Office--Enclosed



Energy and Comfort Results

Thermal Comfort

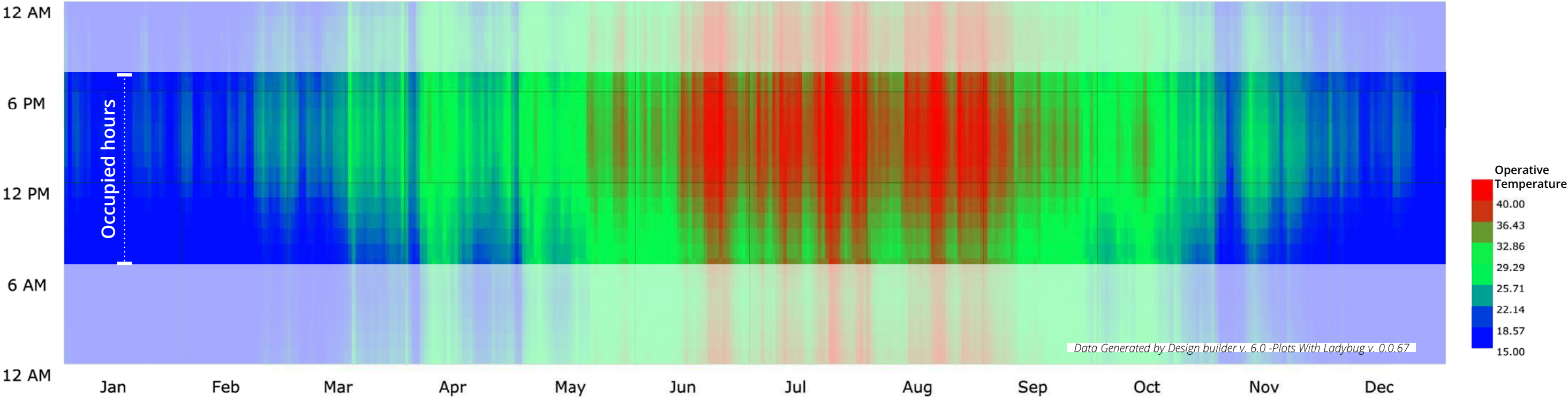
Overheat[% of Occ Times]
Underheat[% of Occ Times]
Comfortable[% of Occ Times]
Weighted Overheat[% of Occ Times]

38.39
32.68
28.93
11371

Energy

Total Cooling Energy[kWh/m2]
Total Heating Energy[kWh/m2]
Energy Use Intensity[kWh/m2]
Primary Energy[kWh/m2]

61
37
197
618



03 | Design Development



Glazings

Ground Floors

PCM-Coated Partition Walls

PCM-Coated Partition Walls

Green Roof

Other Modelling Assumptions = Concept Stage Final Model Assumptions

Walls

- AAC Block Wall

AAC Block Insulation L.1

Roofs

Roof -Insulation L.1

Roof-Insulation I 2

Cool Roof+ Insulation L1Cool Roof+ Insulation L2Green Roof+Insulation L.1Green Roof+ Insulation Level249

Constructions Selection (Step1)		Base Case	Insulation1	Insulation2	Cool Roof + Insulation1	Window + Insulation 2	Green Roof +Insulation1	Green Roof +Insulation2	PCM+Insulation1	PCM+Insulation2
Roof walls Ground Floors Partition Walls Glazings	Roof	Roof-Base	Roof -Insulation L.1	Roof -Insulation L. 2	Cool Roof+ Insulation L.1	Cool Roof+ Insulation L.2	Green Roof+Insulation L.1	Green Roof+ Insulation Level2	Roof -Insulation L.1	Roof -Insulation L. 2
	walls	Brick Wall	AAC Block Wall	AAC Block Insulation L.1	AAC Block Wall	AAC Block Insulation L.1	AAC Block Wall	AAC Block Insulation L.1	AAC Block Wall	AAC Block Insulation L.1
	Ground Floors	Ground Floor-Base	G.F Insulation L.1	G.F Insulation L.2	G.F Insulation L.1	G.F Insulation L.2	G.F Insulation L.1	G.F Insulation L.2	G.F Insulation L.1	G.F Insulation L.2
	Partition Walls	-	-	-	-	-	-	-	PCM Coating	PCM Coating
	Glazings	Window-Base	Dbl Low-E	V-Cool Low-E	Dbl Low-E	V-Cool Low-E	Dbl Low-E	V-Cool Low-E	Dbl Low-E	V-Cool Low-E
Thermal Comfort	Overheat[% of Occ Times]	38.39	39.66	40.30	39.66	39.26	39.69	40.28	40.00	40.51
	Underheat[% of Occ Times]	32.68	19.93	16.66	19.93	18.88	19.37	16.39	18.72	15.37
	Comfortable[% of Occ Times]	28.93	40.41	43.04	40.41	41.87	40.95	43.33	41.28	44.12
	Weighted Overheat[% of Occ Times]	11371.97	9639.56	9737.24	9639.56	9388.02	9642.13	9746.07	9804.00	9935.75
Energy	Total Cooling Energy[kWh/m2]	61.51	65.50	64.32	65.50	61.15	65.35	64.24	64.79	64.29
	Total Heating Energy[kWh/m2]	37.13	19.13	15.21	19.13	17.12	18.52	14.90	18.60	13.58
	Energy Use Intensity[kWh/m2]	194.6589648	194.6589648	175.0976451	181.8942886	173.8388976	181.1384741	174.7209388	180.078035	172.7625924
	Primary Energy[kWh/m2]	607.9408819	606.9913805	592.3348687	606.9913805	583.018026	605.7875893	591.733734	601.7929637	588.0070681
Cost Analysis	Total Material Cost[T]	1565476.333	2227491.333	2027704.733	1800877.333	2062764.733	2572197.333	2816554.733	2756997.333	3001354.733
	Total Material Cost-BaseCase Cost[T]	0	662015	462228.3999	235401	497288.3999	1006721	1251078.4	1191521	1435878.4
	Annual Natural Gas Cost[T]	4780459.615	2462961.058	1957711.442	2462961.058	2204138.654	2384834.135	1918831.442	2393951.25	1747690.096
	Annual Electricity Costs[T]	572568472	591251496	580946584	591251496	569618104	590719656	580679864	586888376	578226440
	Total Energy Cost- Base Energy Cost[T]	0	16365525.44-	5555363.827-	16365525.44-	5526688.962	15755558.52-	5249763.827-	11933395.63-	2625198.481-
	Pay-back Period[years]	0	-40	-83	-14	90	-63	-238	-100	-546

Constructions Selection (Step2)		Base Case	Base+CoolRoof	Base+GreenRoof	Base+PCM	Window Dbl Low-E + CoolRoof	Window Dbl Low-E + GreenRoof	Window Dbl Low-E + PCM
Roof	Roof-Base	Roof -Insulation L. 1	Roof -Insulation L. 2	Cool Roof+ Insulation L.1	Cool Roof+ Insulation L.2	Green Roof+Insulation L.1	Green Roof+ Insulation Level2	
	Brick Wall	AAC Block Wall	AAC Block Insulation L.1	AAC Block Wall	AAC Block Insulation L.1	AAC Block Wall	AAC Block Insulation L.1	
	Ground Floor-Base	G.F Insulation L.1	G.F Insulation L.2	G.F Insulation L.1	G.F Insulation L.2	G.F Insulation L.1	G.F Insulation L.2	
	-	-	-	-	-	-	-	
	Window-Base	Dbl Low-E	V-Cool Low-E	Dbl Low-E	V-Cool Low-E	Dbl Low-E	V-Cool Low-E	
Thermal Comfort	Overheat[% of Occ Times]	38.39	35.81	36.98	35.81	36.68	37.94	37.89
	Underheat[% of Occ Times]	32.68	33.01	30.55	32.32	30.13	26.41	35.21
	Comfortable[% of Occ Times]	28.93	31.19	32.47	31.87	33.19	35.66	26.90
	Weighted Overheat[% of Occ Times]	11371.97	8178.91593	8774.26067	8857.21531	8607.53653	9217.68164	11693.35722
Energy	Total Cooling Energy[kWh/m2]	61.51	56.07	60.91	60.51	58.19	63.35	63.14
	Total Heating Energy[kWh/m2]	37.13	40.11	35.69	31.58	34.82	30.60	28.06
	Energy Use Intensity[kWh/m2]	194.6589648	192.4455434	192.7686444	187.7215909	189.29324	128.9144432	186.7969594
	Primary Energy[kWh/m2]	607.9408819	592.5303733	604.7479848	596.8569922	594.4147161	607.9149442	602.3253166
Cost Analysis	Total Material Cost[T]	1565476.333	1583006.333	2354326.333	1787236.333	1644807.933	2416127.933	1849037.933
	Total Material Cost-BaseCase Cost[T]	0	17530	788850	221760	79331.6	850651.6	283561.6
	Annual Natural Gas Cost[T]	4780459.615	5163703.846	4594442.308	4065144.231	561755776	3939536.538	3612132.692
	Annual Electricity Costs[T]	572568472	554112440	571024520	567517056	566238050	579715160	576779272
	Total Energy Cost- Base Enegy Cost[T]	0	18072787.77	1729969.308	5766731.385	550644894.4-	6305764.923-	3042473.077-
	Pay-back Period[years]	0	1	455	38	7	-134	-93

HVAC Selection



CAV



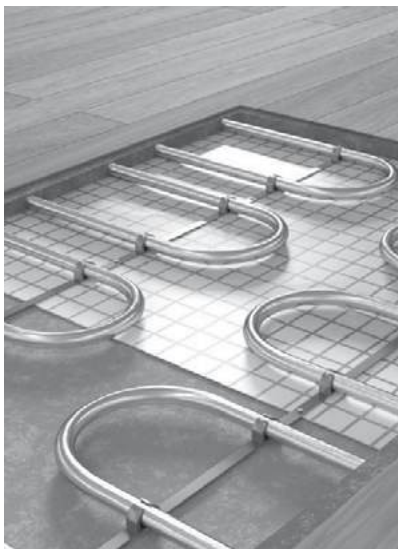
VAV



Fan Coil



Forced Air



Radiant

General Aspects	Noise	4	44	44	44	33	55		
	Initial Cost	4	16	674	16	24	24	8	
	Oppering Cost	2	32	284	40	32	32	24	
	Occupancy	4	50	314	50	40	245	50	248
	Frequency of Use	4	21	21	21	21	21	21	
	User Control	4	36	45	45	36	9		
	Aesthetics	2	33	33	11	33	55		
Energy Use and Technical aspects	Energy Consumption	4	52	65	39	26	26		
	Cooling Capacity	5	65	65	39	65	26		
	Heating Capacity	5	65	65	65	65	52		
	Humidity	3	30	30	20	20	10		
	Pressure Control	4	40	374	16	24	338	8	204
	Efficiency	4	56	56	42	42	28		
	Envriomental Impact	2	30	30	30	30	30		
	Life Span	4	48	48	36	36	36		
	Saftey and health	5	56	56	28	56	14		



HVAC Selection Results

	Base Case	Package 1	Package 2	Package 3	Package 4	Package 5	Package 6	Package 7
Hot Water Supply	Gas Boiler	Gas Boiler	Gas Boiler	Gas Boiler	Gas Boiler	Gas Boiler	Gas Boiler	Gas Boiler
Chilled Water Supply	Air Cooled Chiller	Air Cooled Chiller	Air Cooled Chiller	Air Cooled Chiller	Air Cooled Chiller	Air Cooled Chiller	Air Cooled Chiller	Air Cooled Chiller
Terminals	VAV + Reheat /All Spaces	VAV + Reheat / Communal Spaces Fan Coil Units/Small Spaces	VAV + Reheat /All Spaces	VAV + Reheat / Communal Spaces Fan Coil Units/Small Spaces	VAV + Reheat /All Spaces	VAV + Reheat / Communal Spaces Fan Coil Units/Small Spaces	VAV + Reheat /All Spaces	VAV + Reheat / Communal Spaces Fan Coil Units/Small Spaces
Energy Conservation Systems	-	-	Economiser	Economiser	Heat Recovery	Heat Recovery	Heat Recovery + Economiser	Heat Recovery + Economiser

Energy	Gas[kWh/m2]	36.4	28.6	39.5	28.6	36.4	28.6	28.6	28.6
	HVAC Electricity[kWh/m2]	59.9	54.3	57.5	46.8	53.8	46.6	46.7	46.7
	HVAC total Energy[kWh/m2]	96.3	82.9	97.0	75.4	90.2	75.2	75.3	75.3
	Total EUI[kWh/m2]	192.9	179.5	193.5	171.9	186.8	171.8	171.9	171.9
	total primary[kWh/m2]	603.4	574.5	598.1	547.5	581.4	546.9	547.5	547.5



Lighting System

	ASHRAE Reference	Fluorescent	LED	LED+Linear control	LED+ Stepped control	LED+ Linear / off
Normalised power density [W/m2]	5	3.3	2.5	2.5	2.5	2.5
Return Air Fraction	0	0	0	0	0	0
Radiant Fraction	0.42	0.42	0.42	0.42	0.42	0.42
Visible Fraction	0.18	0.18	0.18	0.18	0.18	0.18
Lighting Control	Off	Off	Off	On	On	On
Control type	-	-	-	Linear	Stepped	Stepped

Energy	Lighting Energy[kWh/m2]	90	60.0	45.4	26.2	25.6	24.2
	Total EUI[kWh/m2]	205	140.2	125.4	105.5	104.9	102.6
	Total Primary Energy[kWh/m2]	699	429.9	375.0	301.7	305.6	299.6

System Sizing

1.25	Heating Load
Design Margin	653.8 kWh
1.15	Cooling Design Capacity
Design Margin	563.1 kWh
	Design Flow Rate
	35.6 m3/s



HVAC System Diagram
And Specifications

Modelling Key Assumptions

Chiller

Reference capacity	Autosize
Reference cop	5.5
Compressor motor efficiency	1
sizing Factor	1
Condenser fan power ratio	0.035
Reference leaving chilled water temperature	6.67
Reference entering condenser fluid temperature	29.4
Leaving chilled water temperature limit	2
Reference chilled water flow rate	autosize
Minimum part load ratio	0.1
Maximum part load ratio	1
Optimum part load ratio	1
Maximum unloading ratio	0.25

Chiller pump

Design power consumption[W]	Autosize
Design Pump head[Pa]	20000
Design minimum flow rate[m3/s]	0
Motor efficiency	0.9
Fraction of Motor Inefficiencies to fluid stream	0
Comp control type	intermittent

Boiler

Fuel type	Natural Gas
Nominal capacity[W]	Autosize
Parasitic electric load[W]	25
Sizing Factor	1
Nominal thermal efficiency	0.89

Boiler pump

Design power consumption[W]	Autosize
Design Pump head[Pa]	20000
Design minimum flow rate[m3/s]	0
Motor efficiency	0.9
Fraction of Motor Inefficiencies to fluid stream	0
Comp control type	intermittent

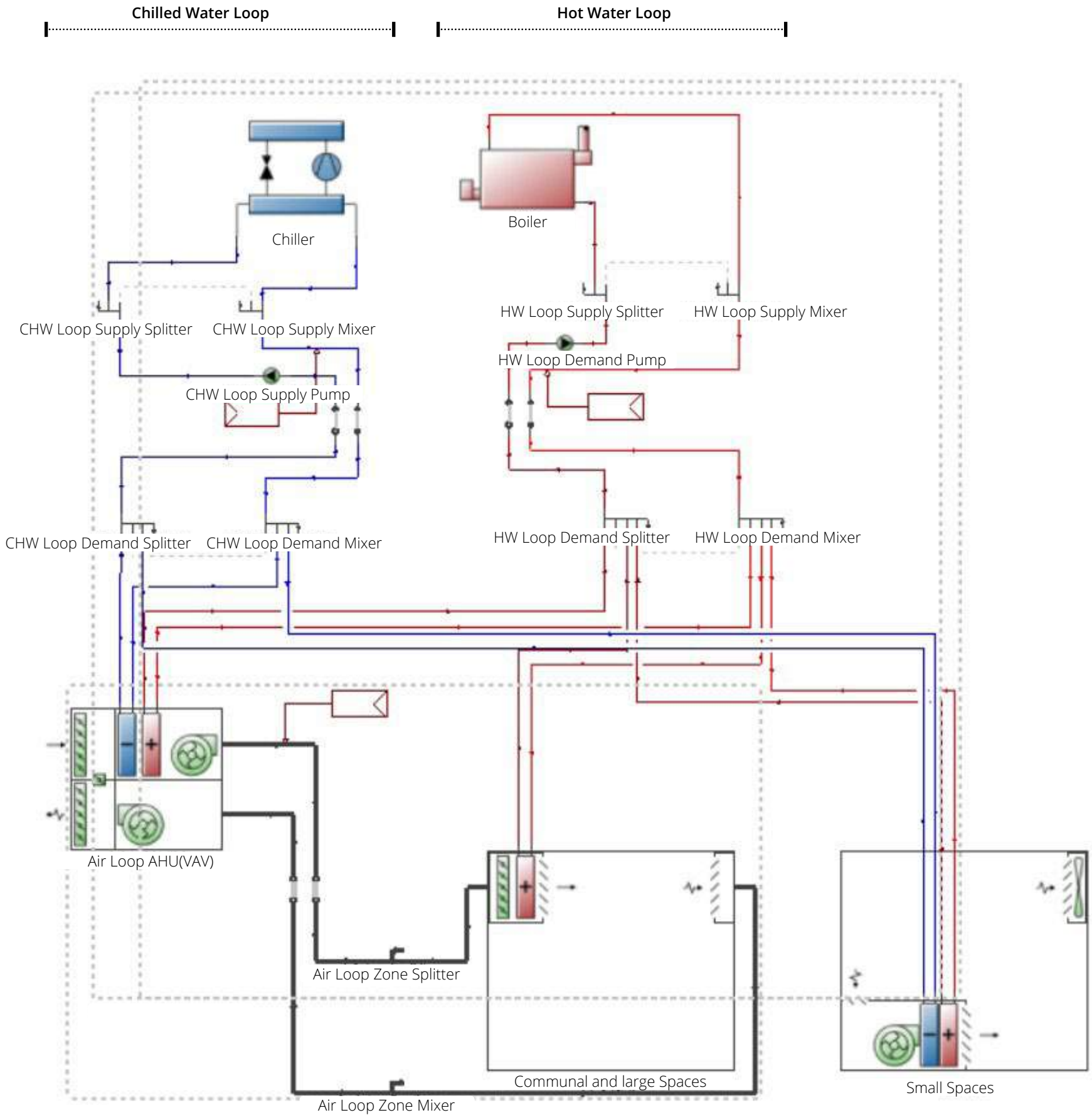
AHU

Availability	On 7/24
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Supply and extract fan

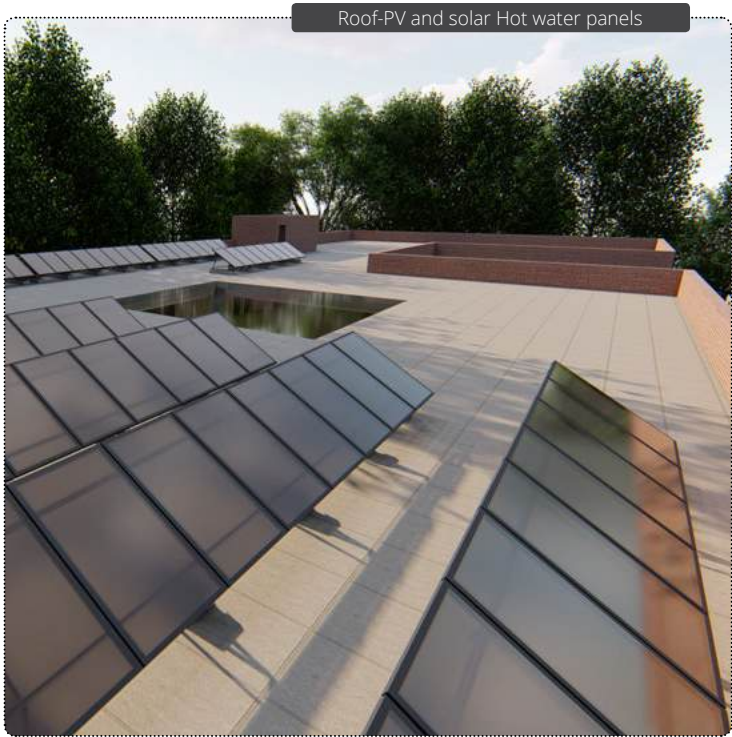
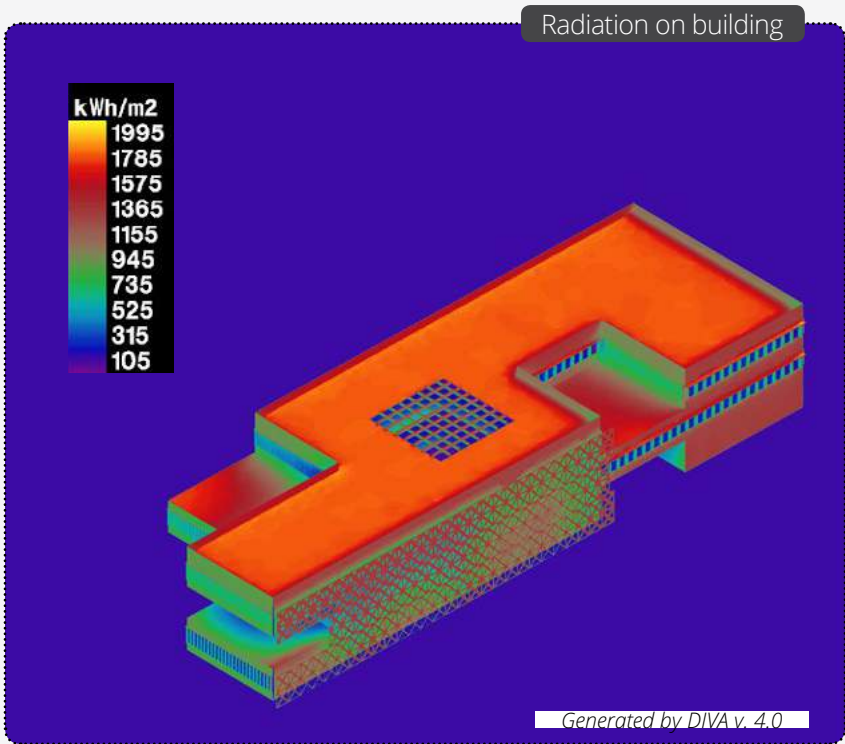
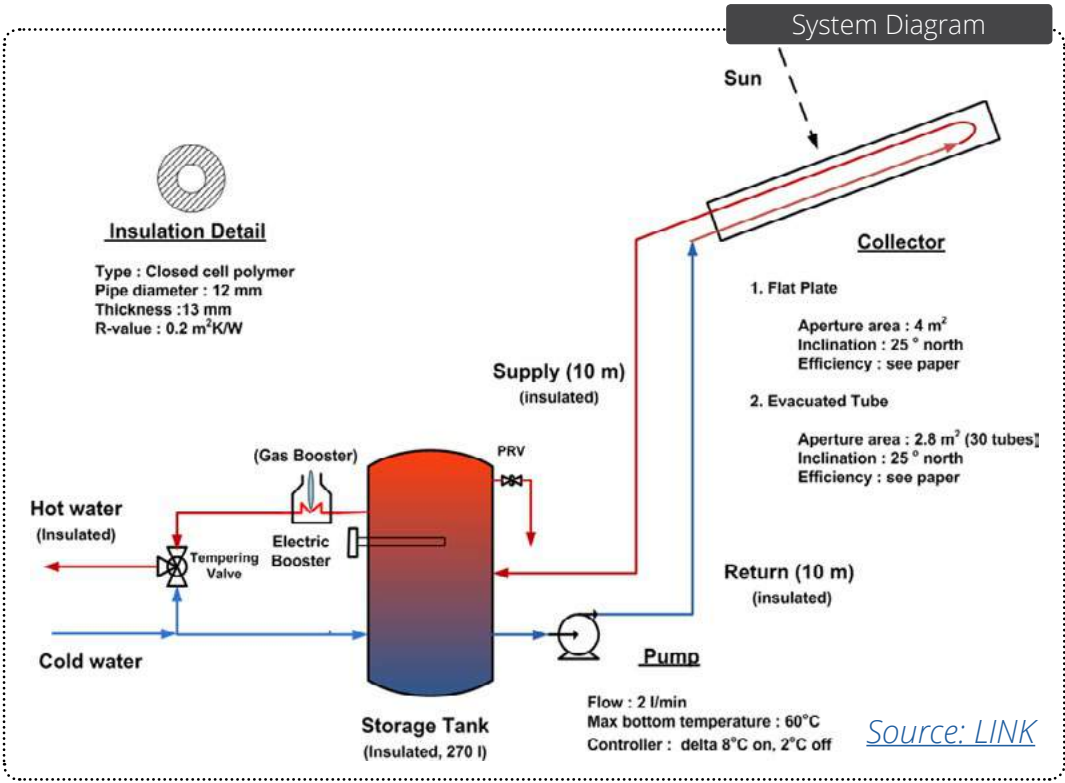
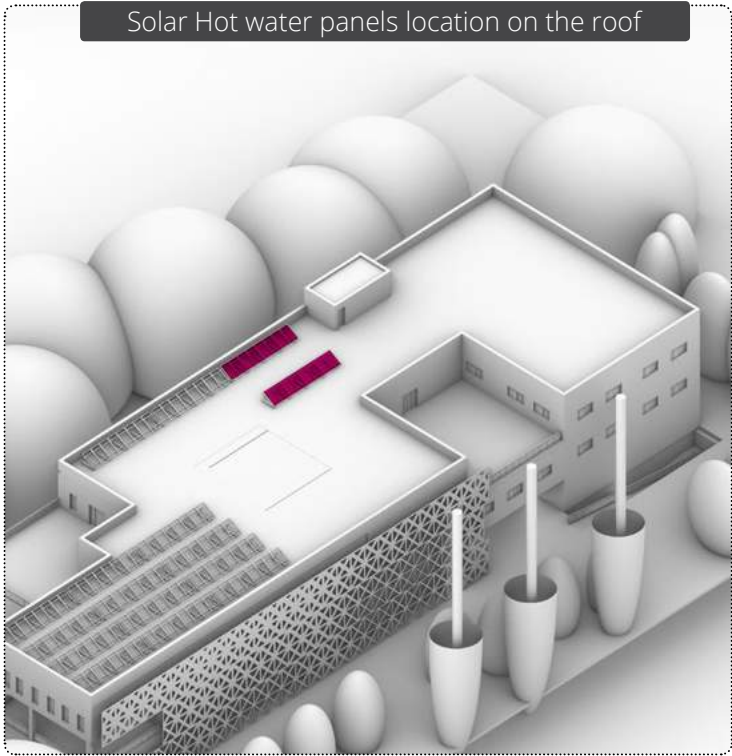
Fan Total Efficiency	0.7
Pressure rise[Pa]	600
Motor efficiency	0.9

HVAC System Diagram



Solar DHW system design

Collector Optical Efficiency	0.7
Collector Thermal Loss[W/m2/C]	4
Collector Active Area Percent[%]	90
Working Fluid Heat Capacity[J/kg/C]	3840
Flow rate per m2[kg/s/m2]	0.012
IAM Coefficient	0.1
Sky exposure factor	1
Beam Index per hour	1
Max Working Temperature[C]	95
Discharge Temperature[C]	3-
Delivary water temperature[C]	60
AVR January Cold Water Temperature[C]	1
Mechanical Room Temperature[C]	20
Pipe Length	50
tank size[lit]	100
tank loss[W/m2/C]	0.3
height diameter tank ratio	2.6
heat exchange effectiveness	0.8



System Sizing

Panels Total Area	DHW annual Load	Annual Solar DHW
32 s.q.m.	1989 kWh	2460 kWh



Construction and Systems Final

Performance Results

Modelling Key Assumptions

Thermal HVAC

Building Airtightness: Medium
VAV + Reheat /Communal Spaces
Fan Coil Units/Small Spaces
Natural Ventilation: On
Natural Ventilation Method: Calculated
Occupancy density[people/s.q.m.]: 0.5
Heating/ cooling Set points[C]: 20/27
Heating/ cooling Set backs[C]: 0/100

Lighting

system: LED+Stepped Control
Two lighting controls per zone

Space

Lounge/Foodcourt seating area/Study halls
Game Room
Multi-functional
Conference hall
Gallery
Elevator shafts/ Staircase/ facilities/parkings,...
Classrooms
Conference room
Prayers room
Administration+ student associations

Category

ASHRAE
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ASHRAE

Name

Educational Facilities - Multi-use assembly (%200 population of conference/meeting room)
Sports and Entertainment - Health club/aerobics room
Educational Facilities - Multi-use assembly (%200 population of conference/meeting room)
Exh/w Assumed Occ/Rp/Ra - WC-public (continous operation>25 - (%50 L/(s*unit)
Public Assembly Spaces - Museum/gallery
Circulation area (corridors and stairways) - non public
Educational Facilities - Classroom (+9)
General - Conference/Meeting
General - Conference/Meeting
Office Buildings - Office--Enclosed

Surface

External walls
Below grade walls
Flat Roof
Partitions
Semi-exposed walls
Semi-exposed Ceilings
Semi-exposed floor
Ground floor
External floor
Internal floor
Glazing

Construction

AAC Block Insulation L.1
Brick air l/w concrete block & l/w plaster
Cool Roof+ Insulation L.2
Project partition
Brick air l/w concrete block & l/w plaster
19mm felt/bitumen on 25mm eps slab on 3mm steel
Floors, Steel-Joist, Spray-On Ins, R0.7) 4-)
G.F Insulation L.2
Floors, Steel-Joist, Spray-On Ins, R0.7) 4-)
Project internal floor
V-Cool Low-E

U-value

0.1
0.95
1.245
1.6
0.95
1.245
0.84
0.47
0.84
2.9
1.5



Energy and Comfort Results

Thermal Comfort

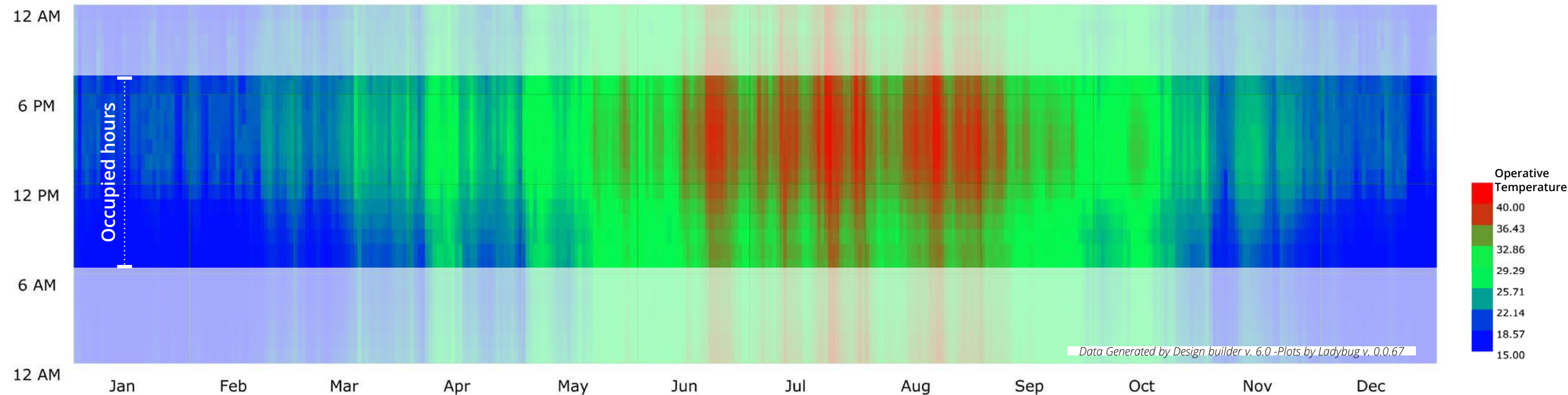
Overheat[% of Occ Times]
Underheat[% of Occ Times]
Comfortable[% of Occ Times]
Weighted Overheat[% of Occ Times]

34.66
23.53
41.81
7534

Energy

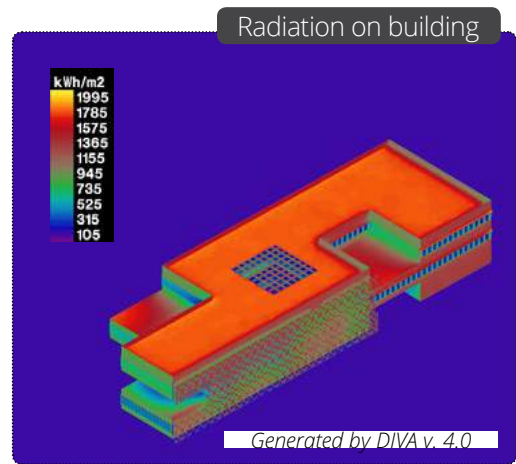
Total Cooling Energy[kWh/m2]
Total Heating Energy[kWh/m2]
Energy Use Intensity[kWh/m2]
Primary Energy[kWh/m2]

23.3
16.35
80.7
250



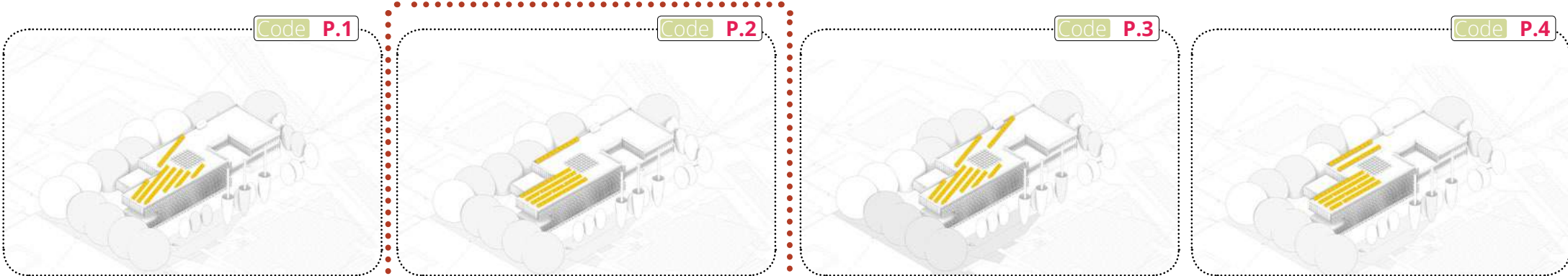
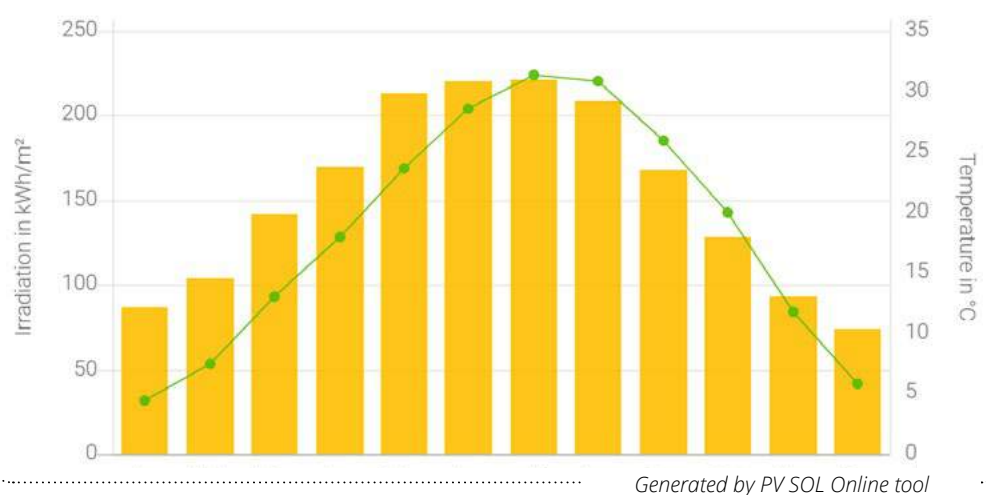
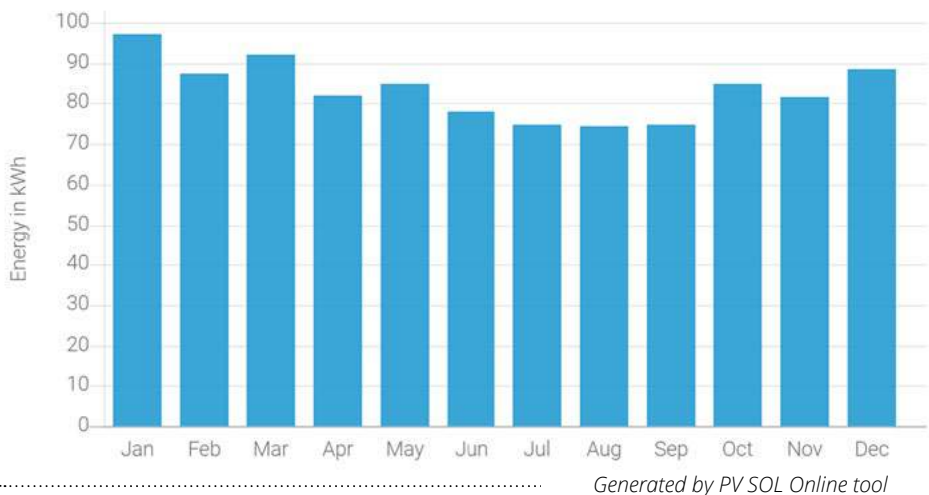


Photovoltaic System design



PV key Assumptions

Efficiency[%]	15
DC to AC derate factor	0.85
Module Active Area[%]	90
module material	Crystalline Silicon (S-ci)
Temperature Coefficient[%]	0.5-/C



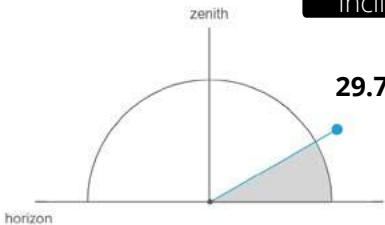
General	PV panels Total Area[m2]	180	180	219	219
	Covered Roof Area[%]	45	52.1	48	55.5
	Orientation	180	160	180	160
	Inclination	29.7	29.7	29.7	29.7
	Nominal Power[kW]	30	30	35	35
Energy	Annual Electricity Generation [kWh]	37006.02	36614.25	43467.39	43007.21
	Annual Electricity Generation [kWh/m2]	8.30	8.21	9.75	9.64
	Generation/ Electricity Consumption [%]	12.86	12.73	15.11	14.95
	Generation/ Primary Energy [%]	11.95	11.82	14.03	13.89
Costs and Gains	Initial Cost [mT]	240	240	280	280
	Annual Gain [mT]	4.07	4.03	4.78	4.73
	Simple Payback Period	7.12	7.2	7.07	7.15

Final System information and Results

Assumptions

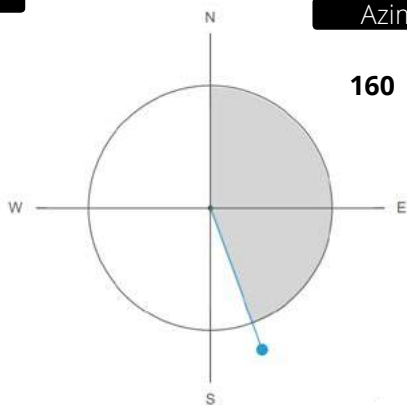
Results	
Annual PV energy[kWh]	44477
Spec. annual yield[kWh/kWp]	1425.55
Performance ratio[%]	8357
Annual PV energy	44477 kWh
General Information	
Latitude	35.74241 °
Longitude	51.57958 °
Annual global irradiation	1828.4 kWh/m²
Average temperature	18.4 °
Load profile	BDEW load profile business (G2)
pv modules total nominal power[kWp]	31.2
Modules Company	Shinsug Solar Eergy Co Ltd.
Model	SS-BM261
Module Power [Wp]	260
Efficiency[%]	16.3
Type	Mono
No. of Modules	120
Total PV power [kWp]	31.2
Inclination	29.7
Orientation	160
Installation type	Mounted-roof
Albedo[%]	20
Shadowing[%]	12
Soiling[%]	5

Inclination



Generated by PV SOL Online tool

Azimuth



Inverter
Manufacturer
No. of Inverters

Configuration info

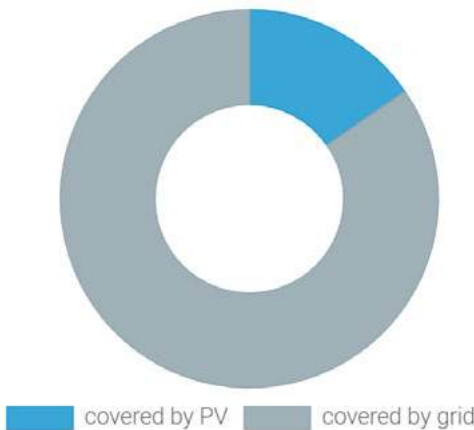
Donfoss Solar Inverters
20

«3 x FLX Pro 9 (Danfoss Solar Inverters)
MPP 1 :1 x 20 - MPP 1 :2 x 20»

Results

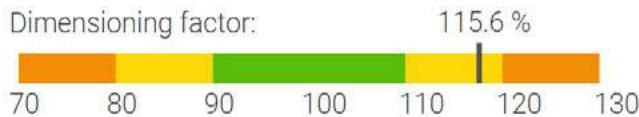
Consumption[kWh]	287670
covered by PV[kWh]	44362
covered by grid	243308
Solar fraction [%]	%15.40
Avoided CO2 emissions [kg/year]	23795

Consumption 287670 kWh

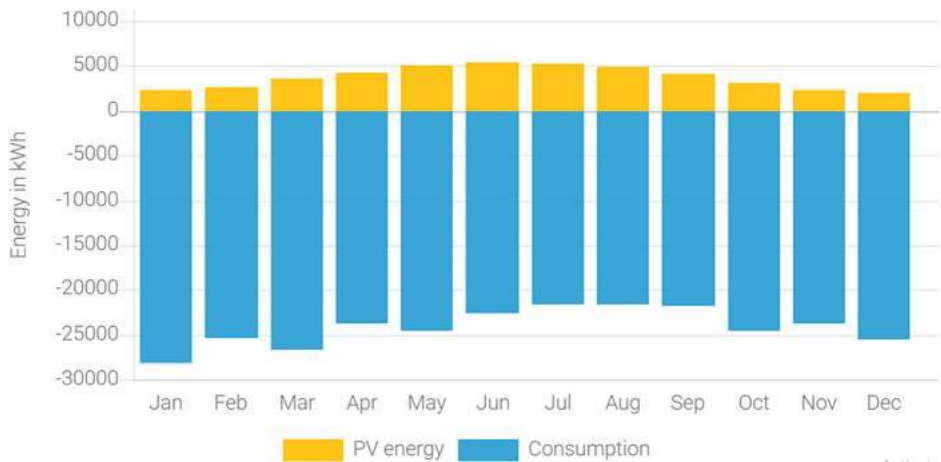


Generated by PV SOL Online tool

Dimentioning factor

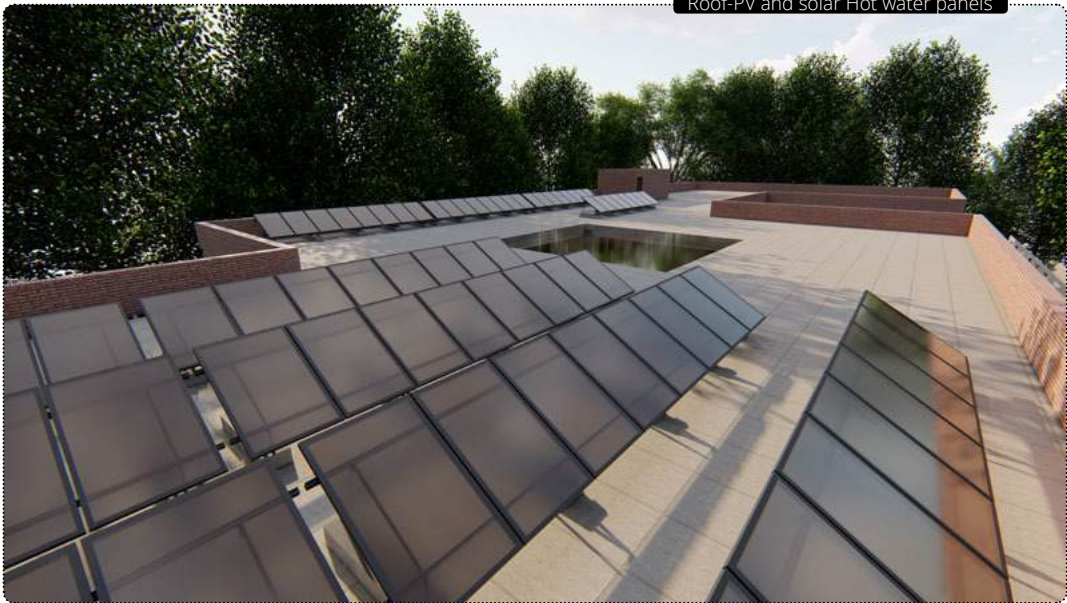


Energy gains and Consumption



Generated by PV SOL Online tool

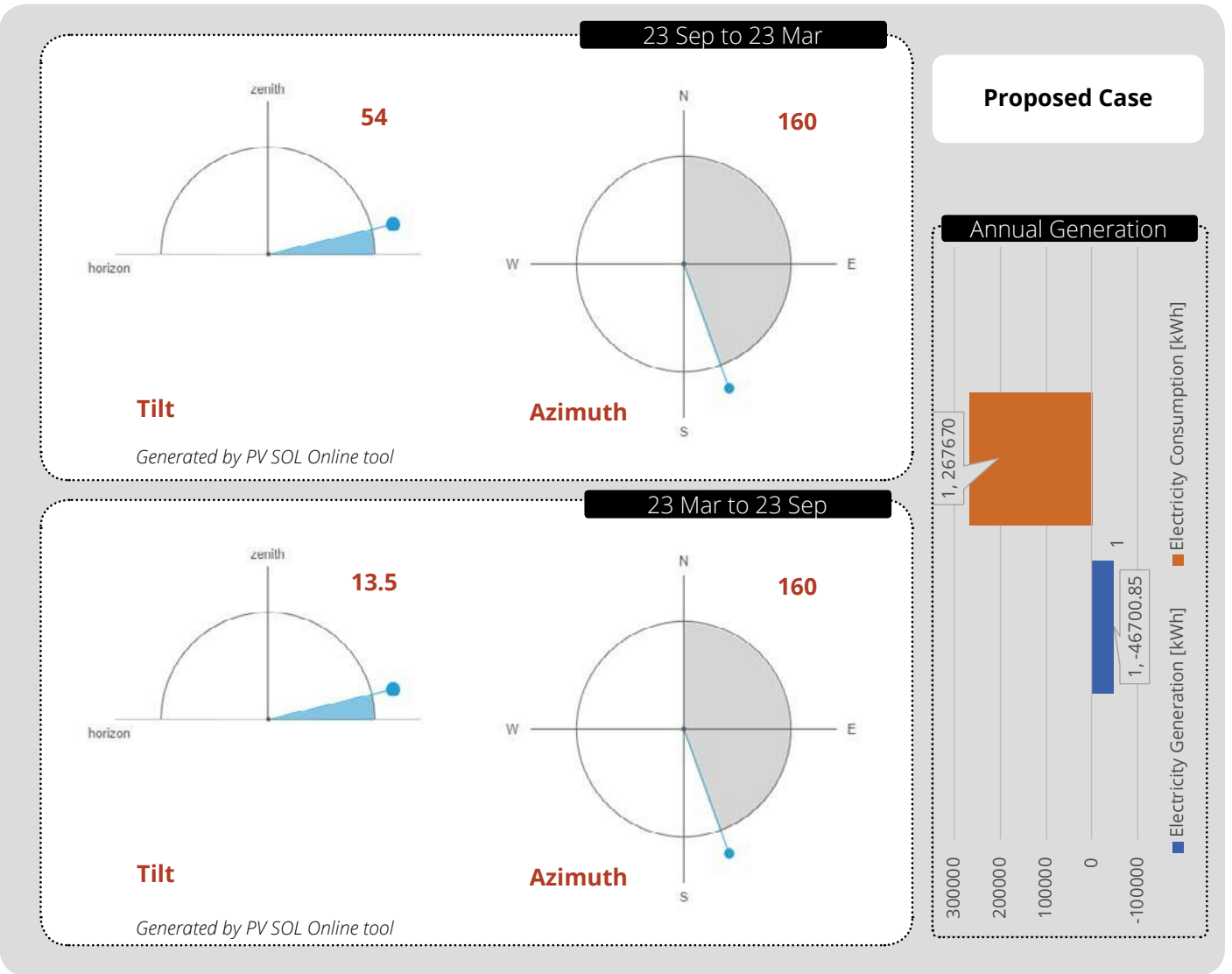
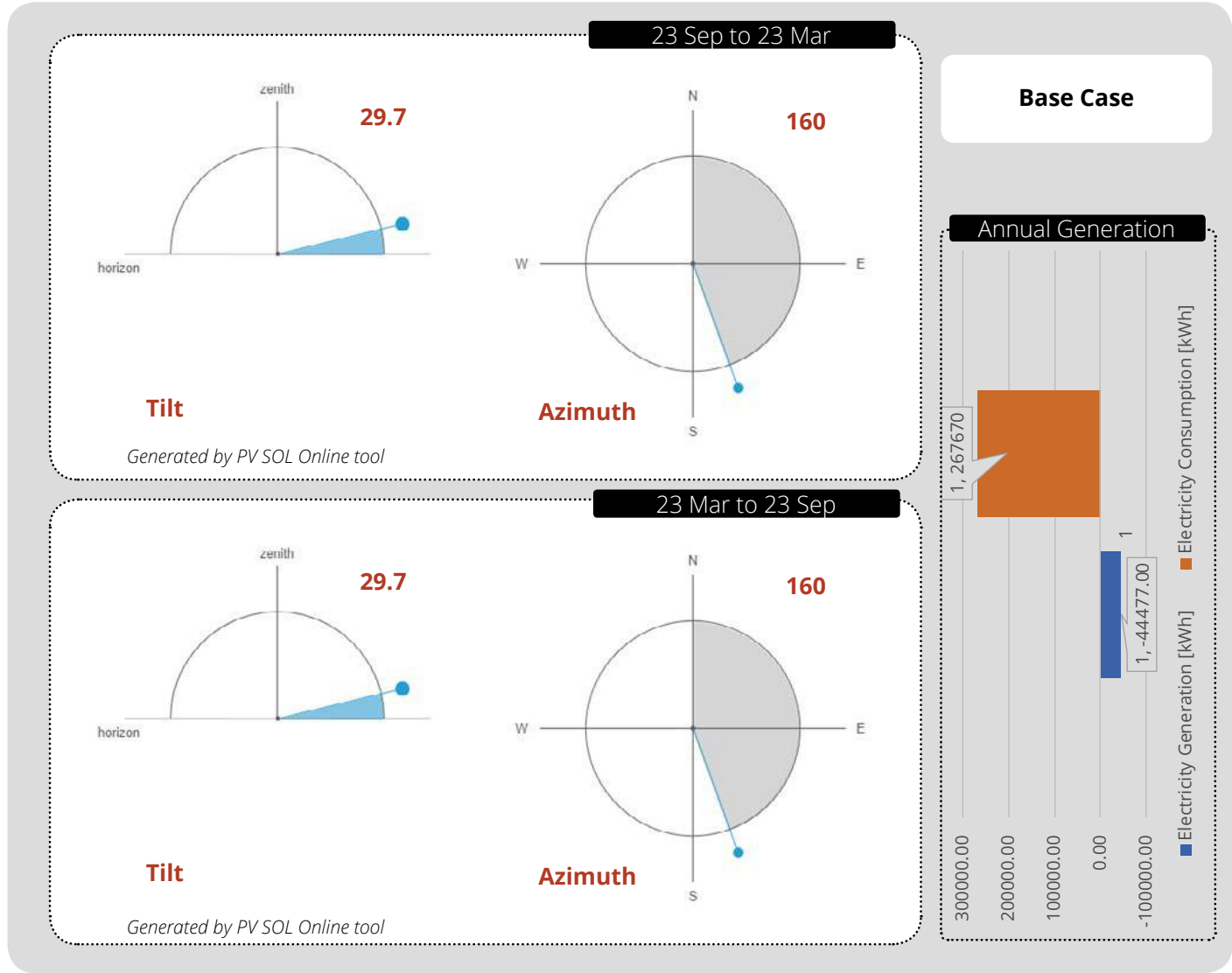
Roof-PV and solar Hot water panels



Final Energy Consumption

Total Cooling Energy[kWh/m2]	23.3
Total Heating Energy[kWh/m2]	16.35
Total Electricity[kWh/m2]	54.6
Energy Use Intensity[kWh/m2]	70.76
Primary Energy[kWh/m2]	220
Annual financial gain[mT]	4.8

Step2 - Orientation Improvement



Final Results	
PV panels Total Area[m2]	180
Covered Roof Area[%]	52.14007782
Orientation	160
Inclination	29.7
Nominal Power[kW]	31.2
Annual Electricity Generation [kWh]	44477.00
Annual Electricity Generation [kWh/m2]	9.972421525
Generation/ Electricity Consumption [%]	16.62
Initial Cost [mT]	240000000
Annual Gain [mT]	40474070.00
Simple Payback Period[years]	5.93
IRR[%]	15
NPV[T]	(IRR 21,450,235.73)
NPV (%25- investment cost)[T]	(IRR 15,995,690.27)
NPV (%25+ investment cost)[T]	(IRR 26,904,781.18)

Final Results	
PV panels Total Area[m2]	180
Covered Roof Area[%]	52.14007782
Orientation	160
Inclination	13.5 hot season- 54 cold season
Nominal Power[kW]	31.2
Annual Electricity Generation [kWh]	46700.85
Annual Electricity Generation [kWh/m2]	10.4710426
Generation/ Electricity Consumption [%]	17.45
Initial Cost [mT]	240000000
Annual Gain [mT]	42497773.50
Simple Payback Period[years]	5.65
IRR[%]	16
NPV[T]	(IRR 21,431,838.42)
NPV (%25- investment cost)[T]	(IRR 16,328,514.24)
NPV (%25+ investment cost)[T]	(IRR 26,886,383.88)

Water Management Case Studies

AbbasPour Student Center

	Name	Type	Location	Climate	Predicted annual consumption of potable water for all uses, excluding process water	Is potable water used for irrigation?	Predicted peak month consumption of potable water for outdoor	Percent of rainwater that can be managed on site	Actual annual consumption of process water (e.g. cooling towers)	Is greywater or blackwater captured for re-use?	Water quality for any stormwater leaving the site	Is rainwater captured for use by the project?
1	Amherst College New Science Center	Education – College/ University (campus-level), Laboratory	Amherst, Massachusetts	Dfb	3,156 gals/ occupant; 41 percent reduction from LEEDv4 baseline	no	n/a	58 %	1,537,700 gallons	no	80 percent of TSS removal	n/a
2	Daniels Building at One Spadina Crescent	Education – College/ University (campus-level)	Toronto, Ontario	Dfb	29 gals/occupant	no	n/a	%100	n/a	n/a	n/a	yes
3	Frick Environmental Center	Education – General, Office – 10,000sf, Public Assembly – General	Pittsburgh, Pennsylvania	Cfa	29,870 gallons; 0.37 gallons per visitor annually (75,000 annual visitors); 40 percent reduction compared with EPA guidelines.	no	n/a	%100	0.6 gallons per visitor annually (44,685 total gallons divided by 75,000 visitors annually)	no	80 percent of TSS removed from stormwater runoff based on the TSS removal percentages from the PA Stormwater BMP Manual located in Chapters 5 and 6.	yes
4	Oregon Zoo Education Center	Education – General	Portland, Oregon	Csb	Indoor water use: 322 kGal/year, 56 percent reduction from federal standards	yes	0.17 gallons/sf	75 %	208,186 gallons	no	84 percent of TSS removed from stormwater runoff	yes
5	Tashjian Bee and Pollinator Discovery Center	Education – General	Chaska, Minnesota	Dfa	39 percent	yes	0.05 gallons/ sq ft	%100	n/a	n/a	97 percent of TSS removed	n/a



Baseline Calculation

	Monthly Rainfall (mm)	Monthly ETo (mm)	Watering Demand (mm)
January	16	1	-15
February	11	2	-9
March	5	4	-9
April	21	6	-9
May	3	8	5
June	5	8	3
July	4	8	5
August	0	7	7
September	14	6	-8
October	7	4	-3
November	18	1	-17
December	26	1	-25

Peak watering month

August

Peak watering demand (mm)

7

Peak watering month ETo (mm)

7

Landscape water allowance (l/month)

274

Landscape water baseline (l/month)

391

LandScape Water Requirements

Average monthly rainfall for the site's peak watering month (mm/month)

0

Zone ID	Hydrozone or Landscape Feature Area (sq m)	Plant Type	Water Requirement	Landscape Coefficient (KL)	Irrigation Type	Distribution Uniformity (DULQ)	LWRH(l/month)
Entrance	40	Trees	Medium	0.5	Drip (press comp)	%90	158
FoodCourt	15	Shrubs	Medium	0.5	Drip (press comp)	%90	59

Total hydrozone or landscape feature area (s.q.m.)

55

Landscape water requirement based on the site's peak watering month (l/month)

217

Summary

Landscape water baseline (l/month)

391

Landscape water requirement (LWR) (l/month)

217

Percentage reduction from baseline (%)

%44

(Must be at least %30)

Outdoor Water Management

Pressure Compensating Drip Irrigation System

PC emitters deliver a precise amount of water regardless of changes in pressure due to long rows or changes in terrain. They can simplify the designing of a system and greatly reduce maintenance since they rarely get plugged. Inside the emitter is a flexible diaphragm that regulates the water flow and tends to flush particles from the system (self-flushing). Using this system, irrigation water usage will be reduced from 391 to 271 liters per month. (%44 reduction %14 above Target)



Rain water Collection Potential Calculation

Roof Area[s.q.m.]

1285

Total Annual Rainfall[mm]

130

Total Rain Water Collection Potential[lit/y]

169,620

Final Results

Landscape water baseline (l/month)

391

Landscape water requirement (LWR) (l/month)

217

Percentage reduction from baseline With Drip Irrigation (%)

%44

Percentage reduction from baseline Using RainWater (%)

%100

0%

100%

reduction from baseline Using Rain water and Drip Irrigation System

44%

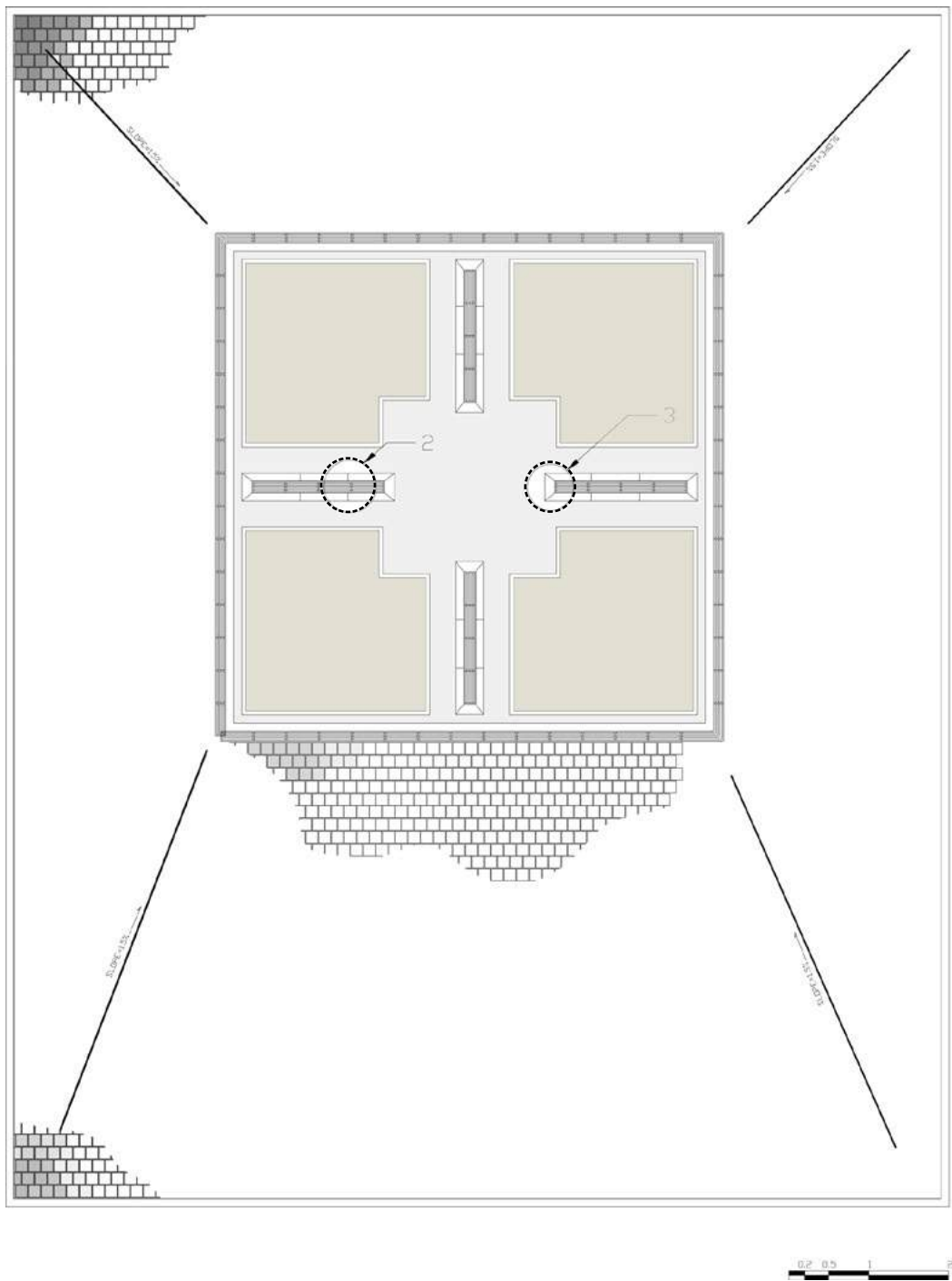
56%

Reduction from baseline With Drip Irrigation

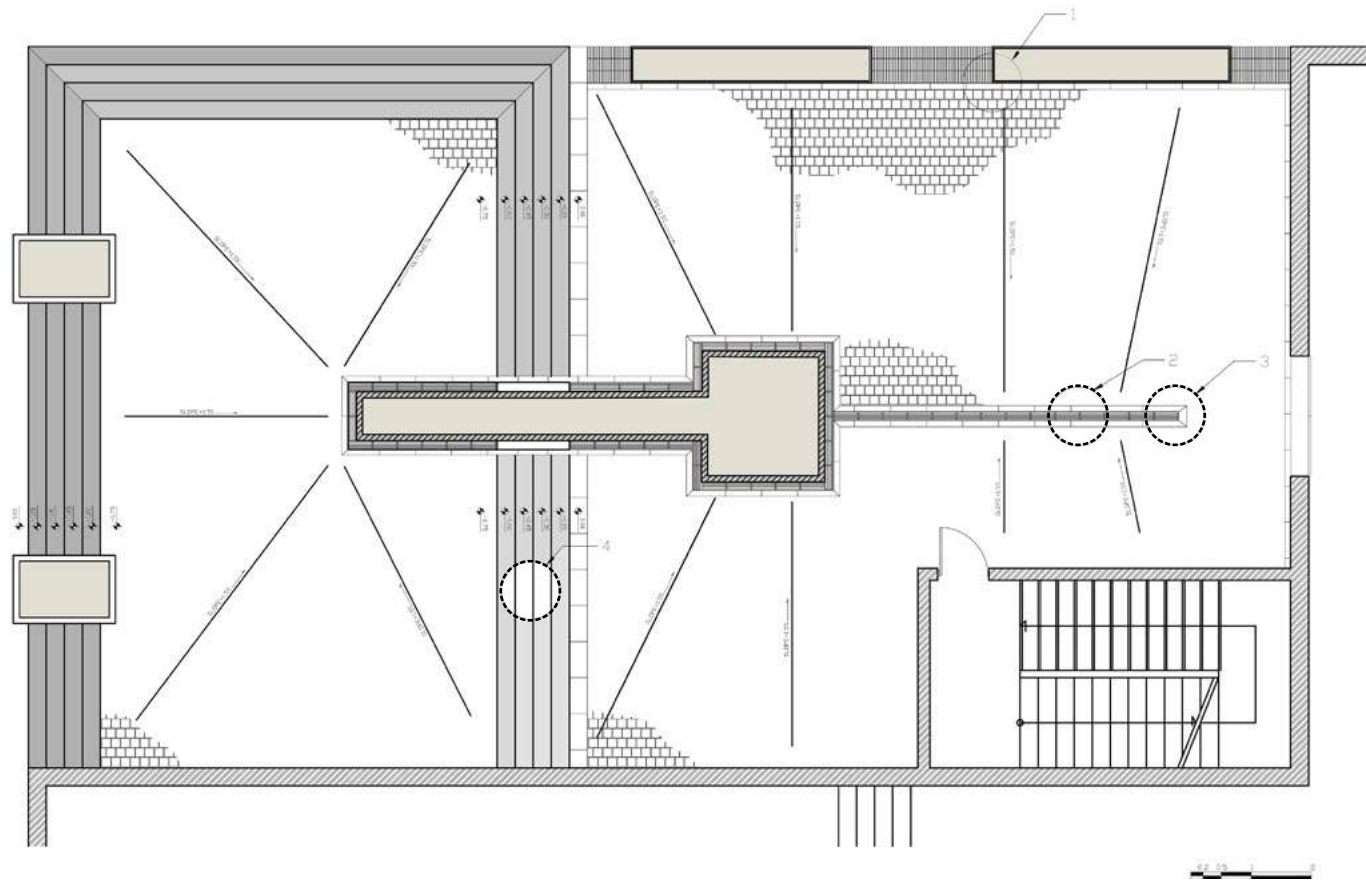
Data Generated with LEED v. 4.0 Water management Spreadsheets -Plots With Excel

 Landscape Layout

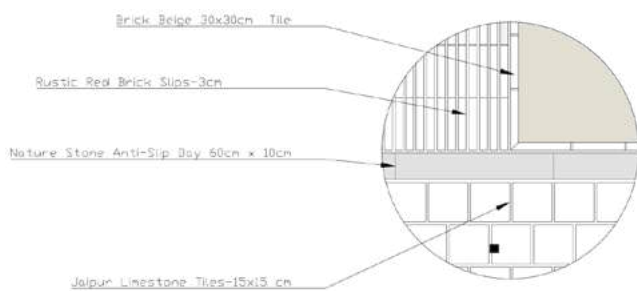
Conference Hall Entrance Area



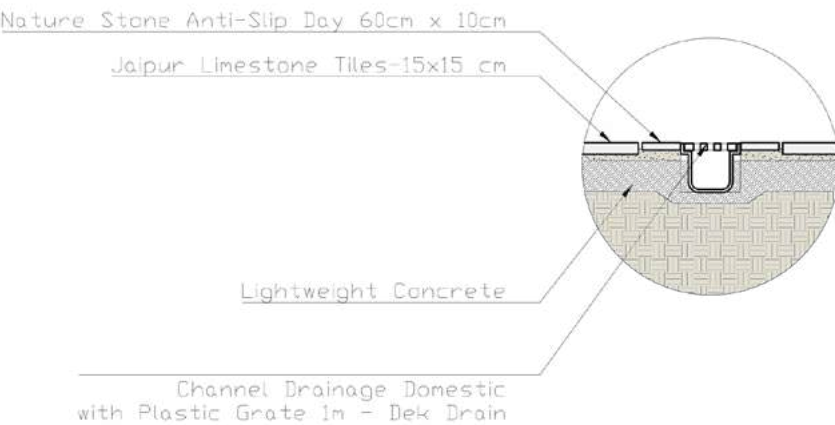
FoodCourt Area



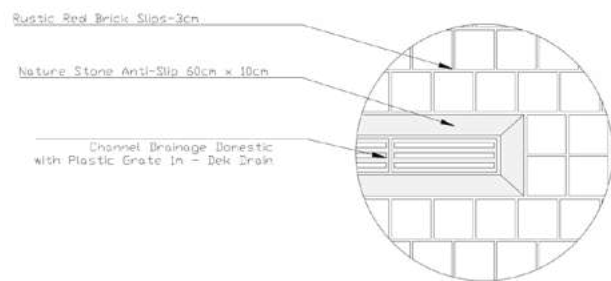
Code Detail.1



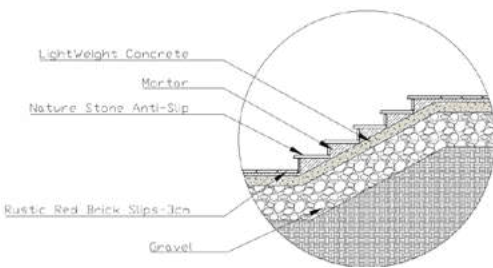
Code Detail.3

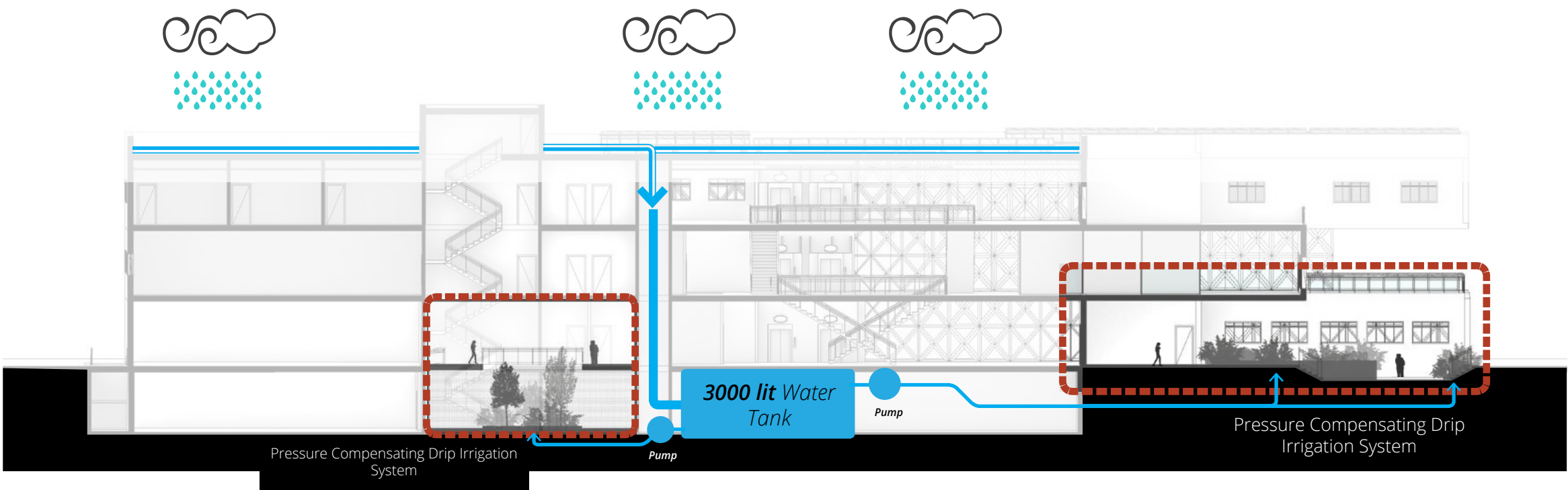


Code Detail.2



Code Detail.4





Indoor Water Management

Fixture Information			Flush Rate		% Occupants
Fixture ID	Fixture Family	Fixture Type	Baseline Flush Rate (lpf)	Design Flush Rate(lpf)	
T-M	Toilet (male)	Low-Flow Water Closet	6.00	3.00	100
T-F	Toilet (female)	Low-Flow Water Closet	6.00	3.00	100
Baseline case annual flush volume (liters/year)			650,310.00		
Design case annual flush volume (liters/year)			325,155.00		

Group Name	Baseline Case (liters/year)			Design Case (liters/year)		
	Annual Flush Volume	Annual Flow Volume	Annual Consumption	Annual Flush Volume	Annual Flow Volume	Annual Consumption
WCs	650,310.00	102,965.75	753,275.75	325,155.00	86,708.00	411,863.00
Annual baseline water consumption (liters/year)			753,275.75			
Annual design water consumption (liters/year)			411,863.00			
Percent water use reduction (%)			%45.32			

Building Information			
Occupancy Type	Employees (FTE)	Visitors	Gender Ratio [%]
Total	28	650	%100
Male	14	450	%68
Female	14	200	%32
Annual days of operation		265	

Total Daily Usage of Water	
Baseline(liters)	Design (liters)
1,602.00	801
52.00	426

Facade Heat Transfer And Details

Detail. 1

Detail .2

Detail .1

Detail .3

1. Venus V-Cool Aluminum curtain wall Profile; fixed to concrete ground floor slab with steel bracket; NBS ref;
2. PPC coated Insulation panel set below curtain wall to prevent thermal bridging, sealant applied between PPC panel and floor slab to prevent water ingress
3. Knauf Thermal floor slab; Glass Wool insulation; thickness 100mm around perimeter of concrete slab to meet the PPC panel for continuous insulation. NBS ref; NBS ref P20/10
4. Visqueen Ecomembrane; 1500 gauge; DPM set below slab to return to internal area; / Roof humidity insulation
5. Sealant applied PPC panel and DPM lapped; Internal face of cut concrete in order to increase watertightness around perimeter.Reinforced concrete Floor slab
6. Gantic Facade-slot drain situated perimeter of the building
7. Permeable Paving
8. Slate Floor Covering, typical dimentions of Slab; 300 x 300 x 12mm, Weight per unit 3.38 kg, Manufacturer: Welsh Slate, Product Code: Colour:Heather Grey Penrhun, Finish: Riven,Characteristics ; non-fading, AA Fire Rating - non combustible
9. Glaster Screed; thickness: 55mm. Type: NHL3.5 or NHL1 5,Ratio: 2.5 Sand *: Hydraulic Lime Premixed,
10. Lime mortar, Thickness: 45 mm, Type : NHL3.5 or NHL5, Ratio 2.5 Sand, 1 Hydraulic Lime Premixed, Colour/ texture: It is the aggregate/sand That precominantly effects the colour and texture.
11. Concrete Ground bearing slab to conform to BS -8500 2 Standardised Prescribed mix ST2. Reinforced
12. XPS thermal insulation to prevent Thermal Bridging
13. Cool-Coating

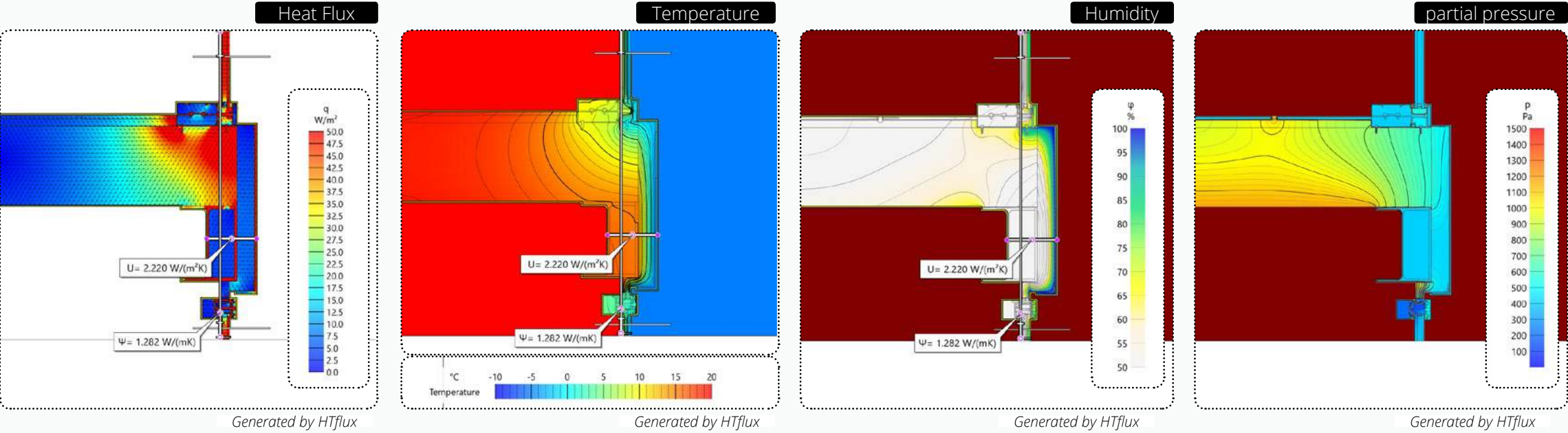
Base Case

U-Value

2.22 W/(m2/K)

Psi-Value

1.28 W/(m/K)



Improved Case

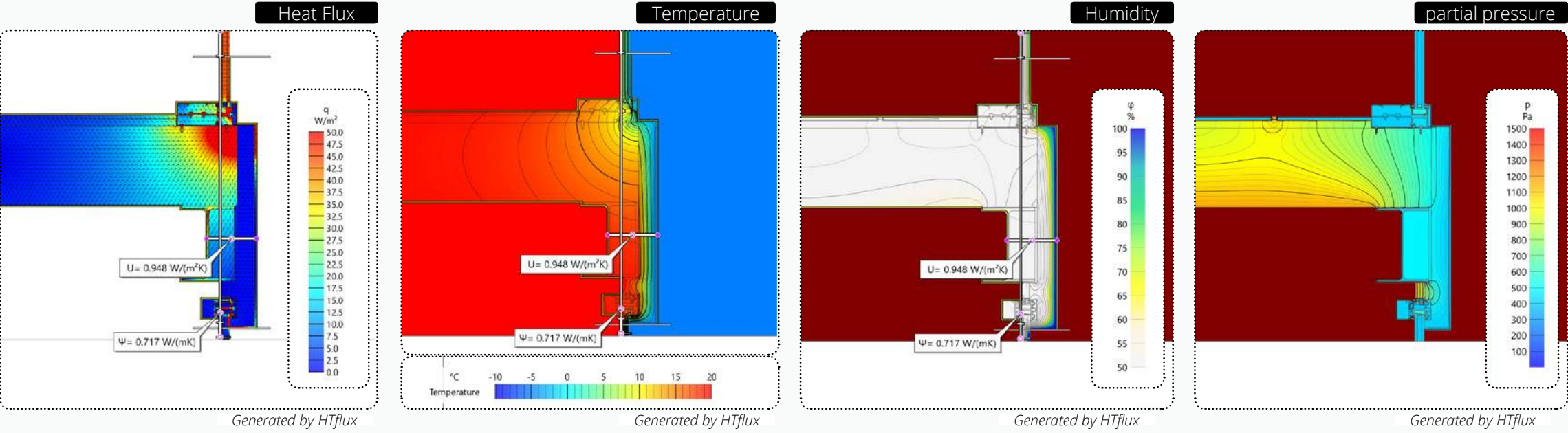
I- Cavities filled with Thermal Insulation
II- Vertical Insulation Extended to the level of curtainwall upper profiles

U-Value

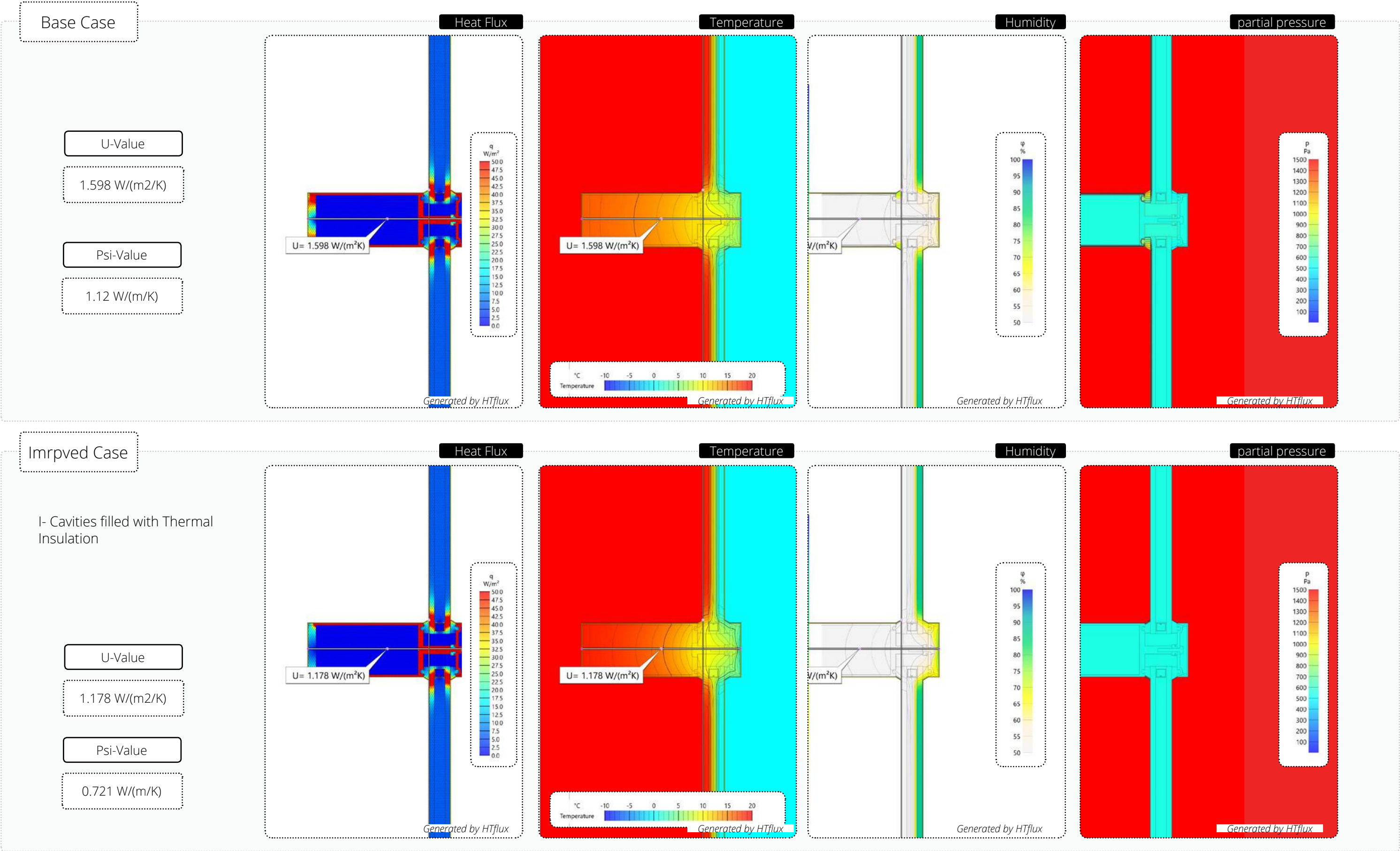
0.948 W/(m2/K)

Psi-Value

0.717 W/(m/K)



 Facade Heat Transfer And Details(Thermal Bridges)



Modelling Key Assumptions

Meshing

Statistics

Setup

Near-Wall Treatment

Reference Values

Edge Sizing

solver preference

size function

Relevance Center

Smoothing

Min size

max size

Max Tet Size

Growth Rate

Nodes

Elements

solver

Time

Velocity formulation

model

type

K-epsilon Model

Model Constants

Epsilon:1.9TKE Prandtl Number: 1»

Area [m2]

Density[kg/m3]

Enthalpy[j/kg]

Length [m]

Pressure [Pascal]

Temperature [k]

velocity[m/s]

Viscosity[kg/m-s]

Ratio of Specific Heats

0.35 m

Fluent

Curvature

Coarse

Medium

0.0267 m

2.67 m

5.34 m

1.2

42133

205898

Pressure-based

Steady

Absolute

Viscous

k-epsilon

Realizable

«C-2

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Natural Ventilation Analysis (CFD)

Base Case

Bigger Inlets

Smaller Inlets

Space Volume (m3)

Inlet Area(Each) (m2)

Outlet Area(Each) (m2)

Total Inlet Area (m2)

Total Outlet Area(m2)

3438

0.58

0.17

8.65

2.16

3438

1.23

0.17

18.45

2.16

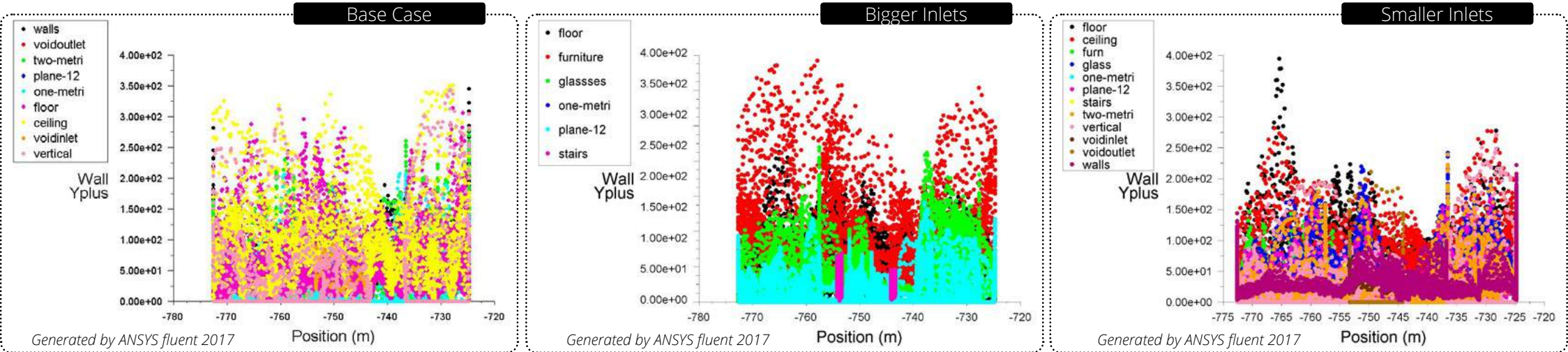
3438

0.29

0.17

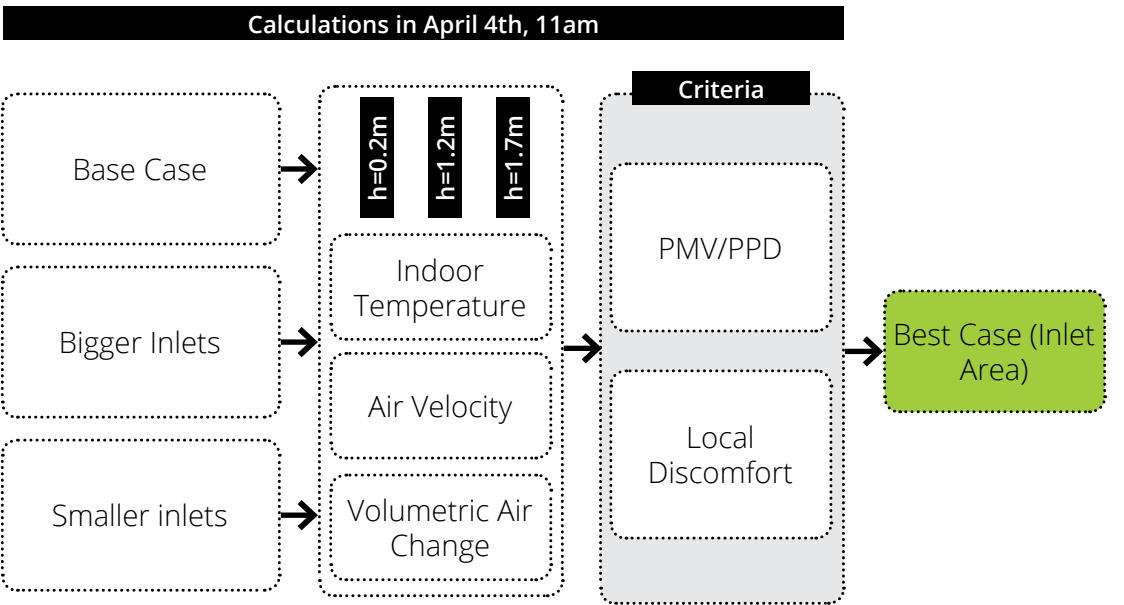
4.35

2.16



Modelling Key Assumptions

Surface	Temperature	Surface Material	Density(kg/m3)	Specific heat(J/kg x K)	Thermal Conductivity(W/m x K)
South inlets	24	-	-	-	-
North Outlets	25	-	-	-	-
Void Inlet	24	-	-	-	-
Void Outlet	24.6	-	-	-	-
Walls	20	Gupsum	800	950	0.17
Floor	22	Timber Flooring	650	1200	0.14
Ceiling	22	Gypsum	800	950	0.17
Glazings	23	Glass	2500	840	0.2
Furniture	22	Wool	198	1360	0.06
Stairs	22	Timber Flooring	650	1200	0.14



Modelling Key Assumptions

Boundary Conditions

Solution Methods

Spatial Discretization

Inlets (South windows) Velocity Magnitude [m/s]

Void inlet Velocity Magnitude [m/s]

Outlets (north windows) Gauge Pressure

Void outlet Gauge Pressure

Other surfaces Roughness constant

Scheme

Simple

Gradient

Pressure

Momentum

Turbulent Kinetic Energy

Turbulent Dissipation Rate

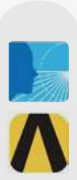
Least Squares Cell Based

Second Order

Second order upwind

First order Upwind

First Order Upwind



Base Case

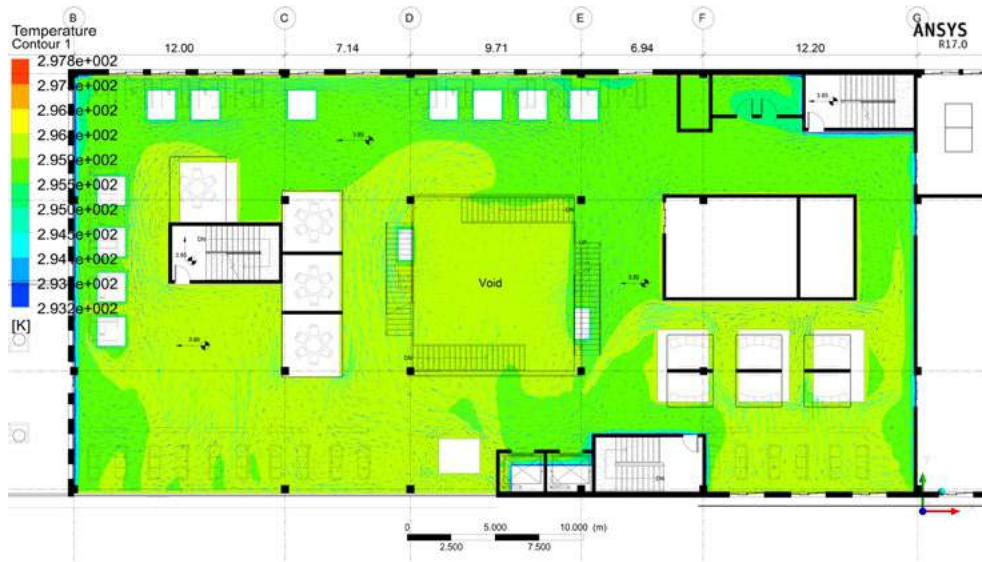
Bigger Inlets

Smaller Inlets

Temperature- 1.2m

Temperature- 1.2m

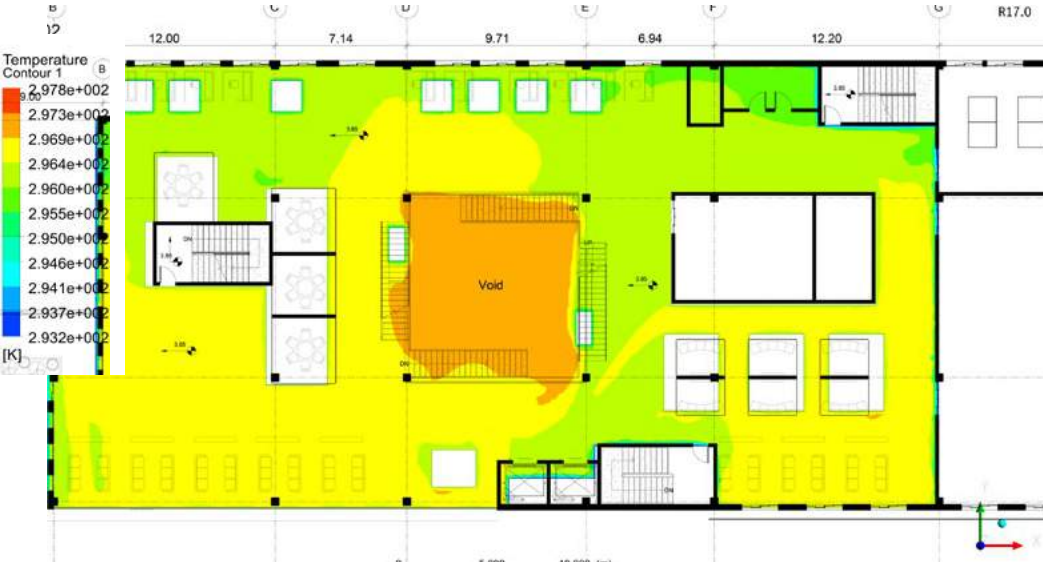
Temperature- 1.2m



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Generated by ANSYS fluent 2017

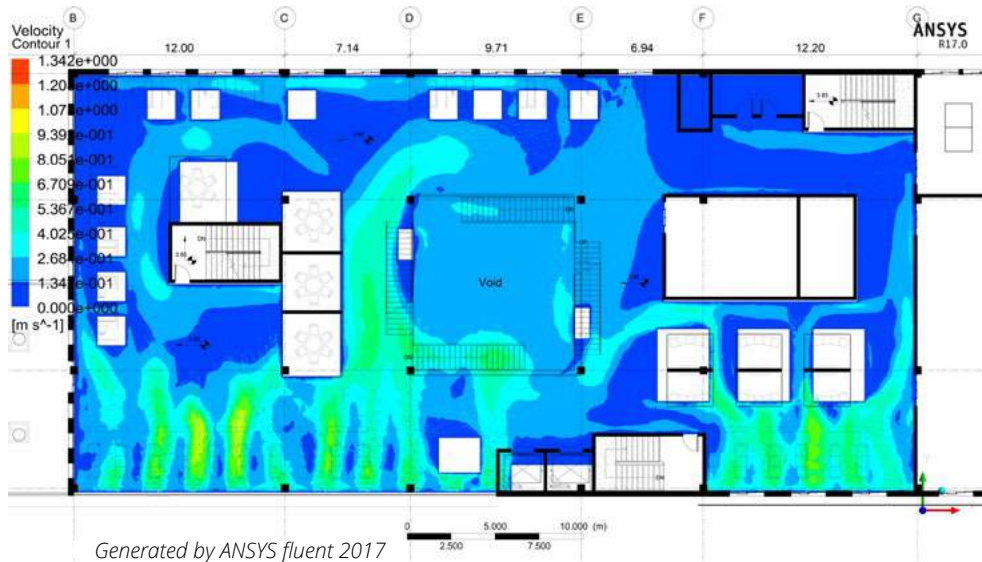


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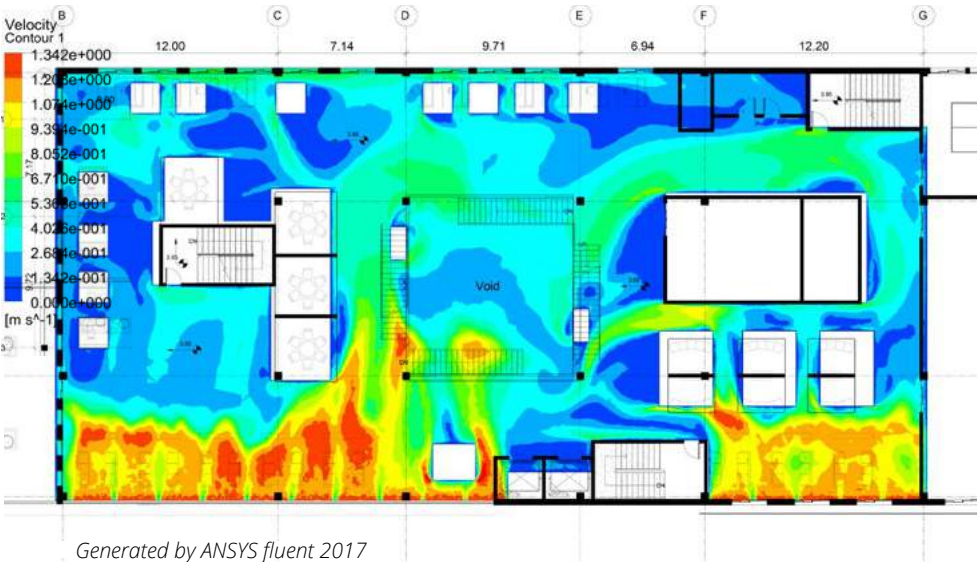
Velocity- 1.2m

Velocity- 1.2m

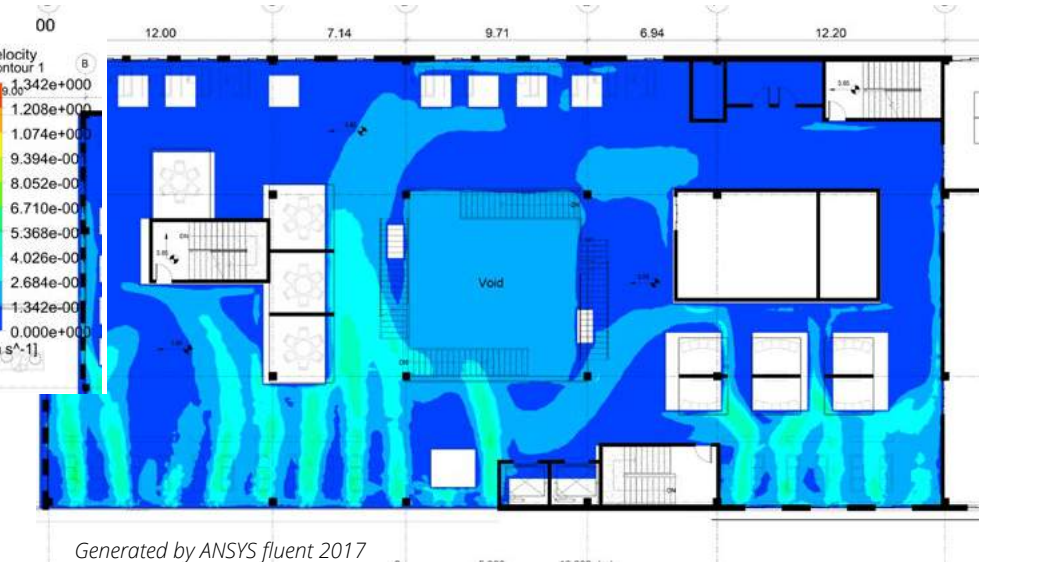
Velocity- 1.2m



Generated by ANSYS fluent 2017

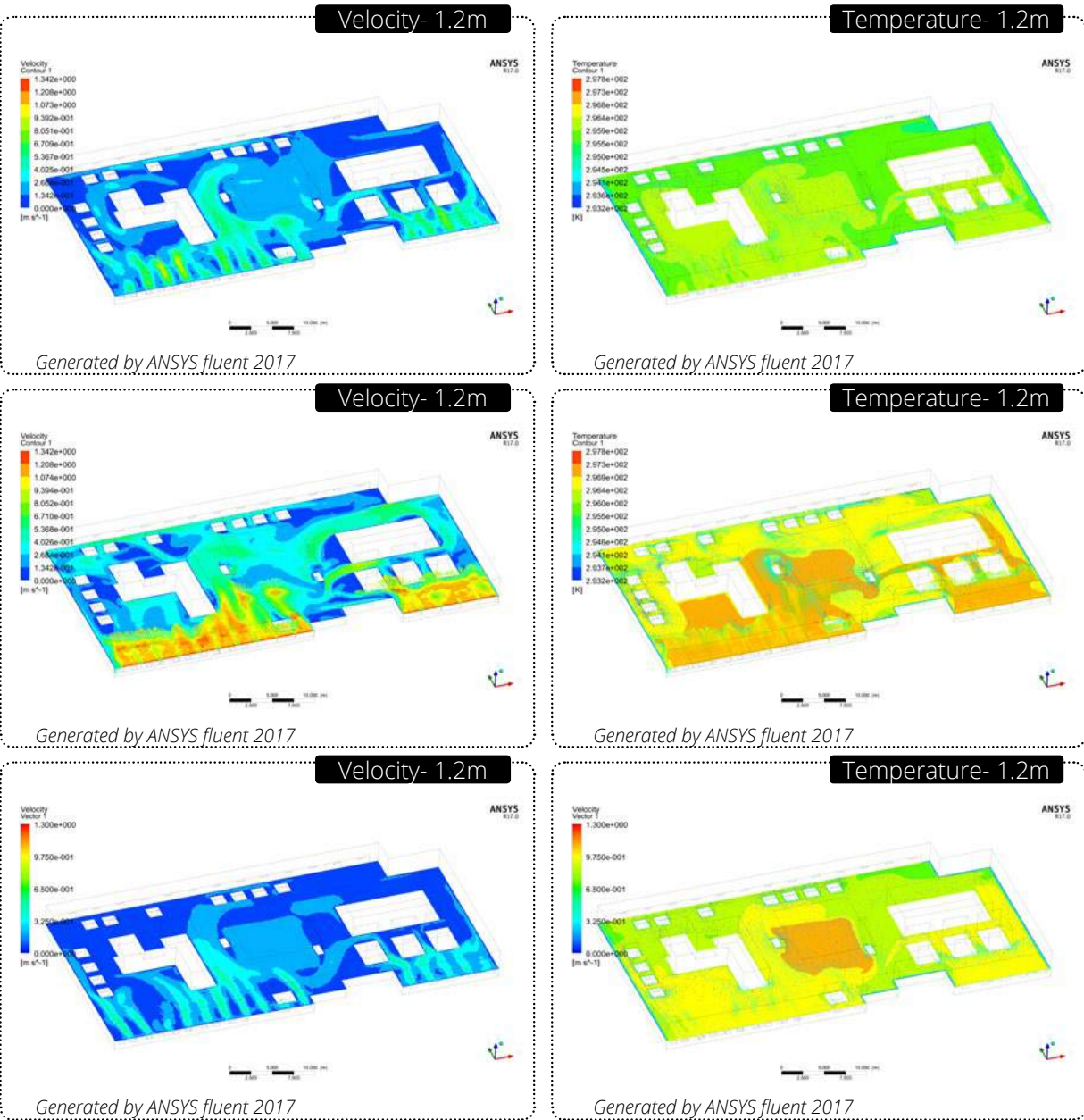


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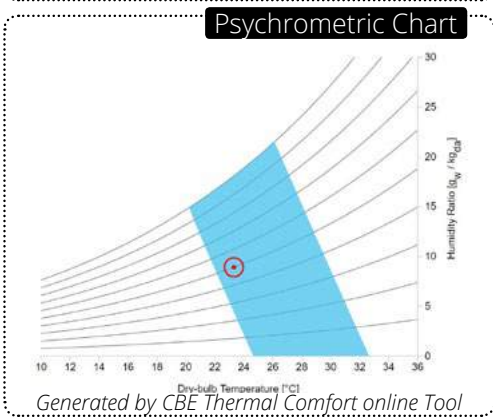
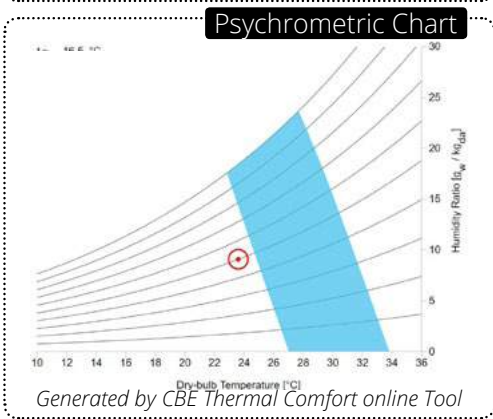
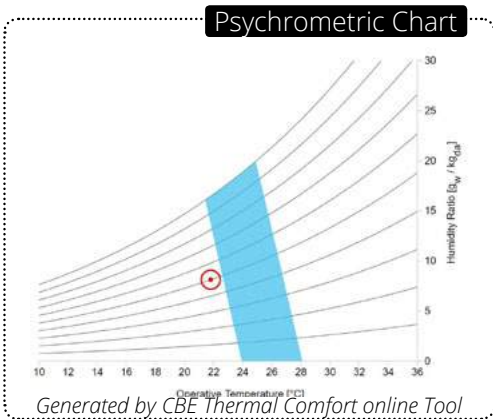


Generated by ANSYS fluent 2017

Natural Ventilation
Analysis Results



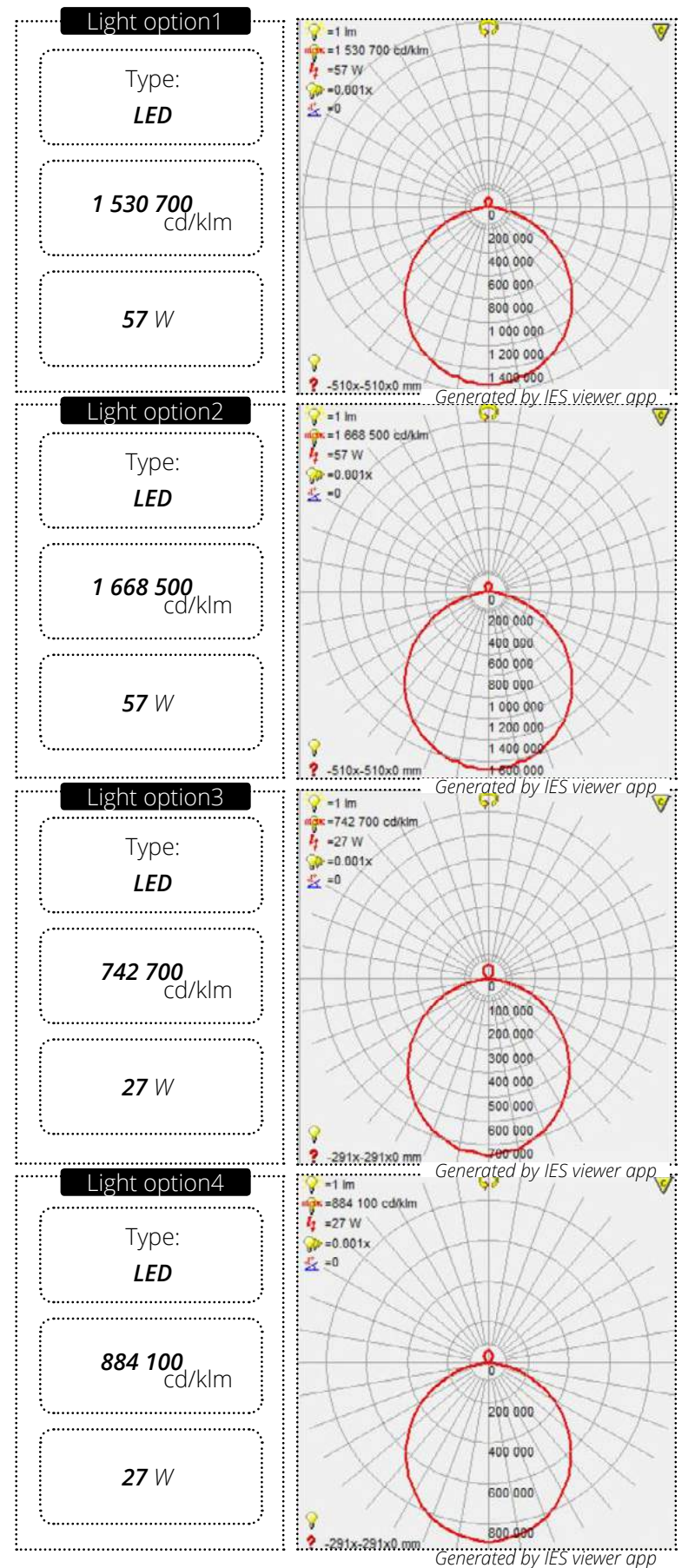
SET	Dry-bulb temp at still air	Cooling effect	Sensation	Compliance	PMV with elevated air speed	PPD with elevated air speed
23.2	21.90	0.8 °C	Neutral		- 0.56	%12
22.6	21.9	1.7 °C	Slightly Cool		- 0.68	%15
24	23.10	0.2 °C	Slightly Cool		- 0.32	%7



Case	Caclucation Height(m)	Relative Humidity(%)	Air Velocity(m/s)	Metabolic Rate (MET)	Clothing Level (Clo)
Base Case	22.68	50	0.20	1.1	0.8
	22.73		0.22		
Bigger Inlets	23.55		0.36		
	23.65		0.48		
Smaller Inlets	23.24		0.15		
	23.35		0.16		
	23.35		0.13		

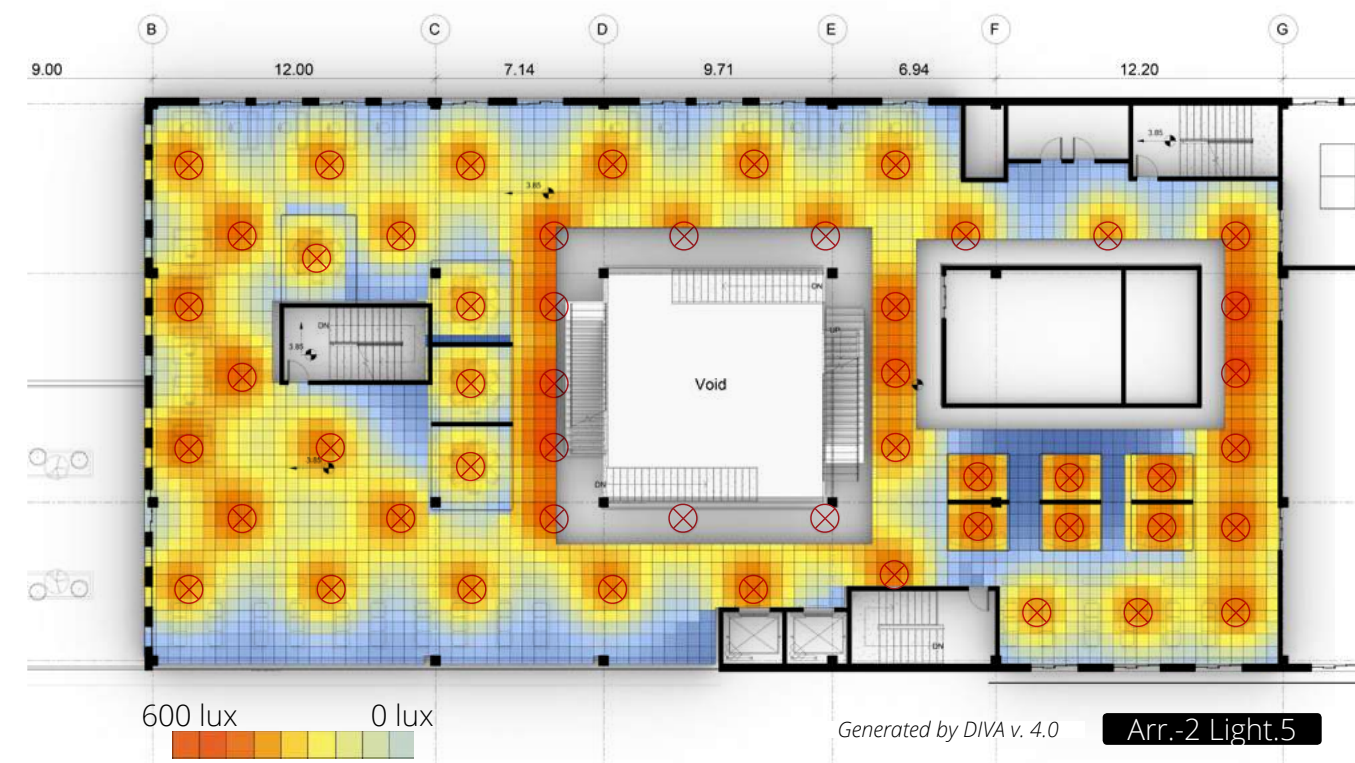
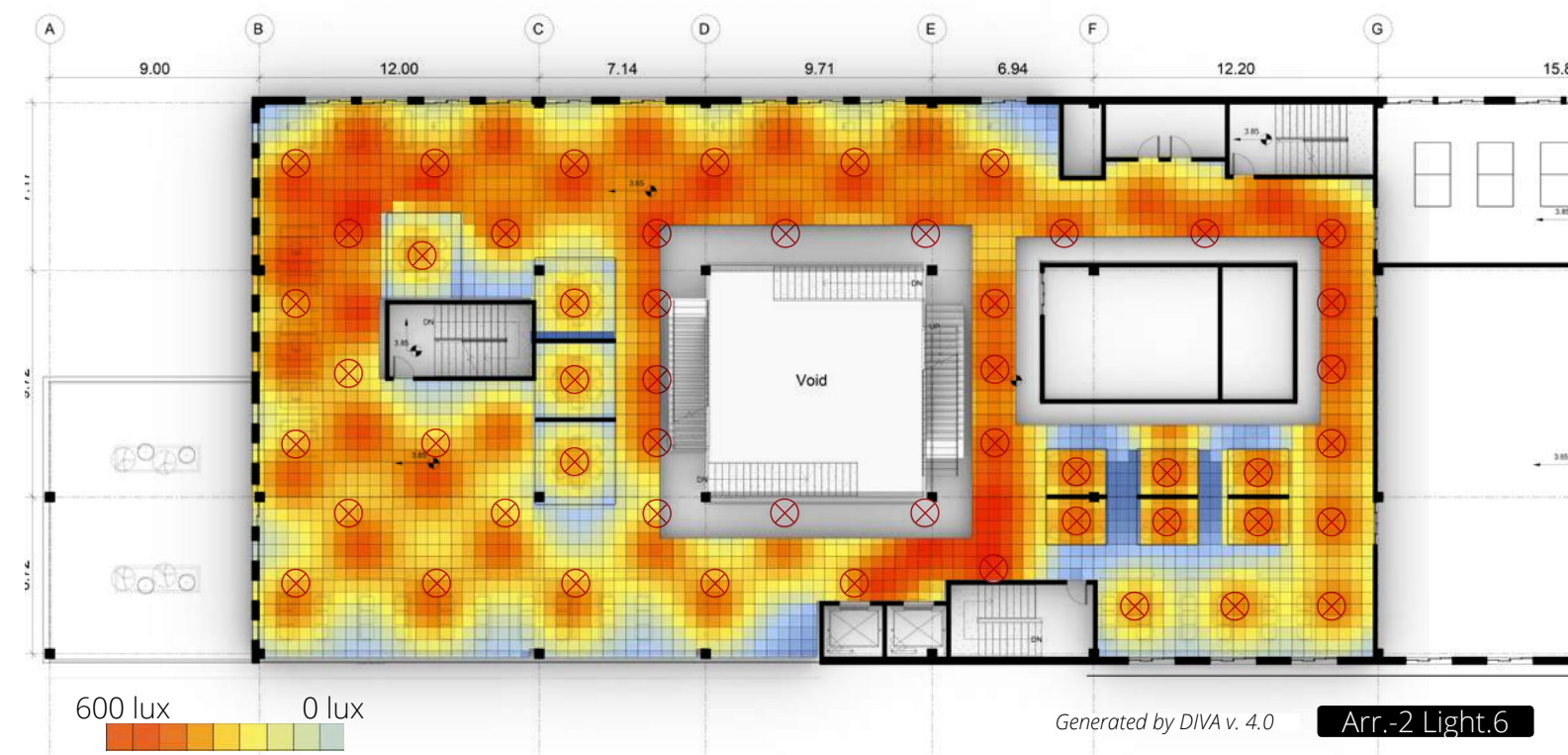
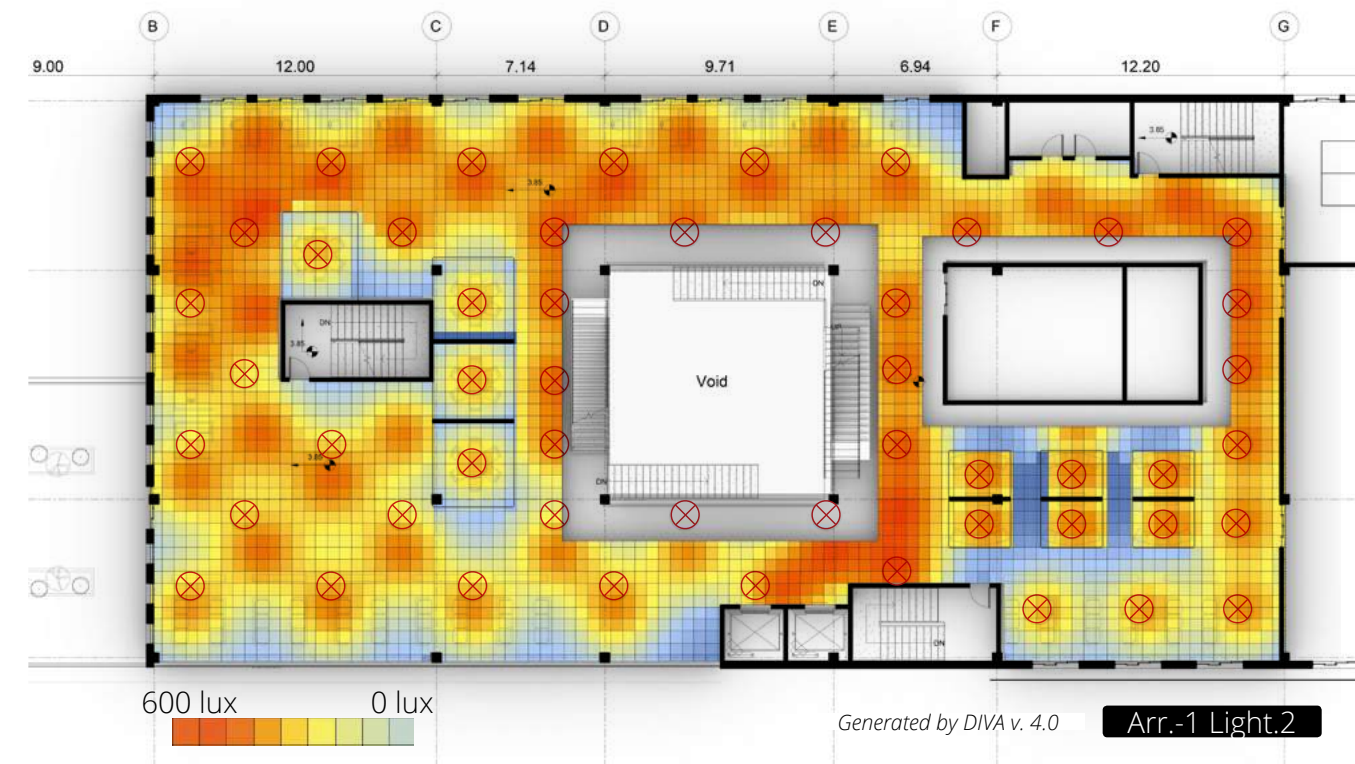
Case	Draft	Ankle Draft			Vertical air temperature difference	Floor surface temperature
		PPD	PMV	OK?		
Base Case		%21	- 0.5			
Bigger Inlets		%26	- 0.4			
Smaller Inlets		%16	- 0.4			

Does not comply with
ASHRAE Standard -55
2017,The air speed value
selected is outside the
range defined by the
ASHRAE 55 standard for
occupants with no local air
speed control





Case	Arr.-1 Light.1	Arr.-1 Light.2	Arr.-1 Light.3	Arr.-1 Light.4	Arr.-2 Light.1	Arr.-2 Light.2	Arr.-2 Light.3	Arr.-2 Light.4
number of Sensors >300lux	581	846	1	1	1418	1743	1	1
number of Sensors <150lux	830	685	2869	2340	341	254	2102	1433
between 150 and 300 lux	1949	1829	490	1019	1601	1363	1257	1926
% of Sensors >300lux	%17.29	%25.18	%0.03	%0.03	%42.20	%51.88	%0.03	%0.03
% of Sensors <150lux	%24.70	%20.39	%85.39	%69.64	%10.15	%7.56	%62.56	%42.65
between 150 and 300 lux	%58.01	%54.43	%14.58	%30.33	%47.65	%40.57	%37.41	%57.32

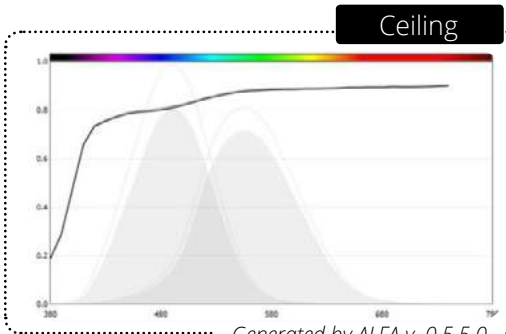
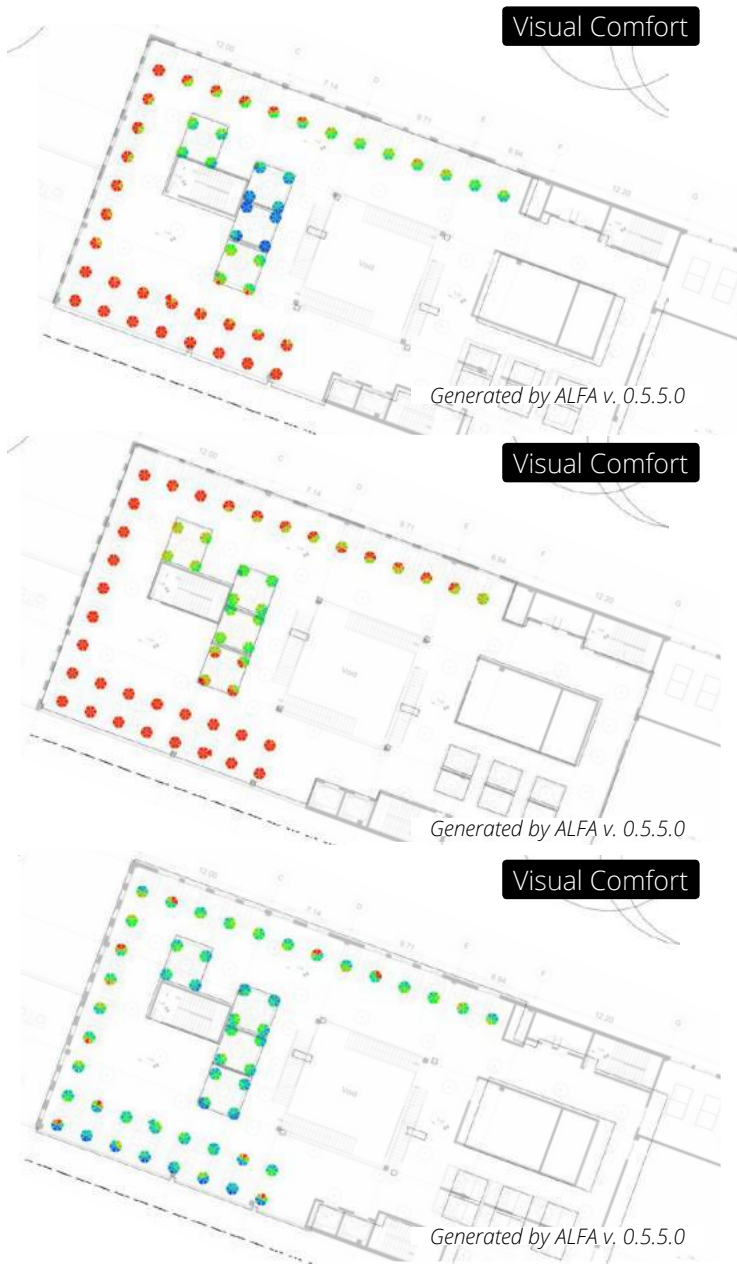




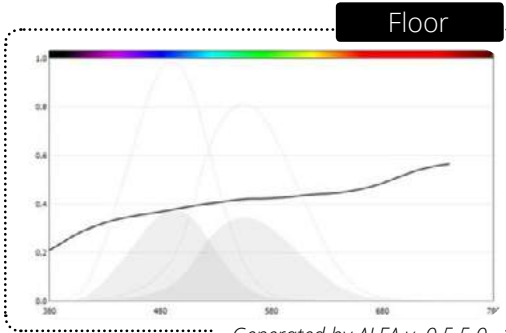
Daytime no lighting(April,4th- 9:00 AM)

Daytime With lighting (April,4th- 9:00 AM)

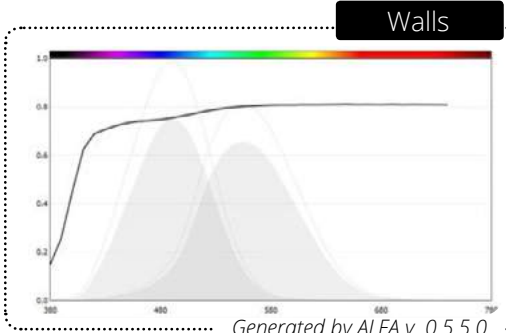
Nighttime With lighting



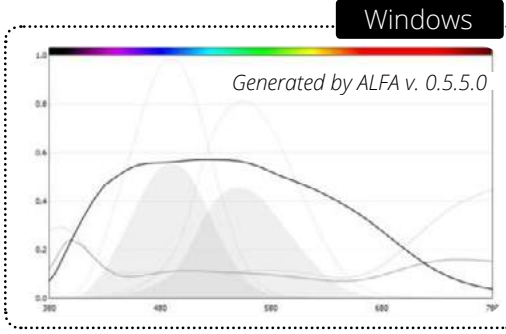
COLOR	WHITE
SPECULARITY	0.5
R(PHOTONIC)	87.2
R(SCOTOPIC)	81.8
M/P	0.94



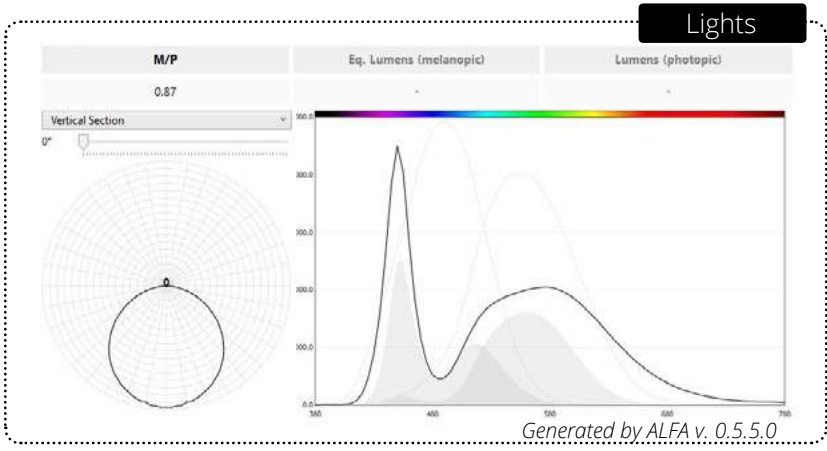
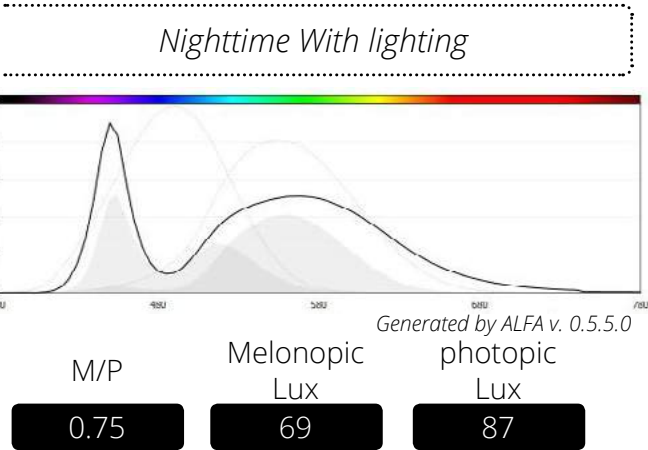
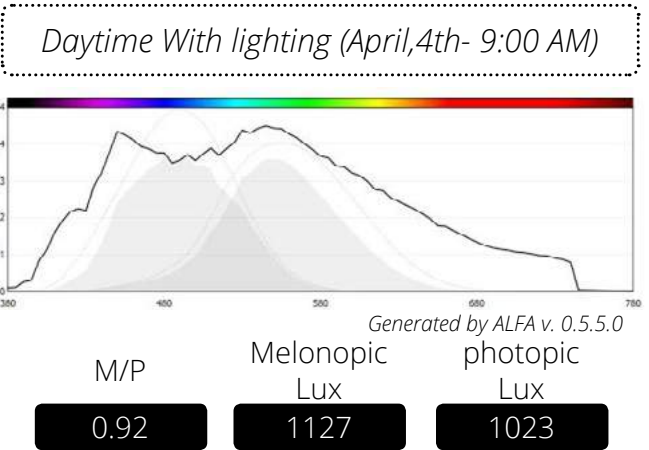
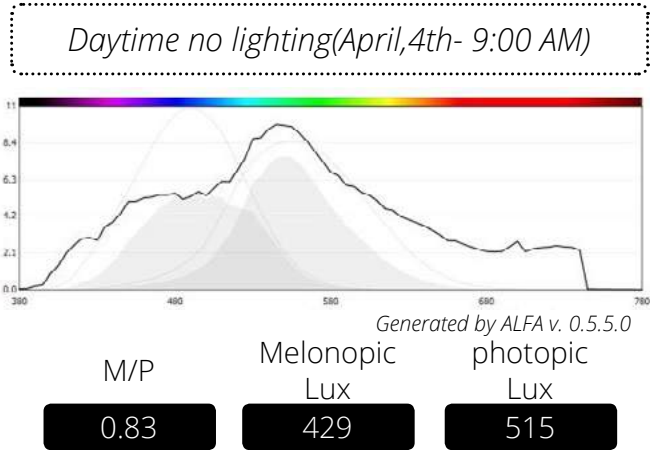
COLOR	GREY
SPECULARITY	0.2
R(PHOTONIC)	41.8
R(SCOTOPIC)	37.6
M/P	0.90

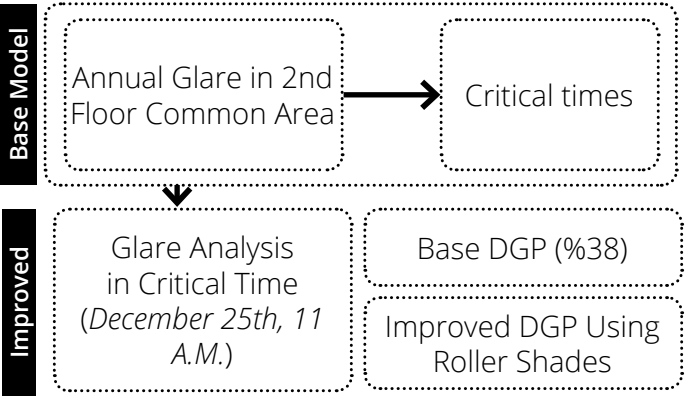
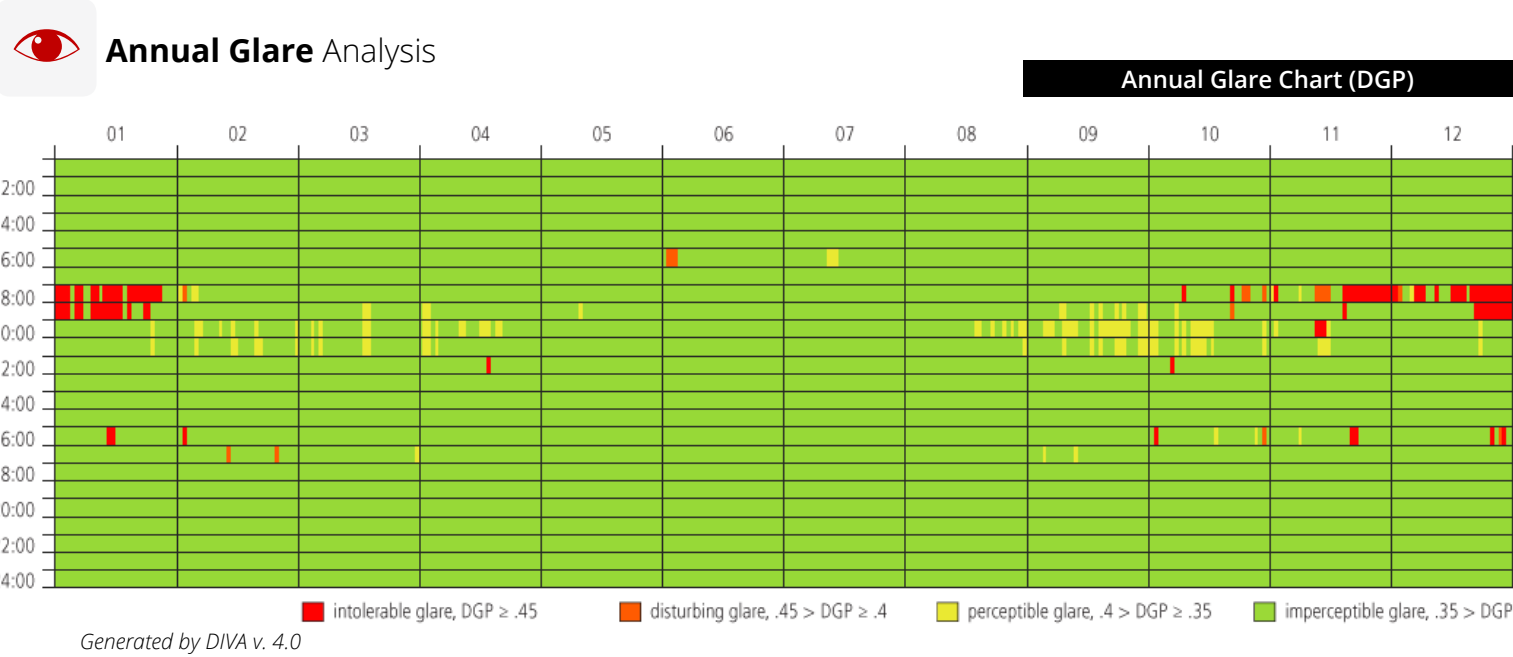
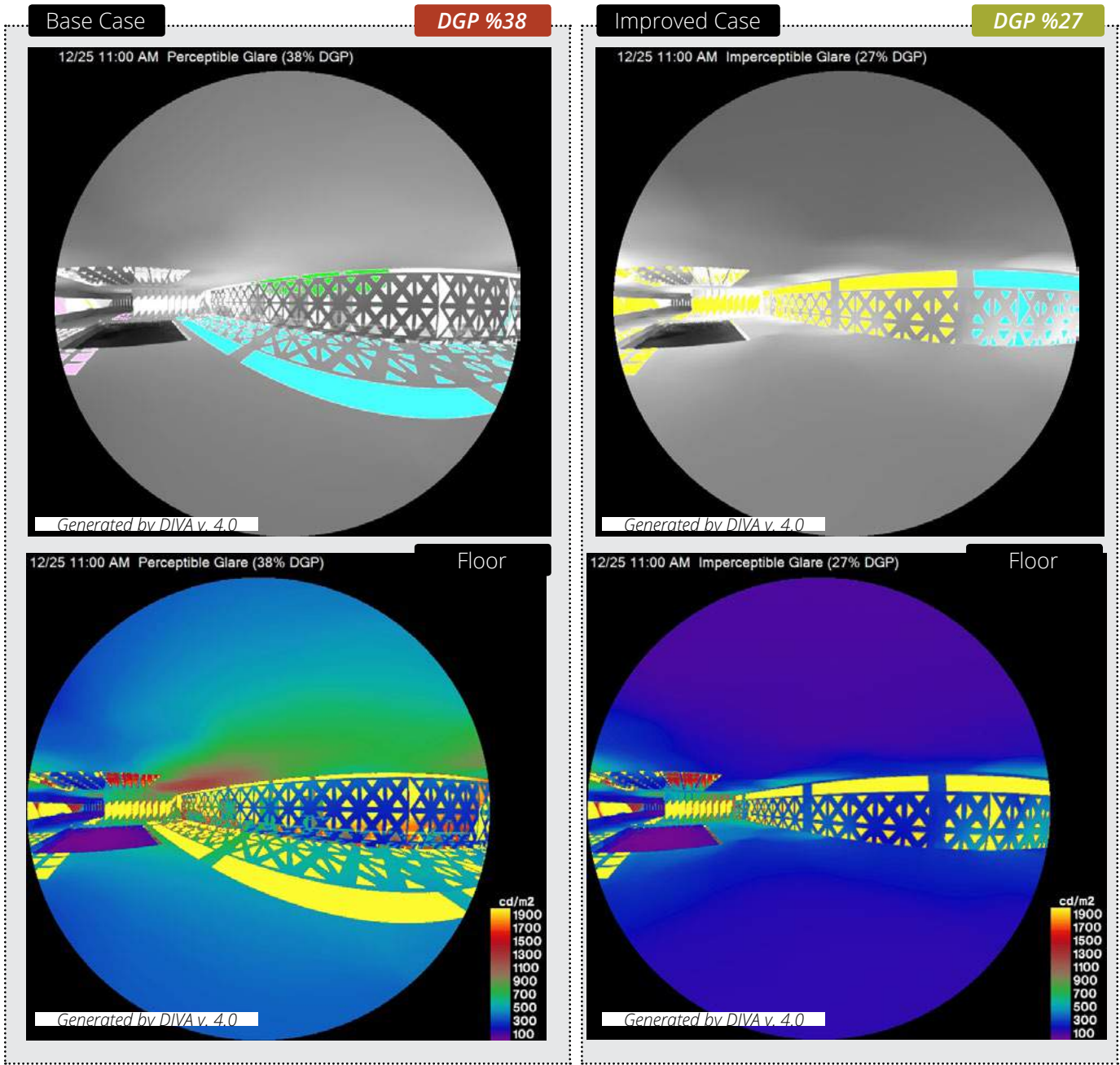


COLOR	WHITE
SPECULARITY	0.4
R(PHOTONIC)	79.8
R(SCOTOPIC)	75.8
M/P	0.95

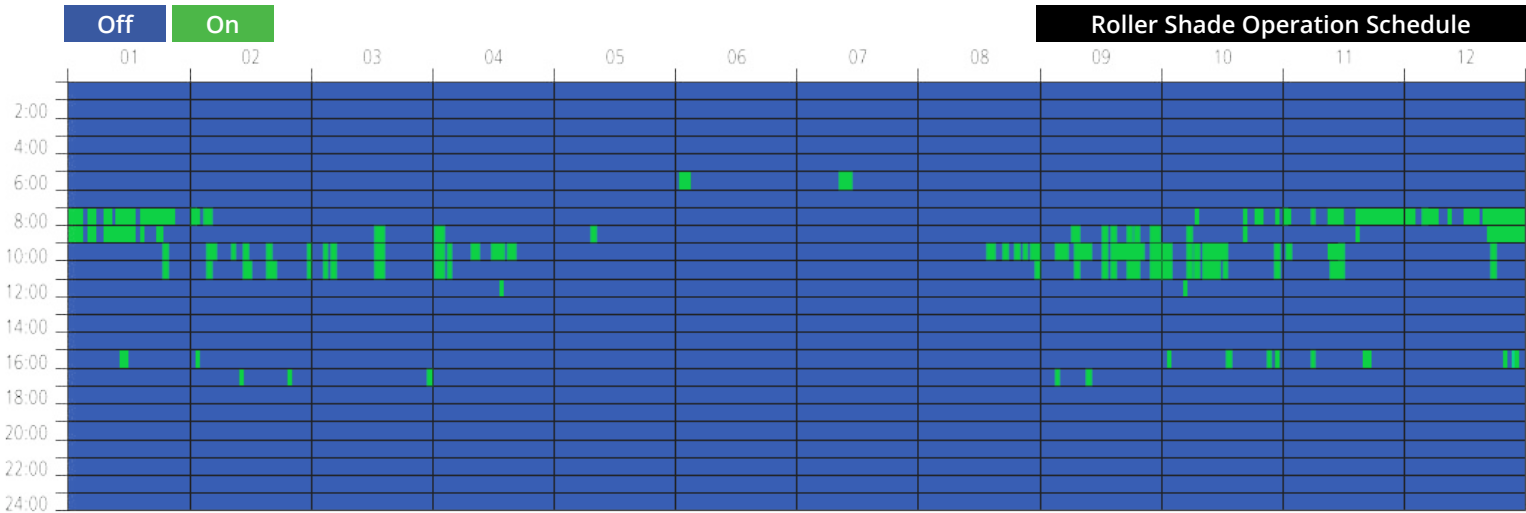
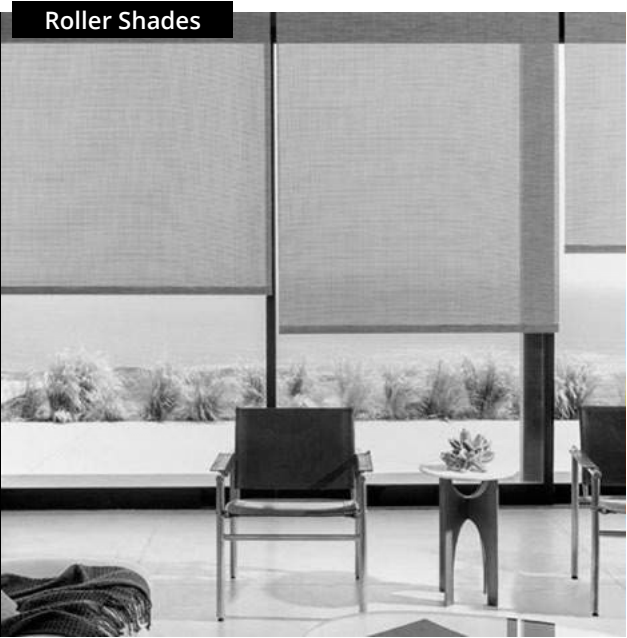


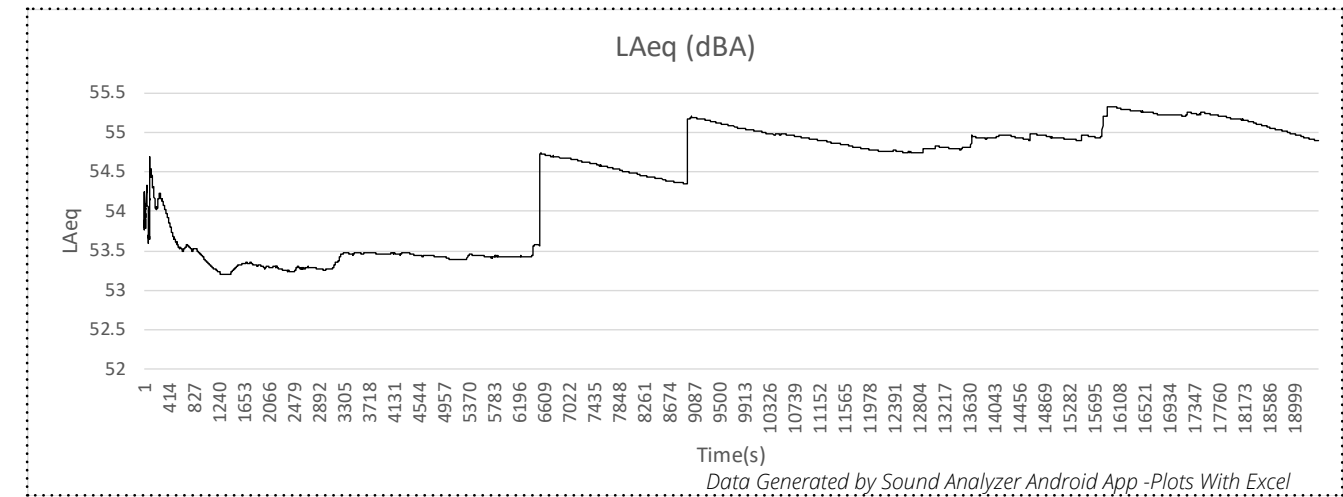
RF(PHO)	9.9
RF(MEL)	10.7
RB(PHO)	%11.00
RB(MEL)	%11.20
T(PHO)	52.90
T(MEL)	54.9
M/P	1.04





Roller Shades Used for Improving Visual Comfort.
Roller Shade Light Transmittance: 0.2.

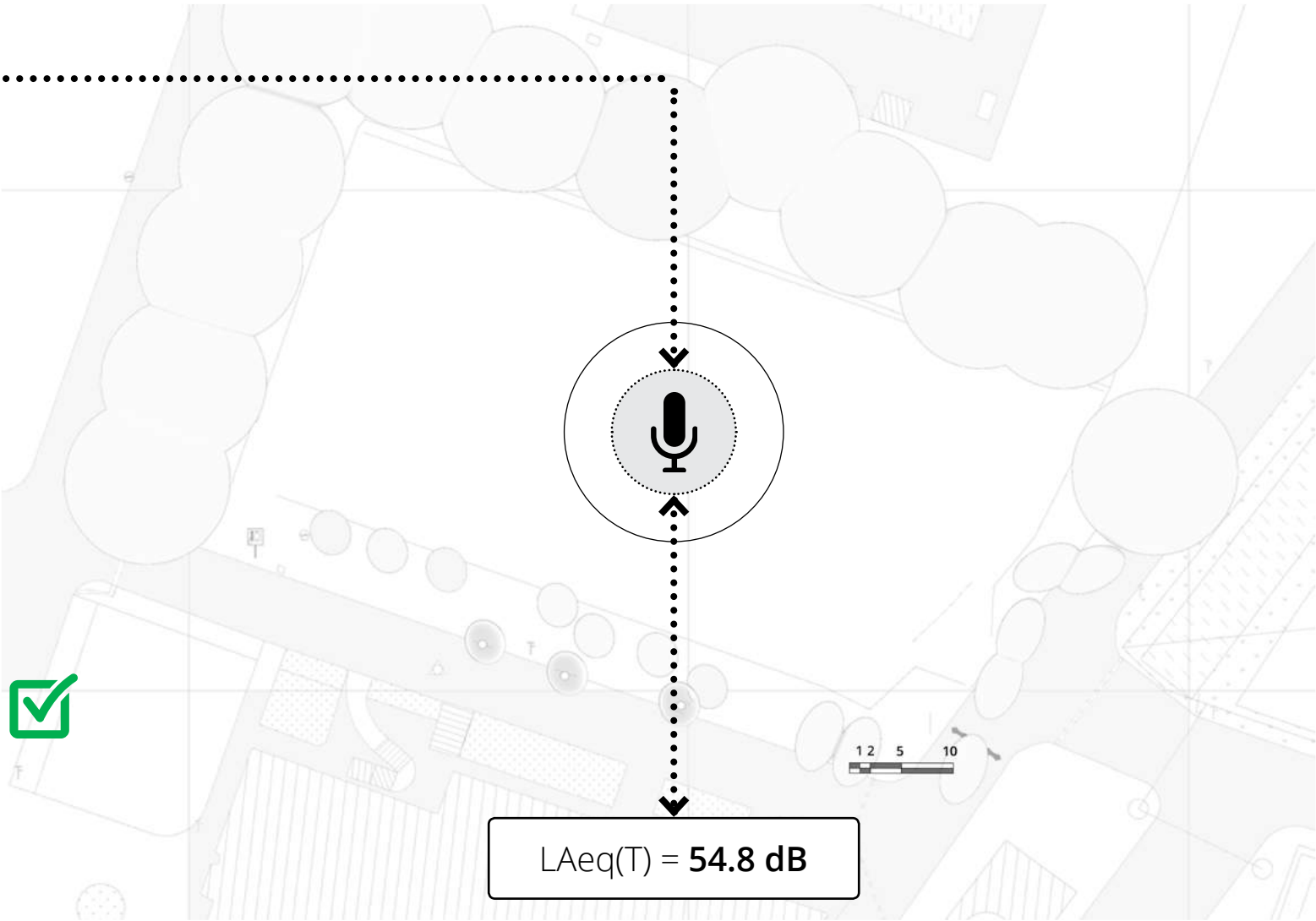




district acoustical type	Background noise: level (LAeqT(30) in dB)		Allowed Building Types
	7 TO 10	10 TO 7	
Low Background noise	55	45	Residential, Tourism Center, Healthcare, Cultural Centers, Sport Centers, Medium Level Buisness Centers
Medium Background noise	65	55	«Educational, Administration, Indoor sports club,Mixed Use buildings(Business, Administration, Residential), Shopping Malls, Bazaars, Exhibition Halls»
High Background noise	75	65	Terminals, Storages, Parkings, Outdoor Sports Club, Industrial Buildings, Military Buildings, Airports



Site Acoustical
Assessment

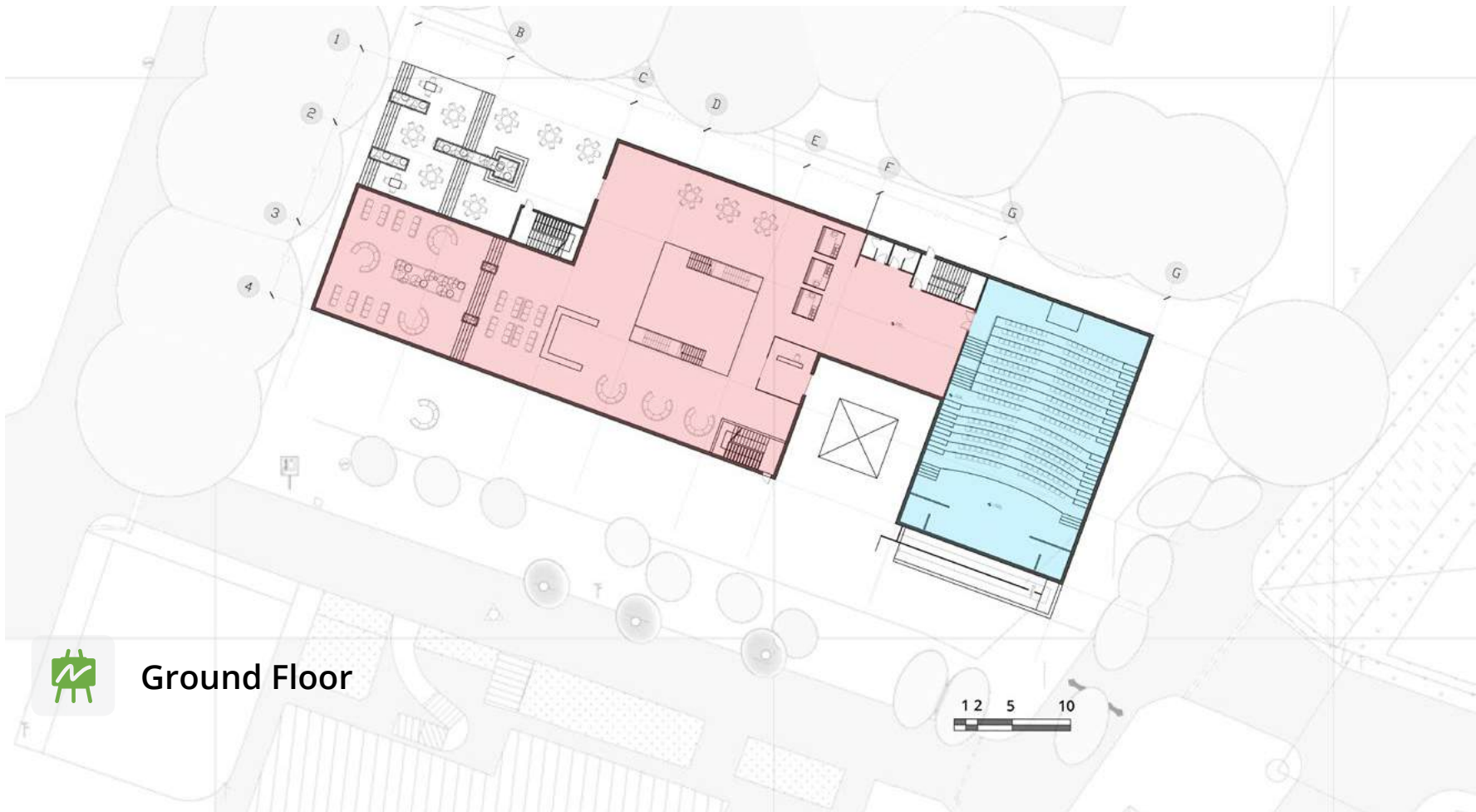
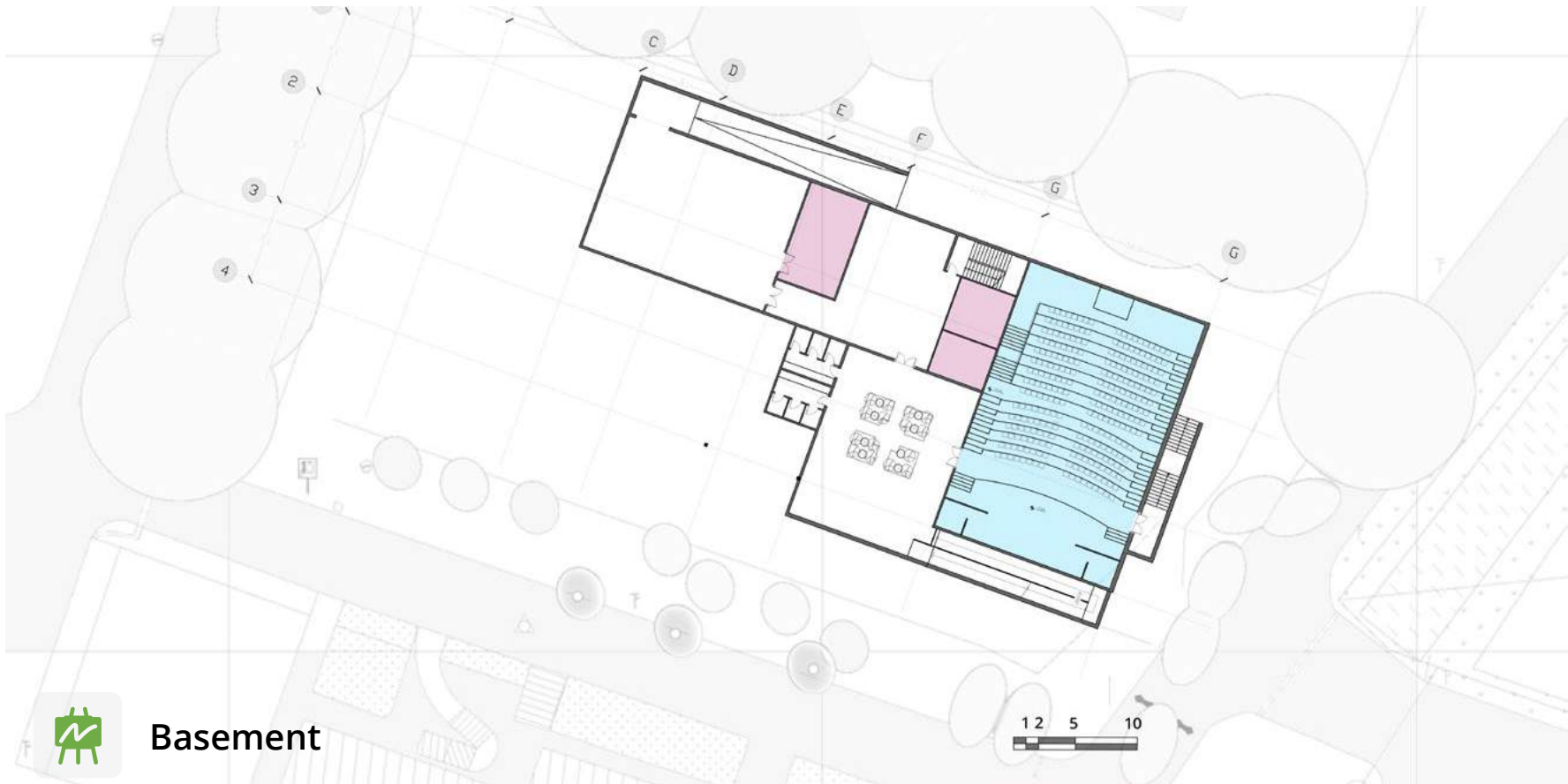


Views From the Site



🔖

Labelling Acoustic Zones

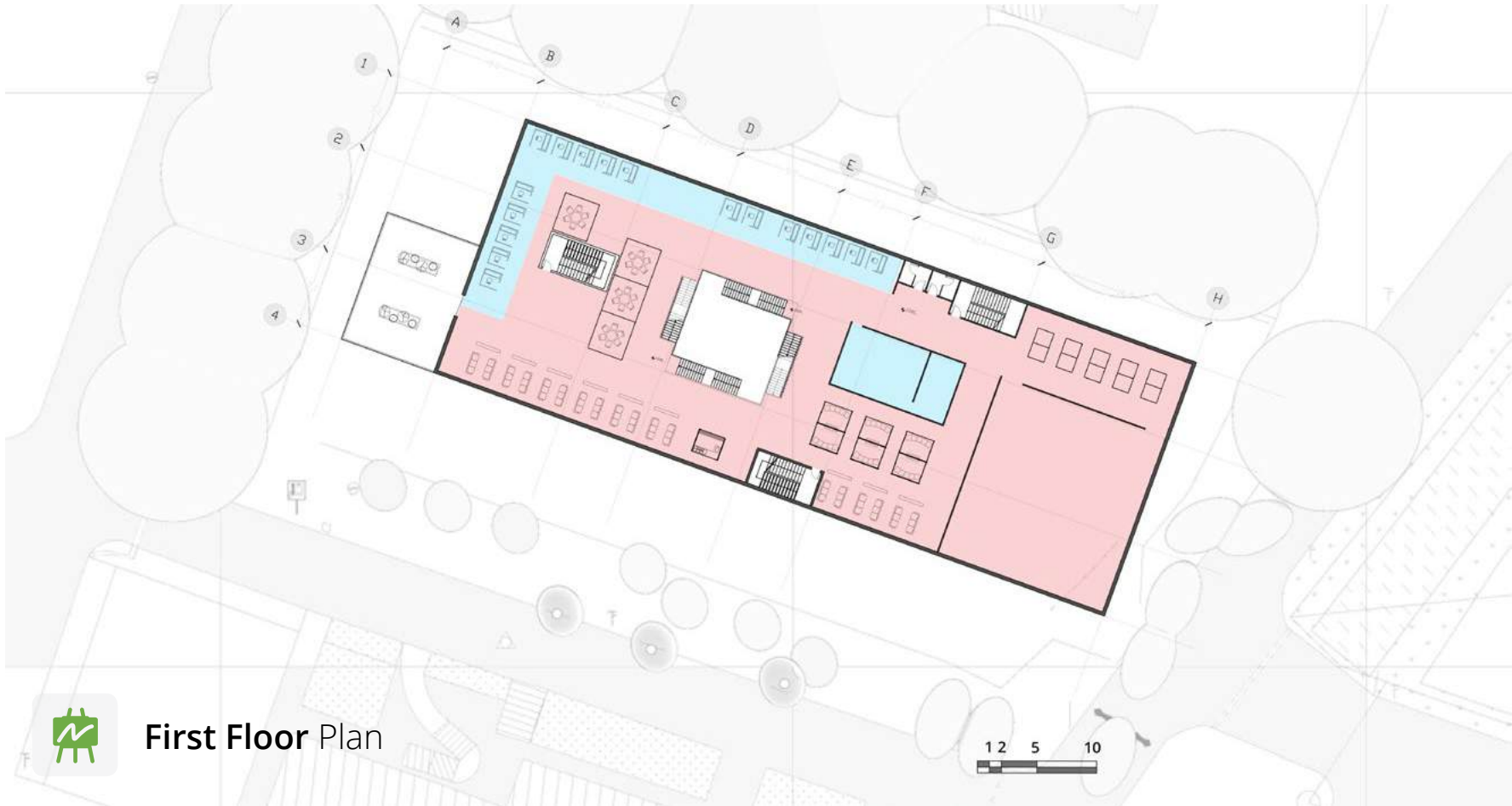


Zone Acoustical Labels

- Loud zones:** includes areas intended for appliances, mechanical equipment or amenities (e.g., kitchens, fitness rooms, social spaces, recreational rooms).
- Quiet zones:** includes areas intended for focused work, wellness, rest, study and/or privacy.
- Mixed zones:** includes areas intended for learning, collaboration and/or presentation.

🔖

Labelling Acoustic Zones



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■ Mixed zones:

includes areas intended for learning, collaboration and/or presentation.



- Acoustic Requirements
- Acoustic Requirements for Building Elements and spaces are listed in the table.
 - The numbers and limits are determined due to Annex 18 and WELL Certification Criteria.

Spaces		Maximum Background noise		Optimum Reverberation Time[s]	Maximum Background noise				β - Impact Sound Isolation Level (IIC[dB])	STI
		L _{Aeq} T(30)	PNC		Partition walls	Exterior Walls/Simple	Exterior Walls/ Multiple	Ceilings/ Floors(simple)		
Basement	W.C.	55	50	-	50	40	35	35	60	-
	Conference Hall	35	35	0.85 in 1000hz	50	55	50	50	60	>0.6
	Fire Control Room	65	60	-	55	40	35	55	60	-
	Electrical/ Systems Control	65	60	-	55	40	35	55	60	-
	Facility Room	65	60	-	55	40	35	55	60	-
Ground Floor	W.C.	55	50	-	50	40	35	35	60	-
	Entrance	50	45	1.5	45	40	35	45	65	-
	Information	50	45	1.5	45	40	35	45	65	>0.6
	Conference Hall	35	35	0.85 in 1000hz	50	55	50	50	60	-
	Food Stall	55	50	-	50	40	35	35	60	-
	Shop	45	40	2	45	40	35	50	45	-
	FoudCourt Sitting Area	50	45	1.5	45	40	35	45	65	-
	Lounge	50	45	1.5	45	40	35	45	65	-
1st Floor	W.C.	55	50	-	50	40	35	35	60	-
	Gallery	35	35	-	50	45	40	50	55	>0.6
	Multi-Functional Space	50	45	1.1	50	50	45	55	60	-
	TV Room	50	45	1.5	45	40	35	45	65	-
	Study Hall (Individual)	35	30	1.5	45	40	35	45	65	-
	Study Hall (Group)	35	30	1.5	45	40	35	45	65	-
	Game Room	50	45	0.8	50	50	45	55	60	-
	Book Cafe	50	45	1.5	45	40	35	45	65	-
Second Floor	W.C.	55	50	-	50	40	35	35	60	-
	Lounge	50	45	1.5	45	40	35	45	65	-
	Study Hall (Individual)	35	30	1.5	45	40	35	45	65	-
	Study Hall (Group)	35	30	1.5	45	40	35	45	65	-
	Class Rooms	35	30	1	35	45	40	50	55	-
	Prayers Room	35	35	-	35	45	40	50	50	>0.6
	Management	40	35	1.2	45	40	35	45	65	-
	Administration	40	35	1.2	45	40	35	45	65	-
	Conference Room	35	35	0.8	35	45	40	50	55	-
	Student Association	40	35	1.2	45	40	35	45	65	-

 Sample Spaces

Space nameConference Room



Space Area[m2]81Space Volume [m3]280

Orientarion	Adjacent Space
South	Outdoors
North	Corridor
East	Lounge
West	Prayers Room
Top	Roof
Bottom	Lounge

Space nameConference Hall



Space Area[m2]371Space Volume [m3]1930

Orientarion	Adjacent Space
South	Ground/Outdoors
North	Ground/Outdoors
East	Ground/Outdoors
West	Outdoors/ Electrical-Systems Control/ Fire Control
Top	Multi-Functional Space
Bottom	Ground

Space nameClass Room



Space Area[m2]30Space Volume [m3]105

Orientarion	Adjacent Space
South	Corridor
North	Outdoors
East	Outdoors
West	Class room
Top	Roof
Bottom	Game Room

Sample Spaces	Spaces	Maximum Background noise		Optimum Reverberation Time[s]	Maximum Background noise				β - Impact Sound Isolation Level (IIC[dB])	STI
		L _{Aeq} T(30)	PNC		Partition walls	Exterior Walls/Simple	Exterior Walls/ Multiple	Ceilings/ Floors(simple)		
	Conference Hall	35	35	0.85 in 1000hz	50	55	50	50	60	>0.6
	Class Rooms	35	30	0.8	35	45	40	50	55	-
	Conference Room	35	35	1	35	45	40	50	55	-

Space name

Conference Hall

Air-borne sound insulation

β -Impact noise Control

Position	Adjacent Space	Surface Type	Design STC	Target STC	
South	Ground/Outdoors	External Wall	55.0	55.0	✓
North	Ground/Outdoors	External Wall	55.0	55.0	✓
East	Ground/Outdoors	External Wall	45.7	50.0	✗
West	Outdoors/ Electrical-Systems Control/ Fire Control	Partition Wall	40	50.0	✗
Top	Multi-Functional Space	Floor/ceiling	51	50.0	✓
Bottom	Ground	Ground	-	50.0	✓

position

Bottom

Adjacent Space

Multi-Functional Space

Separator Surface Type

Mosaic-mortar-concrete
Slab-false ceiling

Base IIC

34

target IIC

55

Target Δ IIC

21

Floating Floor Δ IIC

22

Final IIC

56

Constructions

Glazings

Dbl Low-E

Name	SHGC[%]	VT[%]	VT/SHGC	Total U-value	STC
Dbl Blue 6mm/13mm Air	0.598	0.789	1.29	1.786	45

450,000 T

External Walls

AAC Block Insulation L.1

Layer	Thickness	Conductivity	R-Value	Total U-value	STC
Inside Air			0.11		
Gypsum plaster	0.005	0.31	0.02		
Gypsum-sand plaster	0.02	0.59	0.03		
AAC Block	0.22	0.158	1.39		
XPS	0.030	0.004	7.50		
Cement-Sand Plaster	0.025	0.44	0.06		
Light Weight Brick	0.03	0.08	0.38		
Outside Air			0.06		

160,000 T

Ground Floors

G.F Insulation L.2

Layer	Thickness	Conductivity	R-Value	U-value	STC
Mosaic	0.22	1.65	0.13		
XPS	0.060	0.04	1.50		
Concrete	0.1	0.25	0.40		
Hardcore	0.35	3.5	0.10		

260,000 T

Other Surfaces

Ceilings

Layer	Thickness	Total STC	Total IIC
CoolCoating	0.002		
Polyurethane	0.010		
Mortar	0.090	51	34
Concrete	0.220		

430,000 T

Roofs

Cool Roof+ Insulation L.2

Layer	Thickness	Conductivity	R-Value	Total U-value	STC	IIC
Outside Air			0.05			
CoolCoating	0.002	1	0.00			
Polyurethane	0.010	0.25	0.04			
XPS	0.090	0.04	2.20			
Concrete	0.220	2.5	0.09			
Inside Air			0.09			

520,000 T

Partition Walls

Layer	Thickness	Total STC
Gypsum	0.02	
Bricks	0.10	42
Gypsum	0.020	

90,000 T



Air-borne sound insulation

Position	Adjacent Space	Surface Type	Design STC	Target STC	
South	Outdoors	Curtainwall	45.0	45.0	✓
North	Corridor	Partition Wall	39.3	35.0	✓
East	Lounge	Partition Wall	42.0	35.0	✓
West	Prayers Room	Partition Wall	42.0	35.0	✓
Top	Roof	Roof	53.0	50.0	✓
Bottom	Lounge	Floor/ceiling	51.0	50.0	✓

β-Impact noise Control

<i>position</i>		<i>Adjacent Space</i>	<i>Separator Surface Type</i>		
Top		Roof	Concrete-XPS- polyurthane-coolcoating		
<i>Base IIC</i>	<i>target IIC</i>	<i>Target ΔIIC</i>	<i>Floating Floor ΔIIC</i>	<i>Final IIC</i>	
36	55	19	22	58	

<i>position</i>		<i>Adjacent Space</i>	<i>Separator Surface Type</i>		
Bottom		Lounge	Mosaic-mortar-concrete Slab-false ceiling		
<i>Base IIC</i>	<i>target IIC</i>	<i>Target ΔIIC</i>	<i>Floating Floor ΔIIC</i>	<i>Final IIC</i>	
34	55	21	22	56	

Constructions

Glazings

450,000 T

Name	SHGC[%]	VT[%]	VT/SHGC	Total U-value	STC
Dbl Low-E					
Dbl Blue 6mm/13mm Air	0.598	0.789	1.29	1.786	45

External Walls

160,000 T

Layer	Thickness	Conductivity	R-Value	Total U-value	STC
Inside Air			0.11		
Gypsum plaster	0.005	0.31	0.02		
Gypsum-sand plaster	0.02	0.59	0.03		
AAC Block	0.22	0.158	1.39		
XPS	0.030	0.004	7.50	0.10	42
Cement-Sand Plaster	0.025	0.44	0.06		
Light Weight Brick	0.03	0.08	0.38		
Outside Air			0.06		

Ground Floors

260,000 T

G.F Insulation L.2

Layer	Thickness	Conductivity	R-Value	U-value	STC
Mosaic	0.22	1.65	0.13		
XPS	0.060	0.04	1.50		
Concrete	0.1	0.25	0.40	0.47	42
Hardcore	0.35	3.5	0.10		

Roofs

520,000 T

Cool Roof+ Insulation L.2

Layer	Thickness	Conductivity	R-Value	Total U-value	STC	IIC
Outside Air			0.05			
CoolCoating	0.002	1	0.00			
Polyurethane	0.010	0.25	0.04			
XPS	0.090	0.04	2.20	0.406	53	36
Concrete	0.220	2.5	0.09			
Inside Air			0.09			

Other Surfaces

430,000 T

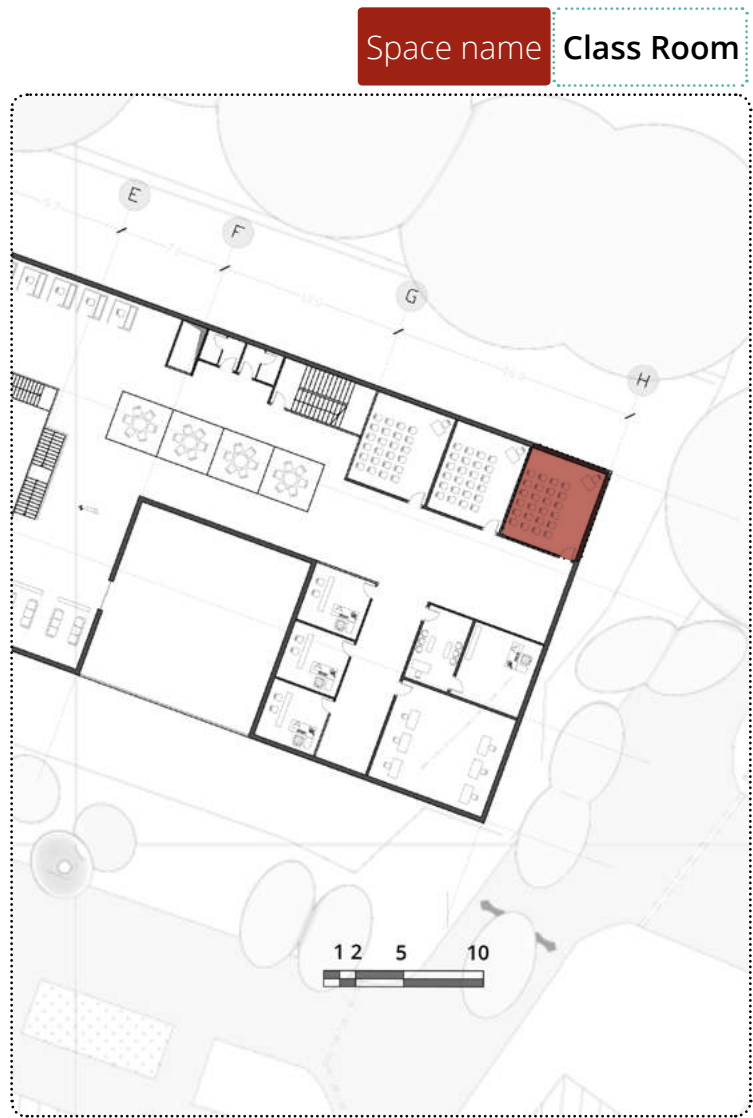
Ceilings

Layer	Thickness	Total STC	Total IIC
CoolCoating	0.002		
Polyurethane	0.010		
Mortar	0.090	51	34
Concrete	0.220		

Partition Walls

90,000 T

Layer	Thickness	Total STC
Gypsum	0.02	
Bricks	0.10	42
Gypsum	0.020	



Air-borne sound insulation

Position	Adjacent Space	Surface Type	Design STC	Target STC	
South	Corridor	Partition Wall	38.6	35.0	✓
North	Outdoors	External Wall	44.7	40.0	✓
East	Outdoors	External Wall	44.7	40.0	✓
West	Class room	Partition Wall	42	35.0	✓
Top	Roof	Roof	53	50.0	✓
Bottom	Game Room	Partition Wall	51	50.0	✓

β-Impact noise Control

position		Adjacent Space		Separator Surface Type	
Top		Roof		Concrete-XPS- polyurthane-coolcoating	
Base IIC	target IIC	Target ΔIIC	Floating Floor ΔIIC	Final IIC	
36	55	19	22	58	

position		Adjacent Space		Separator Surface Type	
Bottom		GameRoom		Mosaic-mortar-concrete Slab-false ceiling	
Base IIC	target IIC	Target ΔIIC	Floating Floor ΔIIC	Final IIC	
34	55	21	22	56	

Constructions

Glazings

Name	SHGC[%]	VT[%]	VT/SHGC	Total U-value	STC
Dbl Low-E					
Dbl Blue 6mm/13mm Air	0.598	0.789	1.29	1.786	45

450,000 T

External Walls

Layer	Thickness	Conductivity	R-Value	Total U-value	STC
Inside Air			0.11		
Gypsum plaster	0.005	0.31	0.02		
Gypsum-sand plaster	0.02	0.59	0.03		
AAC Block	0.22	0.158	1.39		
XPS	0.030	0.004	7.50	0.10	42
Cement-Sand Plaster	0.025	0.44	0.06		
Light Weight Brick	0.03	0.08	0.38		
Outside Air			0.06		

160,000 T

Ground Floors

Layer	Thickness	Conductivity	R-Value	U-value	STC
Mosaic	0.22	1.65	0.13		
XPS	0.060	0.04	1.50		
Concrete	0.1	0.25	0.40	0.47	42
Hardcore	0.35	3.5	0.10		

260,000 T

Roofs

Layer	Thickness	Conductivity	R-Value	Total U-value	STC	IIC
Cool Roof+ Insulation L.2						
Outside Air			0.05			
CoolCoating	0.002	1	0.00			
Polyurethane	0.010	0.25	0.04			
XPS	0.090	0.04	2.20	0.406	53	36
Concrete	0.220	2.5	0.09			
Inside Air			0.09			

520,000 T

Other Surfaces

Ceilings

Layer	Thickness	Total STC	Total IIC
CoolCoating	0.002		
Polyurethane	0.010		
Mortar	0.090	51	34
Concrete	0.220		

430,000 T

Partition Walls

Layer	Thickness	Total STC
Gypsum	0.02	
Bricks	0.10	42
Gypsum	0.020	

90,000 T

 Sound Transmission Class Calculation for Multiple Partitions

Spaces	Surface location	Adjacent Space	Separator Surface Type	Sub-surface type	Main Surface STC (dB)	Total surface area (m2)	main surface area(m2)	Sub-surface STC (dB)	Sub-surface type	Sub-surface area (m2)	Main surface τ	sub-surface τ	total τ	Total STC	Target STC		Base
Conference Room	East	Lounge	Partition Wall+ door	door	42	26.95	24.97	31	Wooden door-thickness=4.5cm-solid-Surface Density=24.5 kg/m2	1.98	6.31E-05	7.94E-04	1.17E-04	39.3	35		
Conference Hall	East	Ground/ Outdoors	Partition Wall+ door	door	53	144	140.04	31	Wooden door-thickness=4.5cm-solid-Surface Density=24.5 kg/m2	3.96	5.01E-06	7.94E-04	2.67E-05	45.7	50		
	West	Outdoors/ Electrical-Systems Control/ Fire Control	Partition Wall+ door	door	42	144	136.08	31	Wooden door-thickness=4.5cm-solid-Surface Density=24.5 kg/m2	7.92	6.31E-05	7.94E-04	1.03E-04	40	50		
Class Room	South	corridor	Partition Wall+ door	door	42	19.6	17.65	31	Wooden door-thickness=4.5cm-solid-Surface Density=24.5 kg/m2	1.98	6.31E-05	7.94E-04	1.37E-04	38.6	35		
	North	Outdoors	External Wall+ Windows	Windows	53	19.6	13.76	40	Laminated double clear windows-sealed	5.9	5.01E-06	1.00E-04	3.35E-05	44.7	40		
	East	Outdoors	External Wall+ Windows	Windows	53	23.1	16.17	40	Laminated double clear windows-sealed	6.93	5.01E-06	1.00E-04	3.35E-05	44.7	40		
Conference Hall	East	Ground/ Outdoors	Partition Wall+ door	door	57	144	140.04	60	All Steel Acoustic Swing Doors	3.96	7.94E-06	1.00E-06	7.75E-06	51.1	50		Improved
	West	Outdoors/ Electrical-Systems Control/ Fire Control	Partition Wall+ door	door	51	144	136.08	60	All Steel Acoustic Swing Doors	7.92	2.00E-06	1.00E-06	1.94E-06	57.1	50		

Improvement options details

Floating floor

SoundBuffer
Universal Sound Reducing Flooring Underlayment

Product Information
Made from %100 virgin latex rubber, SoundBuffer is CRI (Carpet and Rug Institute) certified and residential and commercial rated for under a variety of overlying flooring products. Featuring excellent acoustical ratings, SoundBuffer exceeds the standard for high-rise flooring installations. Thanks to its specific density and compression set characteristics, SoundBuffer helps minimize impressions and indentations that can mainly occur with luxury vinyl.

- 

Can be used with a variety of floating floor types.
- 

Allows fasteners to penetrate through it.
- 

Allows adhesives to adhere for glue down application.
- 

Provides excellent acoustical performance, when compared to traditional polyethylene foam underlayment.
- 

Dense material will minimize the amount of denting and telescoping conformity the planks or tile can have to the subfloor.



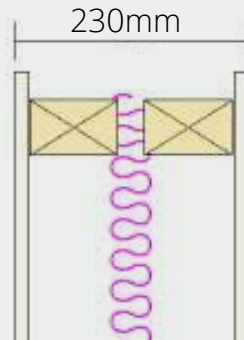
Δ IIC: 22dB

Source: mpglobalproducts.com

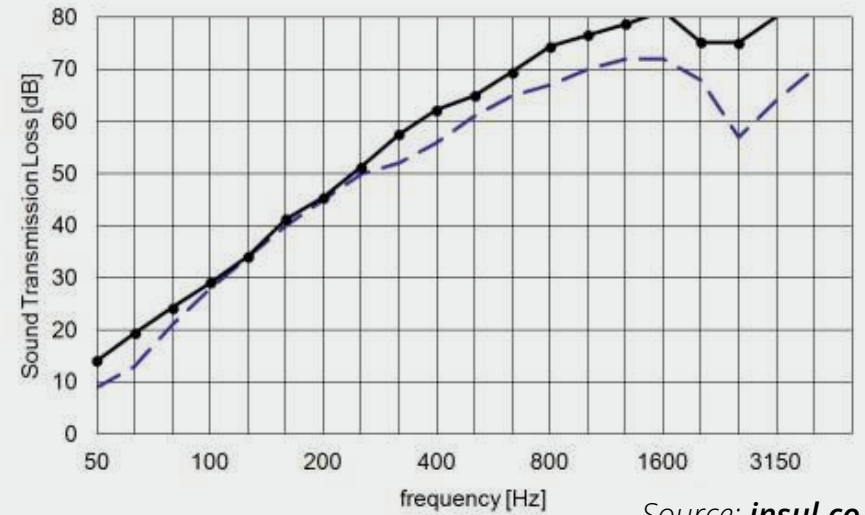
Conference Hall-west Wall

Twin framed wall

1 layer of 16mm fire rated plasterboard each side of separate 90mm studs. Cavity depth of 200mm. 90mm fibreglass blanket in cavity (16kg/m3).



STC: 57 dB

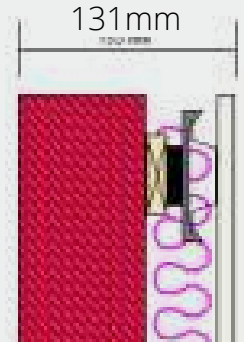


Source: insul.co.nz

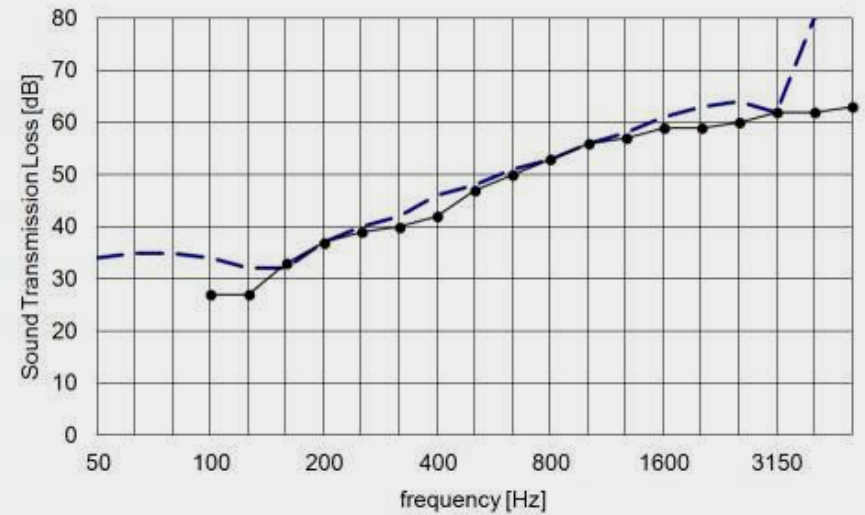
Conference Hall-east Wall

Lined Brickwork

Clay brick (90mm thick) with 1 layer of 16mm plasterboard fixed to acoustic clip and 25mm fibreglass blanket in cavity.



STC: 51 dB



Source: insul.co.nz

Conference Hall-east Doors

All Steel Acoustic Swing Doors

CDL Acoustical metal swing doors are made from steel and provide superior strength and insulating properties, making them suitable as sound barriers (acoustic doors) up to STC57. Available as single or double leaf door, CDL steel hinged acoustic swing doors come complete as a door set with door leaf, frame and hardware factory fitted. Fire ratings are optional up to two hours to UL10c.

Product Features

- Independently tested acoustical doors up to STC 60
- Fire Ratings to UL10c
- Proprietary door hinge with integrated thrust bearing for ADA pull force compliance
- Factory finished with a Polyester Powder Coat Paint, in your choice of RAL color
- Delivered as a complete assembly, including door frame and all finish hardware
- Supervised installation

Configurations

- Single leaf
- Double leaf
- Bi-Fold
- Unequal leaves

Accessories & Custom Features Available

- Vision Lites
- Sports Hall Hardware
- Corten Finish
- Stainless Steel Finish
- Prepared for Cladding

STC: 60 dB



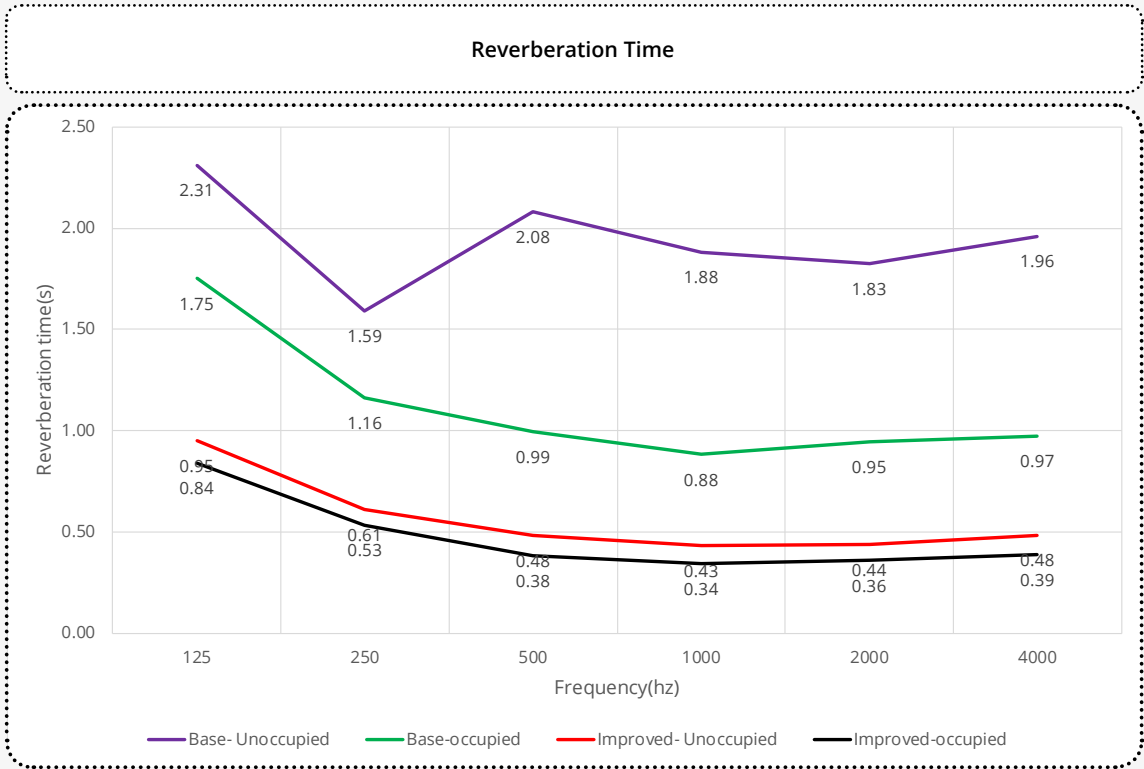
Double Doors with door closer and vision panels

Source: cdldoors.com



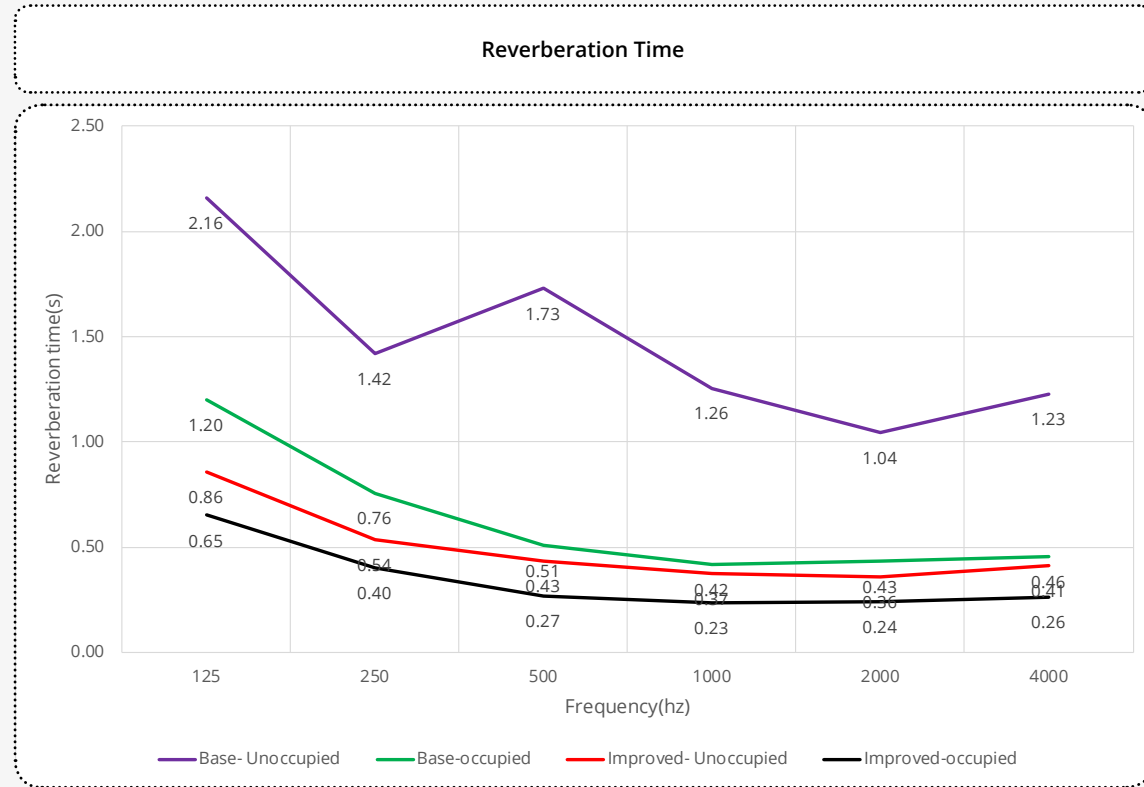
reververation Time
Calculations

Space	Surface	Separator Surface Type	Area/ number	Absorption Coefficient						volume (m3)
				125	250	500	1000	2000	4000	
Conference Room	South	Curtain Wall	42.46	0.15	0.3	0.18	0.05	0.15	0.05	279.45
	North	Partition Wall	40.48	0.01	0.02	0.02	0.03	0.01	0.04	
	East	Partition Wall	29.26	0.01	0.02	0.02	0.03	0.01	0.04	
	West	Partition Wall	29.26	0.01	0.02	0.02	0.03	0.01	0.04	
	Top	Roof	81.2	0.01	0.02	0.02	0.03	0.01	0.04	
	Bottom	Floor/ceiling	81.2	0.01	0.01	0.01	0.02	0.01	0.02	
	Door	wooden door	1.98	0.14	0.1	0.06	0.1	0.14	0.1	
	table	-	28	0.18	0.12	0.1	0.08	0.18	0.07	
	Unoccupied plastic chairs	-	0/30	0.06	0.1	0.1	0.3	0.06	0.2	
	Occupied plastic chairs	-	30/0	0.24	0.4	0.78	0.96	0.24	0.87	
	Wall Absorbers	-	30/0	0.45	0.75	0.8	0.7	0.65	0.6	
	Ceiling Absorbers	-	0/35.3	0.24	0.4	0.78	0.98	0.96	0.87	
	other furniture	-	-							



Data and plot Generated with Excel Code

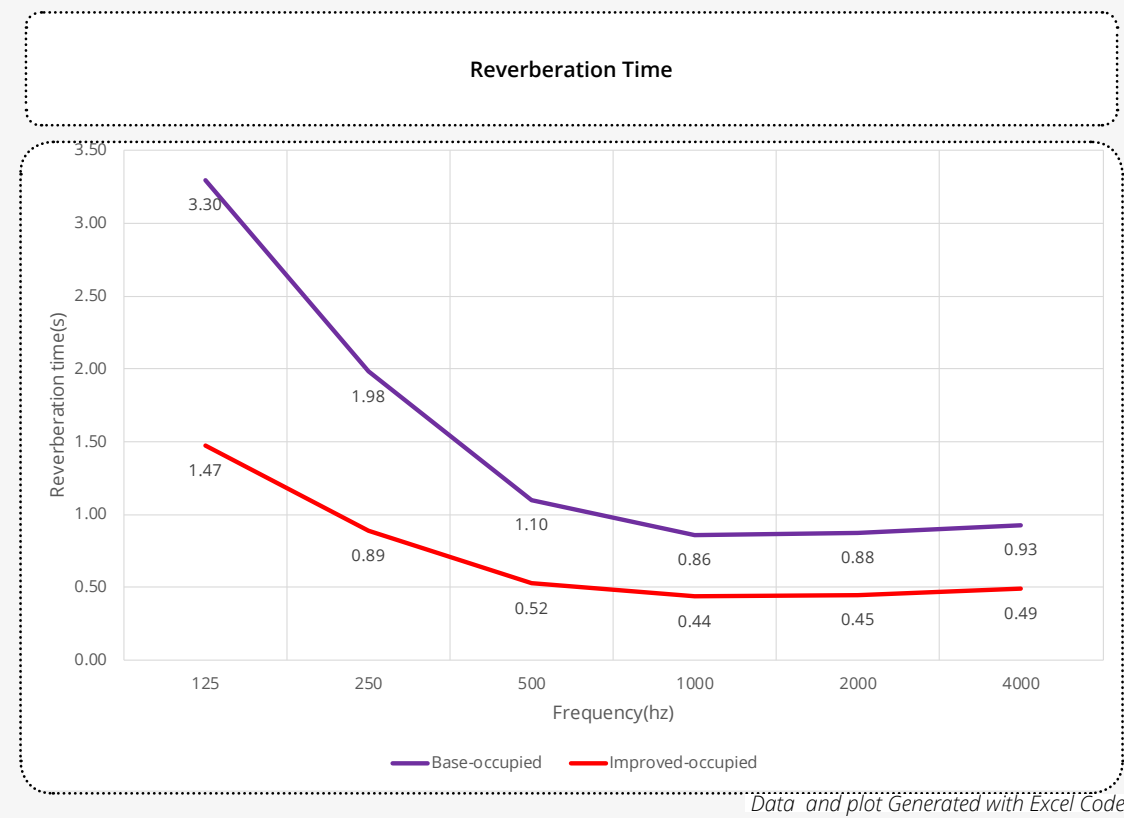
Space	Surface	Separator Surface Type	Area/ number	Absorption Coefficient						volume (m3)
				125	250	500	1000	2000	4000	
Class Room	South	Curtain Wall	6.97	0.01	0.02	0.02	0.03	0.03	0.04	105
	North	Partition Wall	13.7	0.01	0.02	0.02	0.03	0.03	0.04	
	East	Partition Wall	16.2	0.01	0.02	0.02	0.03	0.03	0.04	
	West	Partition Wall	23.1	0.01	0.02	0.02	0.03	0.03	0.04	
	Top	Roof	30.6	0.01	0.02	0.02	0.03	0.03	0.04	
	Bottom	Floor/ceiling	30.6	0.01	0.01	0.01	0.02	0.02	0.02	
	Door	wooden door	1.98	0.14	0.1	0.06	0.08	0.1	0.1	
	Windows	-	12.8	0.15	0.3	0.18	0.1	0.05	0.05	
	Whiteboard	-	2	0.42	0.21	0.1	0.08	0.06	0.06	
	Unoccupied plastic chairs	-	30	0.06	0.1	0.1	0.2	0.3	0.2	
	Occupied plastic chairs	-	0	0.24	0.4	0.78	0.98	0.96	0.87	
	Wall Absorbers	-	0/15	0.45	0.75	0.8	0.7	0.65	0.6	
	Ceiling Absorbers	-	0/23	0.24	0.4	0.78	0.98	0.96	0.87	
	other furniture	-	-							



Data and plot Generated with Excel Code

reververation Time Calculations

Space	Surface	Separator Surface Type	Area/ number	Absorption Coefficient						volume (m3)
				125	250	500	1000	2000	4000	
Conference Hall	South	External wall	84	0.01	0.02	0.02	0.03	0.03	0.04	279.45
	North	External wall	46.2	0.01	0.02	0.02	0.03	0.03	0.04	
	East	External wall	108	0.01	0.02	0.02	0.03	0.03	0.04	
	West	Partition Wall	108	0.01	0.02	0.02	0.03	0.03	0.04	
	Top	Floor/ceiling	96	0.01	0.02	0.02	0.03	0.03	0.04	
	Bottom	Ground	138.24	0.01	0.01	0.01	0.02	0.02	0.02	
	Doors	Woodendoor	12	0.14	0.1	0.06	0.08	0.1	0.1	
	Acoustic Chairs	-	288	0.24	0.4	0.78	0.98	0.96	0.87	
	Wall Absorbers	-	0/115	0.45	0.75	0.8	0.7	0.65	0.6	
	Ceiling Absorbers other furniture	-	0/288	0.24	0.4	0.78	0.98	0.96	0.87	



Improvements

Conference Hall

Ceiling
covering %75 of area with Acoustic panels with the NRC of 0.7

Vertical Surfaces
covering %25 of area with Acoustic panels with the NRC of 0.8

Class Room

Ceiling
covering %75 of area with Acoustic panels with the NRC of 0.7

Vertical Surfaces
covering %25 of area with Acoustic panels with the NRC of 0.8

Conference Room

Ceiling
covering %75 of area with Acoustic panels with the NRC of 0.7

Vertical Surfaces
covering %25 of area with Acoustic panels with the NRC of 0.8

Targets		Base		Improved	
Target RT (Annex 18)	Target RT (WELL)	WELL Credit	Annex 18 Credit	WELL Credit	Annex 18 Credit
1	0.6				
0.8	0.6				
0.85	0.7-0.3				



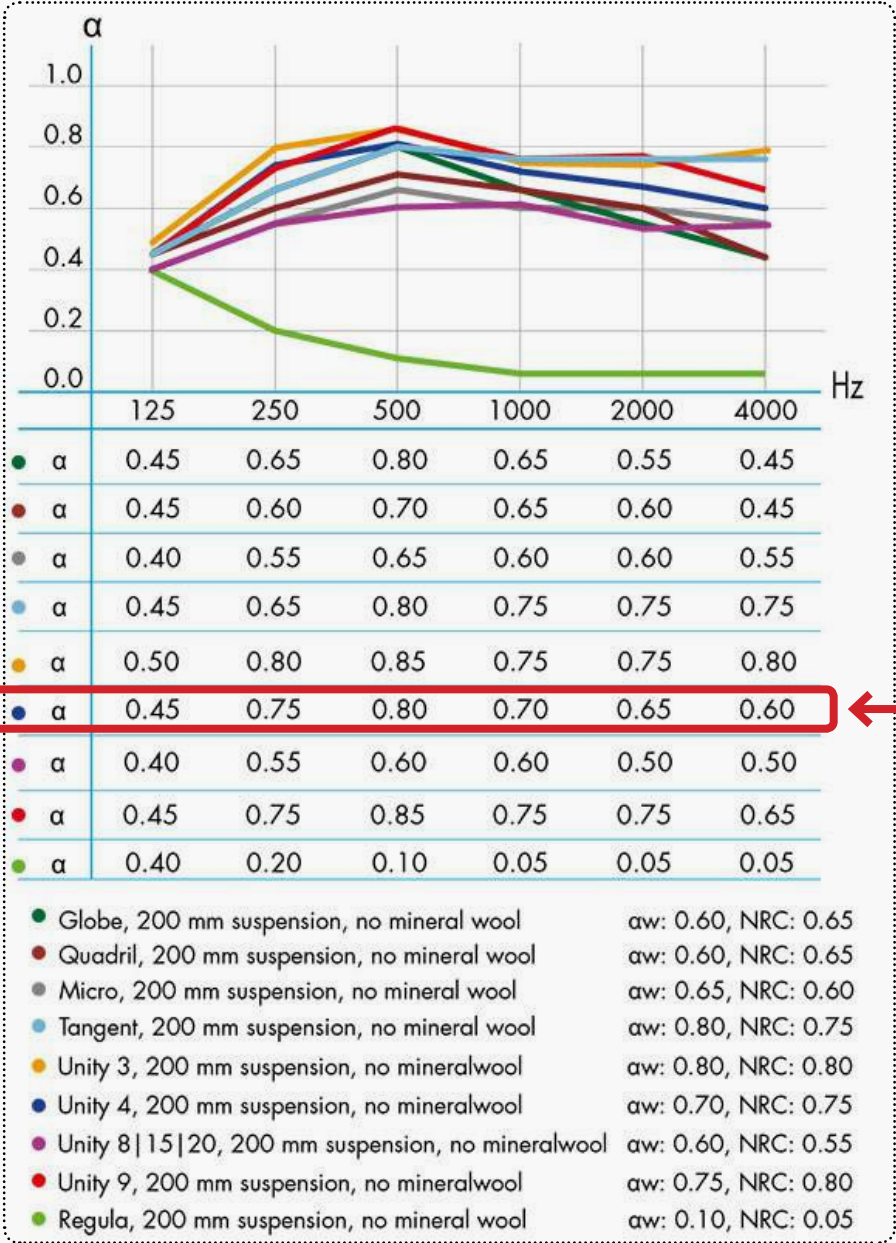
Acoustic tiles
details for Ceilings

INSTALLATION

- Suspension: Recessed grid system
- Distance between hangers ≤ 1200 mm.
 - Distance from wall to first hanger in direction of load-bearing section ≤ 400 mm.
 - Distance between load-bearing sections ≤ 1200 mm.
 - Distance from wall to first load-bearing section ≤ 600 mm.
 - Sub-division executed with cross-sections length 1200 and 600 mm.
 - Screw spacing for attachment of edge section: max. 300mm c/c
 - In the case of fittings, loads must be taken up directly by the concrete deck or via the grid system and extra hangers.

MATERIAL

Material: Perforated plasterboard
Board type: Ceiling tile
Length x width x thickness:
600 x 600 x 12.5
Edge: E+, recessed grid system (8 mm recess)
Perforation: Square 3.5x3.5mm holes, c/c 8.33 mm
Perforation to edge
Perforation percent:
17.2 %
Surface: Water-based acrylic paint
Colour: White NCS 0701 Y24R, RAL 9003; gloss 5
Attachment: 15 mm OR 24 mm T-grid system
Edge section: 20x24 mm angled profile
15+15 x 25+8 mm cornice moulding



SIZE (MODULE)

600 x 600 x 12.5 mm

SURFACE

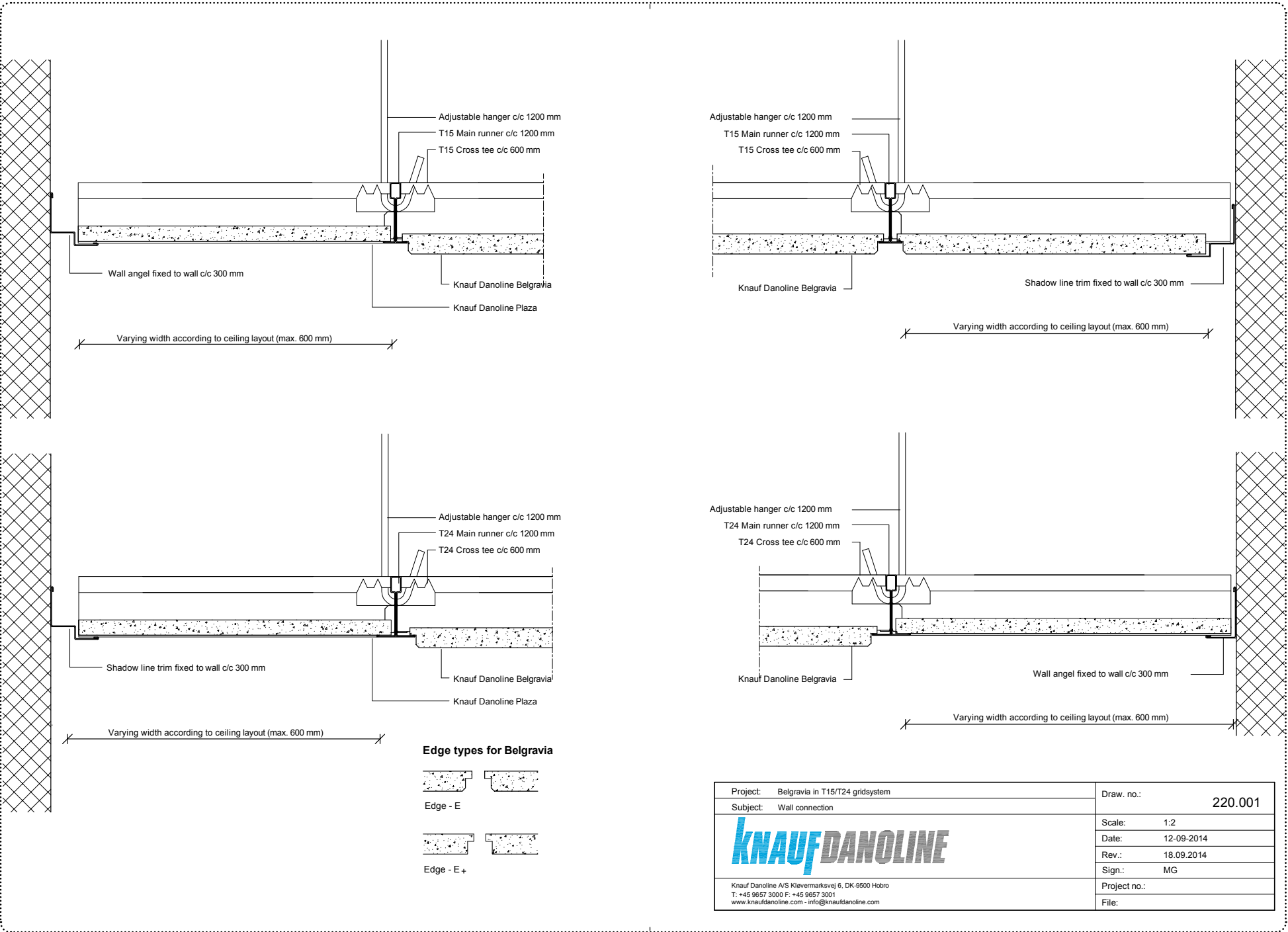
Standard white painted surface NCS 0701 Y24R (RAL 9003).

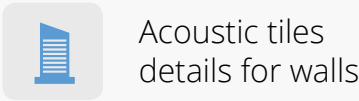
Gloss 5.

Other colours available on request.

WEIGHT KG/M²

Indicative tile weight: 9.90 – 9.20.
All according to type of perforation.





Description
gypsum
screw fixing to a

Colour:
palettes.).

Paint type:
enamel

Properties:
Light Reflection: %5,5

Acoustic properties:
Kino: aw: 0.80; NRC: 0.75
Installation depth: 85 mm with 50 mm mineral wool backing

Fire Class:
A-2s1,d0

Suspension:
600
through the panels or

Characteristics:
reinforced

12.5 x 600 x 600 mm high grade, glass fibre reinforced acoustic, painted boards with acoustic special Kino-perforation.

Kinopanel is with edge B for wall mounting and shall be installed by special support framework with max c/c 300 mm.

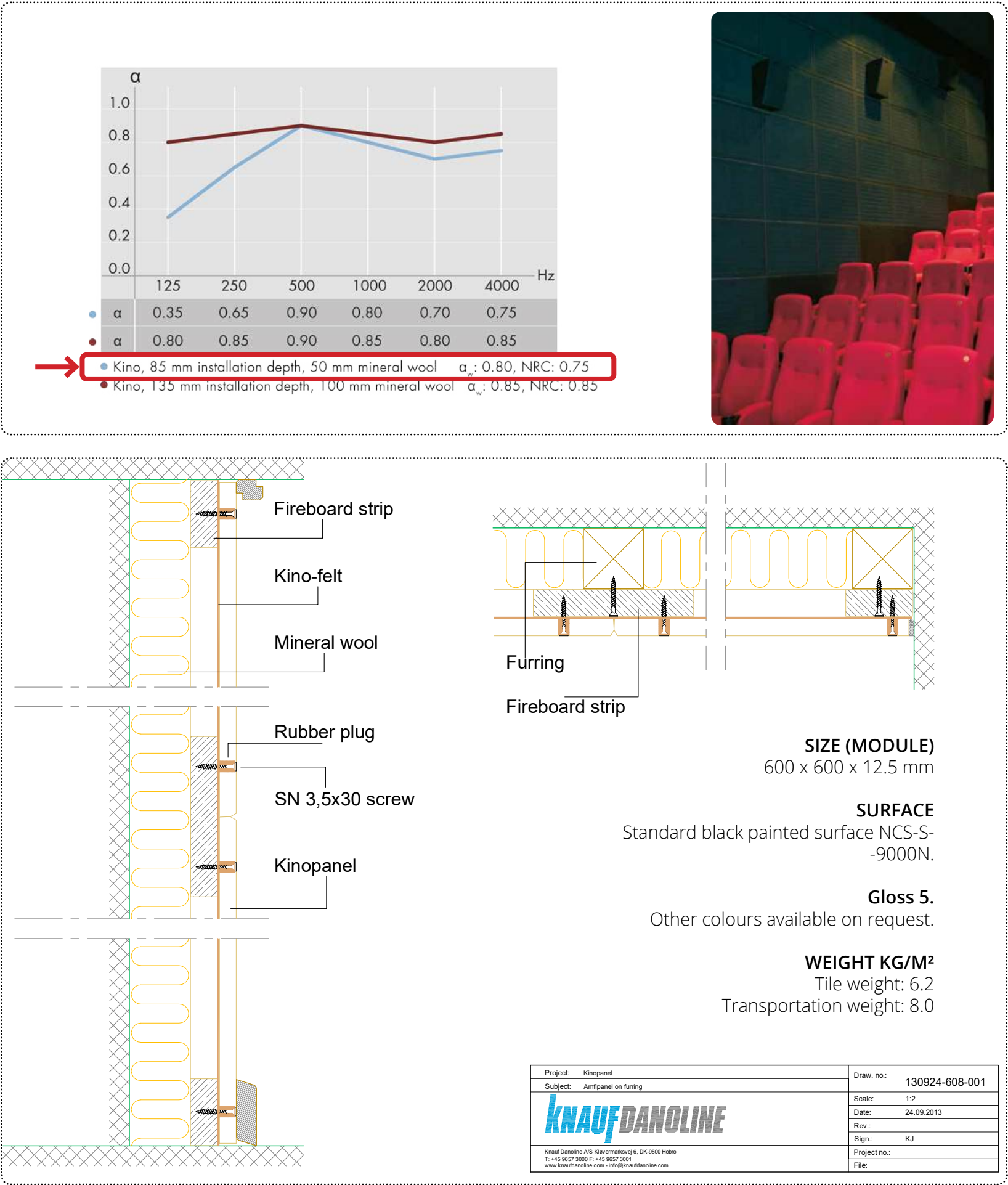
Black RAL 9001 (Other colours available from NCS and RAL colour

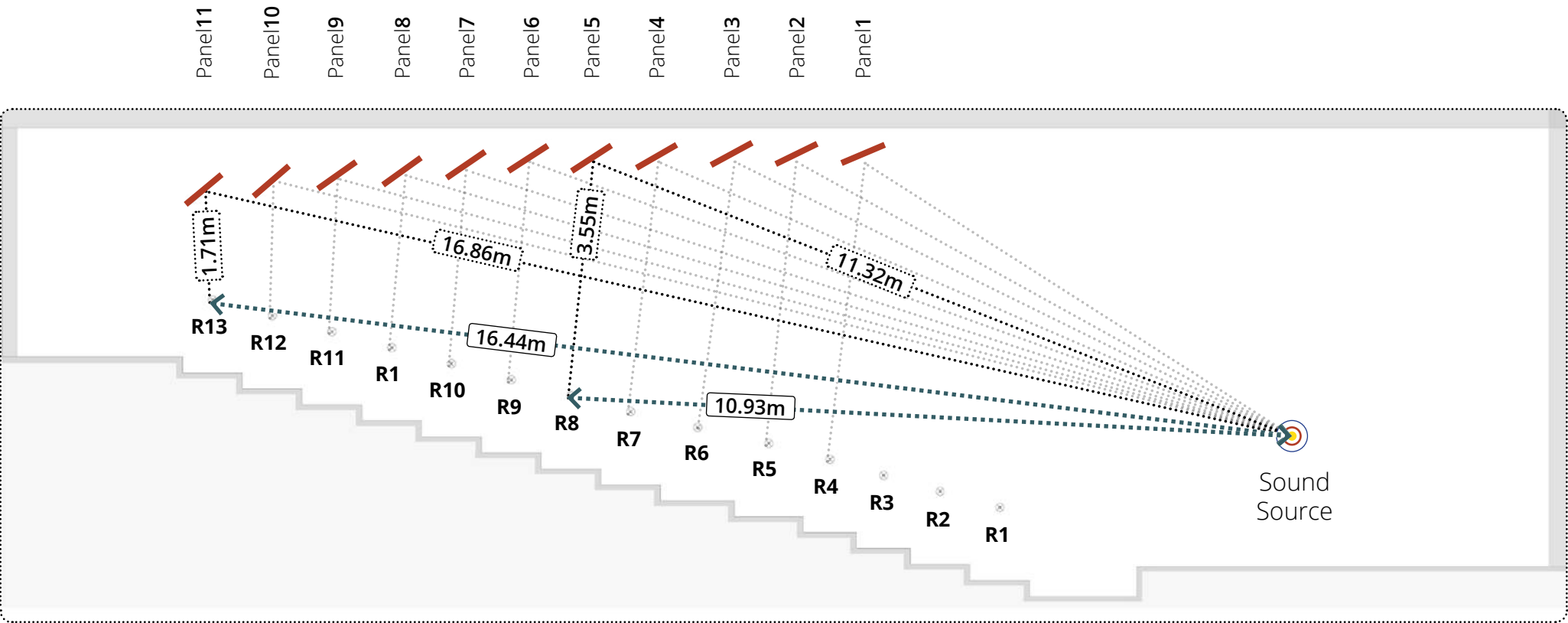
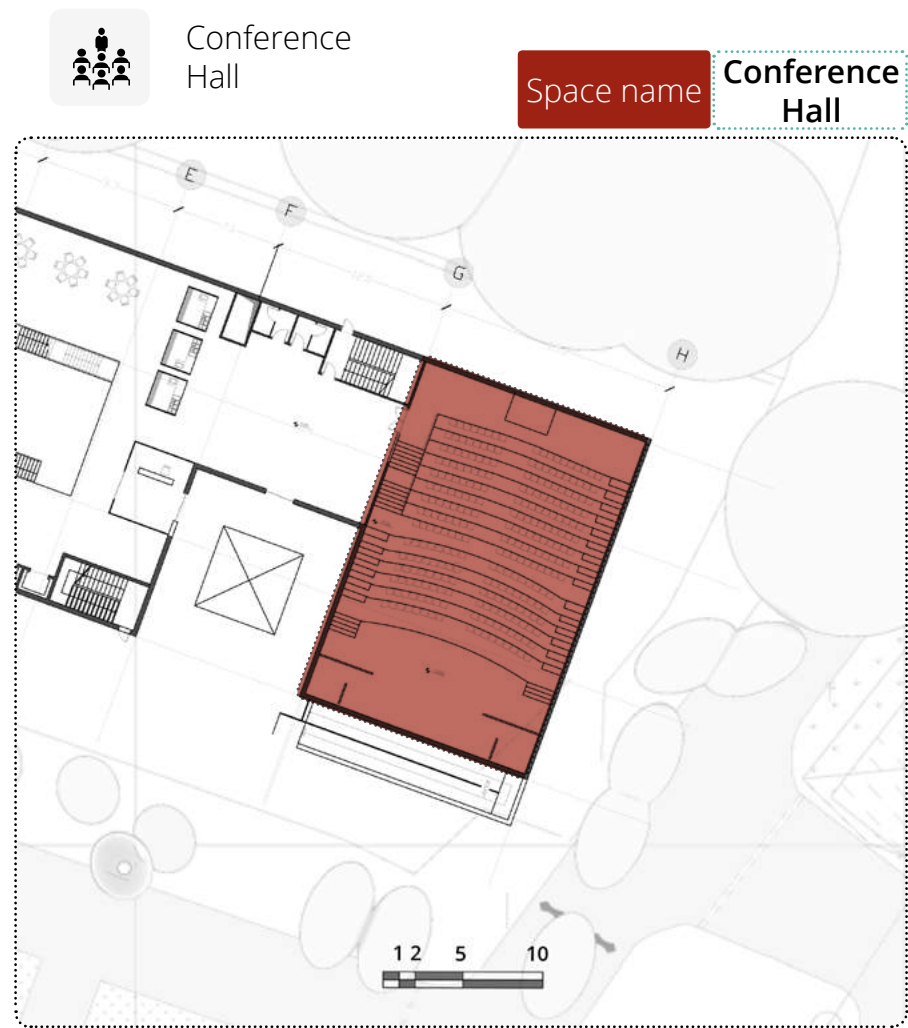
Water based acrylic polyurethane combination paint that hardens to an type finish.

Smooth and hard surface finish that is highly robust and dirt resistant and exceptionally easy to clean.

The panels shall be:

- Manufactured from high grade non-combustible glass fibre gypsum board produced according to EN 520.
- Tested to meet the safety requirements of EN 14190.
- Tested in accordance with EN 1-13501 to achieve minimum fire classification A-2s1,d0.
- Resistant to %90 RH and short term exposure for higher humidity.
- Non hazardous and fully recyclable.





Panel Code	source to panel distance(a) in meters	panel to receiver distance(b) in meters	source to receiver distance(c) in meters	a+b-c (should not be more than 17 meters)
panel 1	7.64	4.53	7.03	5.15
panel 2	8.51	4.28	7.94	4.85
panel 3	9.38	4.06	9.01	4.42
panel 4	10.41	3.80	10.02	4.19
panel 5	11.32	3.55	10.93	3.94
panel 6	12.26	3.31	11.84	3.73
panel 7	13.06	2.97	12.76	3.27
panel 8	13.72	2.65	13.68	2.70
panel 9	14.84	2.42	14.60	2.67
panel 10	15.77	2.10	15.52	2.35
panel 11	16.86	1.71	16.44	2.13

Data Generated with Grasshopper Code



Sound Masking



Open offices, libraries, cafeterias, corridors/hallways: 45 - 48 - dBA



Enclosed offices and quiet zones identified through S01: Sound Mapping, Part 3: Label Acoustic Zones: 42 - 40 dBA

Soundmasking generator, low pass filter, and amplifier – all in one compact housing

- * Internal digital White Noise soundmasking signal
- * -70Volt distributed output
- * Powerful -25Watt Amplifier
- * Low pass filter allows tuning of the soundmasking spectrum
- * Accessible 3A fuse protection
- * Barrier Post and RCA (phono) plug output jacks
- * Can power zones up to 6,000sq.ft.
- * Rated for %100 duty cycle commercial usage
- * Easy to use controls

American Sound Masking ASM1042- Soundmasking Conditioner is a soundmasking generator, low-pass filter, and power amplifier in an easy to use neutral toned enclosure. The ASM1042- is designed for use in soundmasking applications that require centralized control, yet space and/or economy are a consideration. The unit features a simple and clearly labeled controls for ease of use. It offers many of the most popular features usually found on larger centralized soundmasking systems, taking the place of various rack-type components.

The ASM1042- is effective in supplying dedicated soundmasking for up to 8,000 sq.ft. zone.

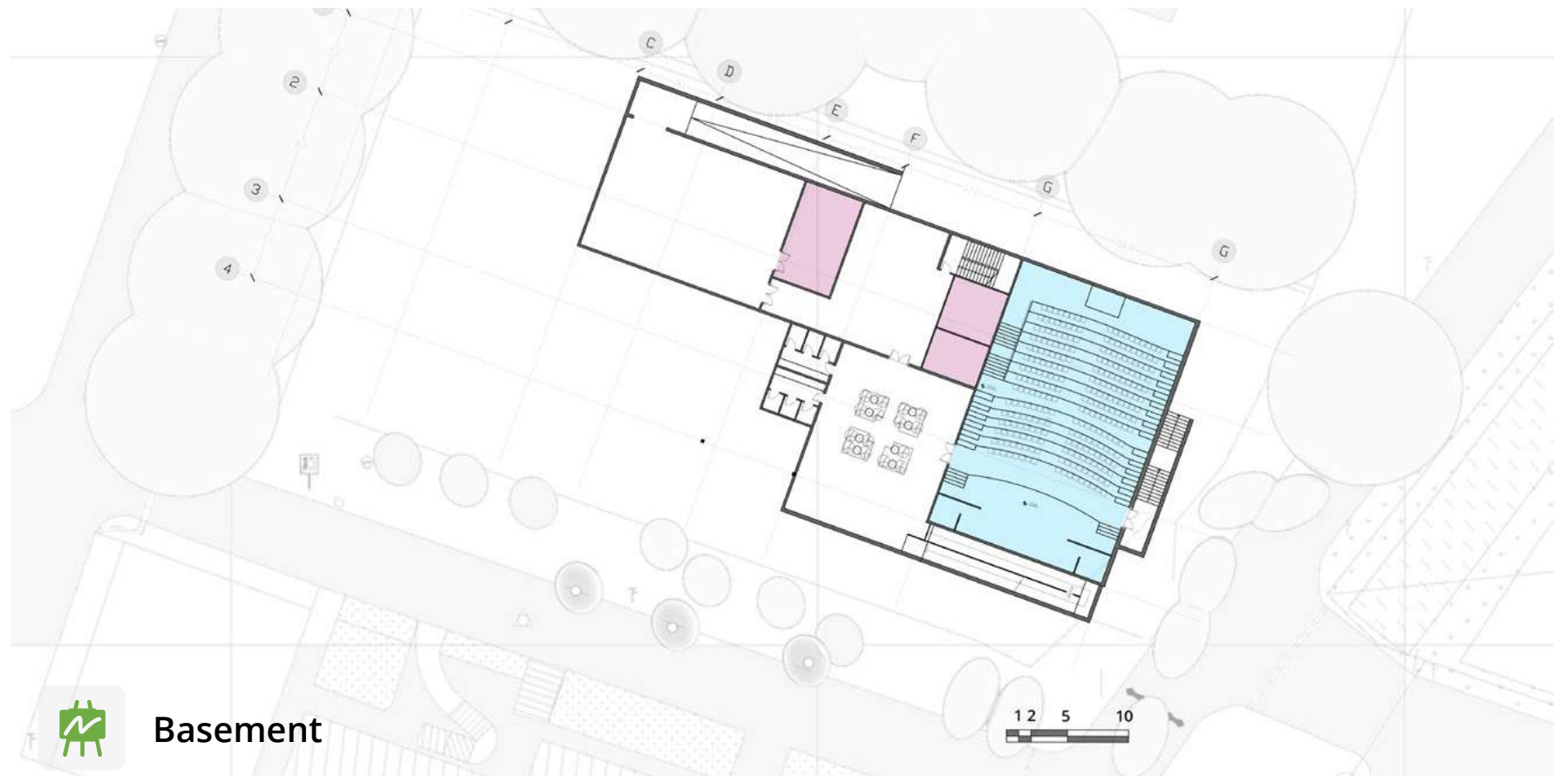
This unit is for Sound Masking only. To include Paging, Emergency Evacuation Paging, Music and/or a Microphone, see the ASM1092- viewed below.



Source: americansoundmasking.com

Zone Acoustical Labels

- **Loud zones:** includes areas intended for appliances, mechanical equipment or amenities (e.g., kitchens, fitness rooms, social spaces, recreational rooms).
- **Quiet zones:** includes areas intended for focused work, wellness, rest, study and/or privacy.
- **Mixed zones:** includes areas intended for learning, collaboration and/or presentation.



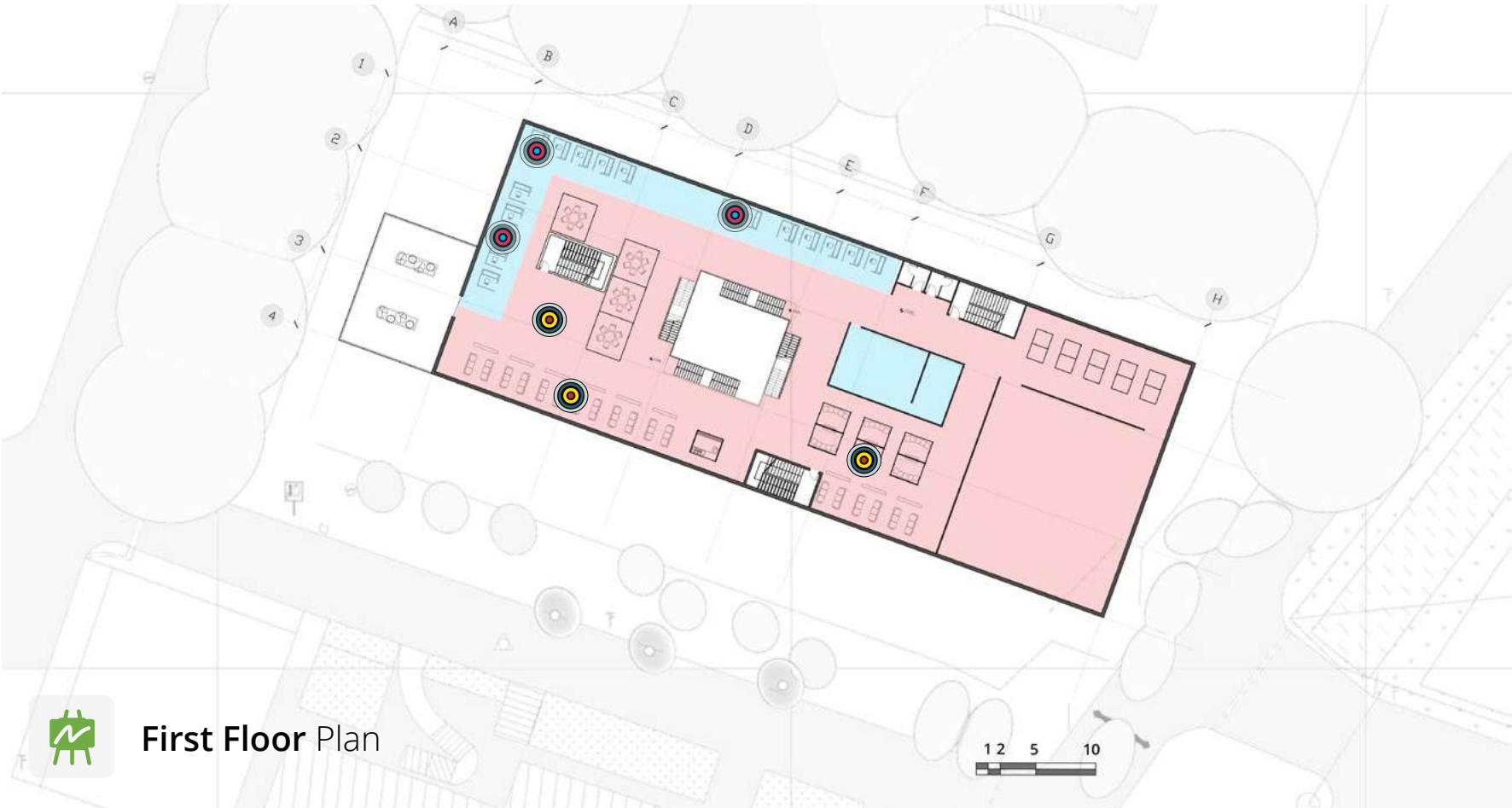
Basement



Ground Floor

 Sound Masking

-  Open offices, libraries, cafeterias, corridors/hallways: 48 - 45 dBA
-  Enclosed offices and quiet zones identified through S01: Sound Mapping, Part 3: Label Acoustic Zones: 42 - 40 dBA



 **First Floor Plan**

Zone Acoustical Labels

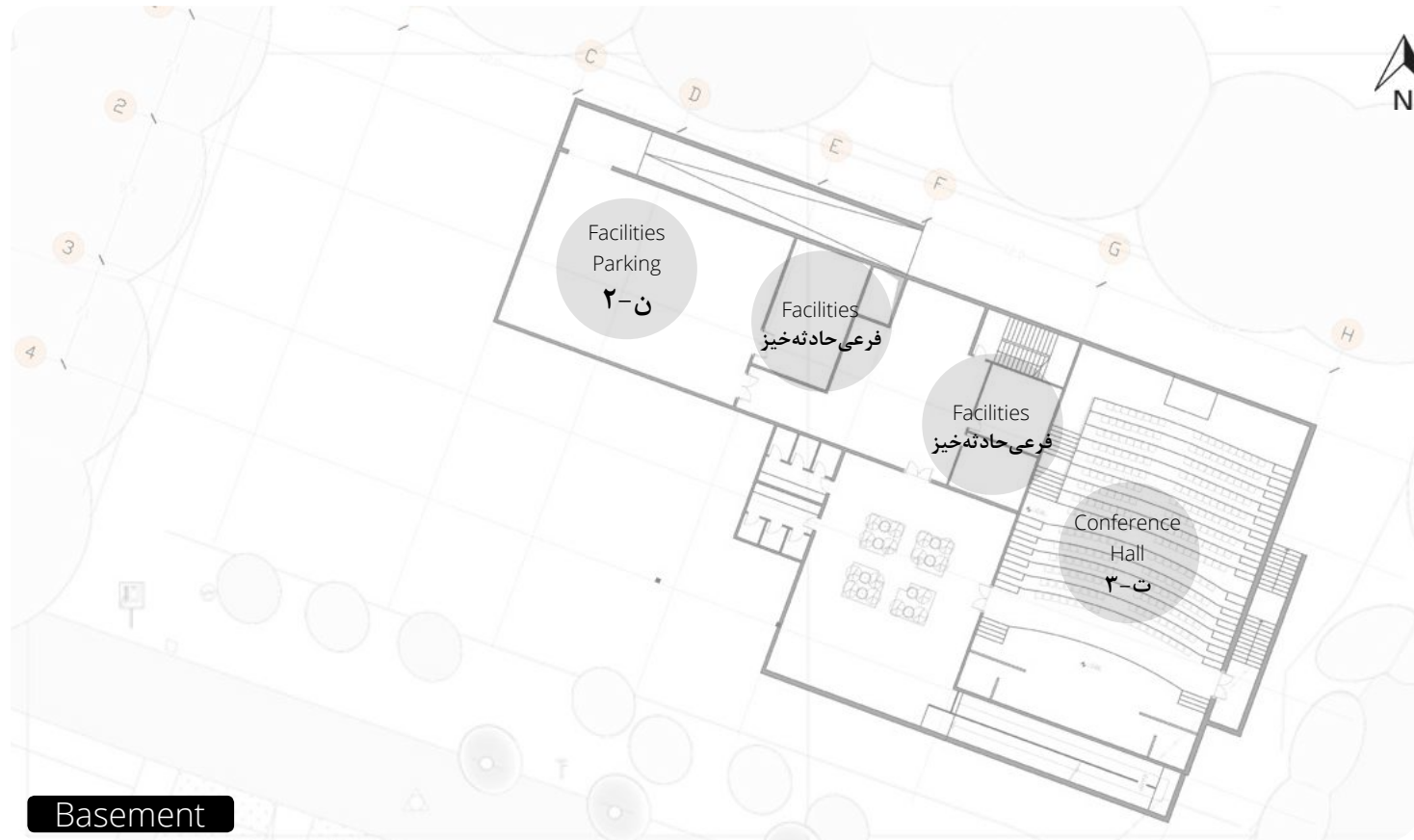
-  **Loud zones:** includes areas intended for appliances, mechanical equipment or amenities (e.g., kitchens, fitness rooms, social spaces, recreational rooms).
-  **Quiet zones:** includes areas intended for focused work, wellness, rest, study and/or privacy.
-  **Mixed zones:** includes areas intended for learning, collaboration and/or presentation.



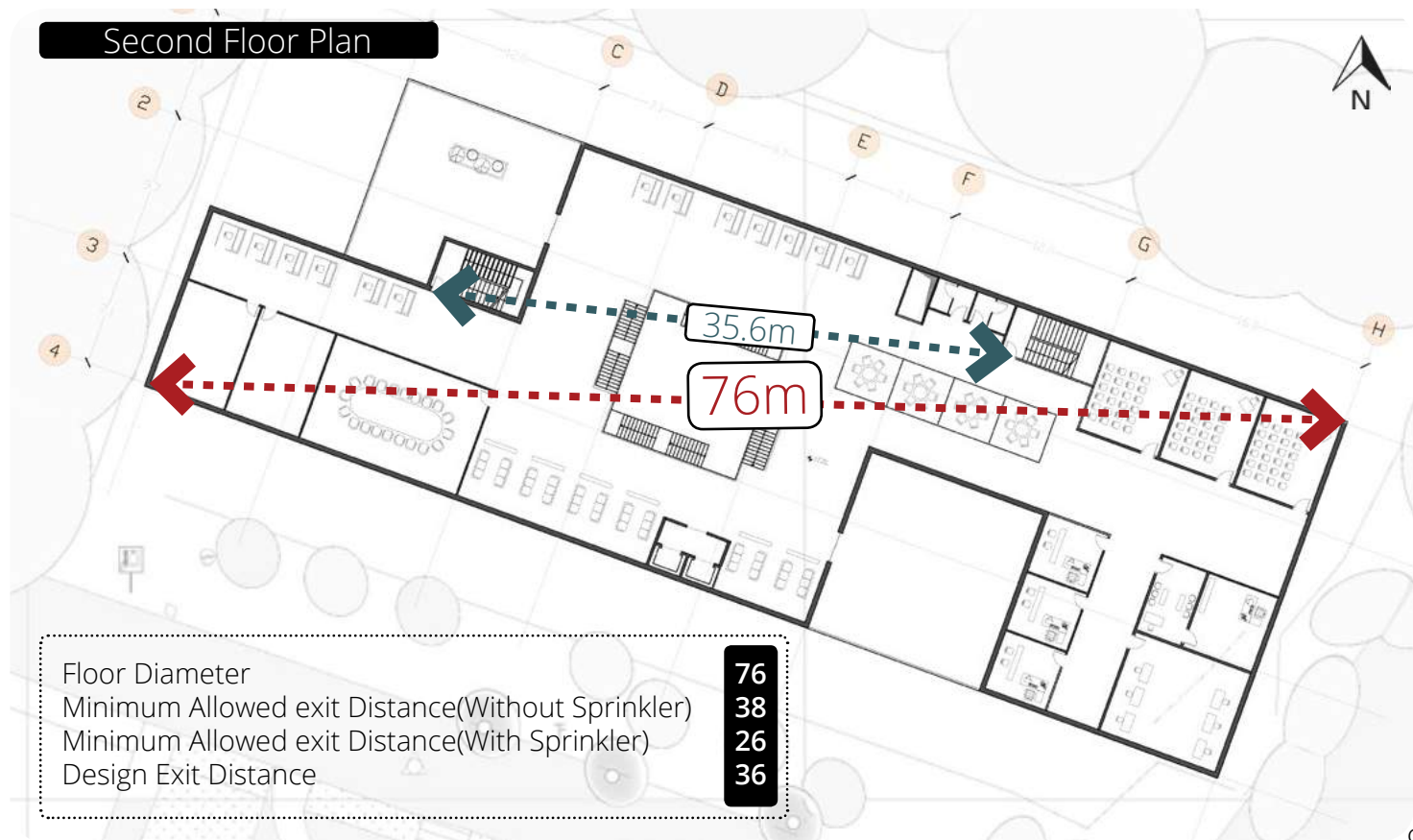
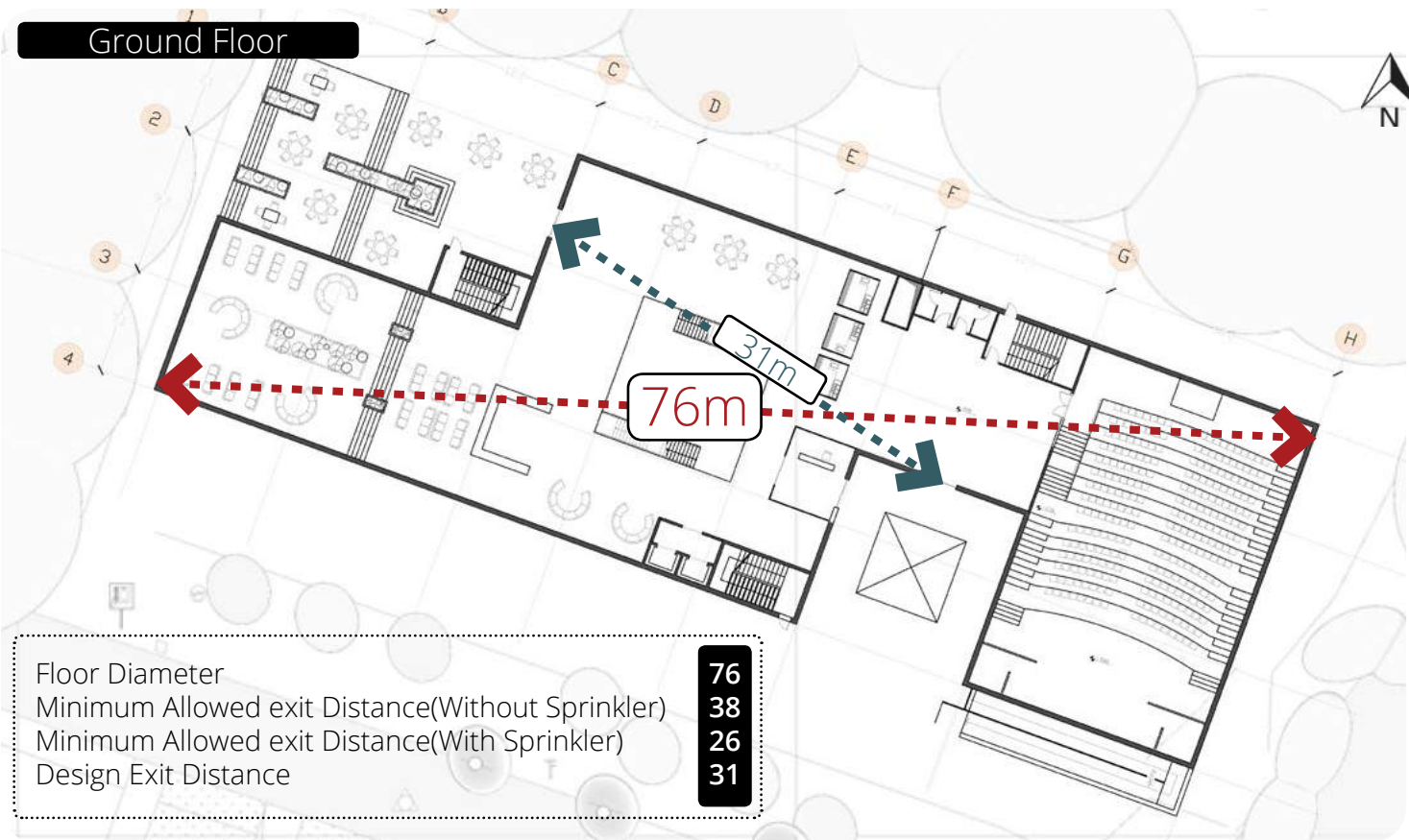
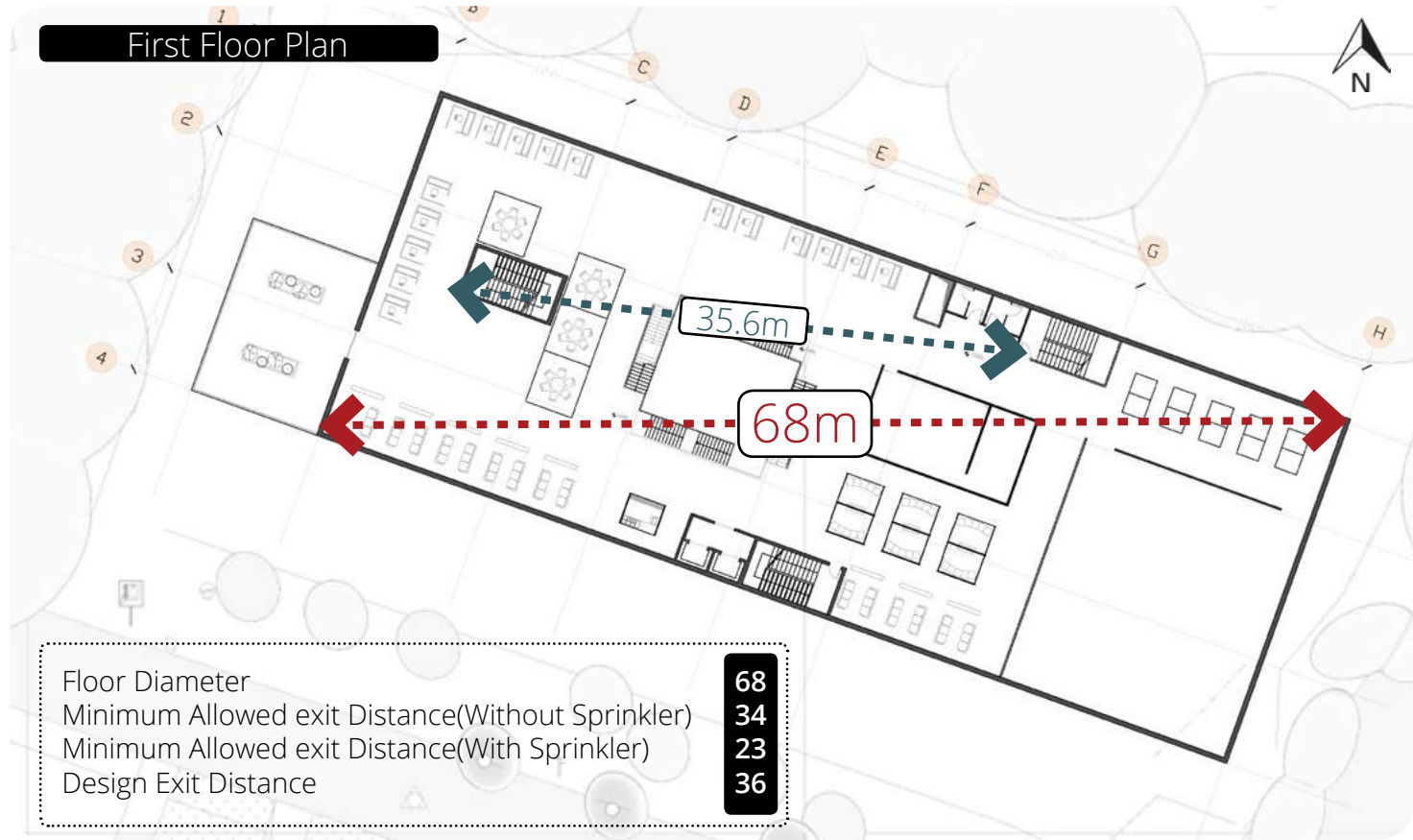
 **Second Floor Plan**



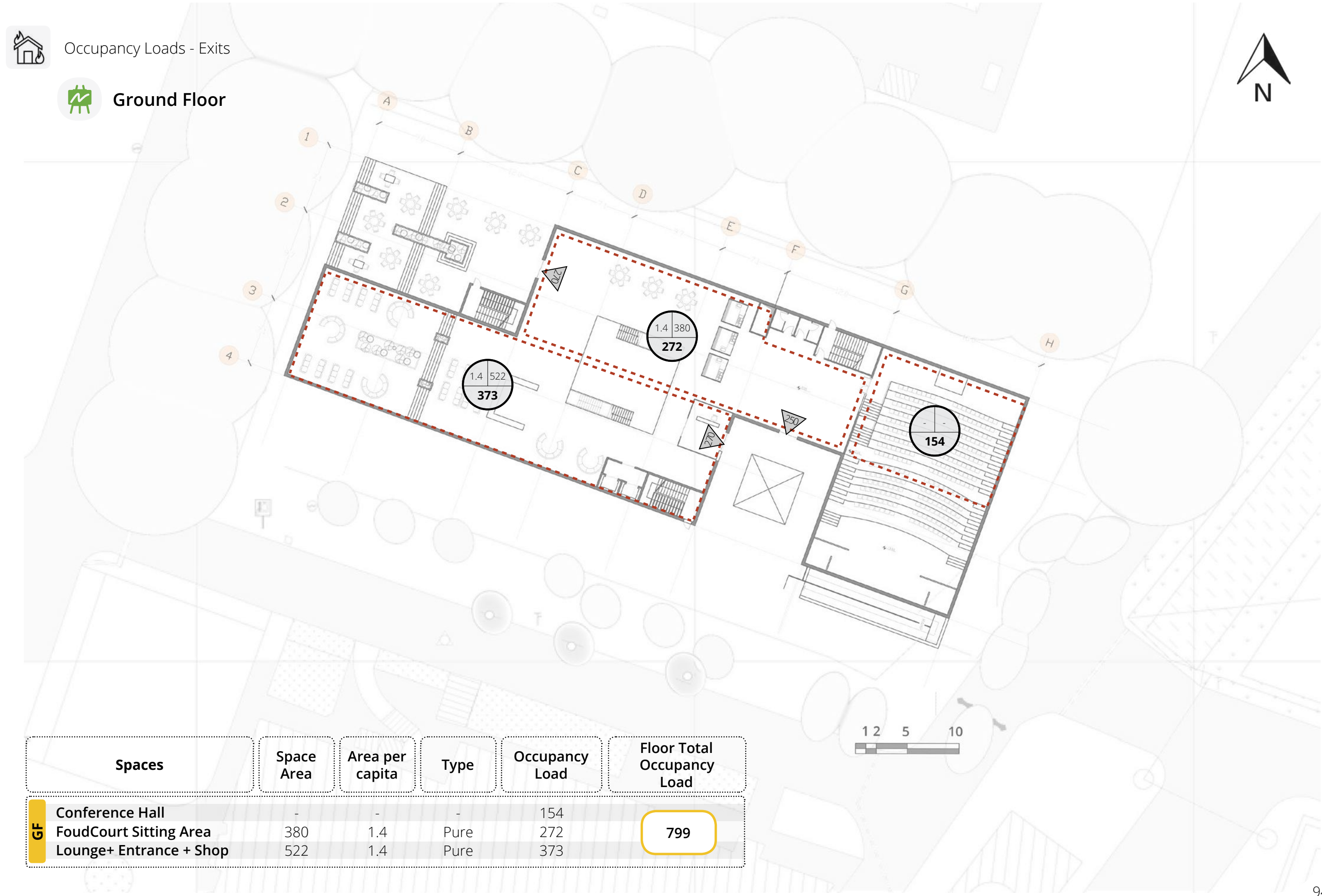
Building Type Determination

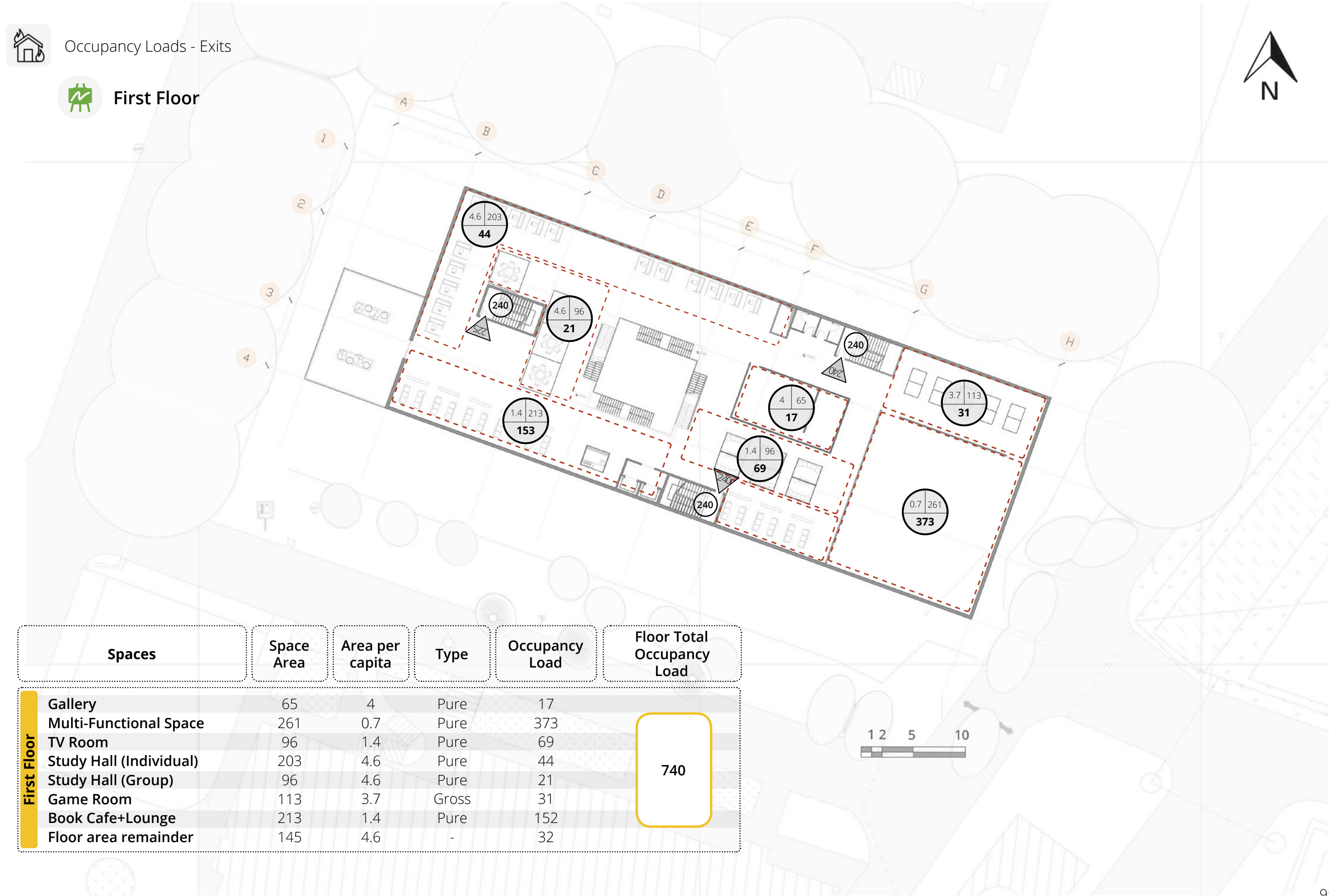


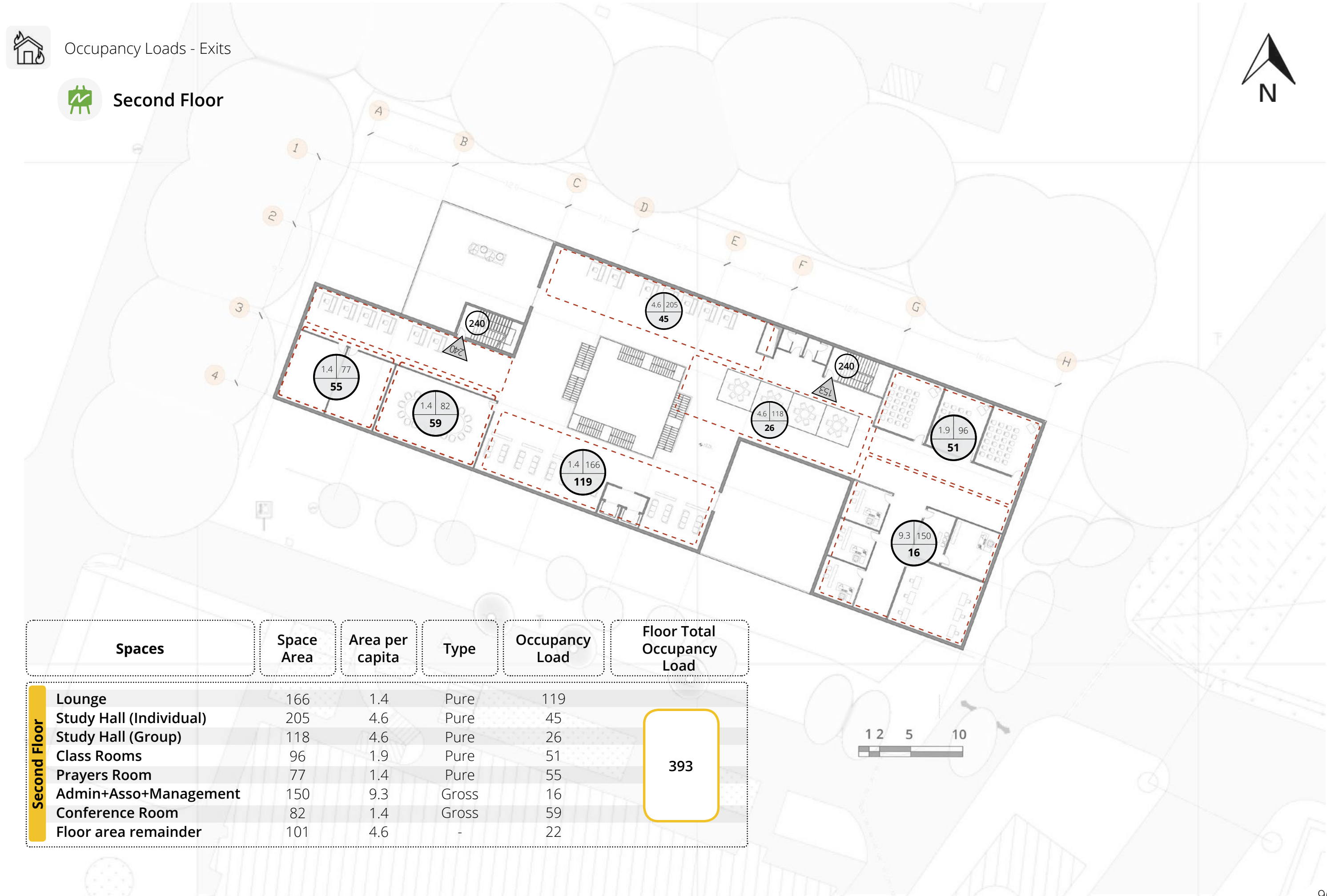
 Allowed **exit Distances**

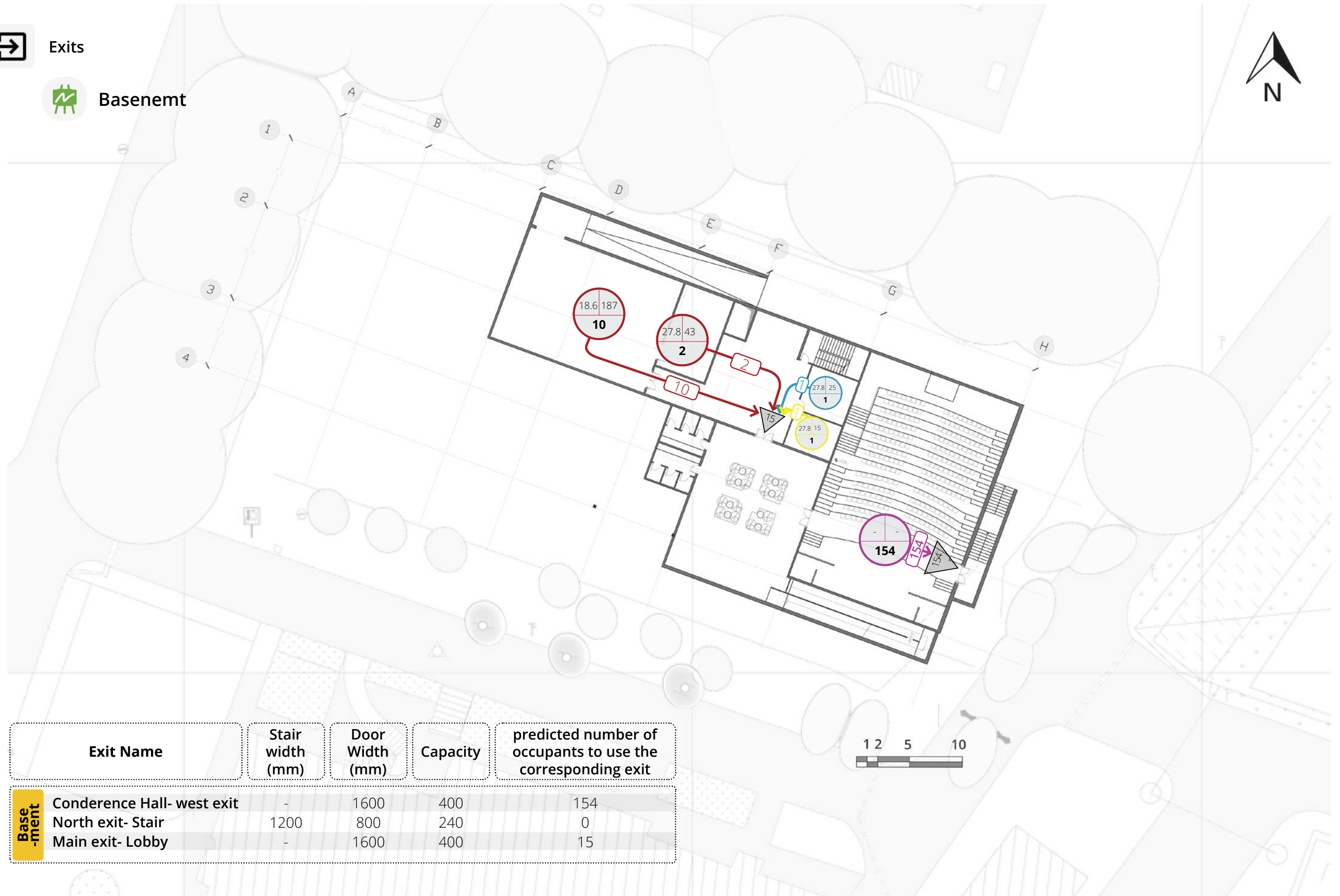
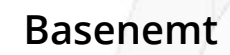


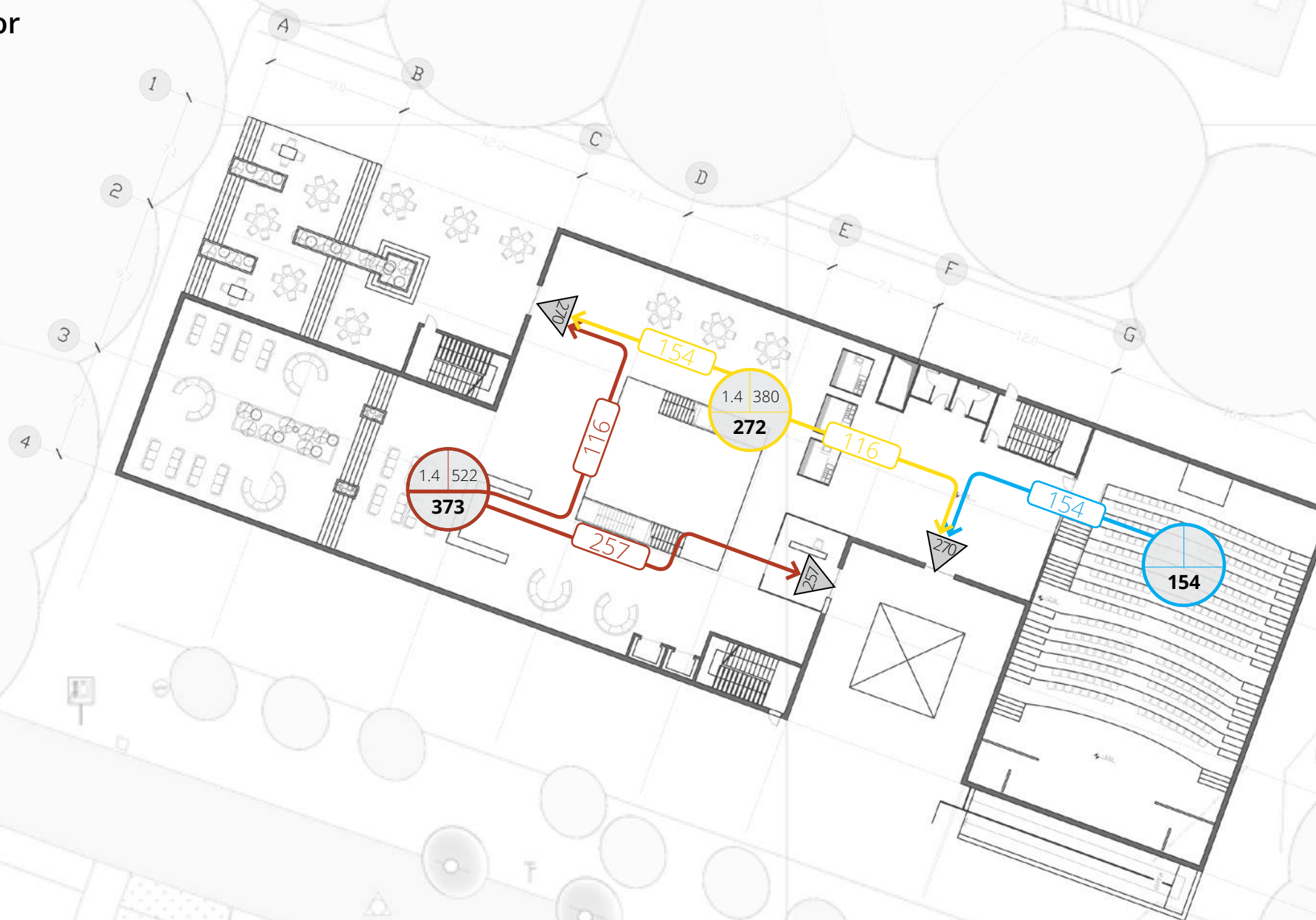
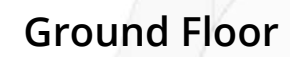




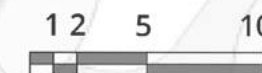


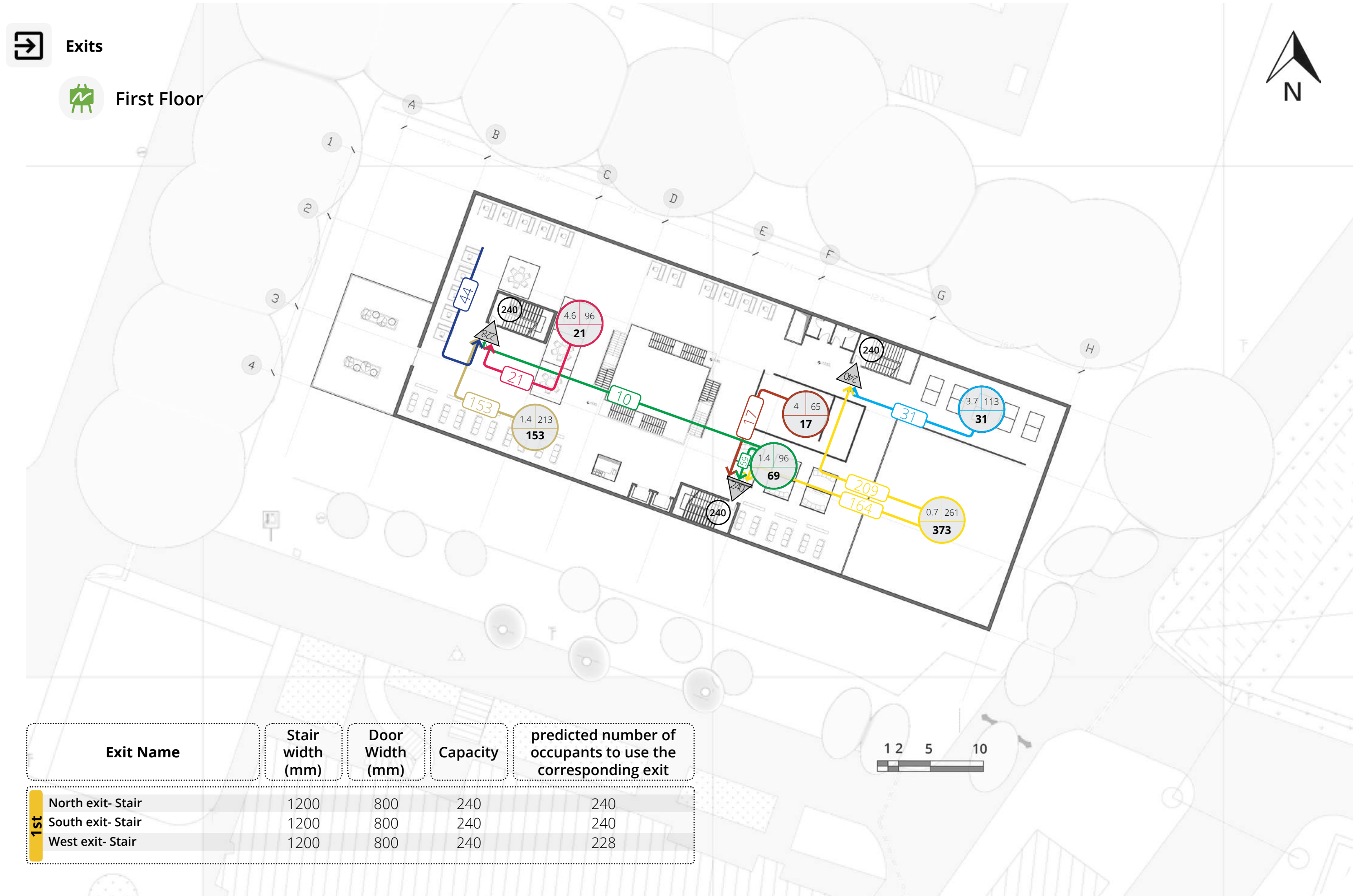


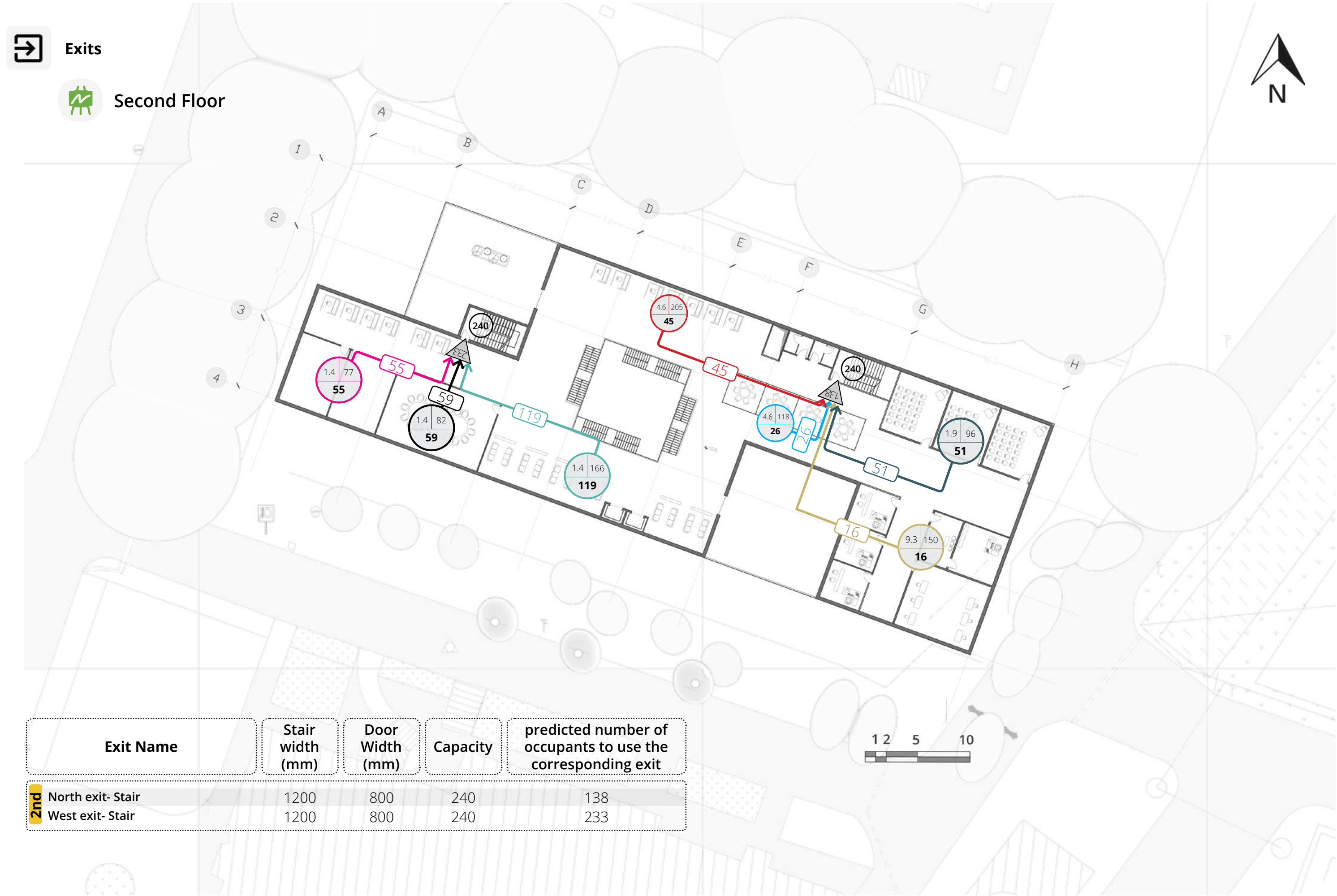


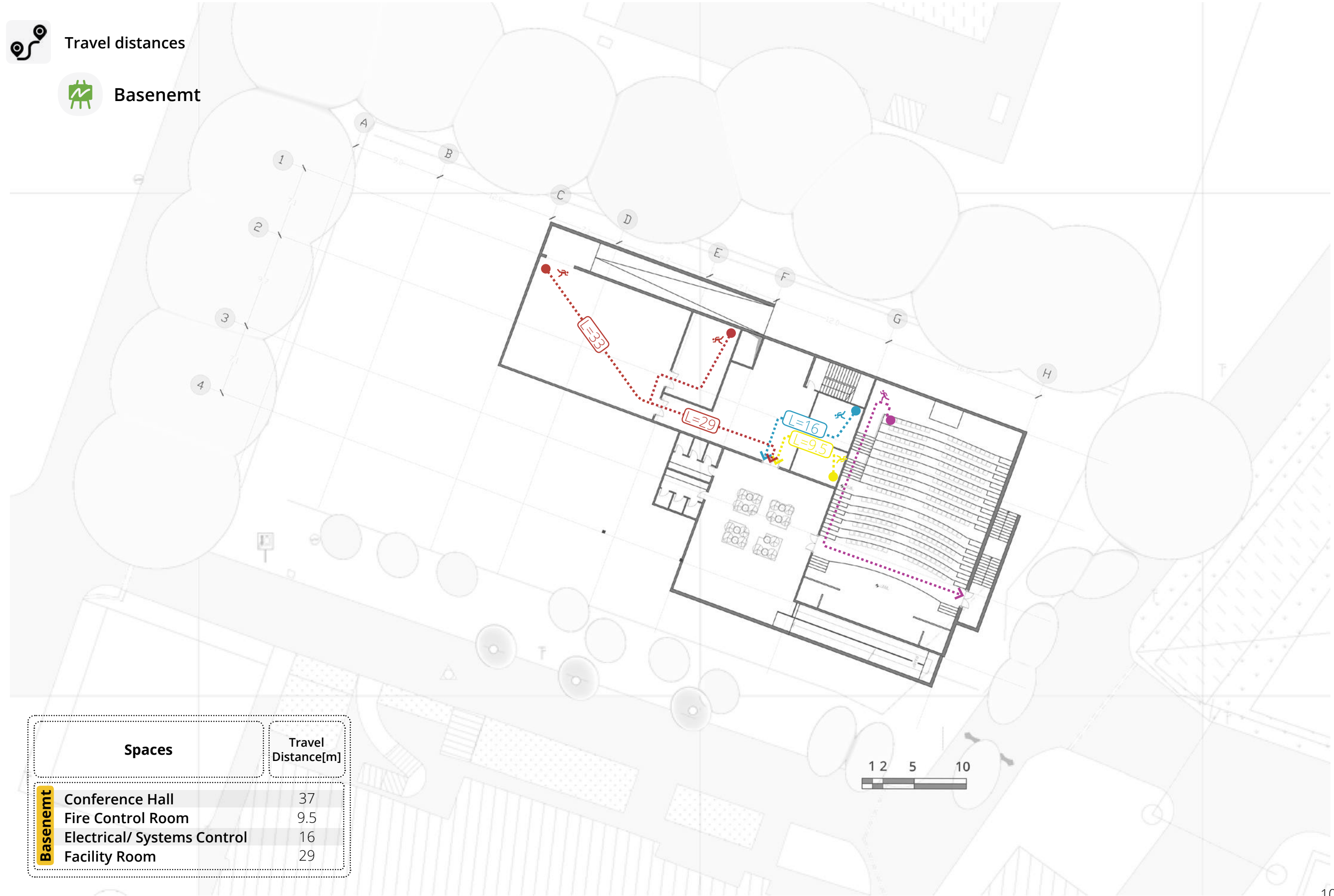


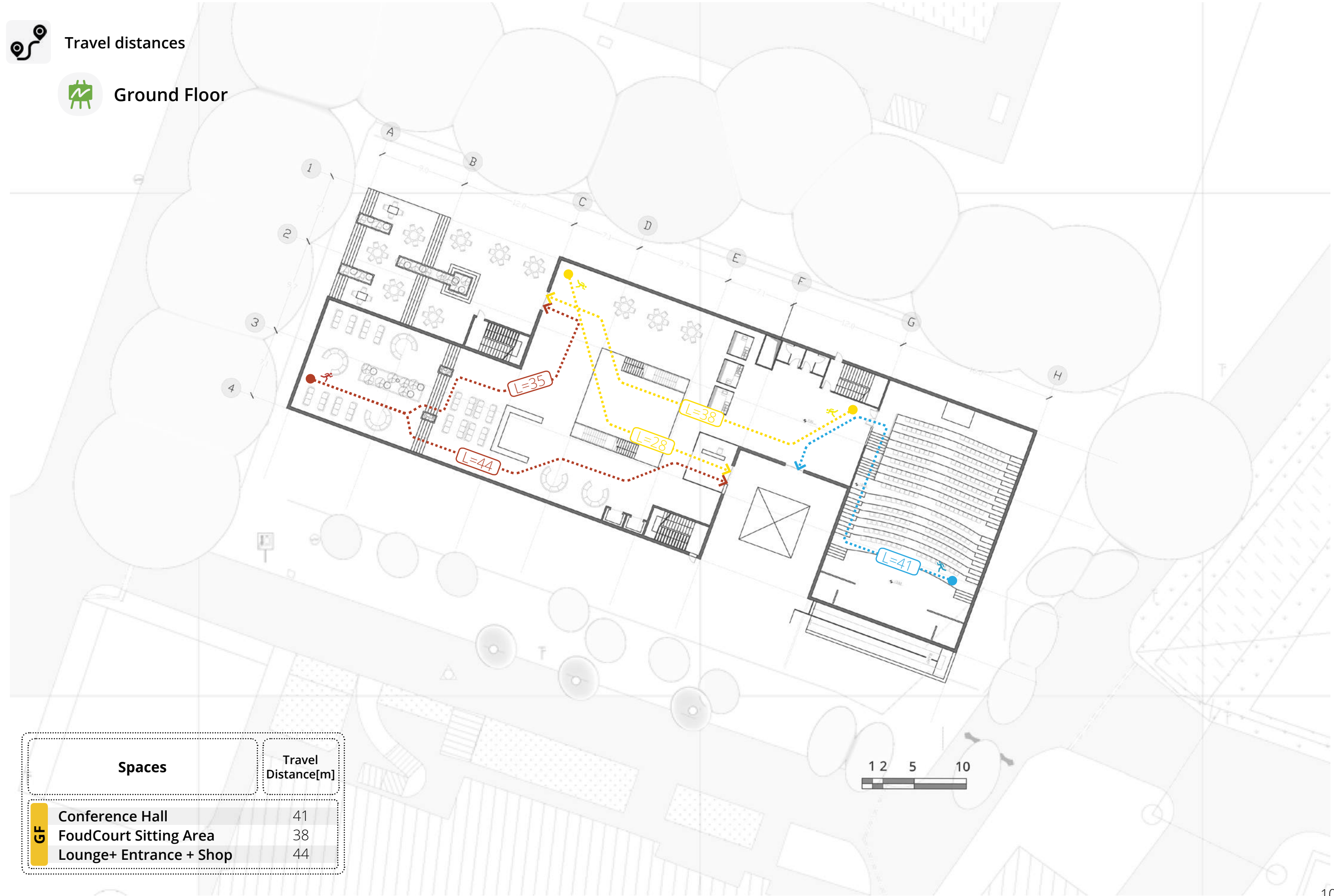
Exit Name		Stair width (mm)	Door Width (mm)	Capacity	predicted number of occupants to use the corresponding exit
GF	Foodcourt exit	-	2000	500	270
	main exit- next to Conference Hall	-	2000	500	270
	main exit- next to lounge and information	-	2000	500	257

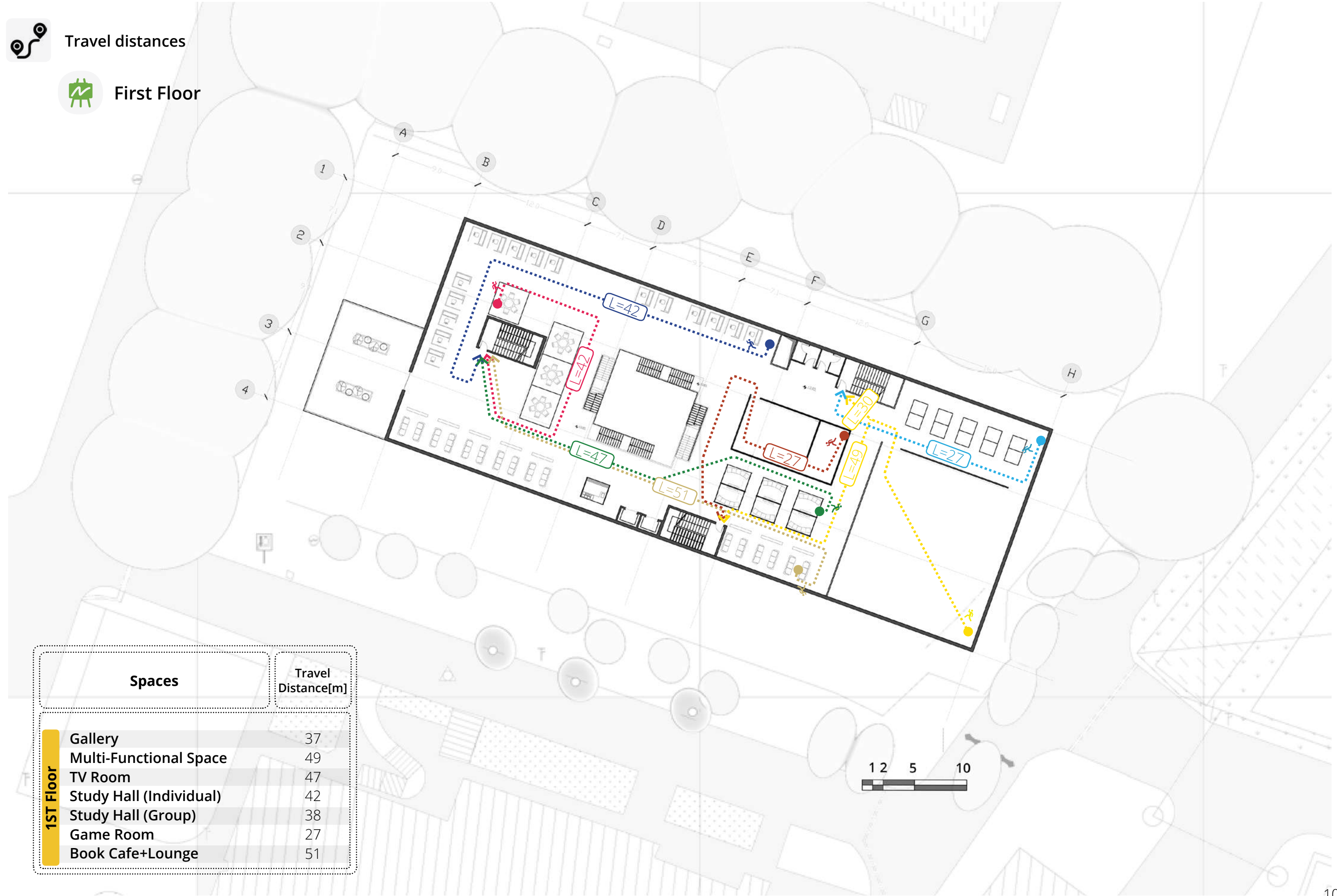


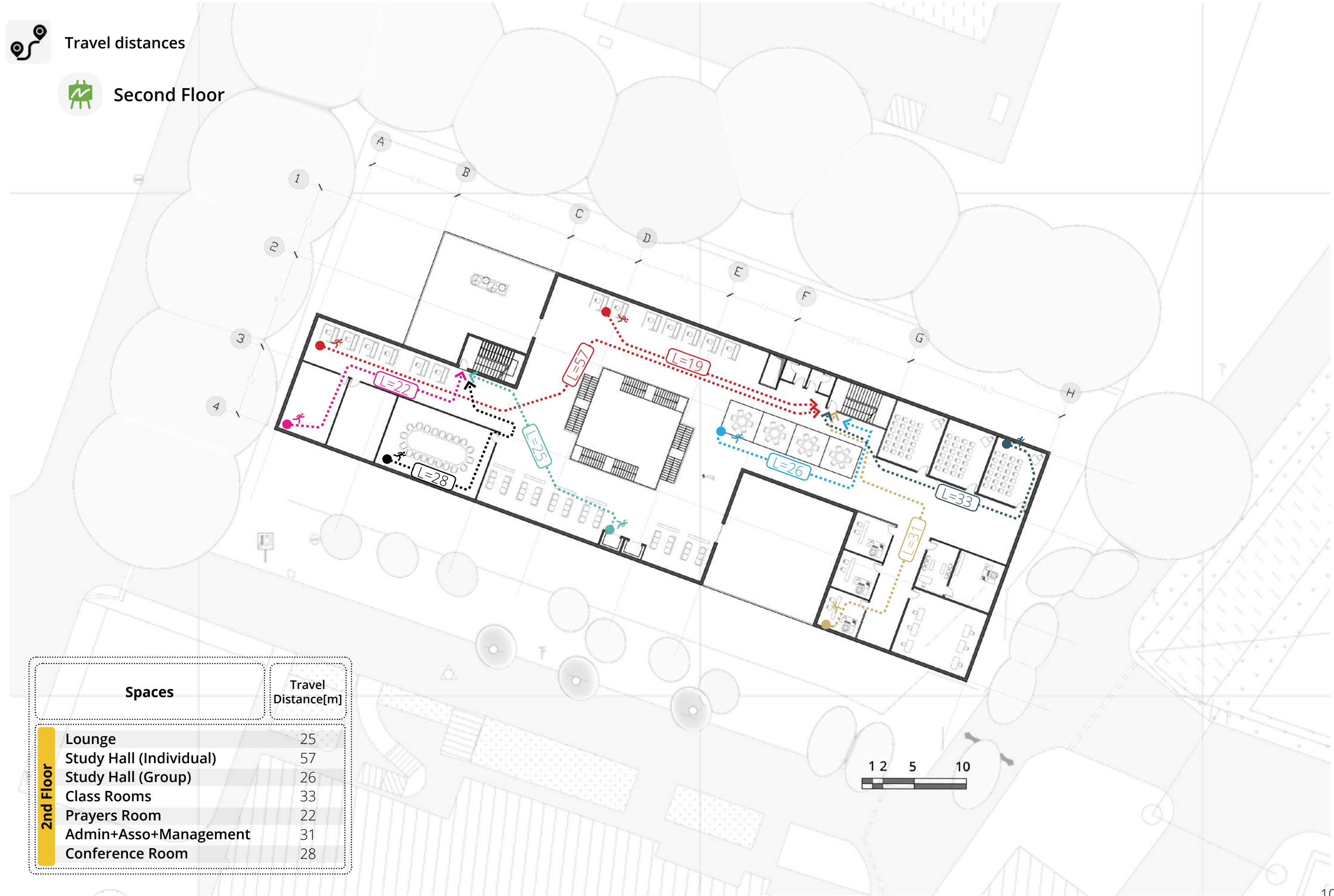


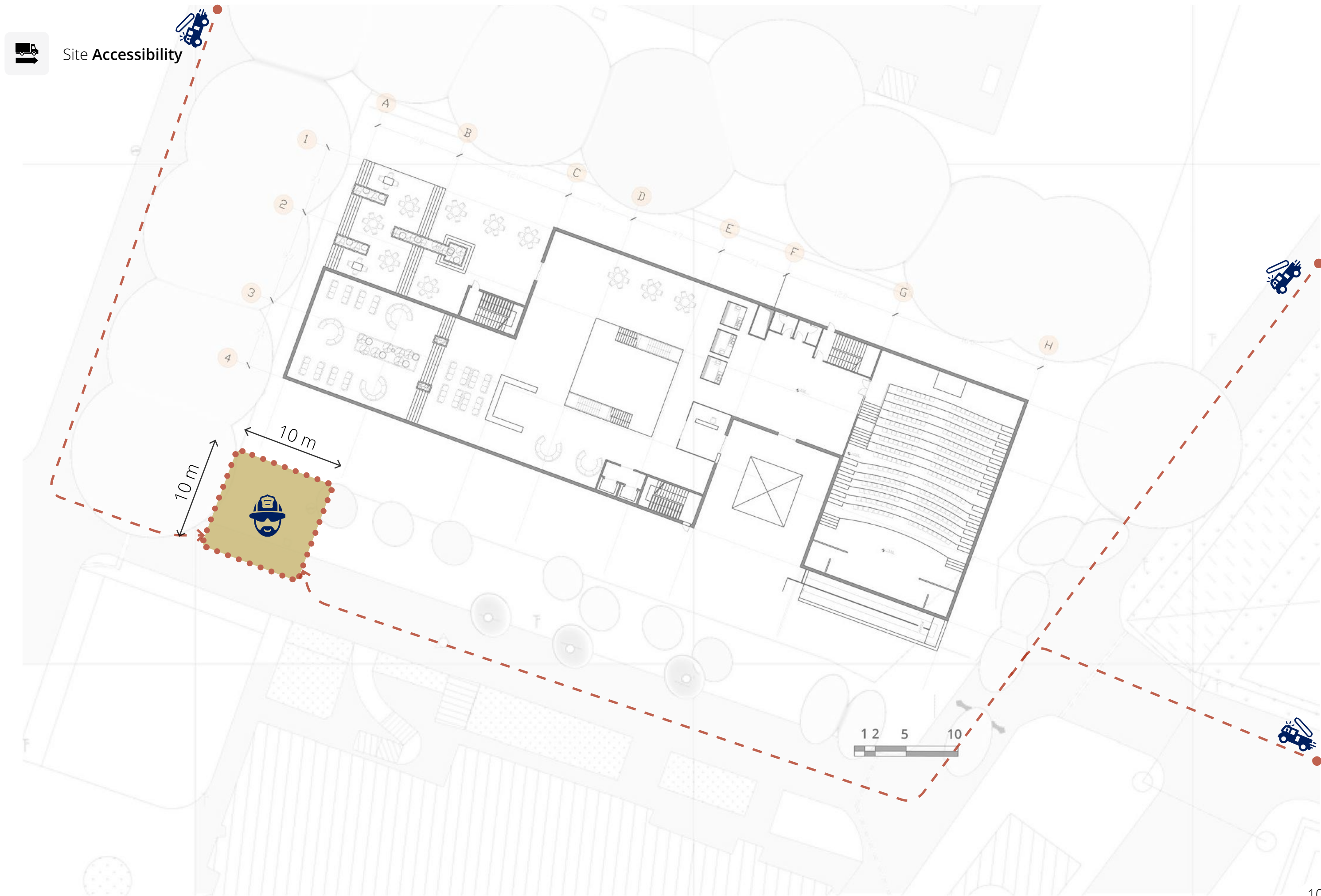








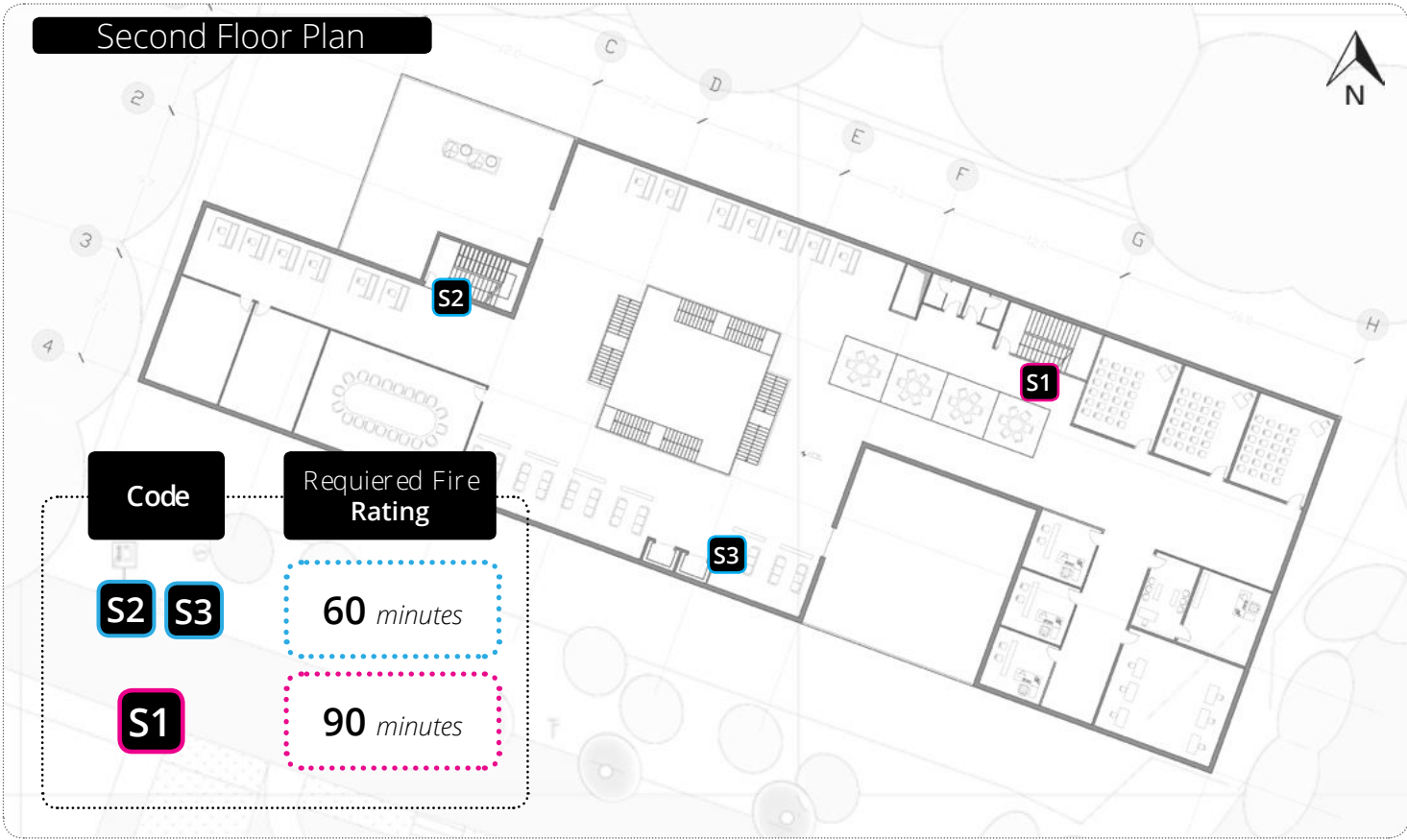
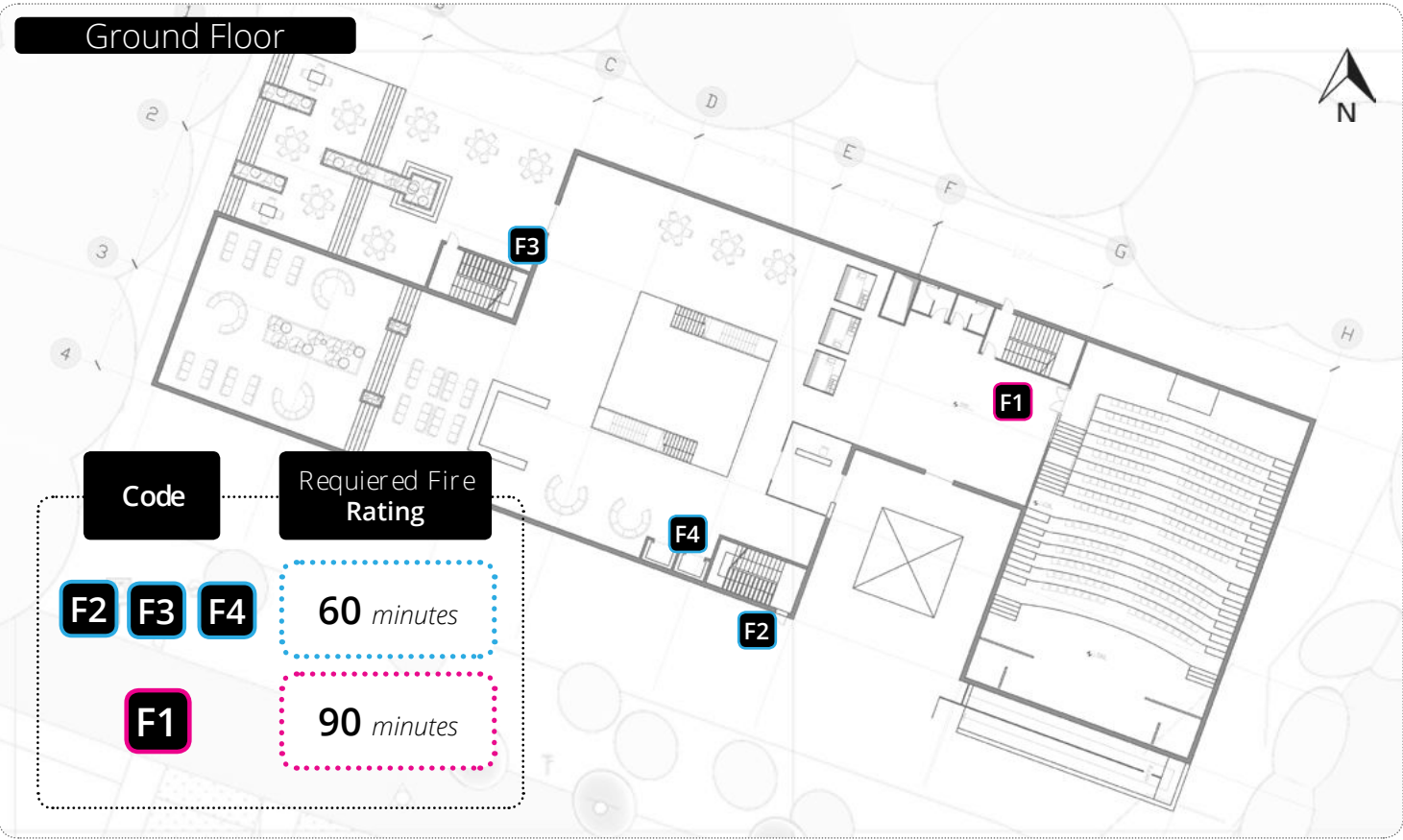
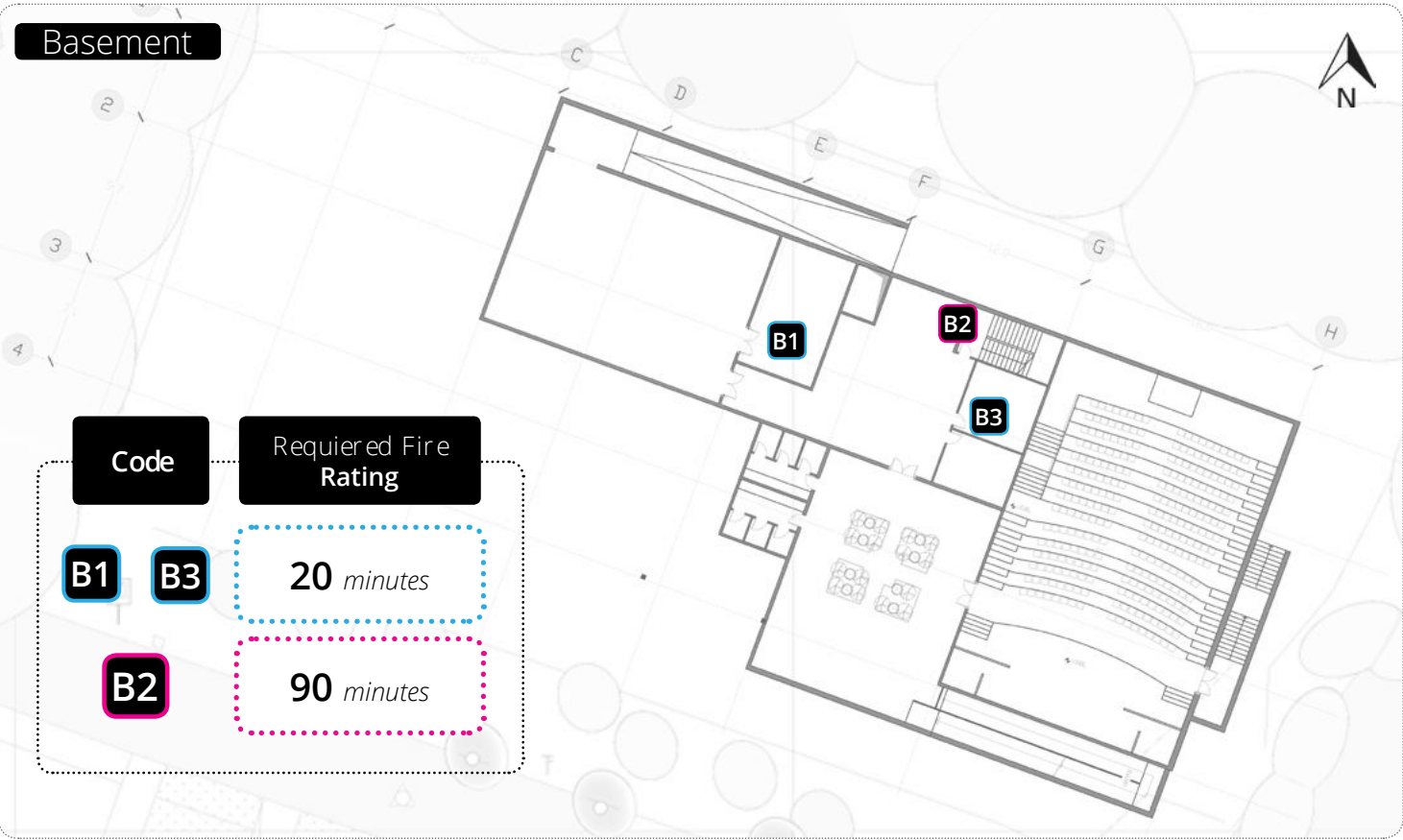




 Required **Fire Ratings-Walls**



 Required **Fire Ratings-Doors**



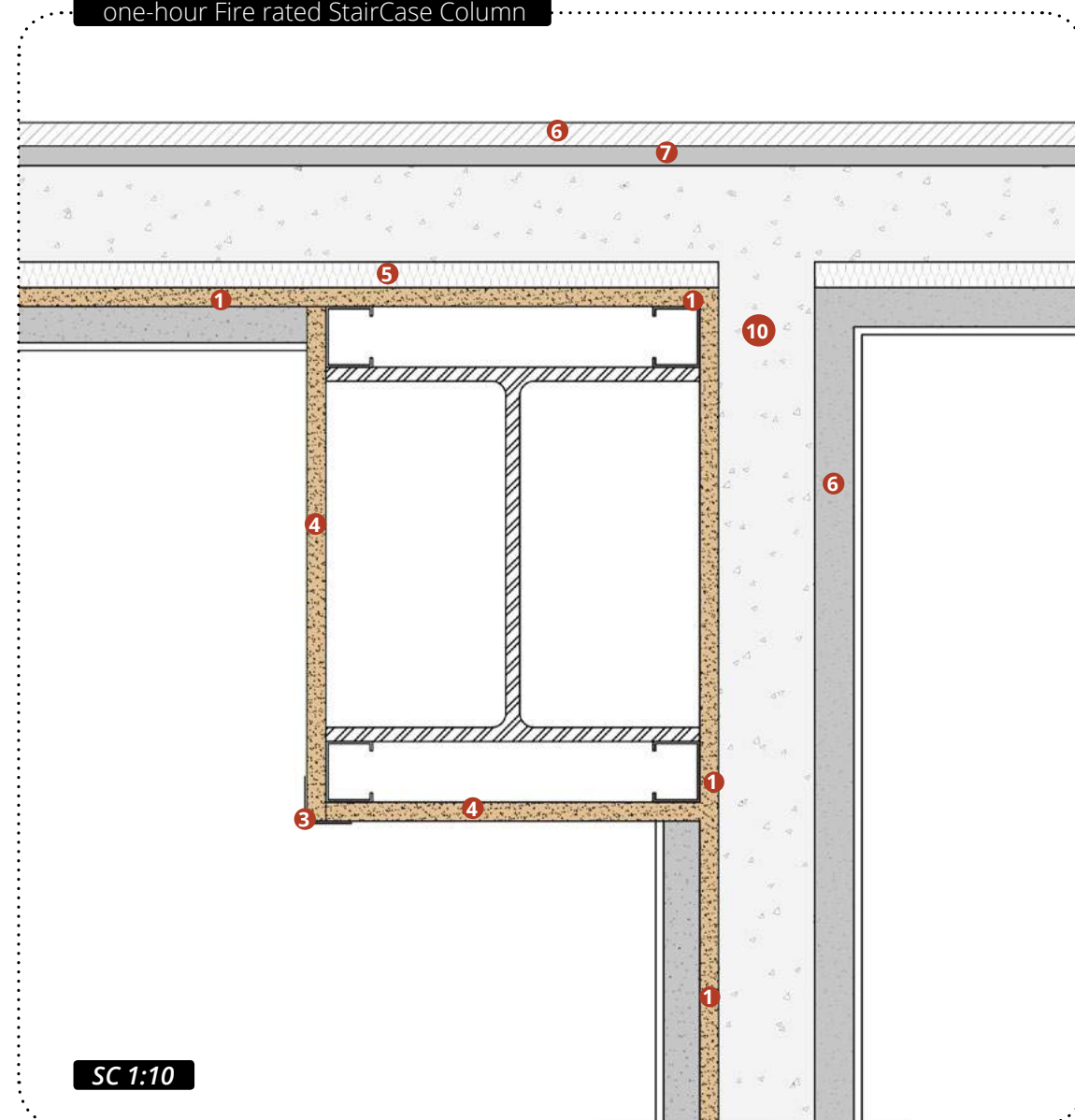


Construction details

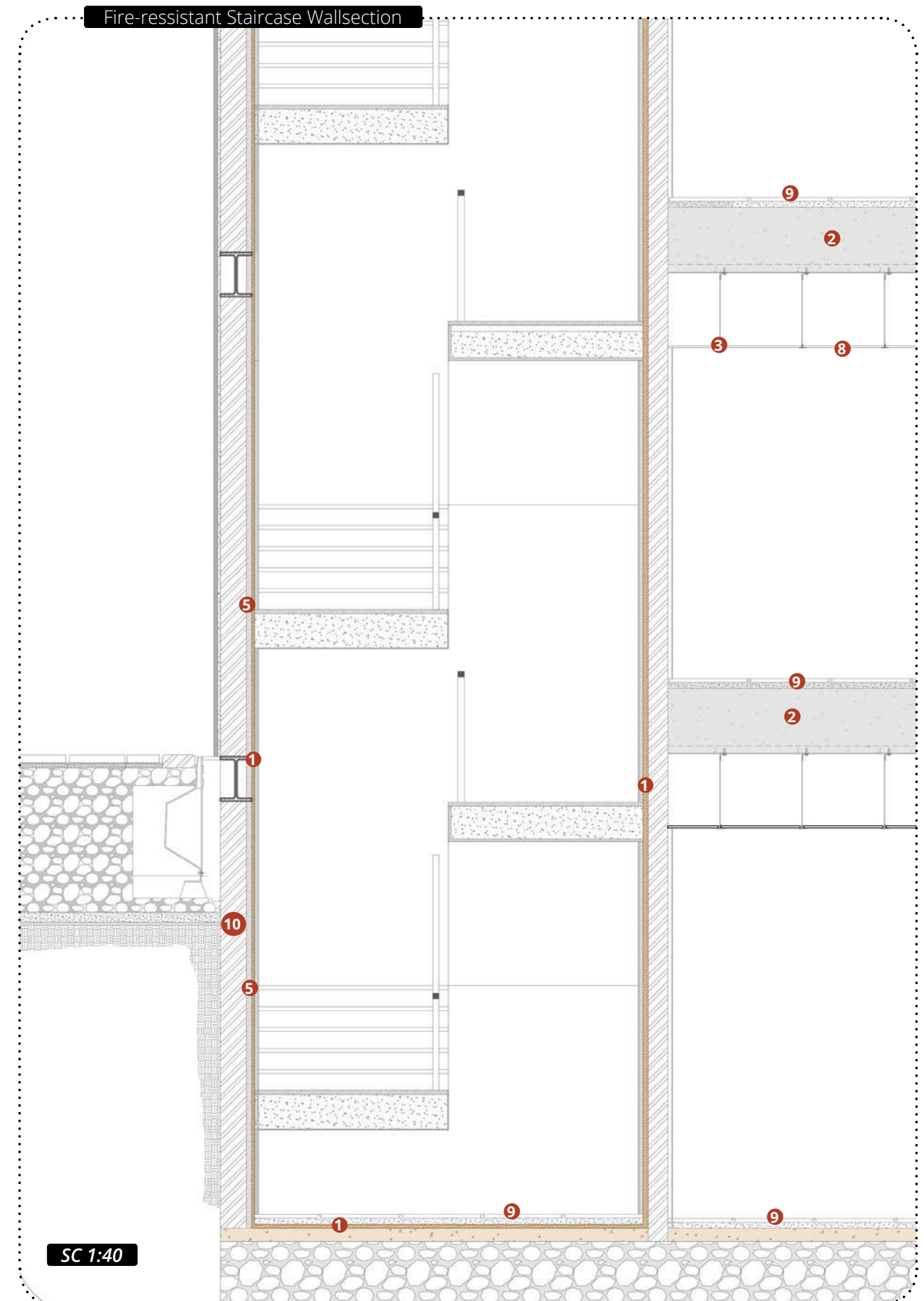
Drawings

1. Spray applied fire resistive material providing flat surface 1cm from bottom of flutes
2. Concrete on Fluted Steel deck
3. 2.5x5 Steel angle
4. 2 layers 1.6cm fire-shield gypsum board
5. XPS thermal insulation
6. Brick Facade
7. Cement-sand Mortar
8. Acoustic tiles for false-ceiling
9. Mosaic Flooring
10. Concrete Wall

one-hour Fire rated StairCase Column



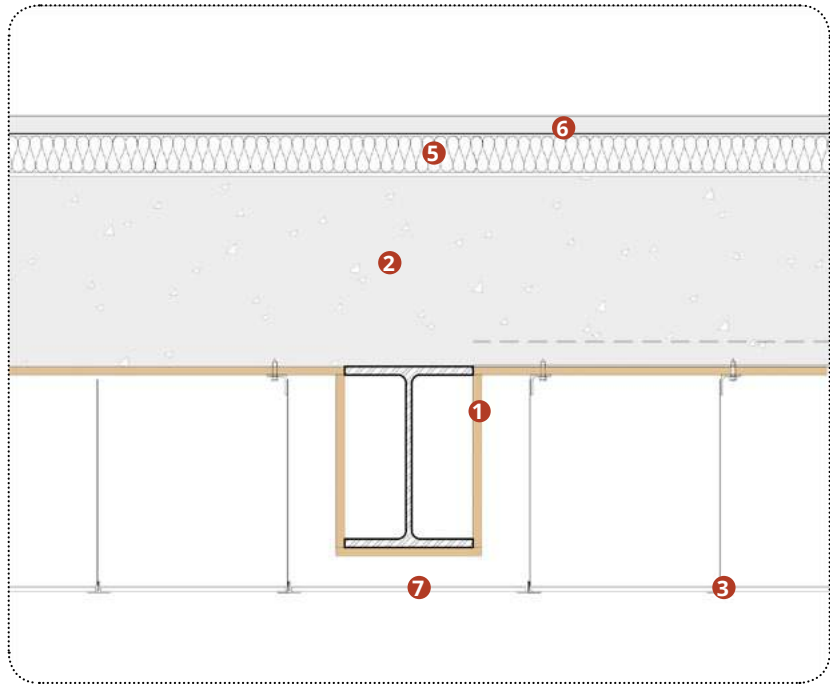
Fire-resistant Staircase Wallsection





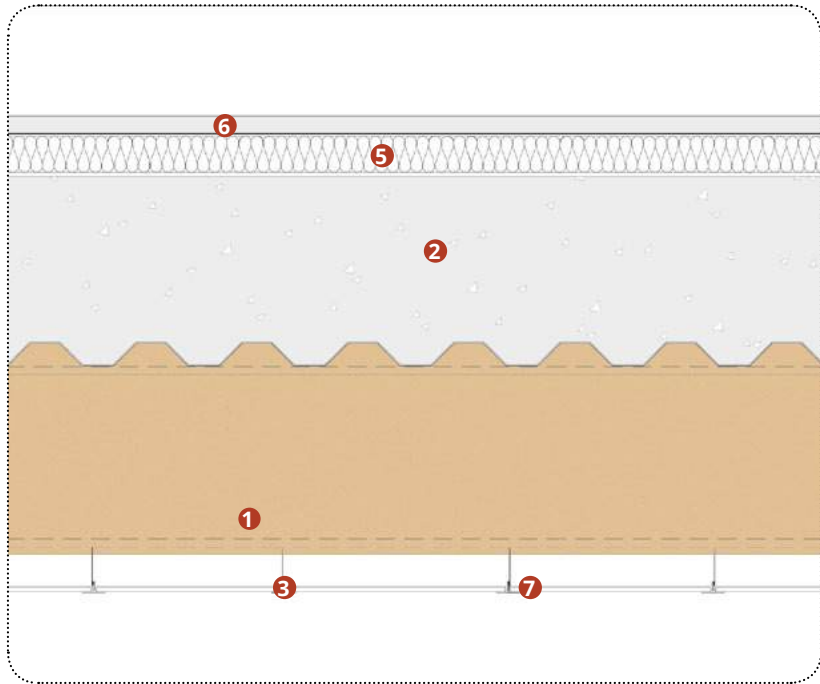
Construction details
Drawings

one-hour Fire rated Beam- Roof



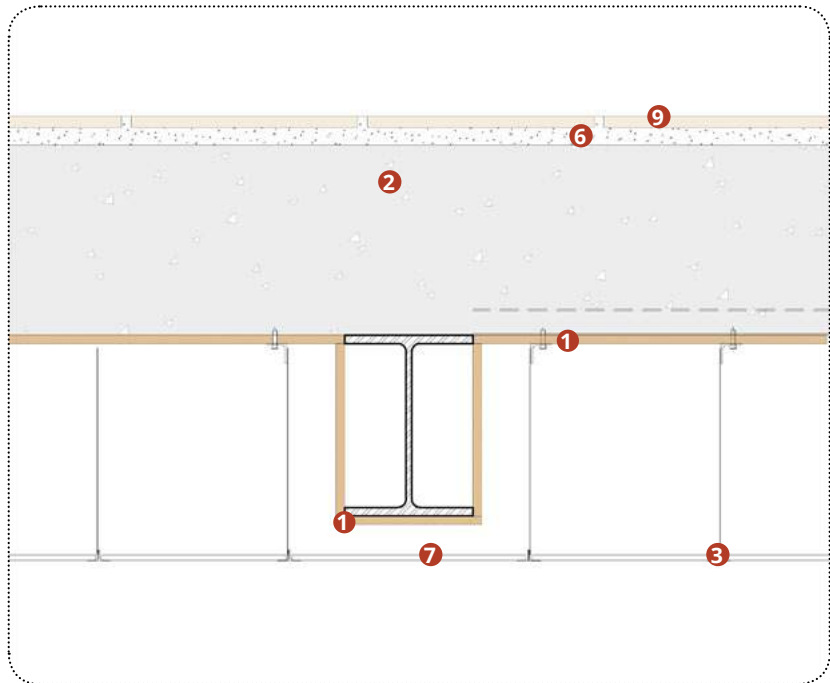
SC 1:5

one-hour Fire rated Beam- Roof



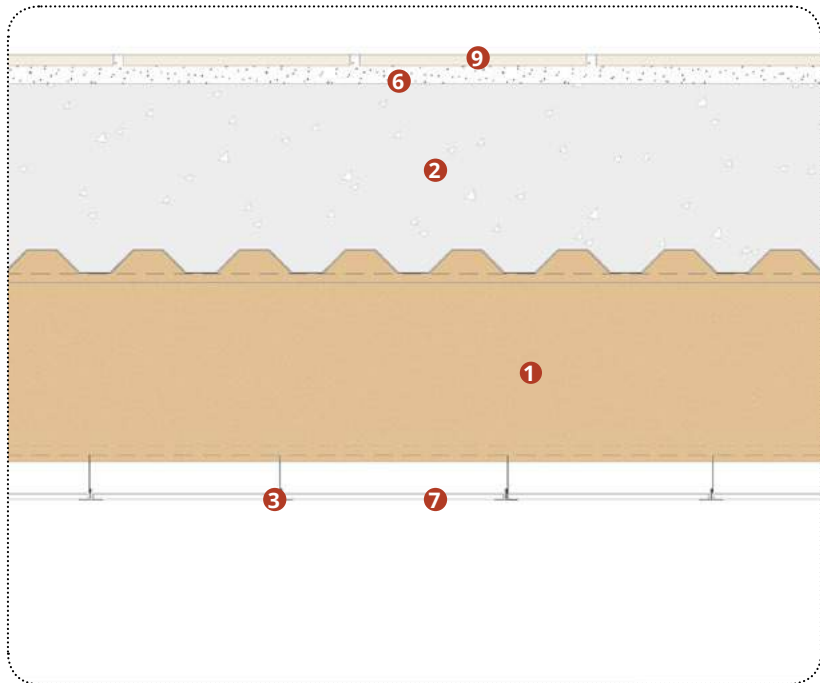
SC 1:5

one-hour Fire rated Beam- Ceiling



SC 1:5

one-hour Fire rated Beam- Ceiling



SC 1:5

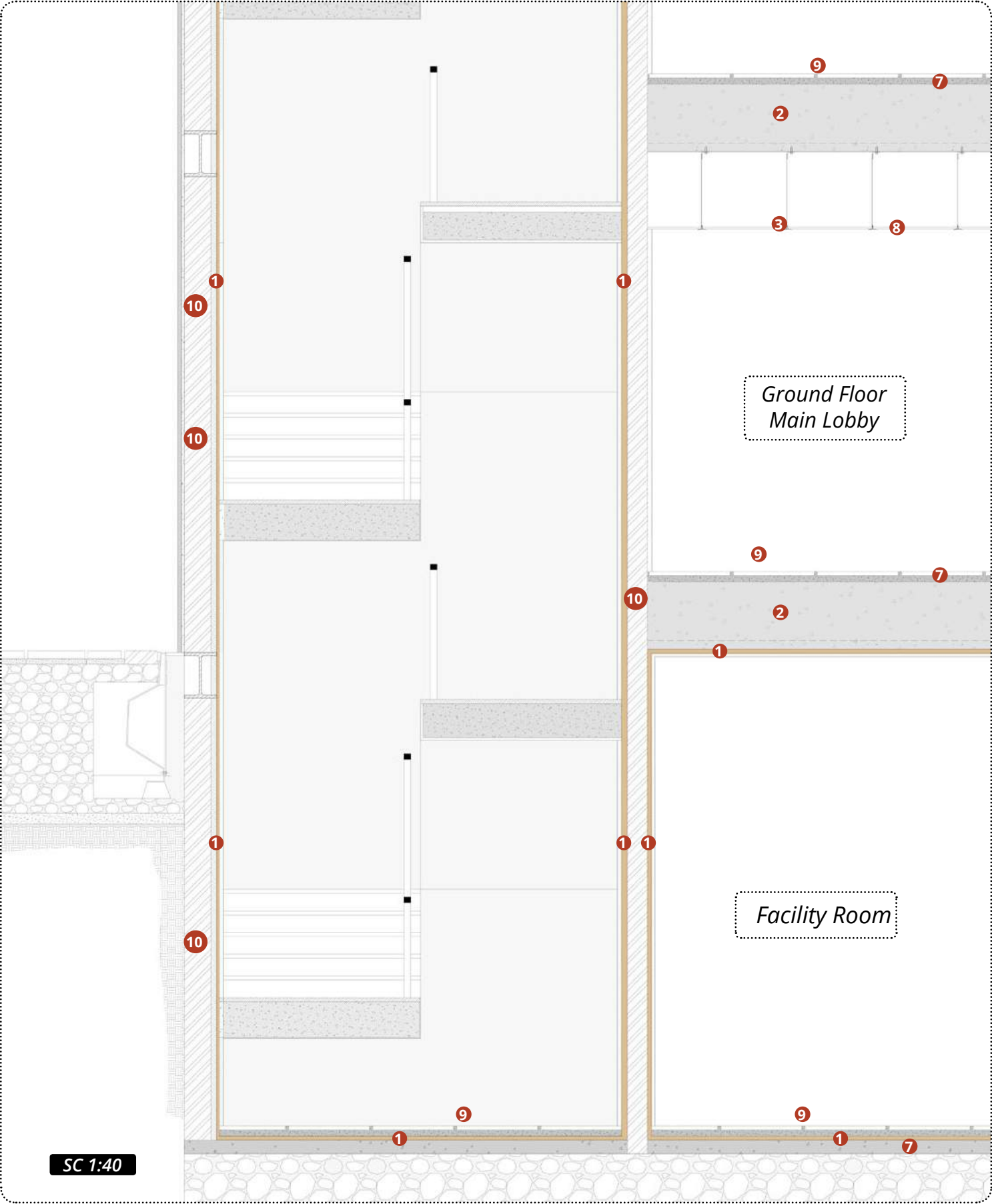
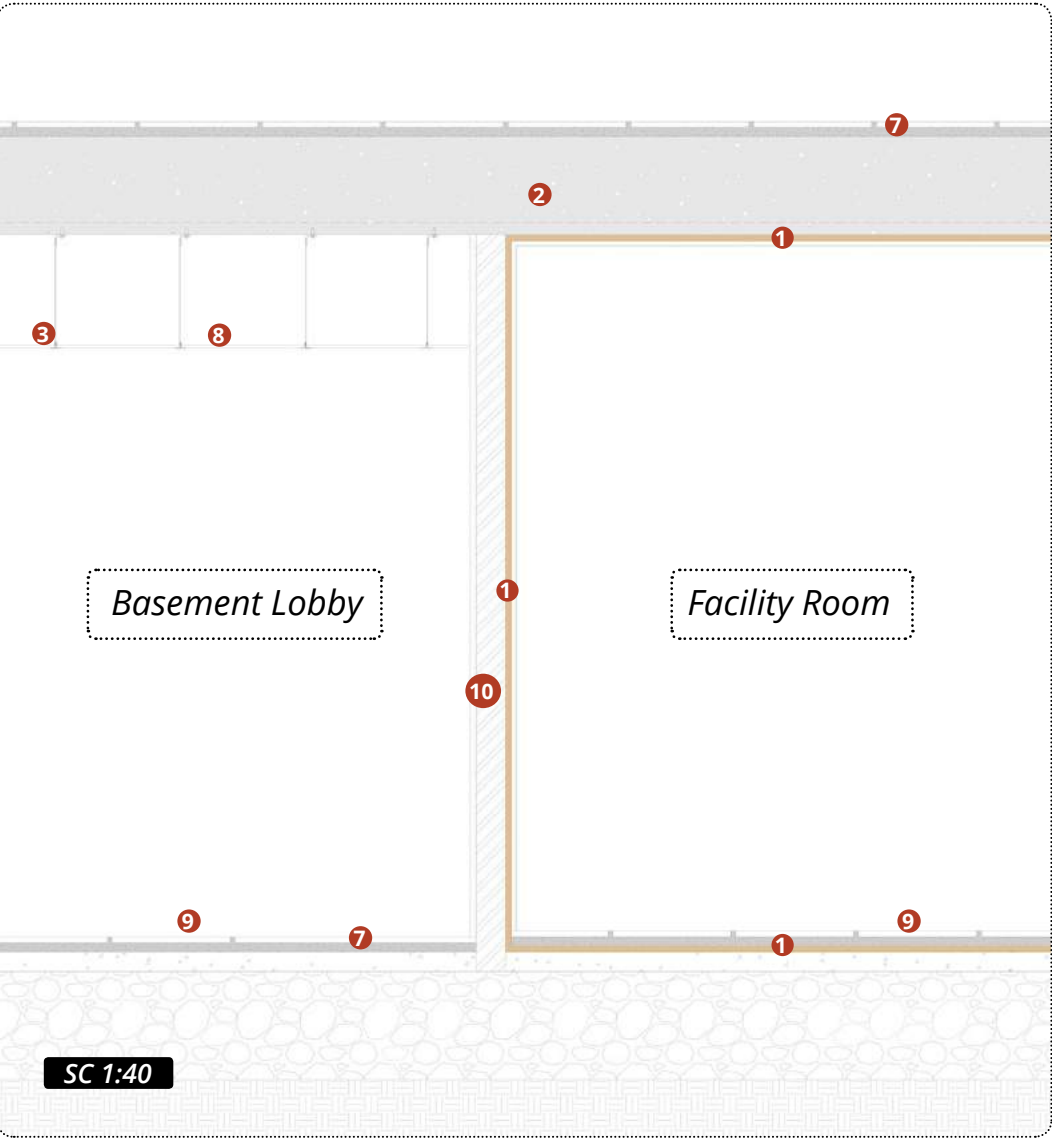
Key

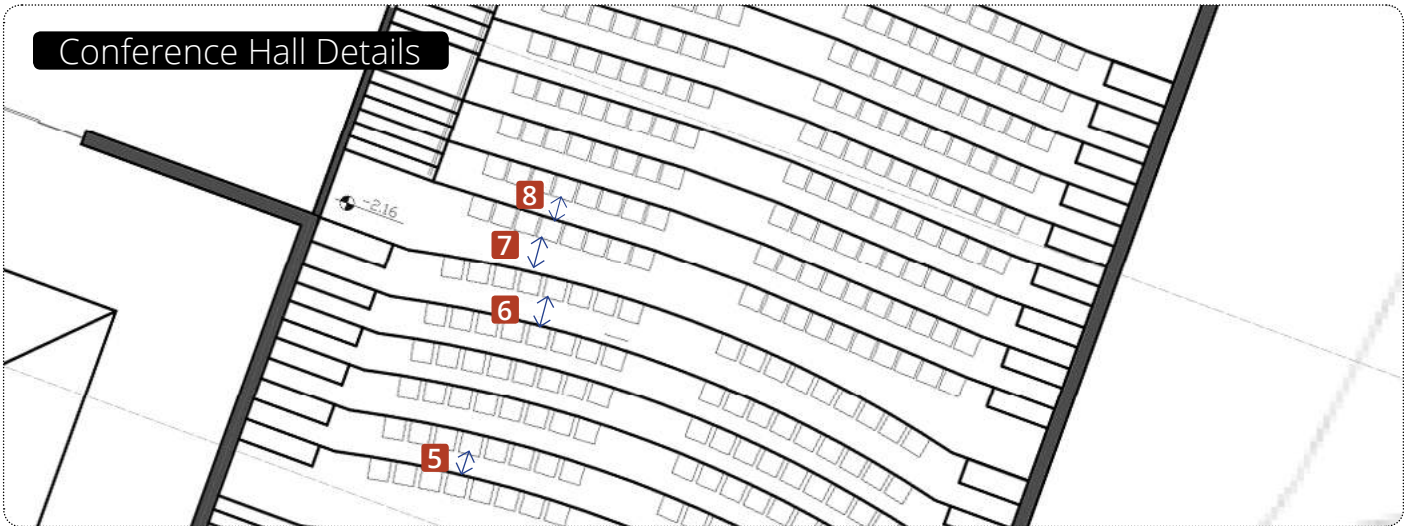
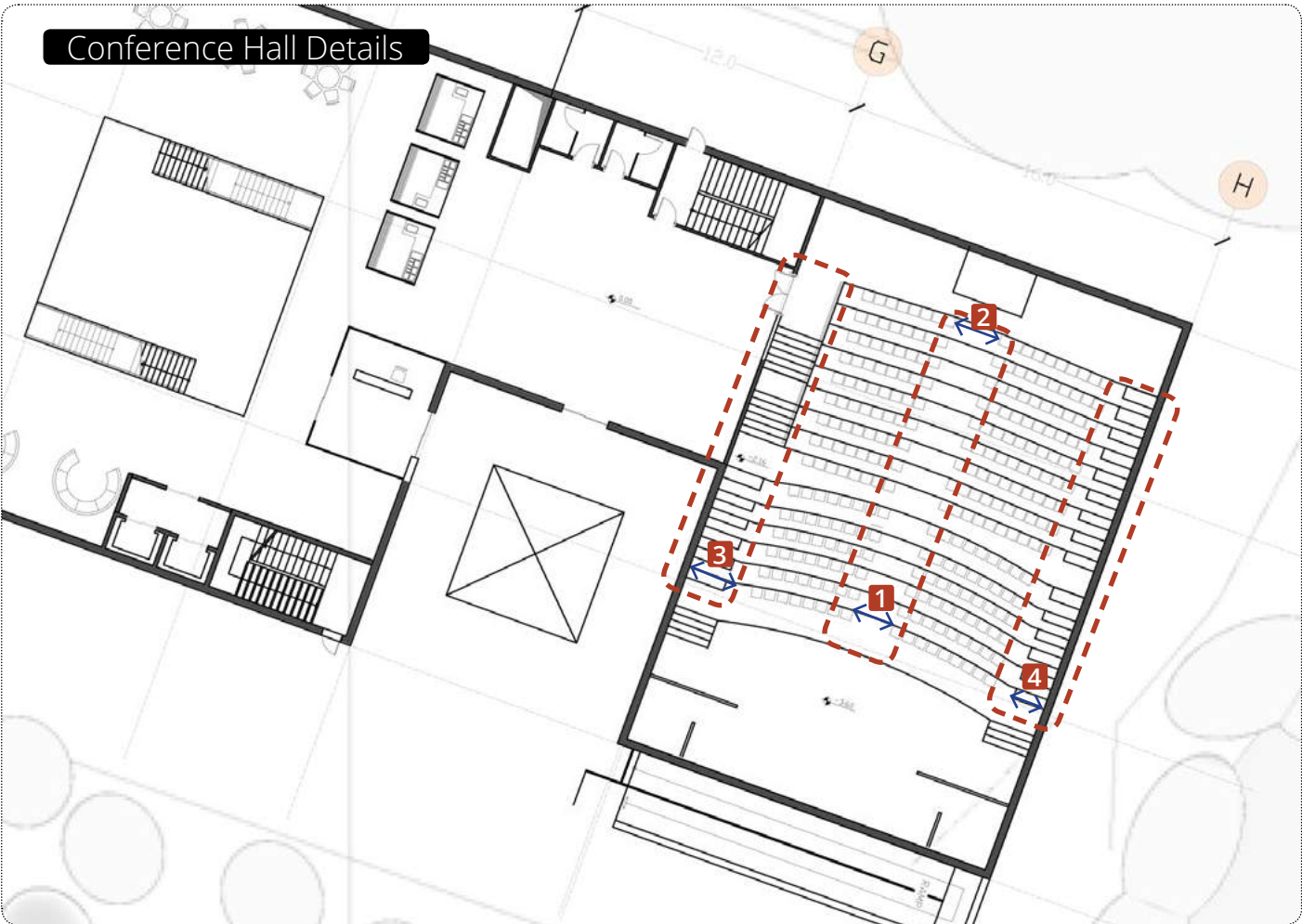
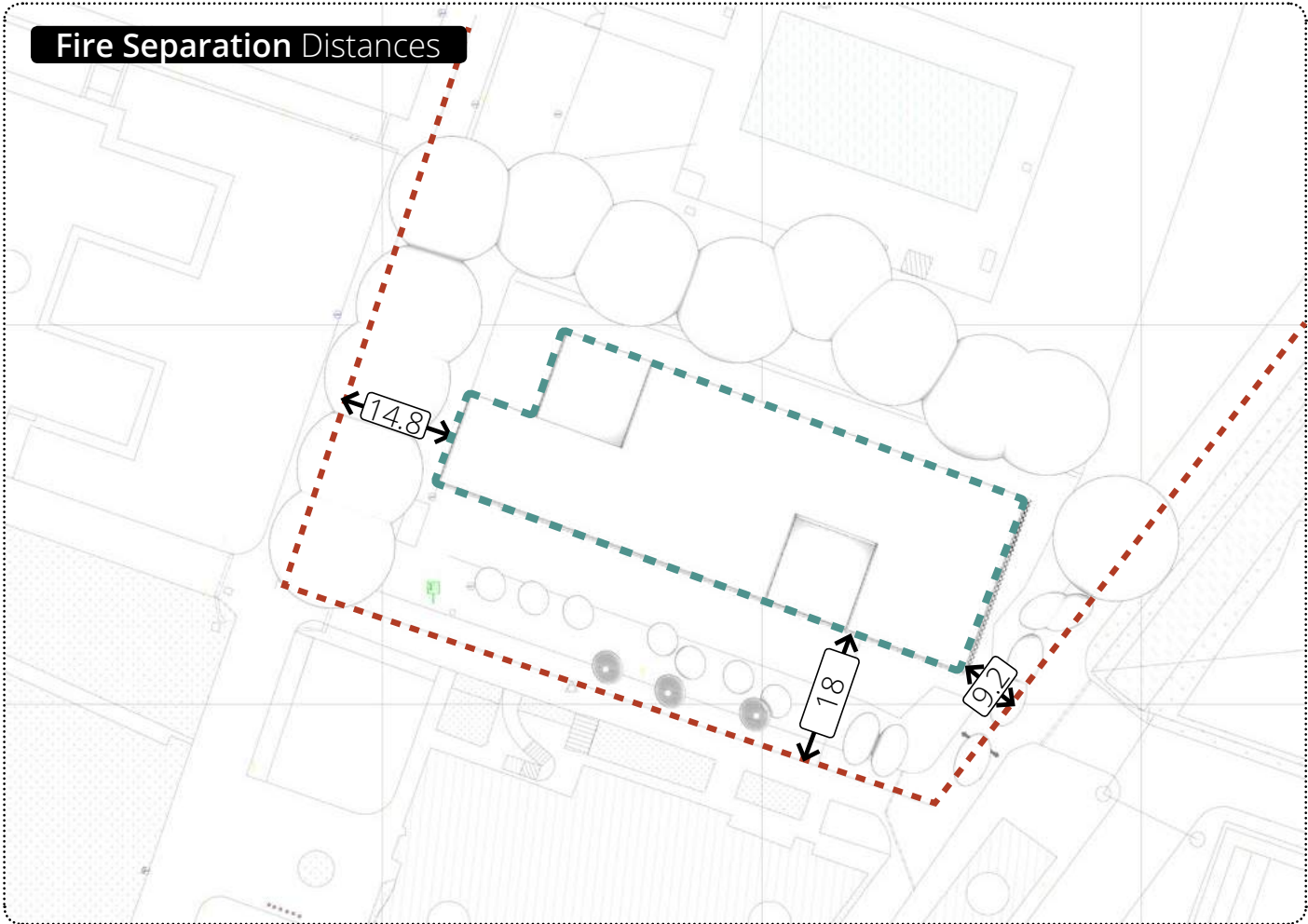
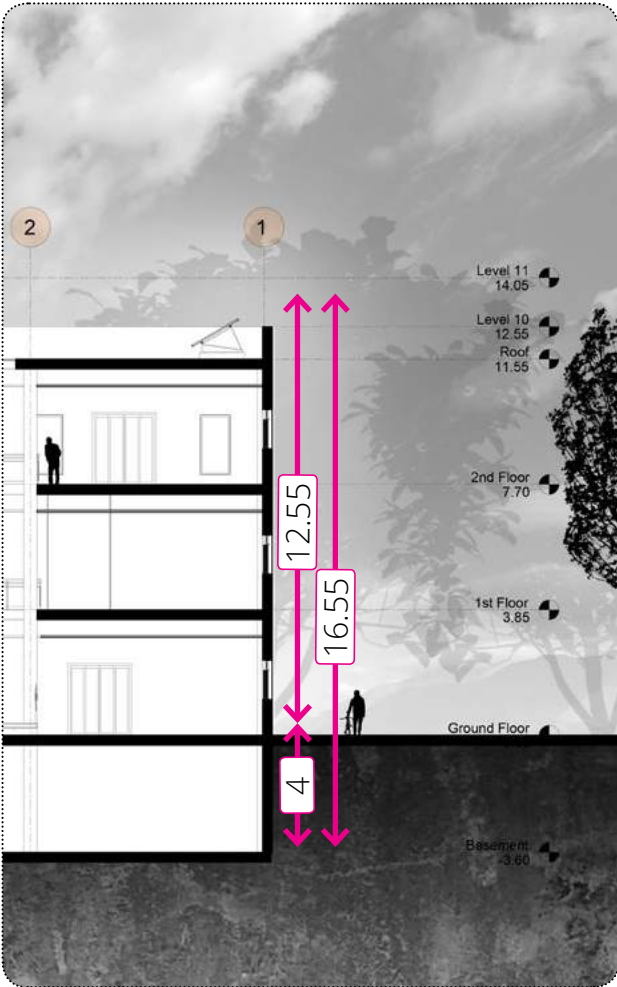
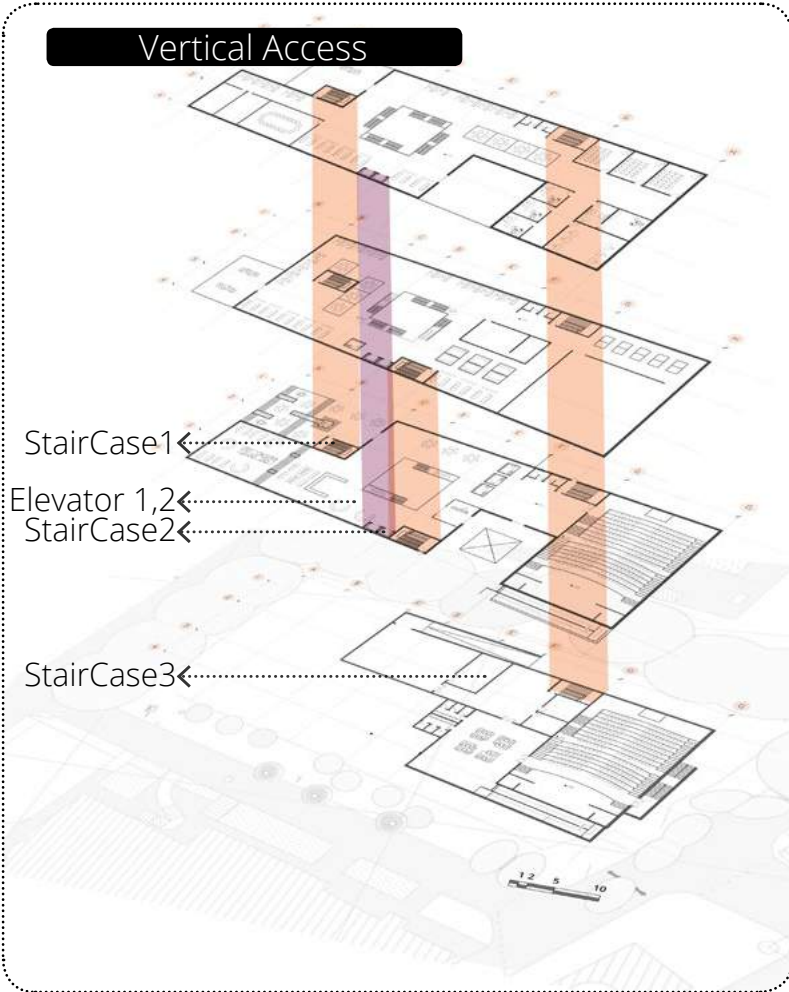
1. Spray applied fire resistive material providing flat surface 1cm from bottom of flutes
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Construction details
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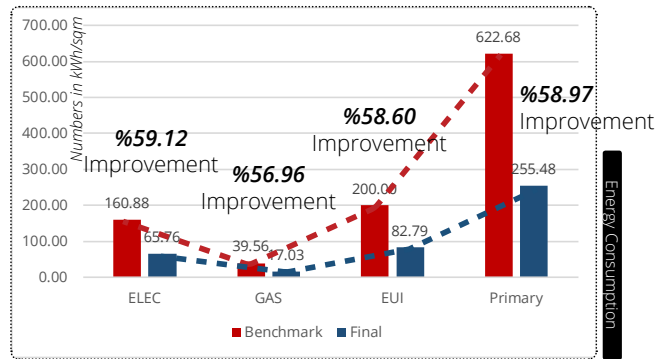




Number	1	2	3	4	5	6	7	8
Required Width(cm)	120	120	90	90	30	30	30	30
Design Width(cm)	170	200	160	129	45	60	77	45

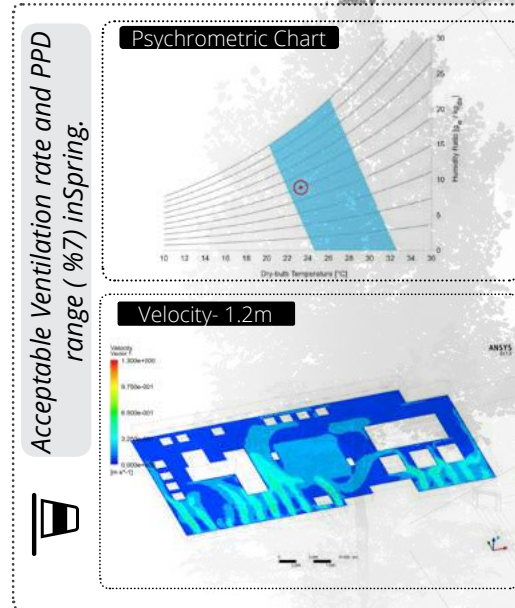
04 | Final **Performance Results, Assessments** and **Documentations**

Indices, Criteria and hights	Tools	Process. Stages	Contents	Report Page
High Performance Building		Introduction	Definition Benefits Case Studies	5
Predesign Studies		Predesign Studies	Location	6
			Public Transportation	6
			Site Dimentions and Adjacent buildings Views From the Site	6
			Location and Accessibility	6
			Temperature and Comfort	7
Energy and atmosphere,IEQ, Carbon, Water, health and Well-being, Daylight Availability, Solar Heat Gainn risk,Site Orientation Study		Programming, Massing and Orientation	Wind Analysis	8
			Radiation Analysisis	8
			Photovoltaic Production Potential	9
			Rainwater Harvesting Potential	9
			Solar Envelope	9
Functional & Aesthetical, Daylighting potential, Thermal Autonomy, Ventilation Potenial, Loads/Energy, Form Aesthetics, Attraction for pedestrians , Structure Challenges, Difficulty of Modelling and Construction, Form Flexibility, Indoor Accessibility		Form Finding and Design	Benchmarking	10
			Goals and Targets	10
			Site Orientation Study	11
			Massing Alternatives	12
			Orientation	12
			Form Alternatives	14
			Form Selection Process	15
			Form Alternatives Analysis	16
			Form Alternatives Analysis Results	19
			Selected Forms From Step 1	20
Thermal Comfort, Daylighting Potential, Glare, Cost, Improvement Potential, Aesthetics, Ventilation Potential, Cone of Vision Analysisis, Weighted Overheat Hours		Façade Design	Form Analysis(Step 2)	21
			Form Analysis(Step 2)	22
			Ventilation Analysisis	23
			Thermal Comfort Analysis	24
			Final Results	25
			Plans	26
			Facade Design Process	30
			Facade Alternatives	32
			Daylight Analysis	34
			Glare Analysis	36
Thermal Comfort, Energy Consumption, Cost Analysisis, Noise, Initial Cost, Operating Cost, Occupancy, Frequency of Use, User Control, Aesthetics,EnergyConsumption, Capacity, Humidity, Pressure Control, Efficiency, Enviromental Impact, Life Span and health		Construction Selection	Facade Alternatives Analysis Results(Step1)	38
			Facade Design Alternatives (Step2)	39
			Daylight Analysis	40
			Glare Potential (Eye Level)	41
			Horizontal 60° Cone of Vision Analysisis (Eye Level)	42
			Natural Ventilation Strategy Selection Process	43
			Natural Ventilation Strategies results	44
			Concept Stage Final Performance Results	47
			Constructions Alternatives	49
			Constructions Selection (Step1)	50
Energy Production, Aesthetics, Simple Payback Period,IRR,NPV		HVAC and Lighting system Selection and design	Constructions Selection (Step2)	51
			HVAC Selection	52
			HVAC Selection Results	53
			HVAC System Sizing	53
			Lighting System	53
			HVAC System Diagram And Specifications	54
			Solar DHW system design	55
			Construction and Systems Final Performance Results	56
			Photovoltaic System design	57
			Final System and Results	58
Outdoor water consumption, indoor Water consumption, water use reduction strategies, baseline and final consumption		Renewable System Design	Step2 - Orientation Improvement	59
			Water Management	60
			Water Management Case Studies	60
			Outdoor Water Management	61
			Landcape Layout	62
			Indoor Water Management	63
			Facade Heat Transfer (Thermal Bridges)	65
			Natural Ventilation Analysis (CFD)	67
			Natural Ventilation Analysis Results	69
			Lighting Design	70
Psi Value for Façade Critical points, Ventilation Rate, PPD/PMV ,optimum Opening Area for Ventilation,Lighting optimum Arrangement and lights,Photopic and melatopic lux in different times, Annual Glare Risk in Critical Views		Water Management	Circadian lighting Analysisis	72
			Annual Glare Analysis	73
			Site Acoustical Assessment	74
			Labelling Acoustic Zones	75
			Acoustic Requirements	77
			Sample Spaces	78
			Noise control	79
			Air-borne sound insulation	79
			β-Impact noise Conrtol	79
			STC Calculation for Multiple Partitions	82
Site Background noise, sound Transmission Class, Reverberation Time, Sound masking		Detailed Design and Analysis	Improvement options details	83
			Room Acoustics	84
			reververation Time Calculations	84
			Acoustic tiles details for Ceilings	86
			Acoustic tiles details for walls	87
			Conference Hall reflector panels design	88
			Sound Masking	89
			Building Type Determination	91
			Allowed exit Distances	92
			Occupancy Loads - Exits	93
StaitsCases and Emergency Exits,Travel Distances		Acoustic Design	Exits	97
			Travel distances	101
			Site Accessibility	105
			Required Fire Ratings-Walls	106
			Required Fire Ratings-Doors	107
			Construction details Drawings	108
			Design Process and Final Performance Data	112
			Project evaluation according to LEED	117
			Final Architecture Documents	118
			References and tools	128
Final Data		Fire Protection		
		Final Performance Results, Assessments and Documentations		



Total CO2 Reduction (annual) [kg] **167,645** 36 X

PV Panels
Consumption[kWh] covered by PV[kWh] 287670
covered by grid 44362
Solar fraction [%] 243308
Avoided CO2 emissions [kg/year] %15.40
23795



Parametric Facade Designed in with regard to the spaces need for Daylight, in addition it controls overheating of the common areas behind it.

EC++ (Annex19)

LEED 3 point for daylight

sDA %81

ASE %23

Certified

Lighting System

Power density [W/m2] 2.5

Return Air Fraction 0

Radiant Fraction 0.42

Visible Fraction 0.18

Lighting Control On

Control type Stepped

LED+ Linear off

AbbasPour **Student Center**
High Performance Building Design
Instructor: Dr.Zomorodian
Student: Nima Forouzandeh
Winter 2020



Roofs

Cool Roof+ Insulation L2

Layer	Thickness	Conductivity	R-Value	Total U-value	STC	HC
Outside Air			0.05			
CoolCoating	0.002	1	0.00			
Polyurethane	0.010	0.25	0.04			
XPS	0.090	0.04	2.20			
Concrete	0.220	2.5	0.09			
Inside Air			0.09			
Total				0.406	53	36

Glazings

Dbl Low-E

Name	SHGC (%)	VT (%)	VT/SHGC	Total U-value	STC
Dbl Blue 6mm/13mm Air	0.598	0.789	1.29	1.786	45

External Walls

AAC Block Insulation L1

Layer	Thickness	Conductivity	R-Value	Total U-value	STC
Inside Air			0.11		
Gypsum plaster	0.005	0.31	0.02		
Gypsum-sand plaster	0.02	0.59	0.03		
AAC Block	0.22	0.158	1.39		
XPS	0.030	0.004	7.50		
Cement-Sand Plaster	0.025	0.44	0.06		
Light Weight Brick	0.03	0.08	0.38		
Outside Air			0.06		
Total				0.10	42

Ground Floor

Layer	Thickness	Conductivity	R-Value	U-value
Mosaic	0.22	1.65	0.13	
XPS	0.060	0.04	1.50	
Concrete	0.1	0.25	0.40	
Hardcore	0.35	3.5	0.10	
Total				0.47

Ceilings

Layer	Thickness	Total STC	Total HC
CoolCoating	0.002		
Polyurethane	0.010		
Mortar	0.090		
Concrete	0.220	51	34

Partition Walls

Layer	Thickness	Total STC
Gypsum	0.02	
Bricks	0.10	
Gypsum	0.020	42

System Sizing and Details

Solar DHW

Panels Total Area 32 sq.m.

DHW annual Load 1989 kWh

Annual Solar DHW 2460 kWh

Hot Water Supply

Gas Boiler

Chilled Water Supply

Air Cooled Chiller

Terminals

VAV + Reheat / Communal Spaces

Fan Coil Units/Small Spaces

Energy Conservation Systems

Heat Recovery

Heating Load 1.25 Design Margin

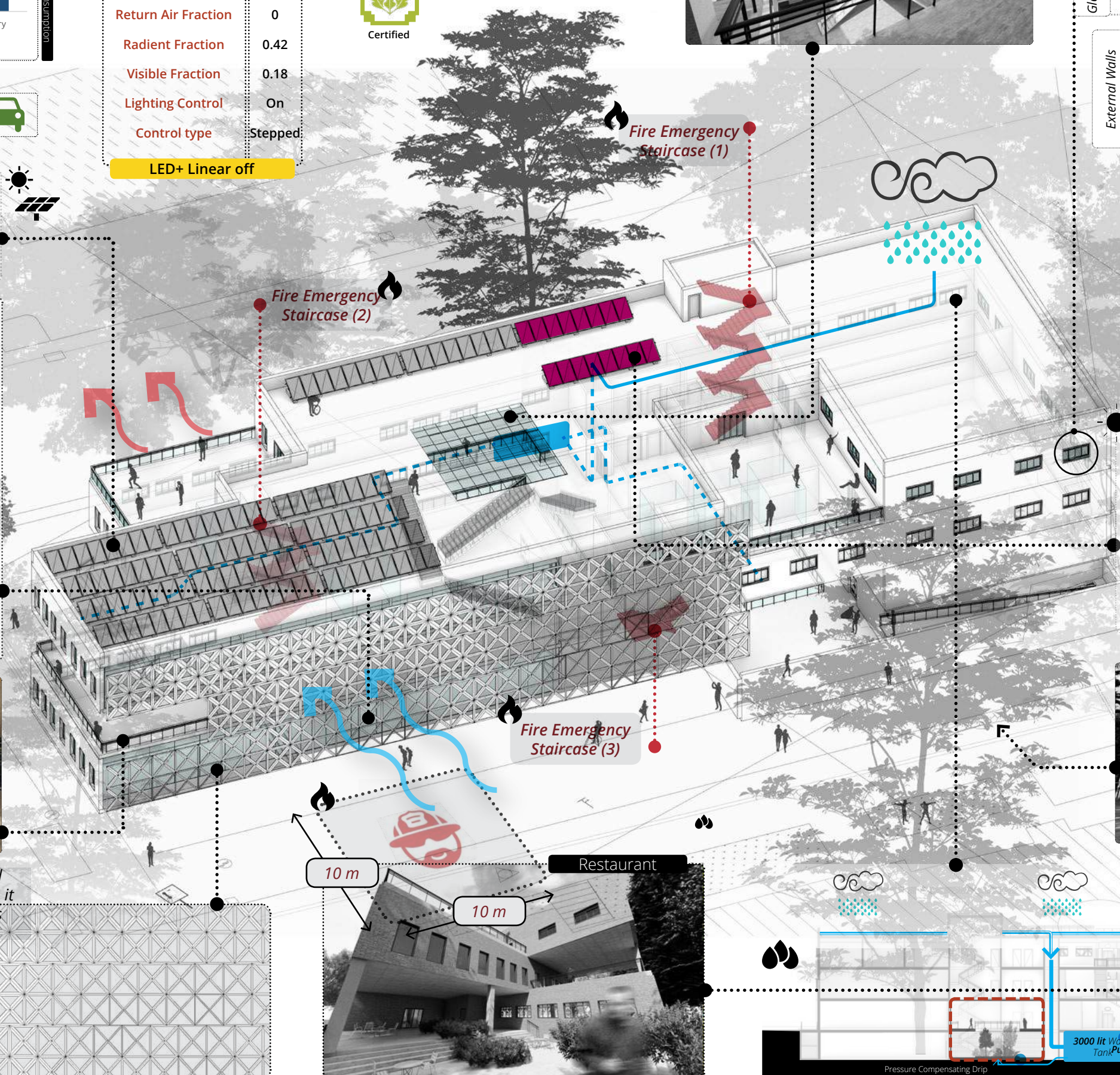
653.8 kWh

Cooling Design Capacity 1.15 Design Margin

563.1 kWh

Design Flow Rate

35.6 m3/s

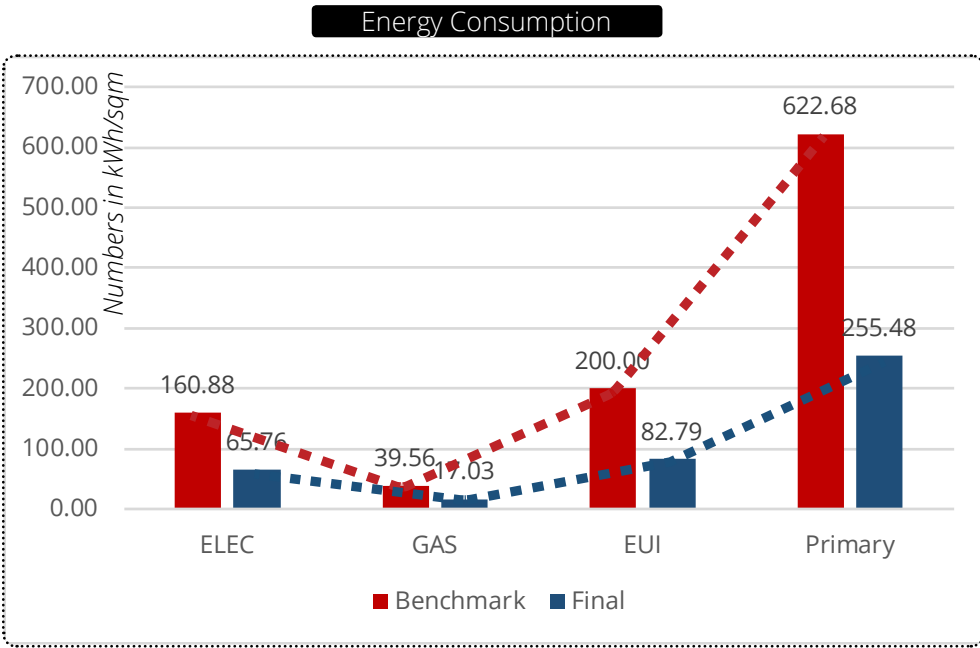
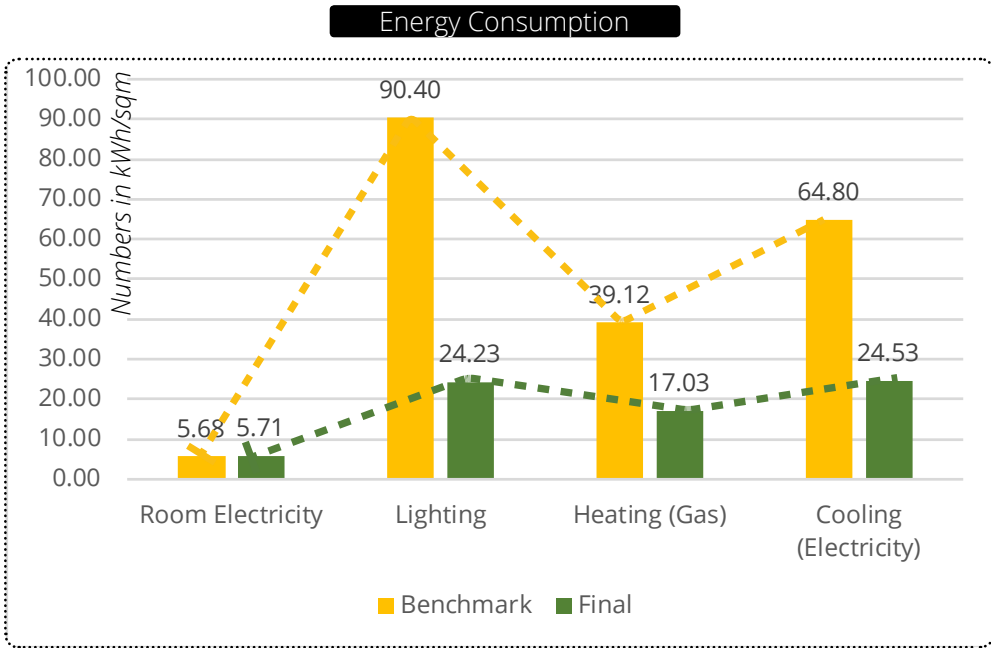


Entrance View

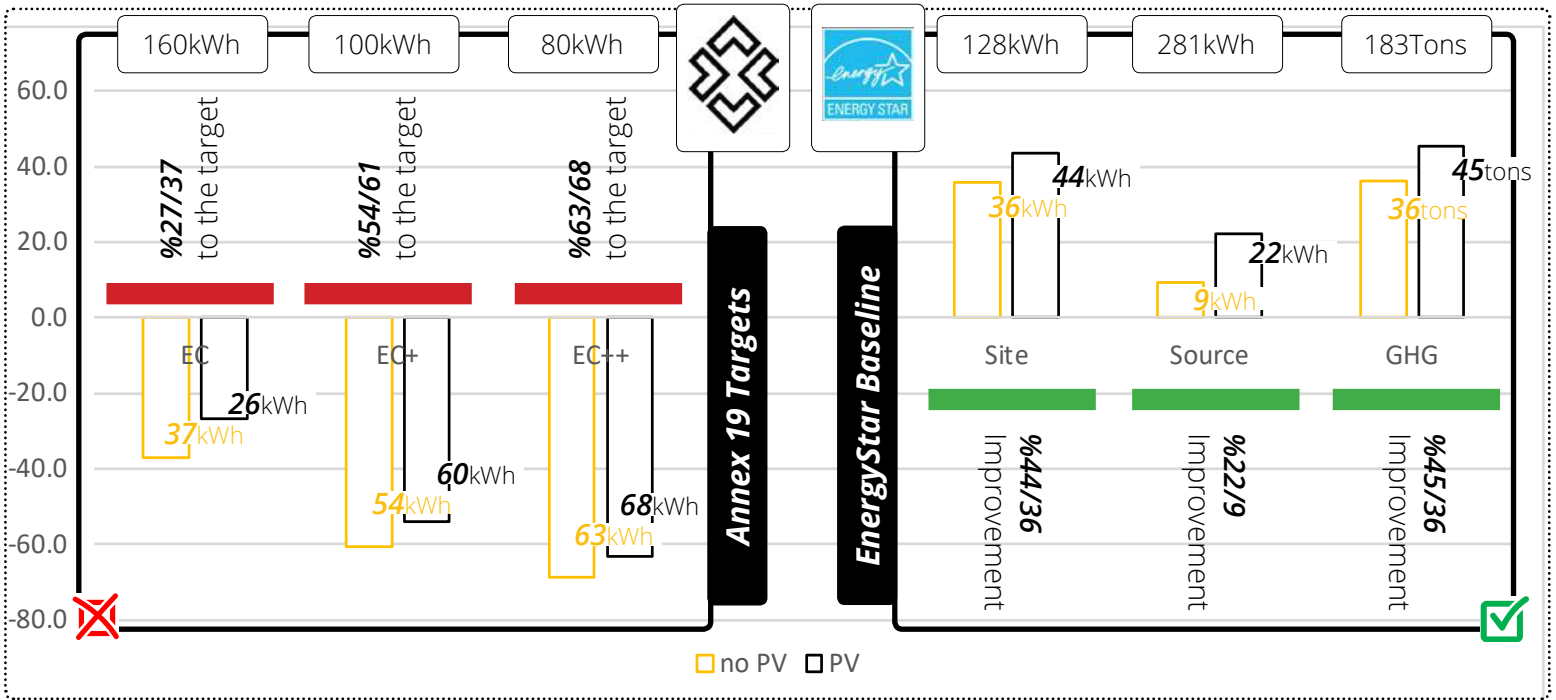
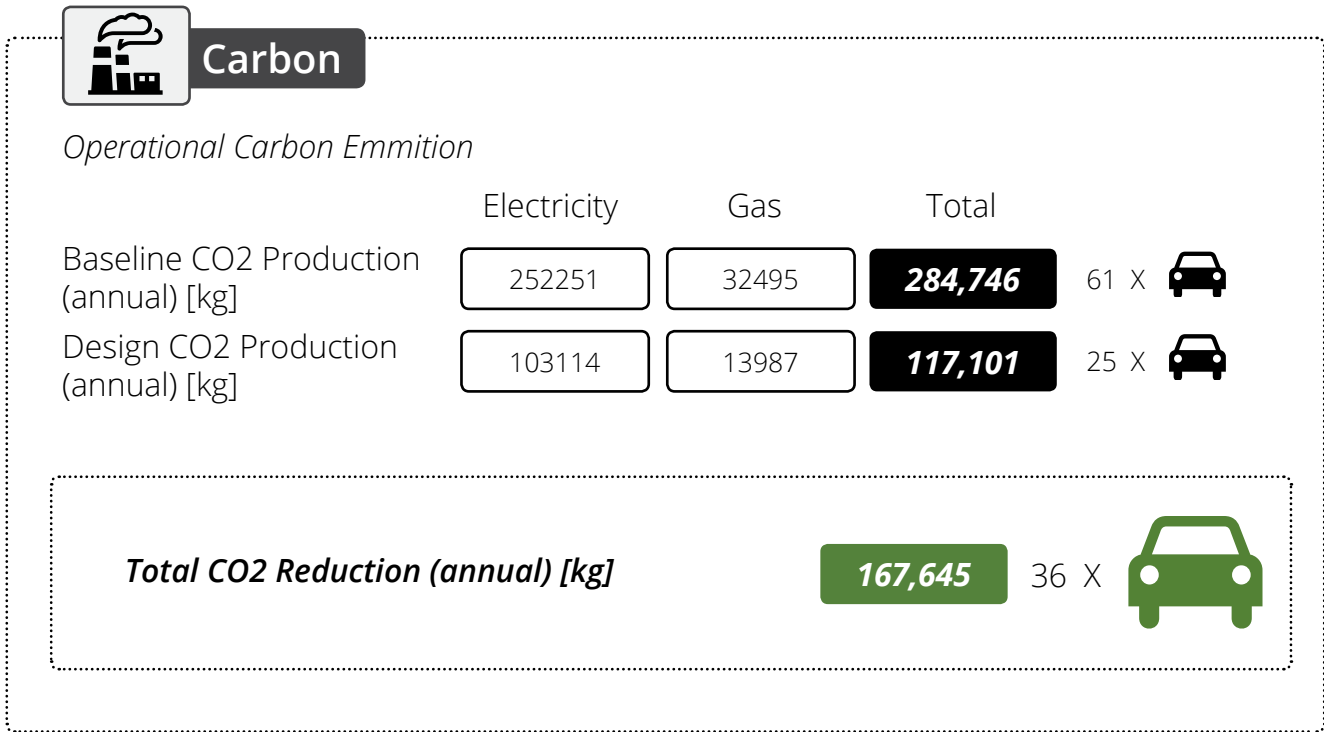
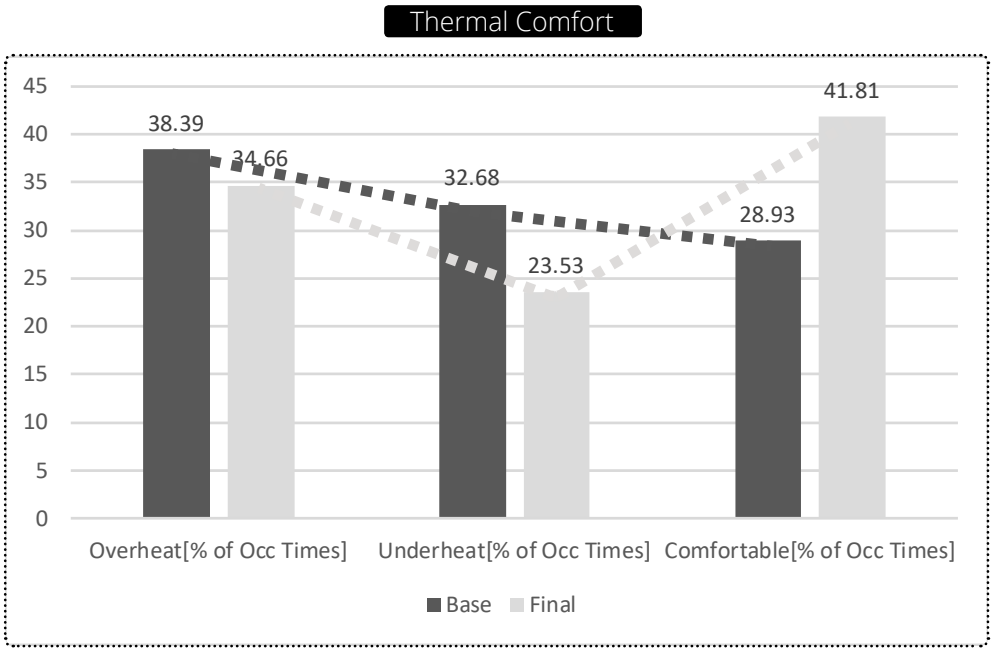
Indoor water use reduction %45.5

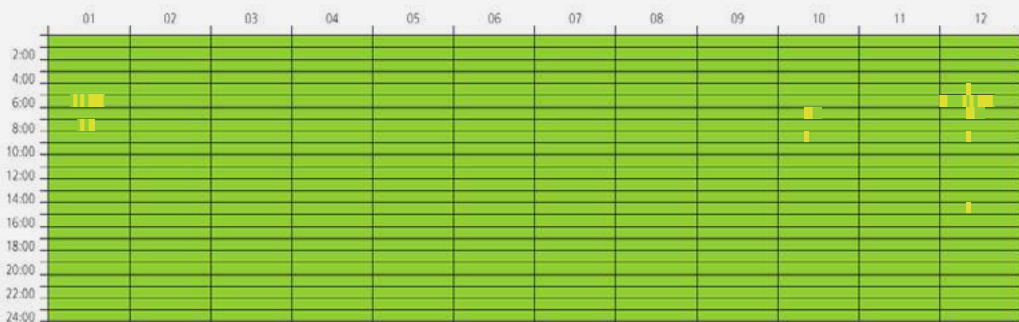
Outdoor water use reduction %100

Energy



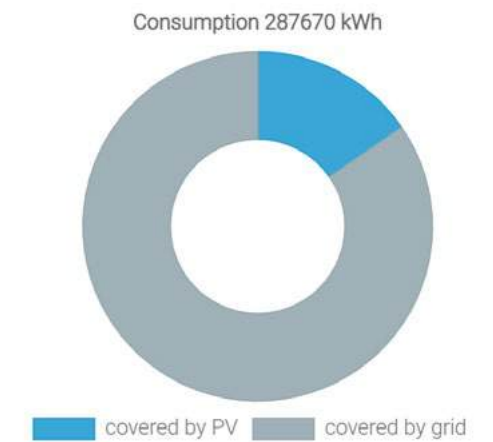
Thermal Comfort





Annual baseline water consumption (liters/year)	753,275.75
Annual design water consumption (liters/year)	411,863.00
Percent water use reduction (%)	%45.32

Consumption[kWh]	287670
covered by PV[kWh]	44362
covered by grid	243308
Solar fraction [%]	%15.40
Avoided CO2 emissions [kg/year]	23795



40	39	19	TOTALS	Possible Points:	110
Certified: 40 to 49 points, Silver: 50 to 59 points, Gold: 60 to 79 points, Platinum: 80 to 110					

Certification Level	Icon	Cost	Points	Requirements
Certified	Green leaf	No Cost	40 points	1. <i>Certified</i> Requirements 2. Light Pollution Reduction 3. Tenant Design and Construction Guidelines 4. Rainwater Management 5. Building Life-Cycle Impact Reduction 6. Cooling Tower Water Use 7. Site Assessment 8. Site Development - Protect or Restore Habitat
Silver	Blue leaf	Low-Cost	54 points	1. <i>Silver</i> Requirements 2. Enhanced Commissioning 3. Water Metering 4. Building Product Disclosure and Optimization - Environmental Product Declarations»
Gold	Yellow leaf	Mid-Cost	64 points	1. <i>Gold</i> Requirements 2. Bicycle Facilities 3. Green Vehicles 4. Indoor Water Use reduction 5. Green Power and Carbon Offsets
Platinum	Grey leaf	Not Eligible	maximum: 69 points	

only the Prerequisites must be provided.

 LEED Checklist

5	1	9	Location and Transportation		20
			Credit	LEED for Neighborhood Development Location	20
1		1	Credit	Sensitive Land Protection	2
		3	Credit	High Priority Site	3
3		3	Credit	Surrounding Density and Diverse Uses	6
	1		Credit	Access to Quality Transit	6
		1	Credit	Bicycle Facilities	1
1		0	Credit	Reduced Parking Footprint	1
		1	Credit	Green Vehicles	1

7	3	1	Water Efficiency		11
Y			Prereq	Outdoor Water Use Reduction	Required
Y			Prereq	Indoor Water Use Reduction	Required
Y			Prereq	Building-Level Water Metering	Required
2			Credit	Outdoor Water Use Reduction	2
5		1	Credit	Indoor Water Use Reduction	6
	2		Credit	Cooling Tower Water Use	2
	1		Credit	Water Metering	1

3	8	0	Sustainable Sites		11
N			Prereq	Construction Activity Pollution Prevention	Required
	1		Credit	Site Assessment	1
	2		Credit	Site Development - Protect or Restore Habitat	2
1			Credit	Open Space	1
	3		Credit	Rainwater Management	3
2			Credit	Heat Island Reduction	2
	1		Credit	Light Pollution Reduction	1
	1		Credit	Tenant Design and Construction Guidelines	1

0	6	0	Innovation		6
	5		Credit	Innovation	5
	1		Credit	LEED Accredited Professional	1

19	11	3	Energy and Atmosphere		33
Y			Prereq	Fundamental Commissioning and Verification	Required
Y			Prereq	Minimum Energy Performance	Required
Y			Prereq	Building-Level Energy Metering	Required
Y			Prereq	Fundamental Refrigerant Management	Required
	6		Credit	Enhanced Commissioning	6
15	3		Credit	Optimize Energy Performance	18
	1		Credit	Advanced Energy Metering	1
	1	1	Credit	Demand Response	2
3			Credit	Renewable Energy Production	3
1			Credit	Enhanced Refrigerant Management	1
		2	Credit	Green Power and Carbon Offsets	2

0	12	2	Materials and Resources		14
Y			Prereq	Storage and Collection of Recyclables	Required
Y			Prereq	Construction and Demolition Waste Management Planning	Required
	4	2	Credit	Building Life-Cycle Impact Reduction	6
	2		Credit	Building Product Disclosure and Optimization - Environmental Product Declarations	2
	2		Credit	Building Product Disclosure and Optimization - Sourcing of Raw Materials	2
	2		Credit	Building Product Disclosure and Optimization - Material Ingredients	2
	2		Credit	Construction and Demolition Waste Management	2

6	4	0	Indoor Environmental Quality		10
Y			Prereq	Minimum Indoor Air Quality Performance	Required
Y			Prereq	Environmental Tobacco Smoke Control	Required
2			Credit	Enhanced Indoor Air Quality Strategies	2
	3		Credit	Low-Emitting Materials	3
	1		Credit	Construction Indoor Air Quality Management Plan	1
3			Credit	Daylight	3
1			Credit	Quality Views	1

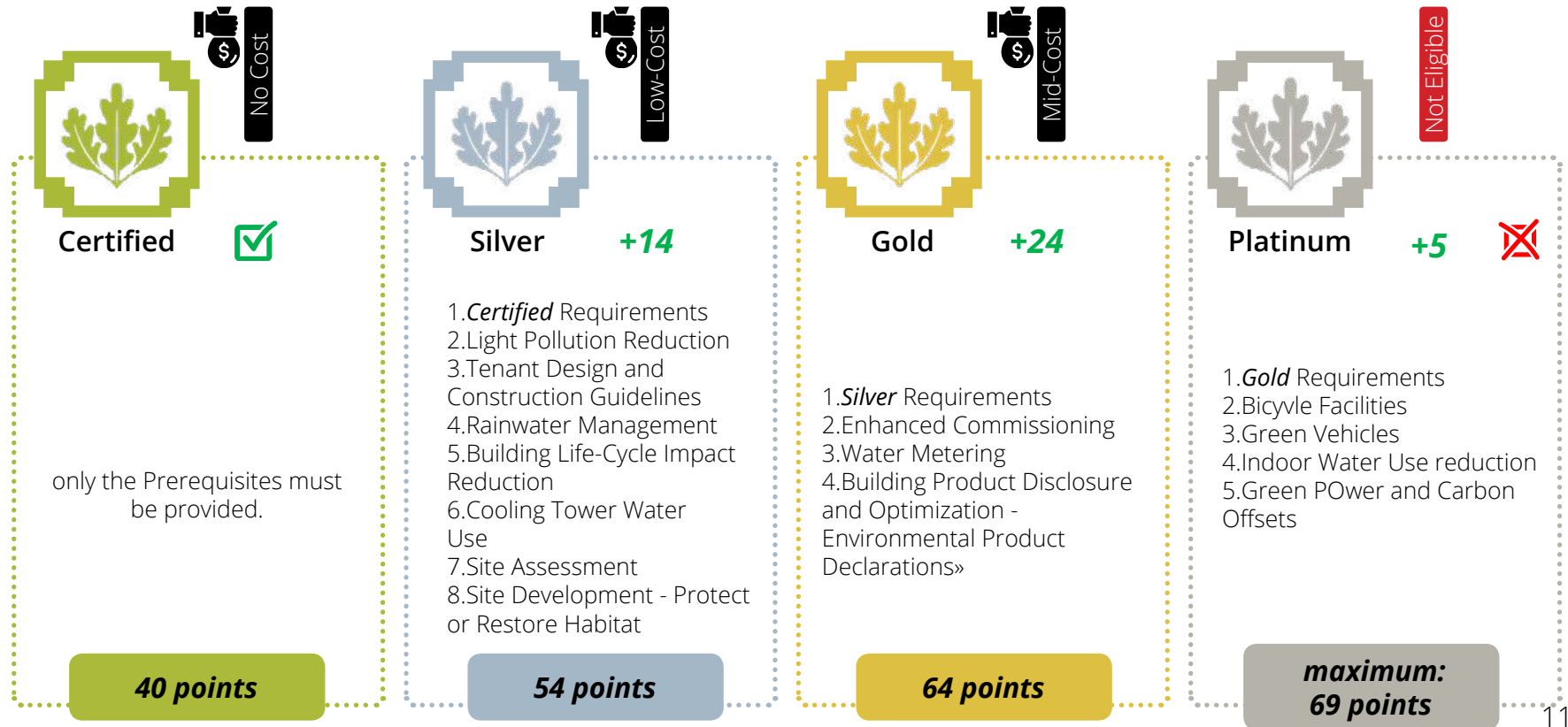
0	0	4	Regional Priority			4
		1	Credit	Regional Priority: Specific Credit	1	
		1	Credit	Regional Priority: Specific Credit	1	
		1	Credit	Regional Priority: Specific Credit	1	
		1	Credit	Regional Priority: Specific Credit	1	

40	39	19	TOTALS	Possible Points:	110
Certified: 40 to 49 points, Silver: 50 to 59 points, Gold: 60 to 79 points, Platinum: 80 to 110					

 Prerequisites

1. Construction Activity Pollution Prevention
2. Outdoor Water Use Reduction
3. Indoor Water Use Reduction
4. Building-Level Water Metering
5. Fundamental Commissioning and Verification
6. Minimum Energy Performance
7. Building-Level Energy Metering
8. Fundamental Refrigerant Management
9. Storage and Collection of Recyclables
10. Construction and Demolition Waste Management Planning
11. Management Planning
12. Minimum Indoor Air Quality Performance
13. Environmental Tobacco Smoke Control

 Improvement Proposals



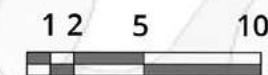


Basement Floor Plan



KEY

- 1 Entrance
- 2 Information
- 3 Conference Hall
- 4 W.C.
- 5 Kitchen
- 6 Food Stall
- 7 Shop
- 8 FoudCourt Sitting Area
- 9 Fire Control Room
- 10 Electrical/ Systems Control
- 11 Facility Room
- 12 Facility Access/Parking
- 13 Gallery
- 14 Gallery-Storage
- 15 Multi-Functional Space
- 16 TV Room
- 17 Lounge
- 18 Study Hall (Individual)
- 19 Study Hall (Group)
- 20 Game Room
- 21 Class Room
- 22 Prayers Room
- 23 Management
- 24 Administration
- 25 Conference Room
- 26 Student Association
- 27 Book Cafe



**Ground Floor Plan****KEY**

- 1** Entrance
- 2** Information
- 3** Conference Hall
- 4** W.C.
- 5** Kitchen
- 6** Food Stall
- 7** Shop
- 8** FoudCourt Sitting Area
- 9** Fire Control Room
- 10** Electrical/ Systems Control
- 11** Facility Room
- 12** Facility Access/Parking
- 13** Gallery
- 14** Gallery-Storage
- 15** Multi-Functional Space
- 16** TV Room
- 17** Lounge
- 18** Study Hall (Individual)
- 19** Study Hall (Group)
- 20** Game Room
- 21** Class Room
- 22** Prayers Room
- 23** Management
- 24** Administration
- 25** Conference Room
- 26** Student Association
- 27** Book Cafe

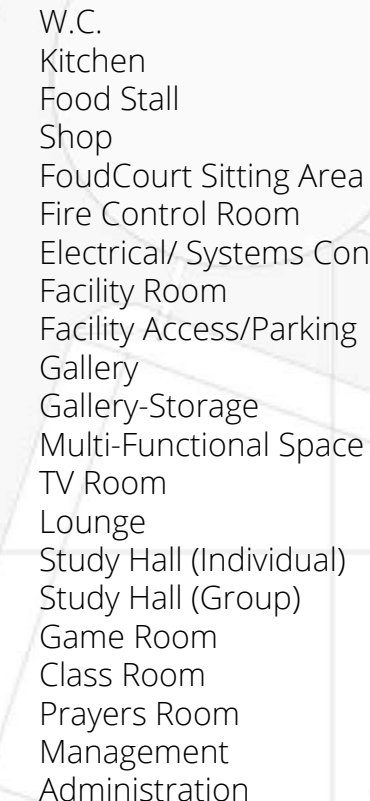




First Floor Plan



KEY

- 
- 1** Entrance
 - 2** Information
 - 3** Conference Hall
 - 4** W.C.
 - 5** Kitchen
 - 6** Food Stall
 - 7** Shop
 - 8** FoudCourt Sitting Area
 - 9** Fire Control Room
 - 10** Electrical/ Systems Control
 - 11** Facility Room
 - 12** Facility Access/Parking
 - 13** Gallery
 - 14** Gallery-Storage
 - 15** Multi-Functional Space
 - 16** TV Room
 - 17** Lounge
 - 18** Study Hall (Individual)
 - 19** Study Hall (Group)
 - 20** Game Room
 - 21** Class Room
 - 22** Prayers Room
 - 23** Management
 - 24** Administration
 - 25** Conference Room
 - 26** Student Association
 - 27** Book Cafe



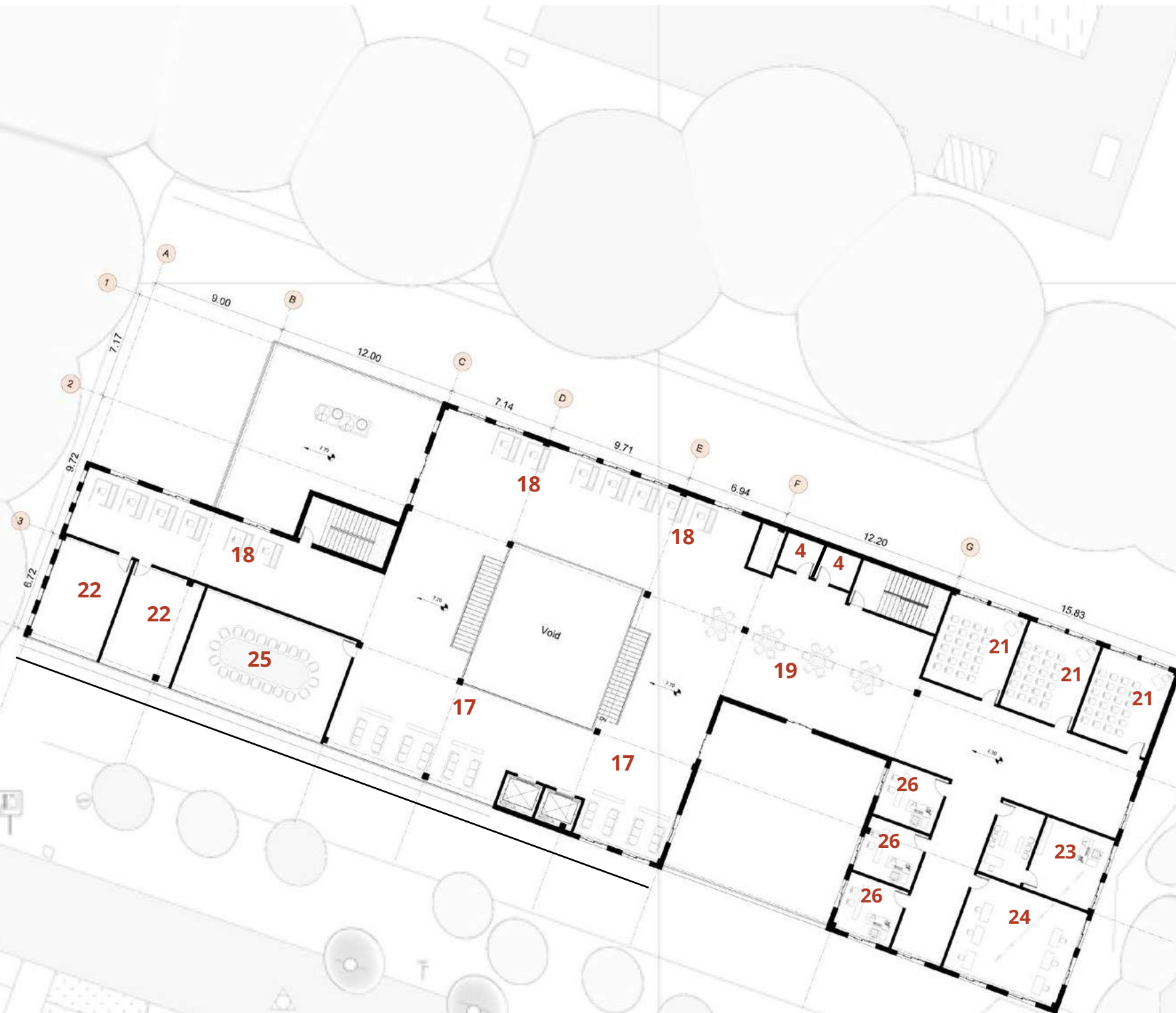


Second Floor Plan



KEY

- 1 Entrance
- 2 Information
- 3 Conference Hall
- 4 W.C.
- 5 Kitchen
- 6 Food Stall
- 7 Shop
- 8 FoudCourt Sitting Area
- 9 Fire Control Room
- 10 Electrical/ Systems Control
- 11 Facility Room
- 12 Facility Access/Parking
- 13 Gallery
- 14 Gallery-Storage
- 15 Multi-Functional Space
- 16 TV Room
- 17 Lounge
- 18 Study Hall (Individual)
- 19 Study Hall (Group)
- 20 Game Room
- 21 Class Room
- 22 Prayers Room
- 23 Management
- 24 Administration
- 25 Conference Room
- 26 Student Association
- 27 Book Cafe





Elevations

East Elevation



South Elevation





Elevations

West Elevation



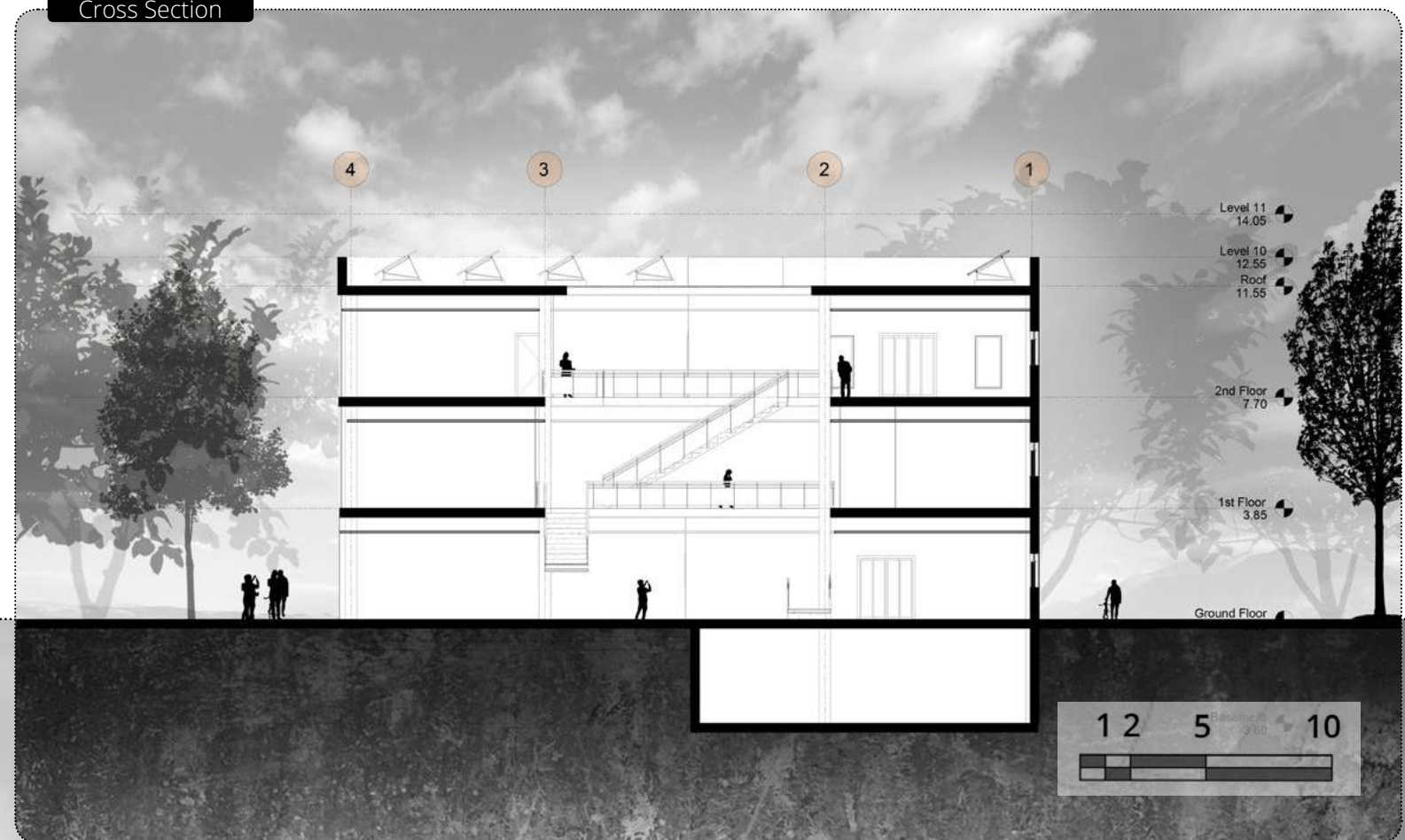
North Elevation



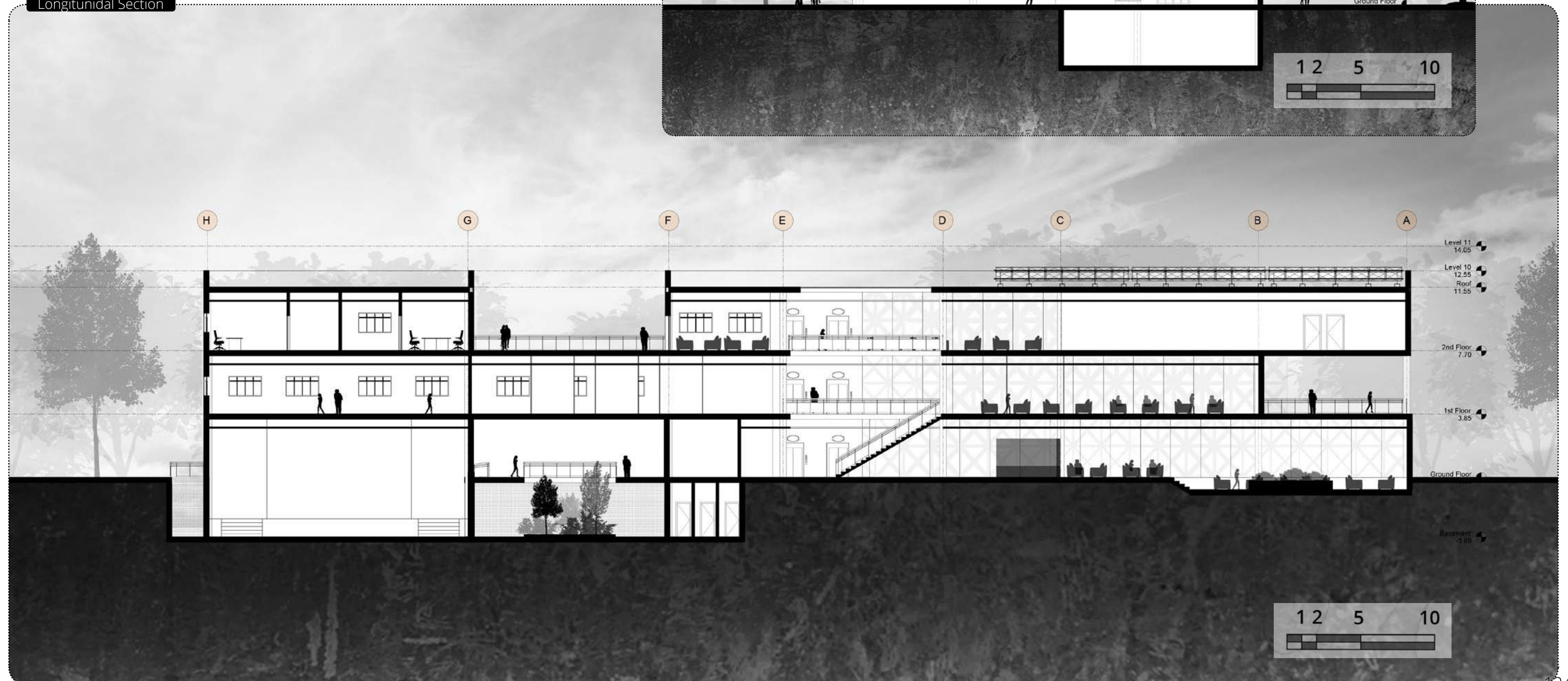


Sections

Cross Section



Longitudinal Section



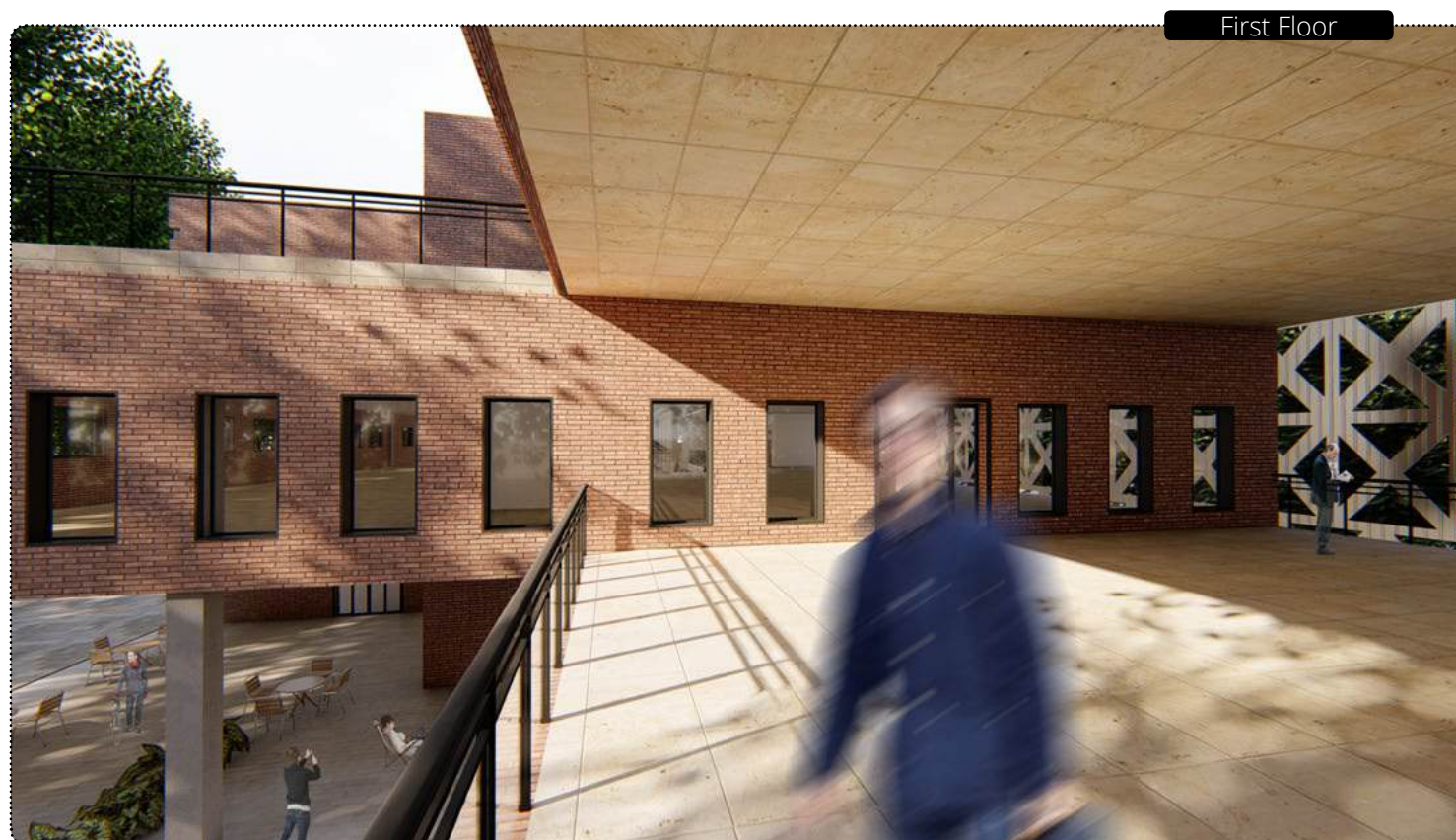


Sections



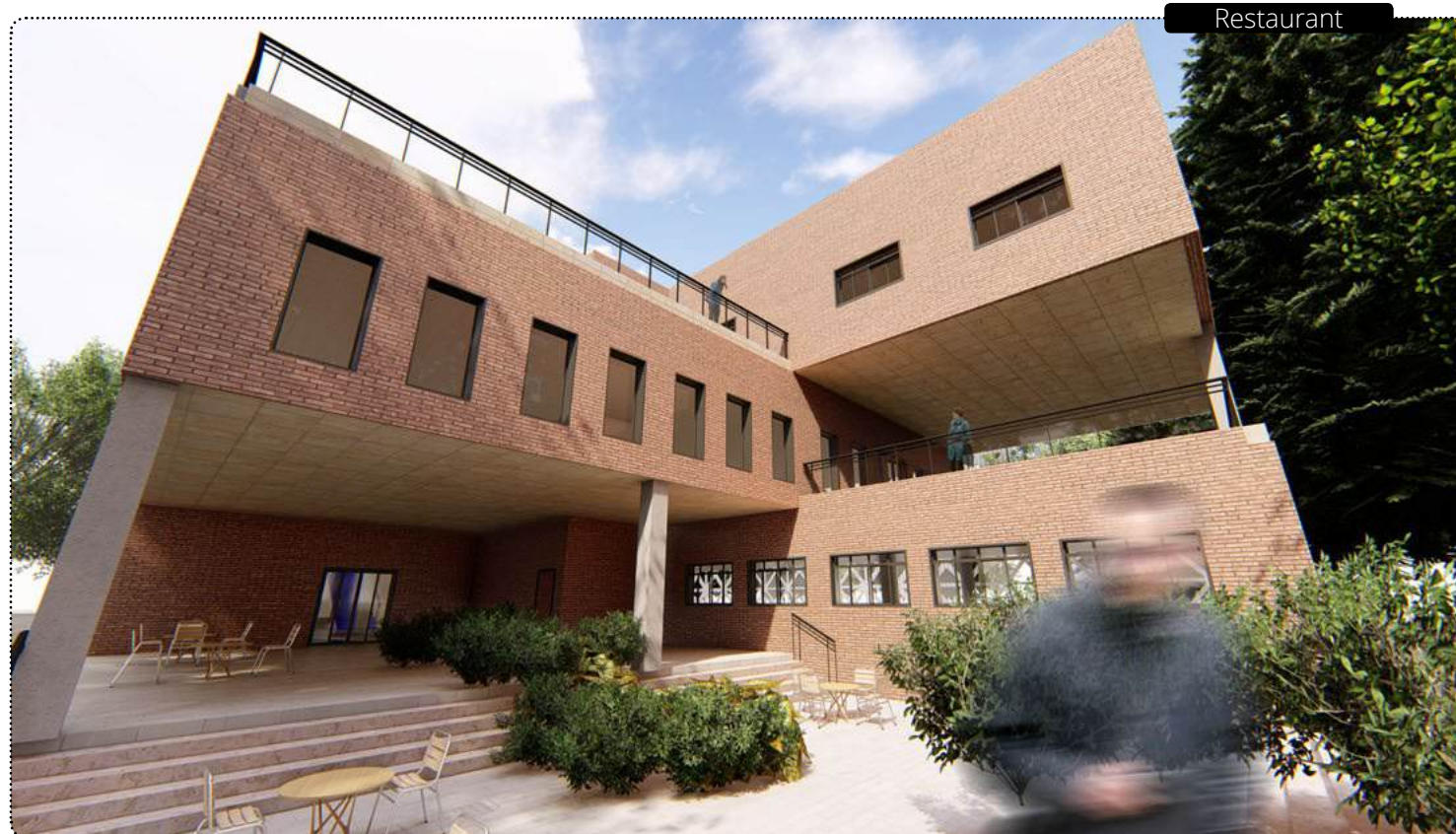
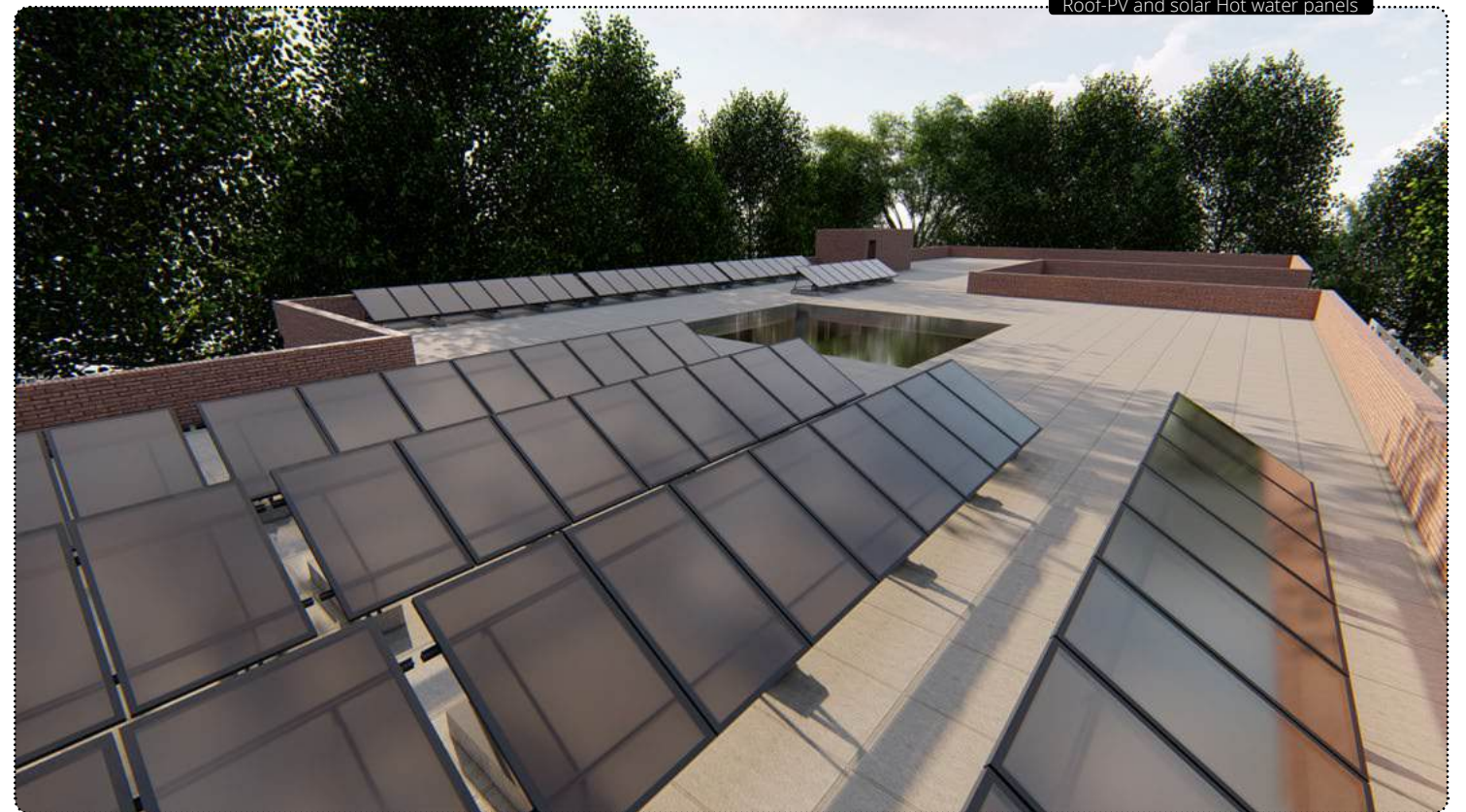


Sections





Sections



List Tools and Software used

Ladybug v. 0.0.67	
RETscreen	
EnergyStar Portfolio Manager	
Archsim	
DIVA v. 4.0	
Honeybee v. 0.0.64	
Design Builder v. 6.0	
PV-Sol Online Tool	
LEED Water Reduction Calculator	
ANSYS Fluent	
HTflux	
Insul	
Sound analyzer app	
CBE Thermal Comfort Tool	
ALFA v. 0.5.5.0	
Climate Consultant V 6.0	

List of Guides, Rating systems, Standards and codes:

- Iranian National Building Codes Annex 3,4,13,14,19
- WELL V.2 Certificate Guidelines
- LEED V.4 Guideline
- ASHRAE %30 and %50 design Guides
- RELi V.2 Guidelines
- ASHRAE 55
- Ashrae 62.1 (Ventilation for Acceptable Indoor Air Quality)
- Ashrae 189.1 (Standard for the design of High Performance Green Buildings)
- LEED v4 for D+C - WE Prerequisite Outdoor Water Use Reduction and WE Credit Outdoor Water Use Reduction
- LEED v4 for O+M - WE Credit Outdoor Water Use Reduction
- LEED for Homes Rating System 2008
- (The Irrigation Association, October 2001) Landscape Irrigation Scheduling and Water Management, IA 2005
- 2009 WaterSense Single-Family New Home Specification
- LEED v4 and v4.1 for D+C and v4 for O+M - WE Prerequisite/ Credit Indoor Water Use Reduction
- LEED v4 for ND - GIB Prerequisite/Credit Indoor Water Use Reduction and GIB Credit Wastewater Management
- ISO 717-1 :2013
- Acoustics — Rating of sound insulation in buildings and of building elements — Part 1: Airborne sound insulation
- ISO 717-2 :2013
- Acoustics — Rating of sound insulation in buildings and of building elements — Part 2: Impact sound insulation

Nomenclature

- DBT: DryBulb Temperature
- TC:Thermal Comfort
- SVF: Surface to Volume Factor
- OT: Operative Temperature
- ACR: Air Change Rate
- HSA: Horizontal Shading Angle
- VSA: Vertical Shading Angle
- mT: million Tomans
- kWh:
- kWh/m2:
- ETo: Evapotranspiration Rate
- DU(LQ) or EU*: Water Distribution Uniformity
- KL:Plant Type Associated Landscape Coefficient
- LWR: Landscape water requirement
- Drip (press comp): Pressure Compensating Drip Irrigation System
- lpf: Liters per flush

The End